

Display CMC's

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00415	155S00391	ALT_PARTS	ALL	CMC, 350SM, 7HGz, MUR

NAND Ultimate

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00340	1	HYNIX, 3DV4, ULTIMATE	U2600	CRITICAL	ULTIMATE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00285	335S00340	ALT_PARTS	U2600	TOSHIBA, BICS3, ULT
335S00286	335S00340	ALT_PARTS	U2600	SANDISK, BICS3, ULT
335S00288	335S00340	ALT_PARTS	U2600	SAMSUNG, 3DV4, ULT

Extreme

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00241	1	HYNIX, 3DV3, Extreme	U2600	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00342	335S00241	ALT_PARTS	U2600	HYNIX, 3DV4, SUPREME
335S00228	335S00241	ALT_PARTS	U2600	TOSHIBA, 3DV4, SUPREME
335S00340	335S00241	ALT_PARTS	U2600	SANDISK, BISC3, SUPREME
335S00276	335S00241	ALT_PARTS	U2600	SAMSUNG, 3DV4, SUPREME
335S00338	335S00241	ALT_PARTS	U2600	SAMSUNG, 3DV4, SUPREME

Max

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00043	1	3DV4, MAX	U2600	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00339	335S00343	ALT_PARTS	U2600	SAMSUNG, 3DV4, MAX

Extreme

TODO: Need to get ALT APN's

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00149		ALT_PARTS	ALL	0402-3T, 10.5uF@1V, Kyocera
138S00150	138S00149	ALT_PARTS	ALL	0402-3T, 10.5uF@1V, SENC0
138S00151	138S00149	ALT_PARTS	ALL	0402-3T, 10.5uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	ALT_PARTS	ALL	0402, 16uF@1V, Kyocera
138S00163	138S00144	ALT_PARTS	ALL	0402, 16uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	ALT_PARTS	ALL	0201, 3uF@1V, Kyocera
138S00164	138S00139	ALT_PARTS	ALL	0201, 3uF@1V, TY

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00221	138S00146	ALT_PARTS	ALL	0402, 5.1uF@3V, Kyocera

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	ALT_PARTS	ALL	0201, 1.1uF@3V, Kyocera
138S00141		ALT_PARTS	ALL	0201, 1.1uF@3V, SENC0
138S00166	138S00141	ALT_PARTS	ALL	0201, 1.1uF@3V, Taiyo

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S0764	118S0717	ALT_PARTS	ALL	RES, 3.92K, 0.1%, 0201
138S0648	138S0652	ALT_PARTS	ALL	CAP, X5R, 4.7UF, 6.3V, 6.69MM, 0402, TAIYO
138S0739	138S0706	ALT_PARTS	ALL	CAP, C0G, X5R, 0.22UF, 20V, 0.2V, 20%
138S00049	138S0831	ALT_PARTS	ALL	CAP, C0G, X5R, 2.2UF, 20V, 0.2V, 20%
132S0025316		ALT_PARTS	ALL	CAP, C0G, X5R, 0.1UF, 20V, 0.2V, 20%

PMIC Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S000721		ALT_PARTS		IND, MLD, 1UH, 3.6A, 62MO, H=0.8MM, 2016
152S000821		ALT_PARTS		IND, MLD, 1UH, 20%, 2.1A, 62MO, H=0.80, 2012
152S00066	152S00021	ALT_PARTS		IND, MLD, 1UH, 20%, 2.1A, 62MO, H=0.65, 2012
152S00078	152S00031	ALT_PARTS	ALL	IND, MLD, 0.22UH, 20%, 5.6A, 69MO, H=0.65, 1608
152S00018	152S00031	ALT_PARTS	ALL	IND, MLD, 0.22UH, 20%, 5.6A, 69MO, H=0.65, 1608
152S00716	152S00075	ALT_PARTS	ALL	IND, MLD, 1.0UH, 20%, 2.5A, 78MO, H=0.8, 2012
152S00024	152S00033	ALT_PARTS	L2740	IND, MLD, 1UH, 20%, 2A, 69MO, H=0.65, 2012
152S000833		ALT_PARTS		IND, MLD, 1UH, 20%, 2A, 69MO, H=0.65, 2012
152S00025	152S00023	ALT_PARTS	ALL	IND, MLD, 1UH, 20%, 3A, 60MO, H=0.65, 2016
998-13888	152S00098	ALT_PARTS	ALL	IND, CPLE, 0.1UH, 0.1A, 50MO, +/-0.20%, 2012
998-13807	152S00097	ALT_PARTS	ALL	IND, CPLE, 0.1UH, 0.1A, 27MO, +/-0.20%, 2010

CRITICAL PART#	COMMENT
152S000721	IND, MLD, 1UH, 3.6A, 62MO, H=0.8MM, 2016
152S00026	IND, MLD, 1UH, 20%, 2.1A, 52MO, H=0.80, 2012
152S00066	IND, MLD, 1UH, 20%, 2.1A, 62MO, H=0.65, 2012
152S00078	IND, MLD, 0.22UH, 20%, 5.6A, 59MO, H=0.65, 1608
152S00018	IND, MLD, 0.22UH, 20%, 5.6A, 49MO, H=0.65, 1608
152S00716	IND, MLD, 1.0UH, 20%, 2.5A, 78MO, H=0.8, 2012
152S00024	IND, MLD, 1UH, 20%, 2A, 69MO, H=0.65, 2012
152S00019	IND, MLD, 1UH, 20%, 2A, 69MO, H=0.65, 2012
152S00025	IND, MLD, 1UH, 20%, 3A, 60MO, H=0.65, 2016

Yangtze Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00047	152S000918	ALT_PARTS	ALL	IND, MLD, 0.47UH, TY
152S00072	152S000918	ALT_PARTS	ALL	IND, MLD, 0.47UH, TDK

CRITICAL PART#	COMMENT
152S00047	IND, MLD, 0.22UH, 20%, 5.6A, 49MOHM, H=-.65, 1608
152S00072	IND, MLD, 0.22UH, 20%, 5.6A, 49MOHM, H=-.65, 1608

XTAL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S000118		ALT_PARTS		XTAL, 24M, 1512
197S00118		ALT_PARTS		XTAL, 24M, 1512

Old EEPROM

PART#	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00234	1 EEPROM_CAS24S128	U1402	CRITICAL	?

NEON Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00721	152S00076	ALT_PARTS	L4100, L4120	TY, 1MO

ANSEL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00716	152S00075	ALT_PARTS	L3700	TY, 1MO

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EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (NLR_TOP_IMOLA,839-4987,ULTIMATE)	EEEE_JCN3	CRITICAL	ULTIMATE
	1	EEEE FOR (NLR_TOP_IMOLA,839-84965,EXTREME)	EEEE_JCN1	CRITICAL	EXTREME
825-7691	1	EEEE FOR (NLR_TOP_IMOLA,839-84966,SUPREME)	EEEE_JCN2	CRITICAL	MAX

Kobol OMIT

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00367	1	KOBOL	U3600	CRITICAL	COMMON

Cyprus OMIT

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S00463	1	CYPRUS 4GB	U1000	CRITICAL	SOC

Kobol Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
338S00304	338S00367	ALT_PARTS	U3600	IC, GRAPHITE, BM1283AA, LGA15
998-12443	338S00367	ALT_PARTS	U3600	IC, GRAPHITE 325, BM1283AA, LGA15

Cyprus ALTs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00464	339S00463	ALT_PARTS	U1000	CYPRUS 4GB
339S00465	339S00463	ALT_PARTS	U1000	CYPRUS 4GB

PMU XTAL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S00102	197S00098	ALT_PARTS	Y3000	PMU XTAL, KDS
197S00103	197S00098	ALT_PARTS	Y3000	PMU XTAL, TXC
197S00173	197S00098	ALT_PARTS	Y3000	PMU XTAL, MDK

Combo Stiffener

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
604-19651	1	Combo Stiffener	ST0401	CRITICAL	ALL

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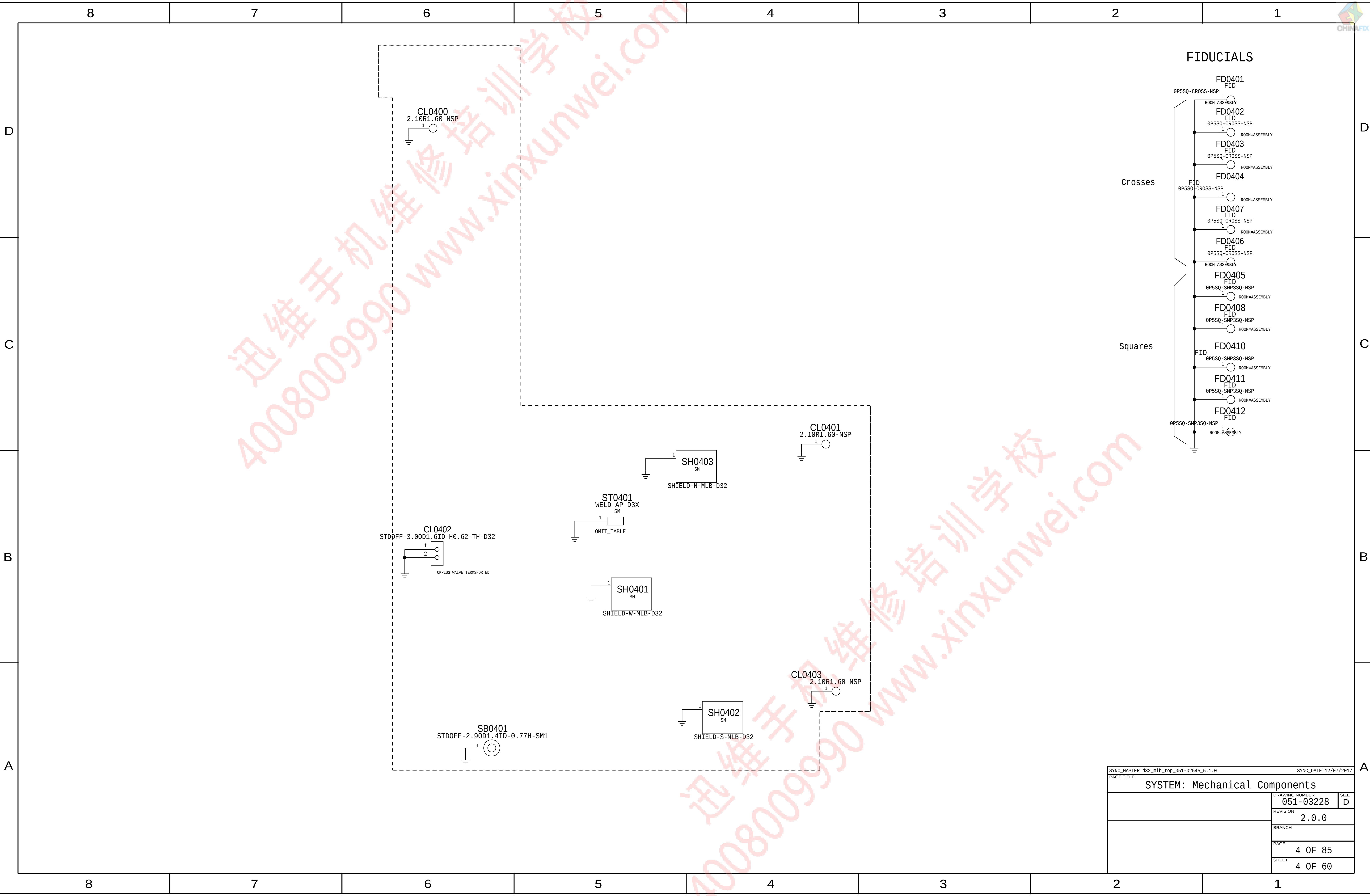
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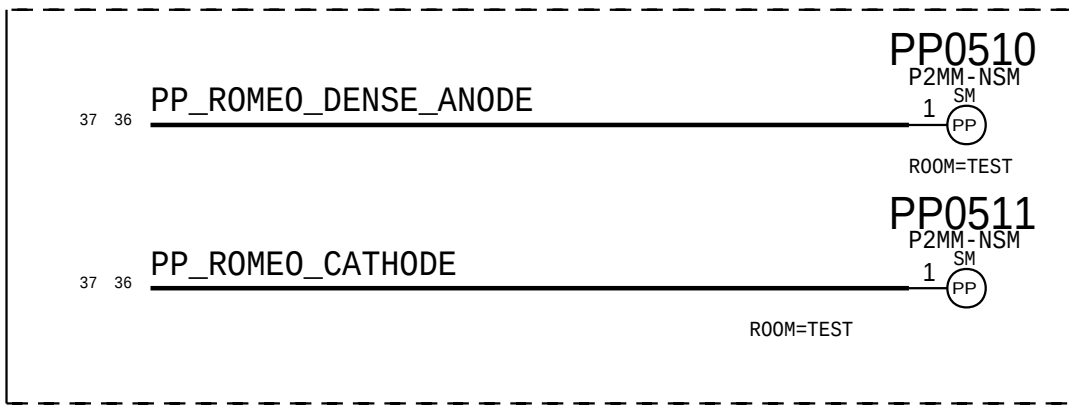
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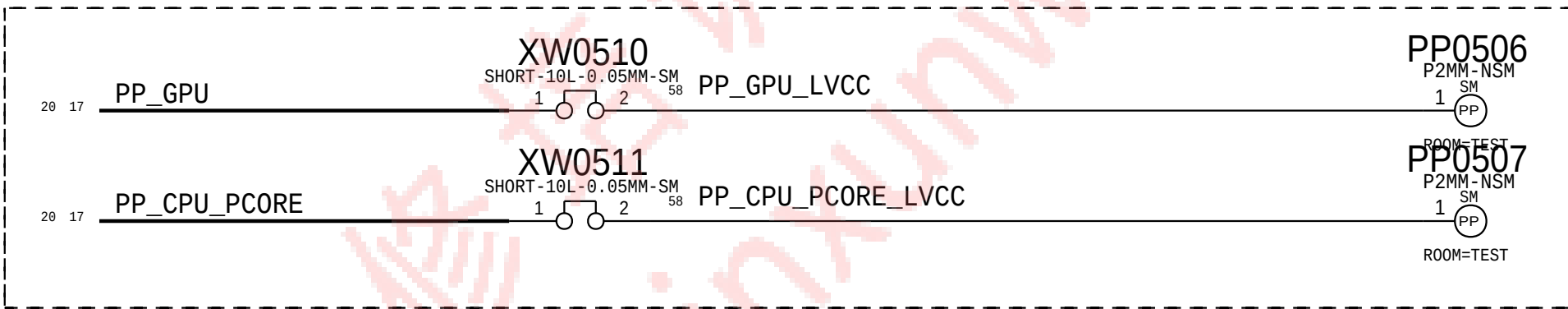


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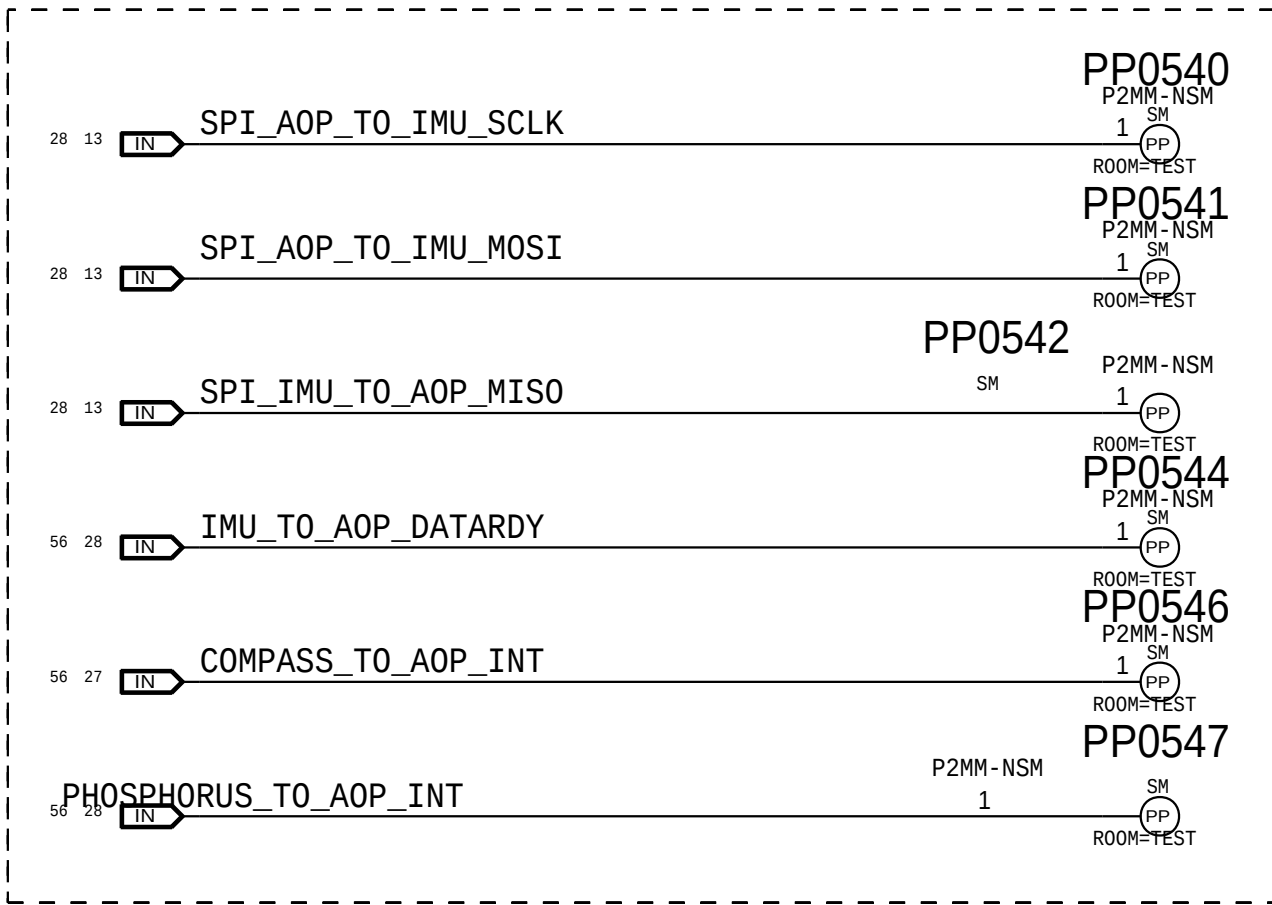
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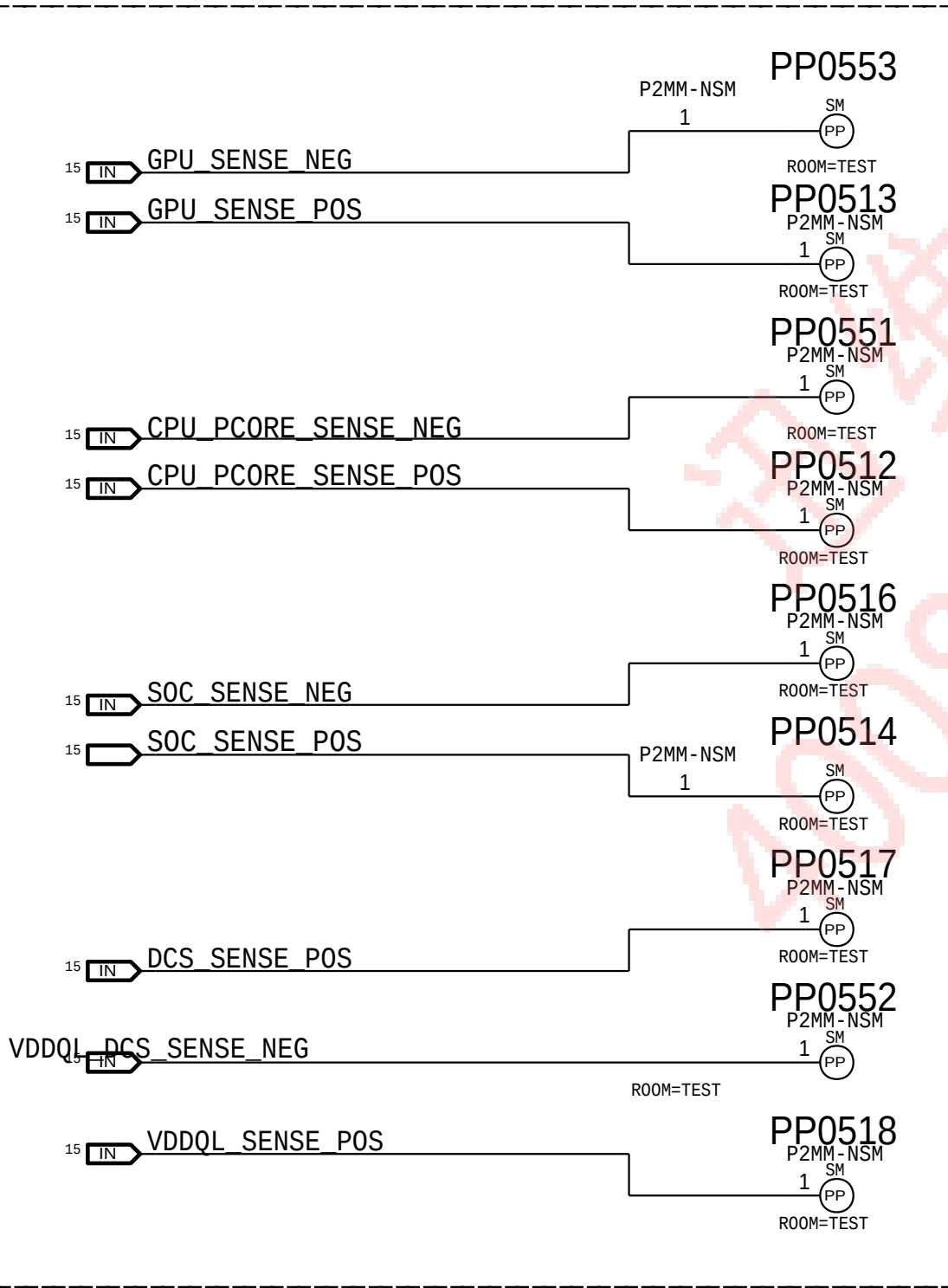
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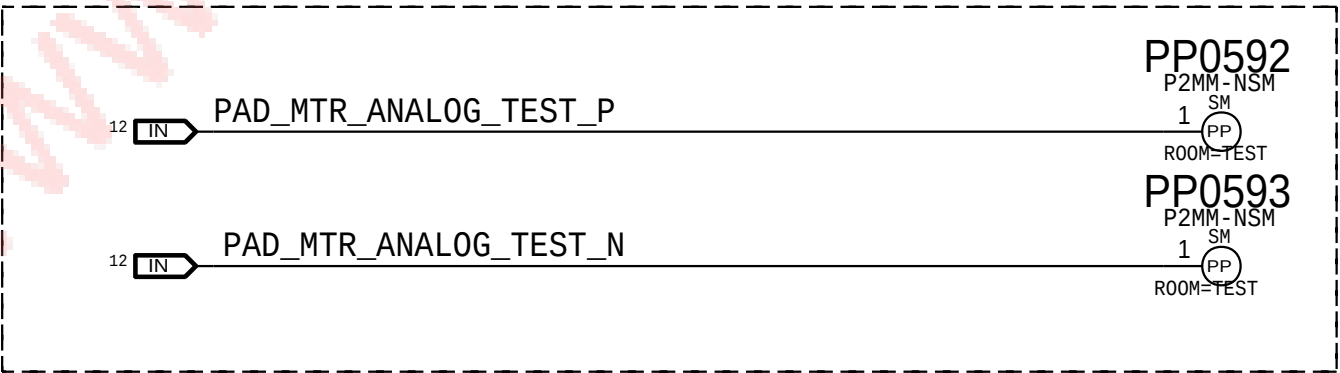
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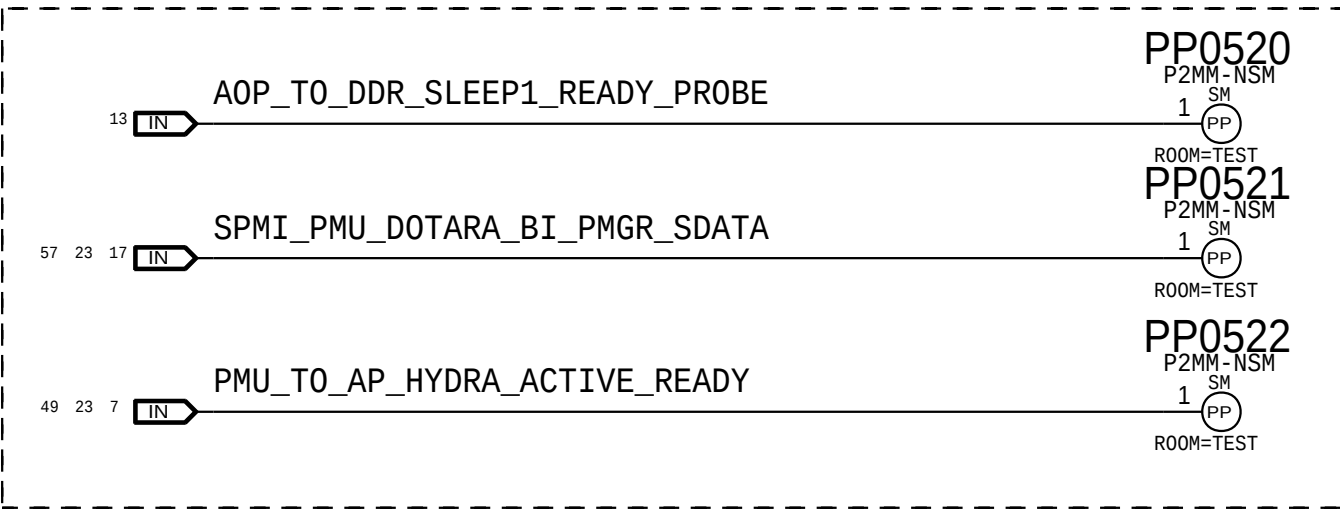
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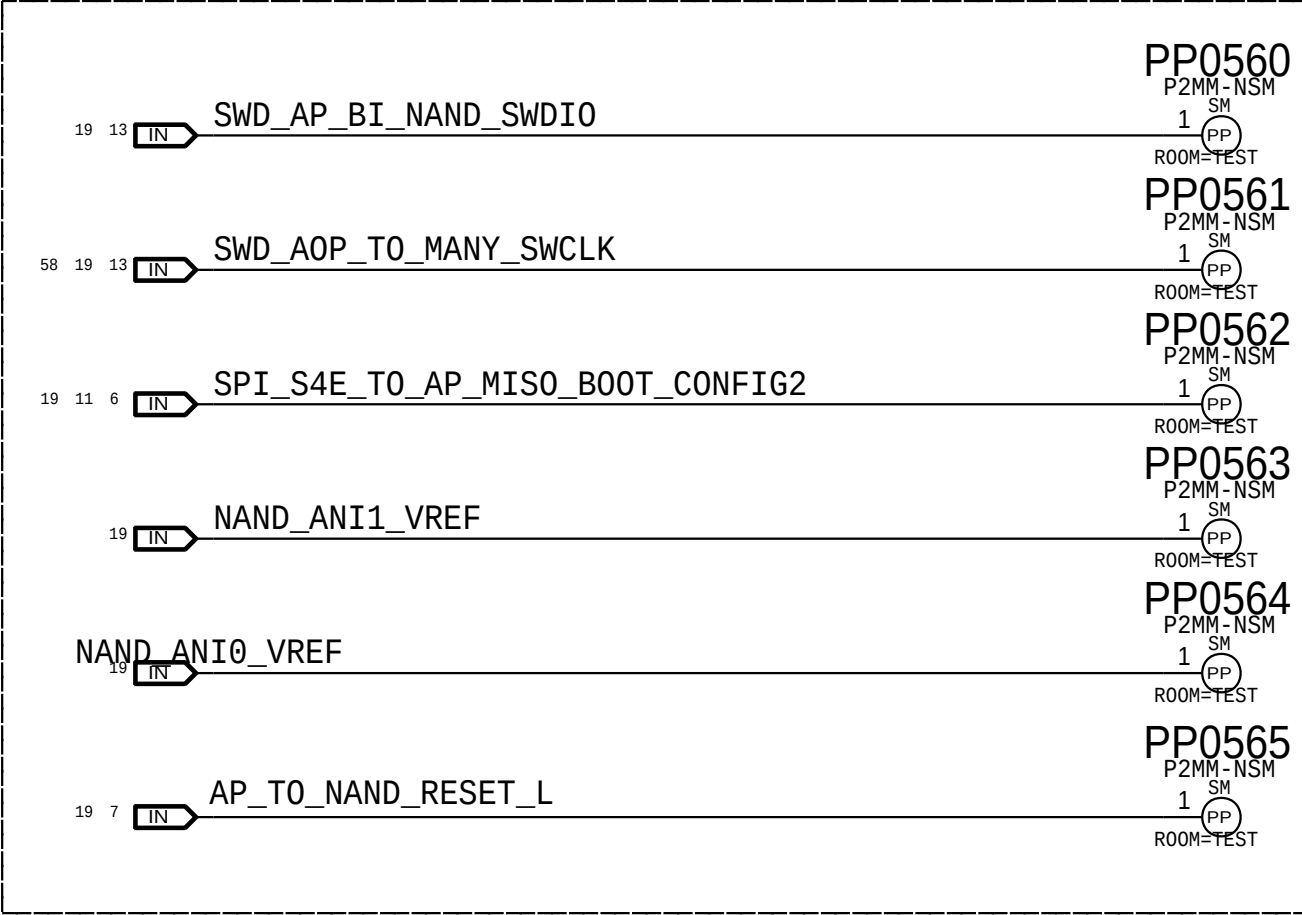
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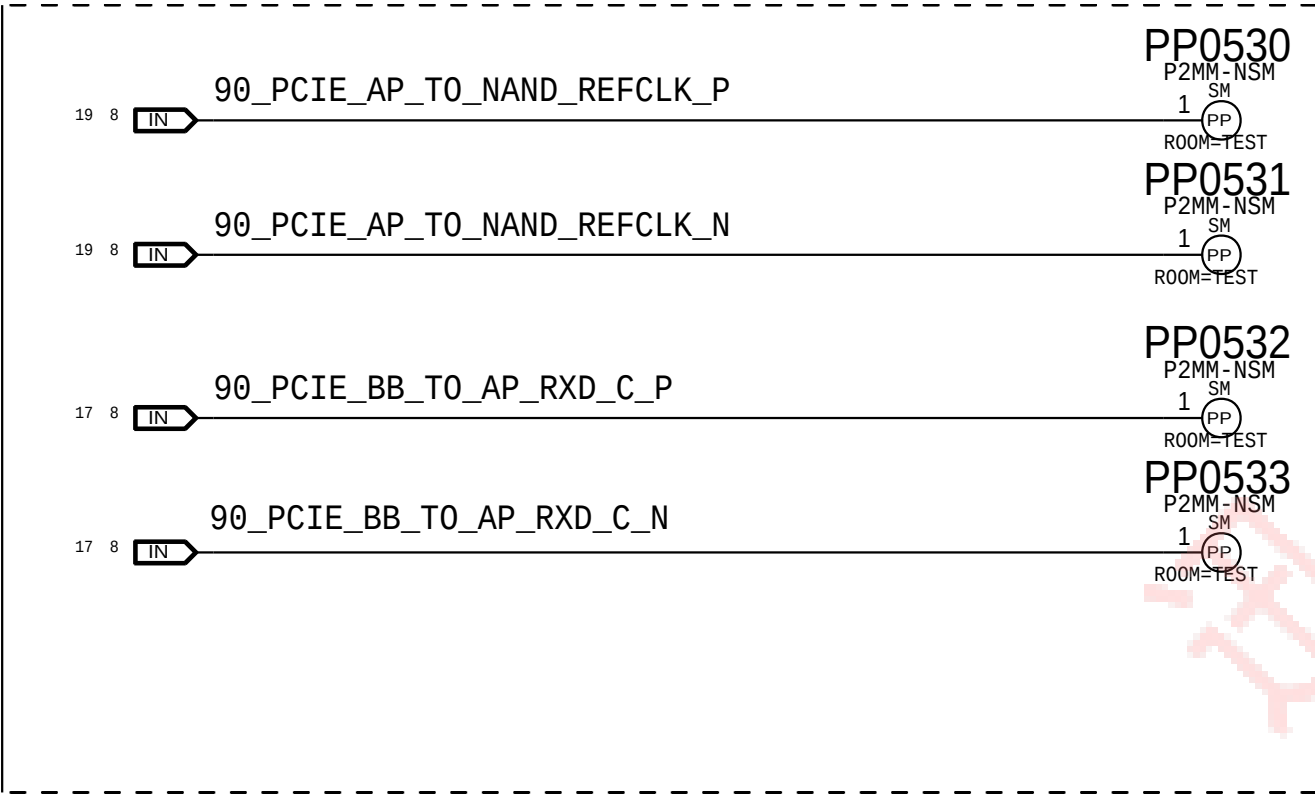
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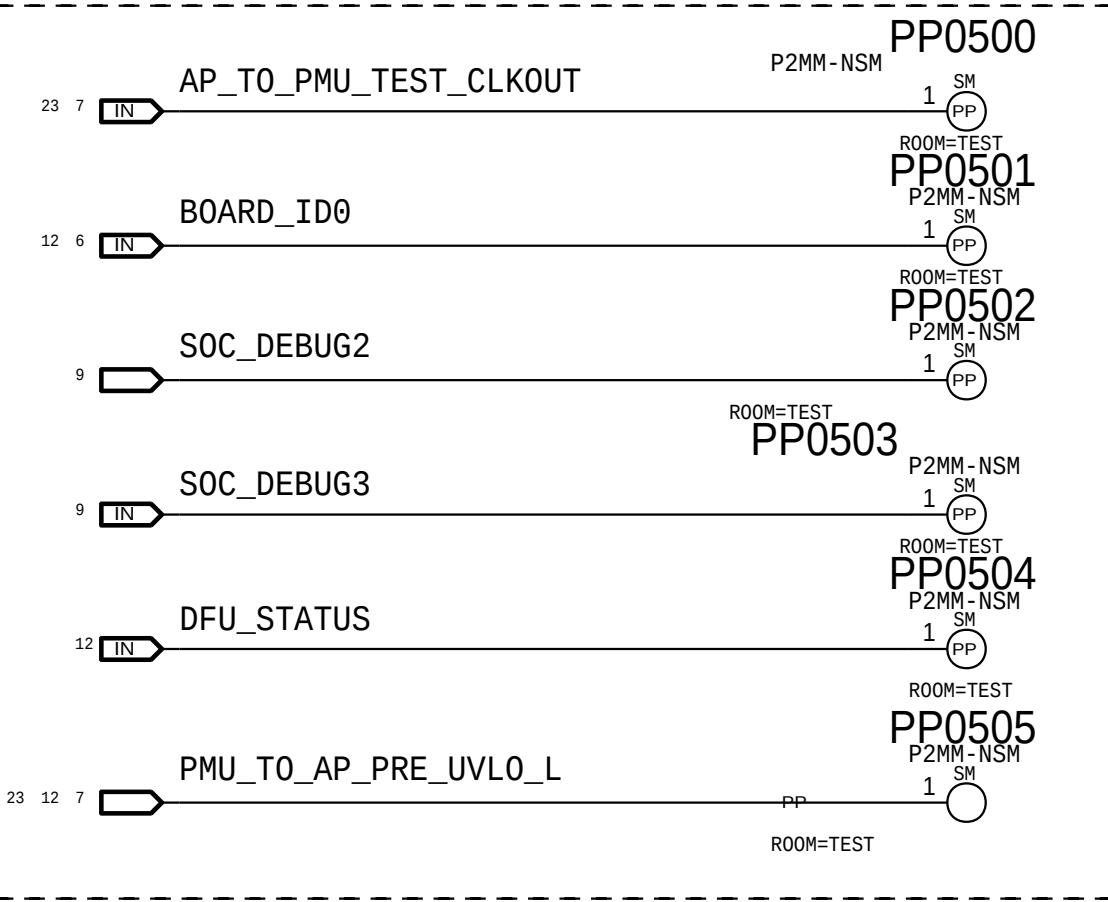
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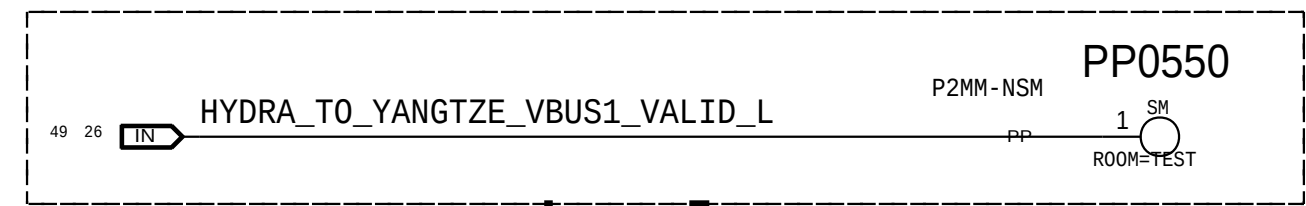
PCIE Refclk



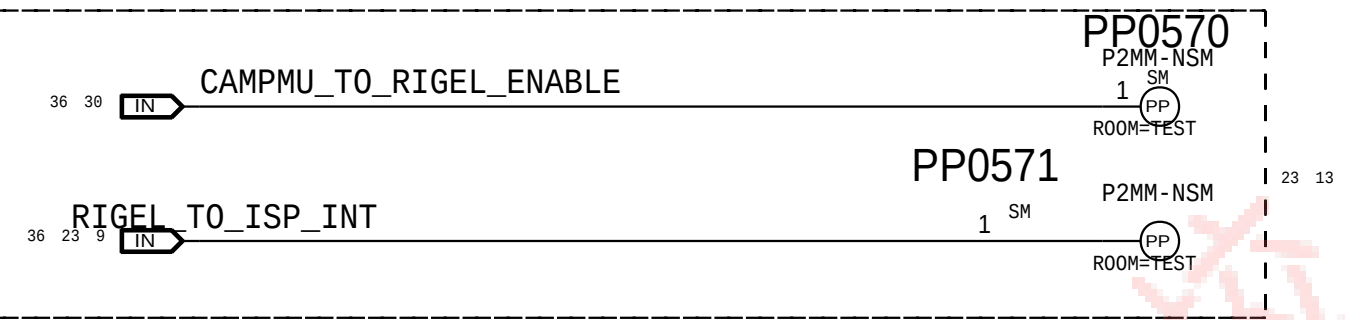
SOC Debug



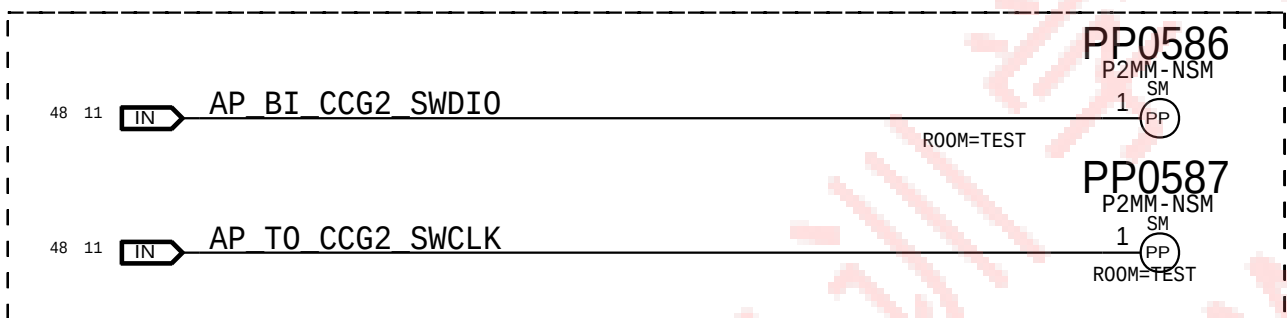
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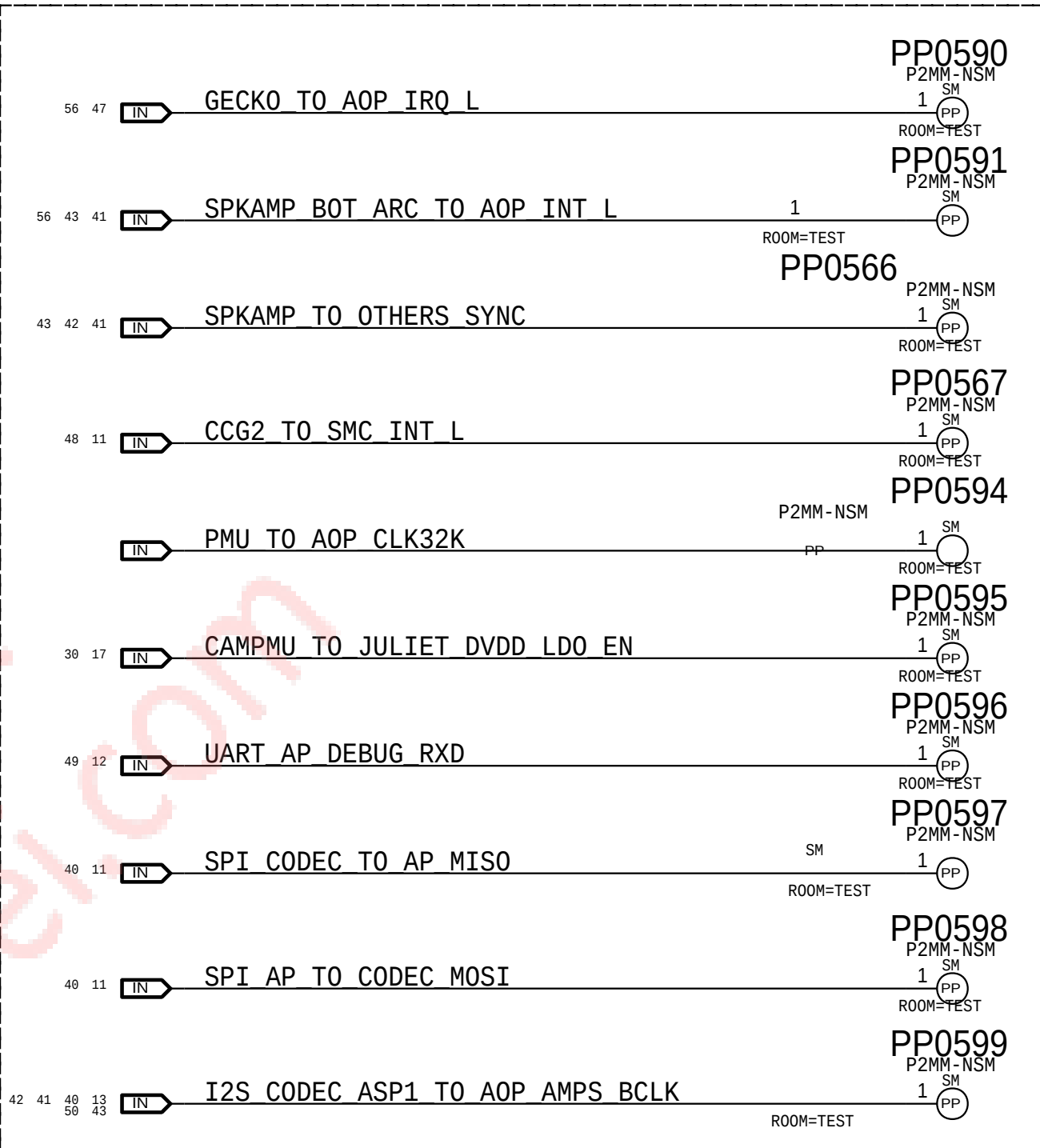
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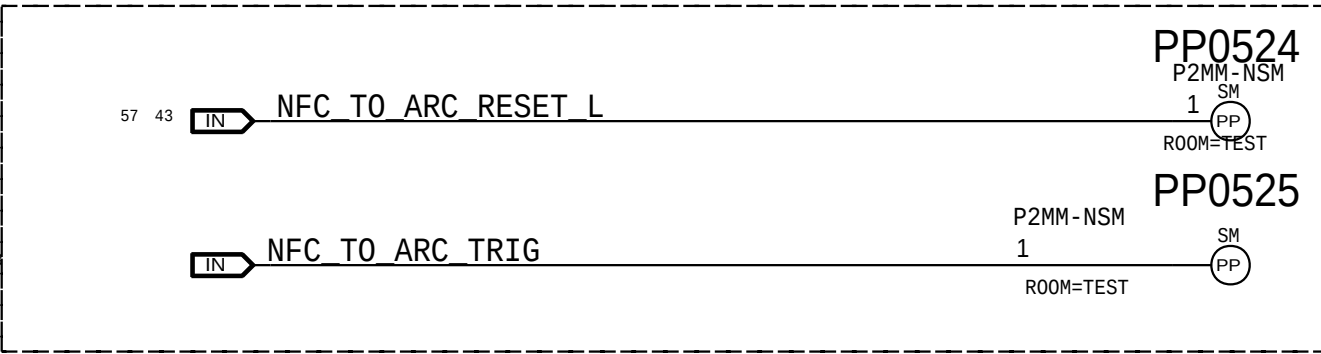
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VALIDATION PP's



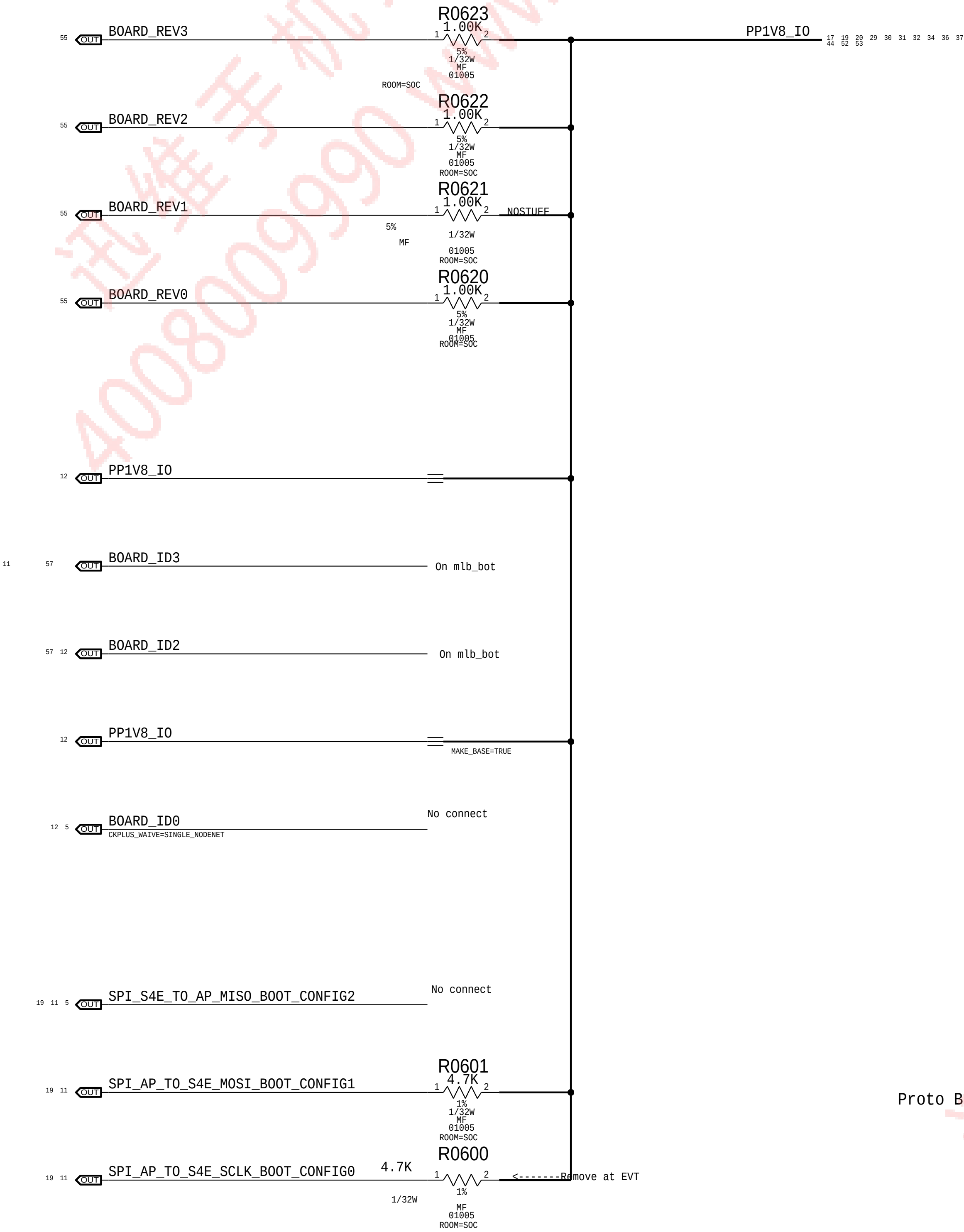
WALLET MODE



SYSTEM: Testpoints (Top)		
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TOP BOARD ONLY CONFIGURATION IS D33 MLB MAV
BOTTOM BOARD SELECTS ICE/MAV and D32/D33

BOOTSTRAPPING:BOARD REV
BOARD ID
BOOT CONFIG



SELECTED -->

Board Rev[3:0]				
Float = Low PU = High				
	3	2	1	0
	Build Major		Build Minor	
Proto 1	1	1	1	1
(Spare)	1	1	1	0
Proto 2	1	1	0	1
(Spare)	1	1	0	0
EVT	1	0	1	1
(Spare)	1	0	1	0
Carrier	0	1	1	1
(Spare)	0	1	1	0
DVT	0	0	1	1
(Spare)	0	0	1	0
(Spare)	0	0	0	1
PVT	0	0	0	0

DEFAULT -->

Board ID[4:0]					
Float = Low PU = High					
	4	3	2	1	0
	Denali = 0, Imola = 1	Mav = 0, Ice = 1	00=Open 01=D33 10=N84 11=D32		MLB = 0, Dev = 1
D33 MLB Mav	0	0	0	1	0
D33p MLB Mav	1	0	0	1	0
D33 Dev Mav	0	0	0	1	1
D33p Dev Mav	1	0	0	1	1
D33 MLB Ice	0	1	0	1	0
D33p MLB Ice	1	1	0	1	0
D33 Dev Ice	0	1	0	1	1
D33p Dev Ice	1	1	0	1	1
D32 MLB Mav	0	0	1	1	0
D32 Dev Mav	0	0	1	1	1
D32 MLB Ice	0	1	1	1	0
D32 Dev Ice	0	1	1	1	1

POR -->
Proto Builds -->

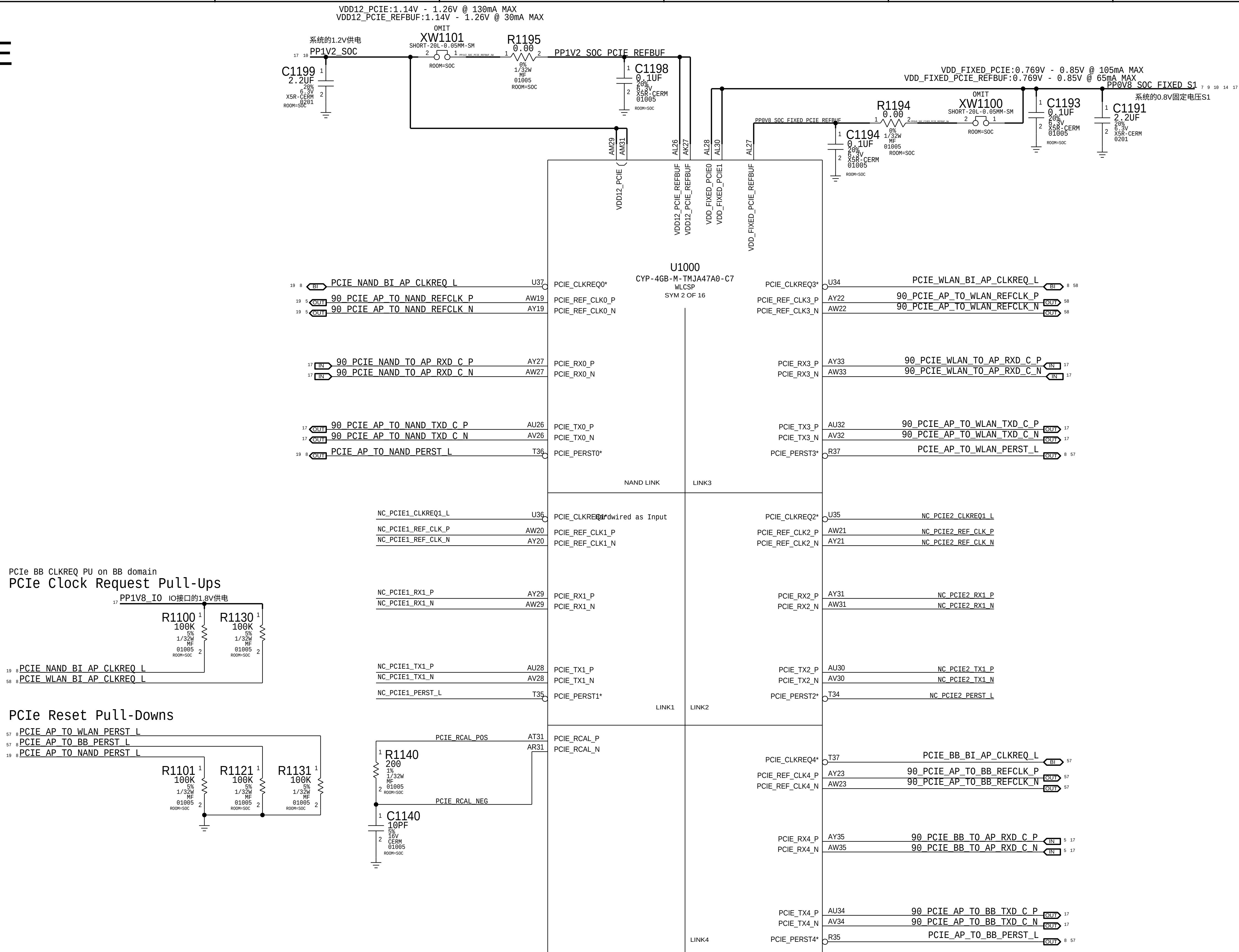
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Float = Low PU = High			
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SPI NOR on SPI0 12 MHz	0	0	0
SPI NOR on SPI0 12 MHz Test	0	0	1
SPI NAND on SPI0 12 MHz	0	1	0
SPI NAND on SPI0 12 MHz Test	0	1	1
SPI NOR on SPI0 40 MHz	1	0	0
SPI NOR on SPI0 40 MHz Test	1	0	1
SPI NOR on SPI0 6 MHz	1	1	0
SPI NOR on SPI0 6 MHz Test	1	1	1

PAGE TITLE		
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D



SOC - PCIE



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SOC - MIPI

NEED MIPI LANE AND POLAIRTY SWAPPING MAP

(Analog)
VDD_FIXED_MIPID 0.769V - 0.85V @ TBDmA MAX
VDD_FIXED_MIPIC 0.769V - 0.85V @ TBDmA MAX
VDD_FIXED_MIPID_PLL 0.769V - 0.85V @ TBDmA MAX

PP0V8_SOC_FIXED_S1 系统的0.8V固定电压S1

VDD18_MIPI*:1.62V - 1.98V @ TBDmA MAX

PP1V8_I0_17

IO接口的1.8V供电

C1291 2.2UF 1 2
X5R 6.3V 0805-SOC

C1290 0.1UF 1 2
X5R 6.3V 0805-SOC

C1296 0.1UF 1 2
X5R 6.3V 0805-SOC

C1295 2.2UF 1 2
X5R 6.3V 0805-SOC

E17

AW9

AP9

G18

AL10

VDD_FIXED_MIPIC

VDD_FIXED_MIPID

VDD_FIXED_MIPID_PLL

VDD18_MIPIC

VDD18_MIPID

MIPI lanes can all flip polarity for routing purposes

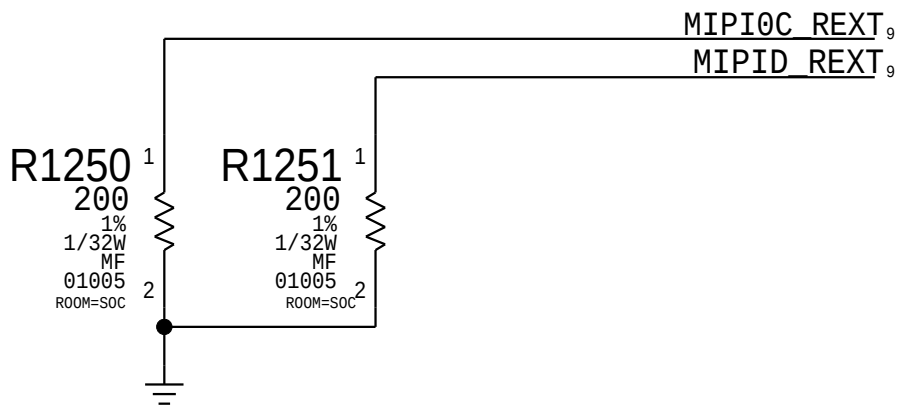
Juliet MIPI

GNDED offpage on MLB

Display MIPI

GNDED offpage on MLB

MIPI Reference



U1000

CYP-4GB-M-TMJA47A0-C7
MLCSP
SYM 3 OF 16

ALT FUNC's

SIO_LEAP_MADI_IN and AOP_LEAP_MADI_IN
SIO_LEAP_MADI_OUT and AOP_LEAP_MADI_OUT

MTR_ADC_DOUT and SOC_DEBUG2

MTR_ADC_CLKOUT and SOC_DEBUG3

SENSOR3_CLK
PLL_DIG0BS_IN_0 and ISP_FCAM_SPMI_SDATA
PLL_DIG0BS_IN_1 and ISP_FCAM_SPMI_SCLK

CANT SWAP DUE TO BiD1

ISP_SPMI_SDATA
ISP_SPMI_SCLK

SENSOR0_CLK
SENSOR1_CLK
SENSOR2_CLK

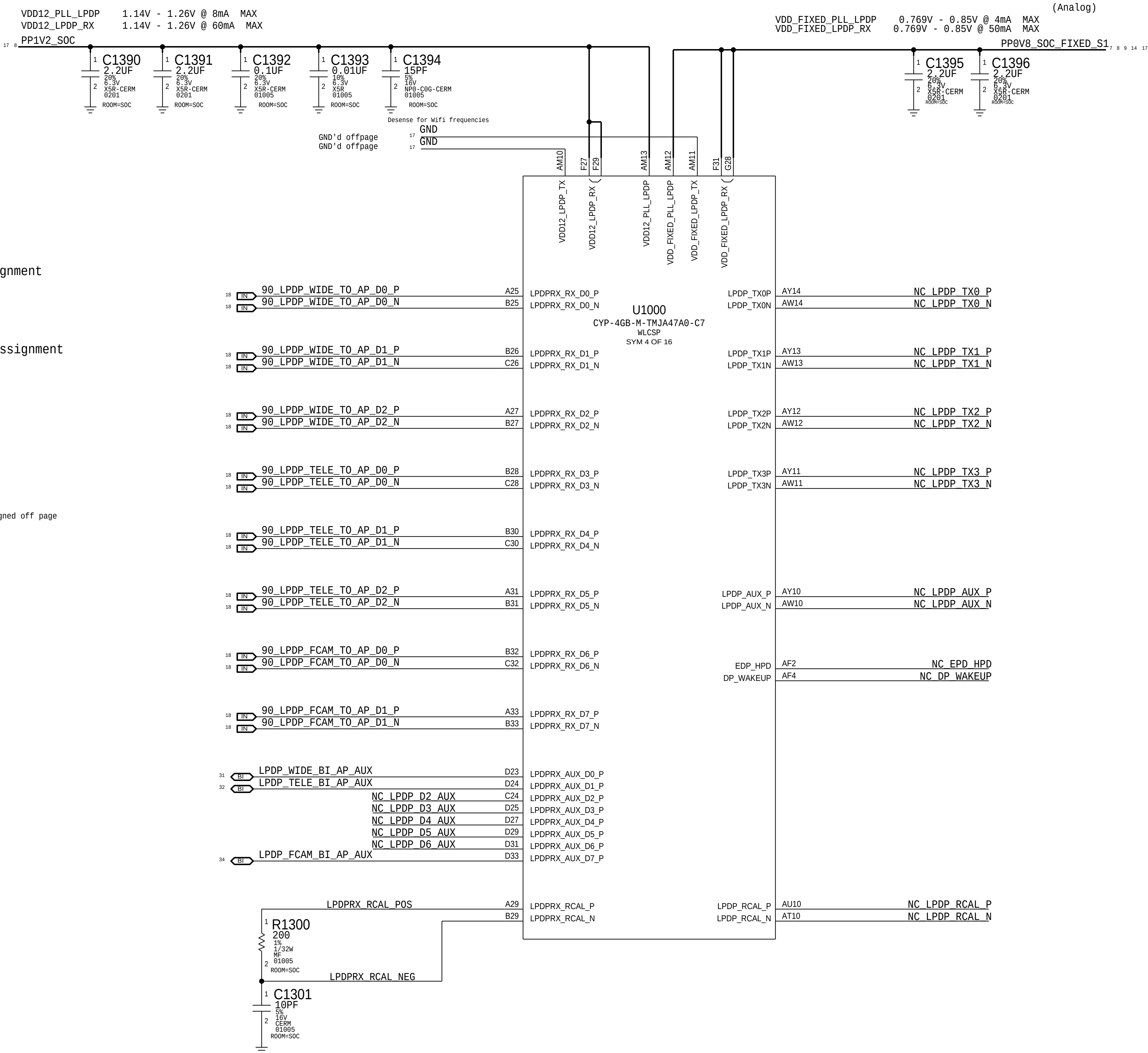
R1240 33.2 1 2
1/32W
01005
ROOM-SOC

Series Terminations Offpage

SOC: MIPI

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SOC - LPDP



Dan LPDP Lane Assignment

Wide: 0-2
Tele: 3-5
Fcam: 6-7

Justin LPDP Lane Assignment

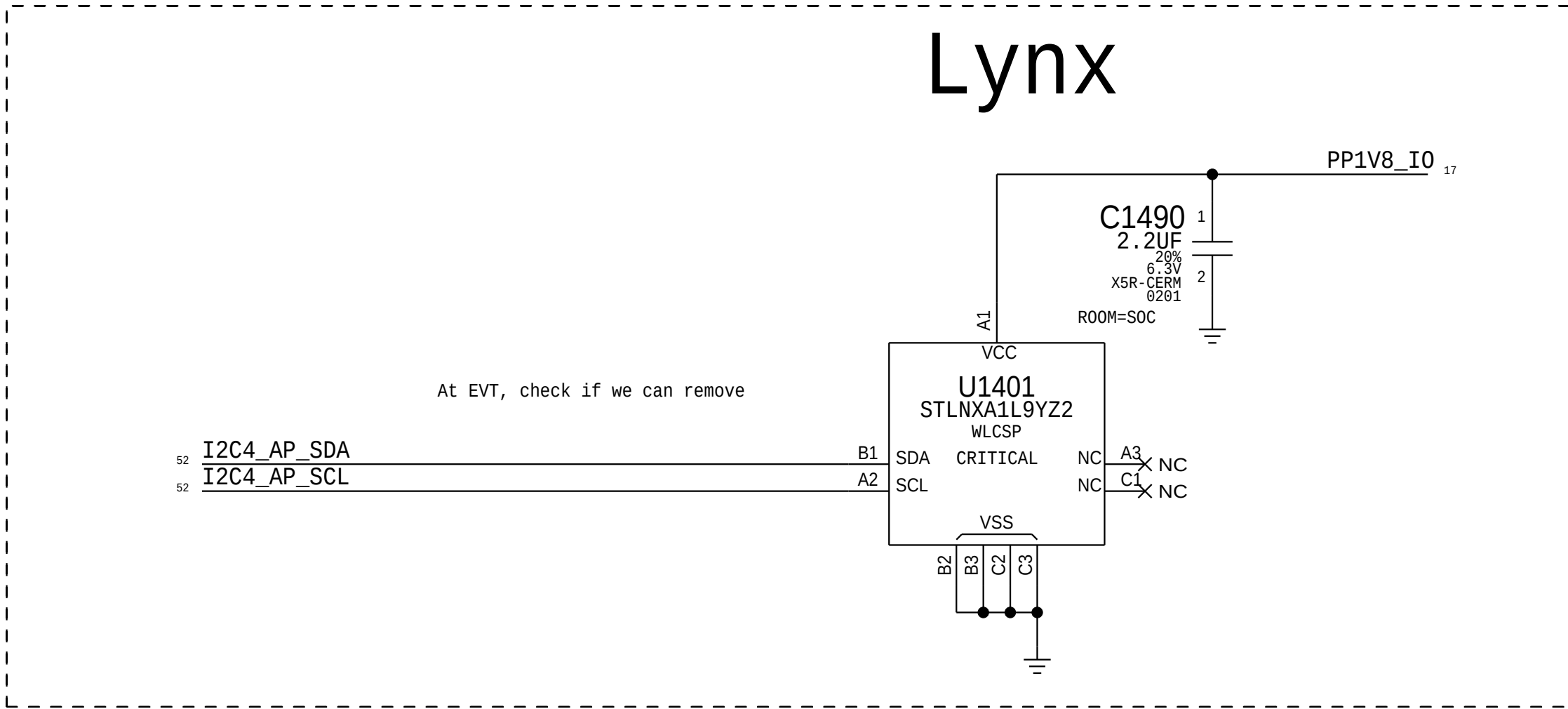
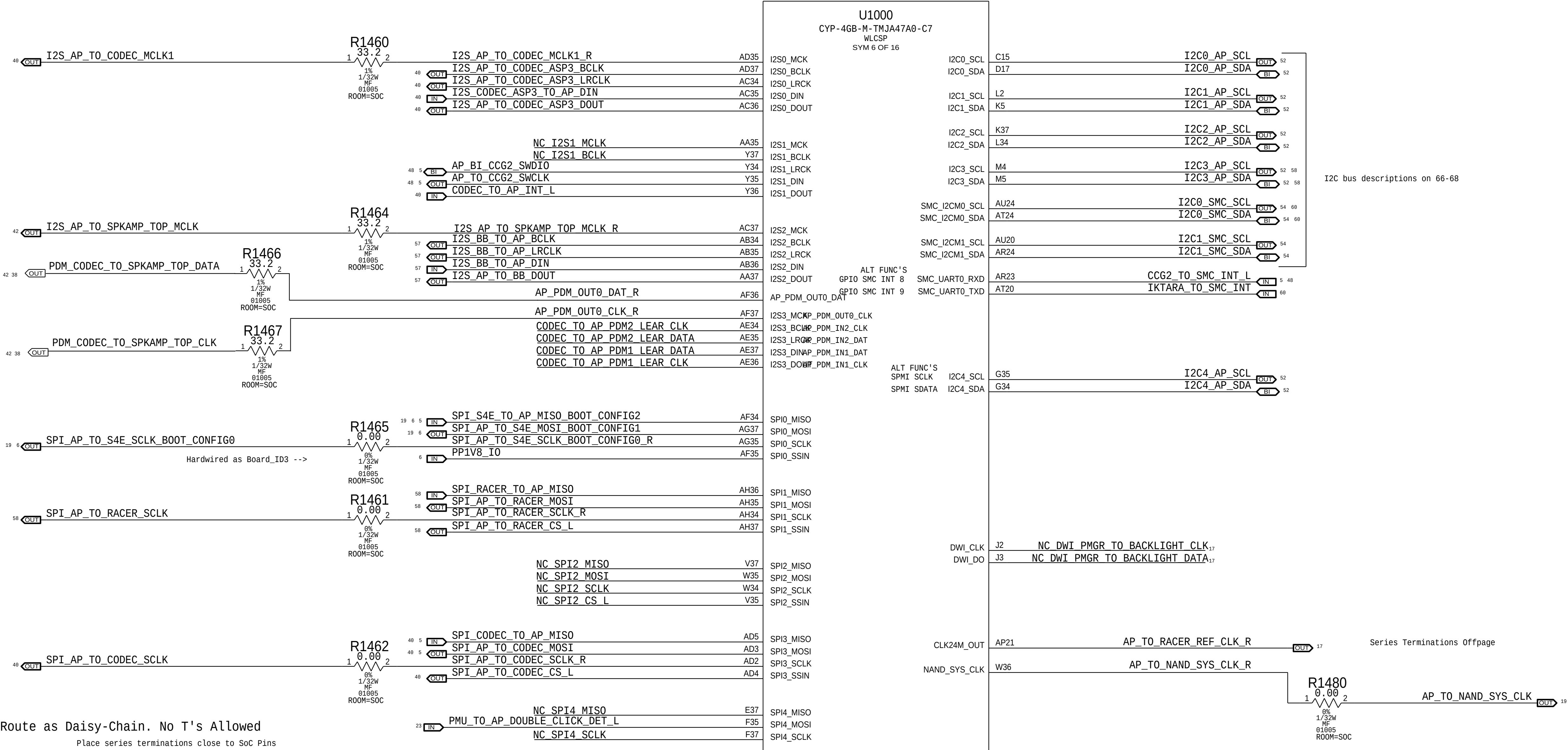
Wide: 2-4
Tele: 5-7
Fcam: 0-1

LPD Assigned off page

SOC: LPDP

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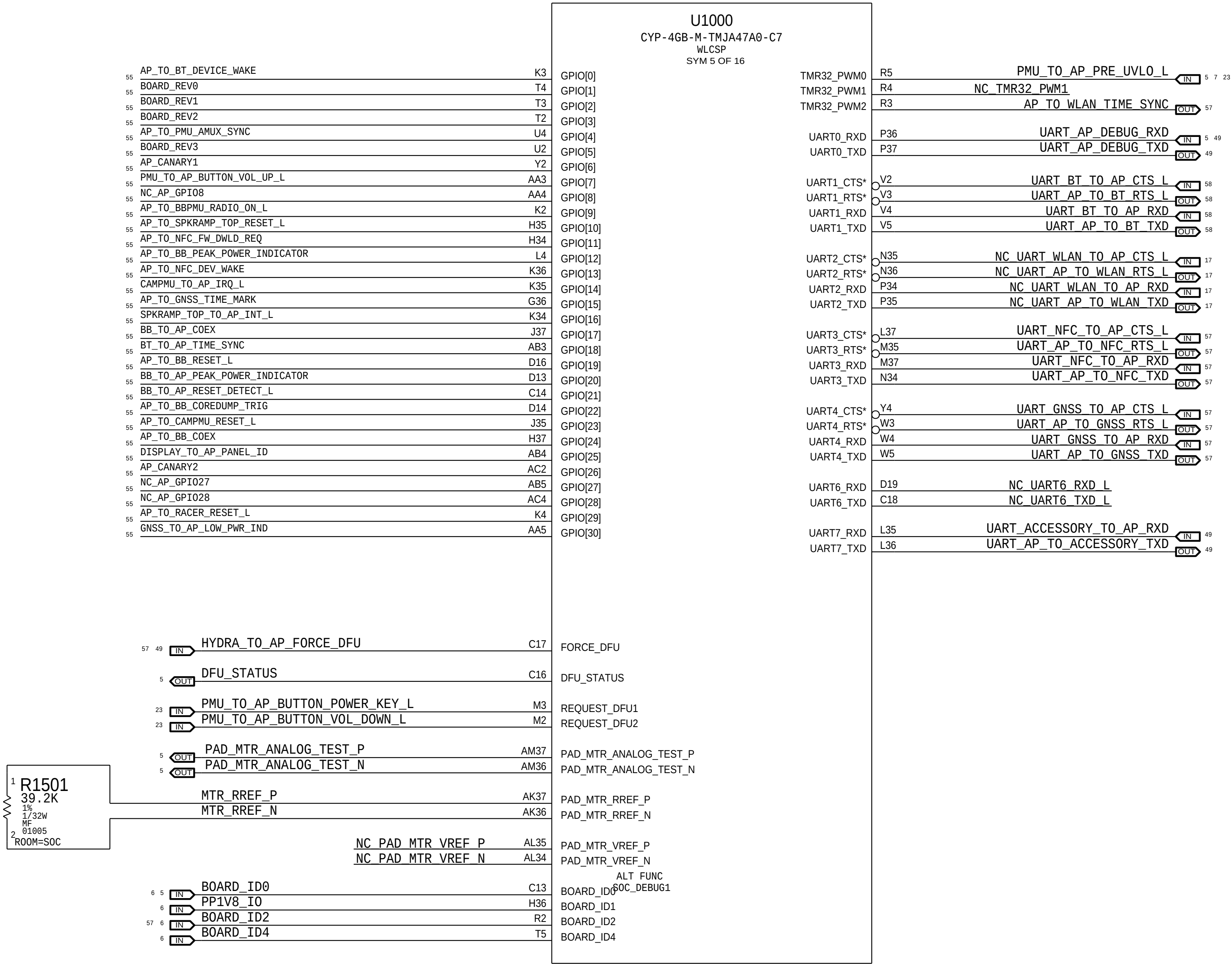
SOC - SERIAL INTERFACES



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SOC: SERIAL		
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SOC - GPIO INTERFACES

GPIOs are wired on page 70



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SOC - AOP

D

C

B

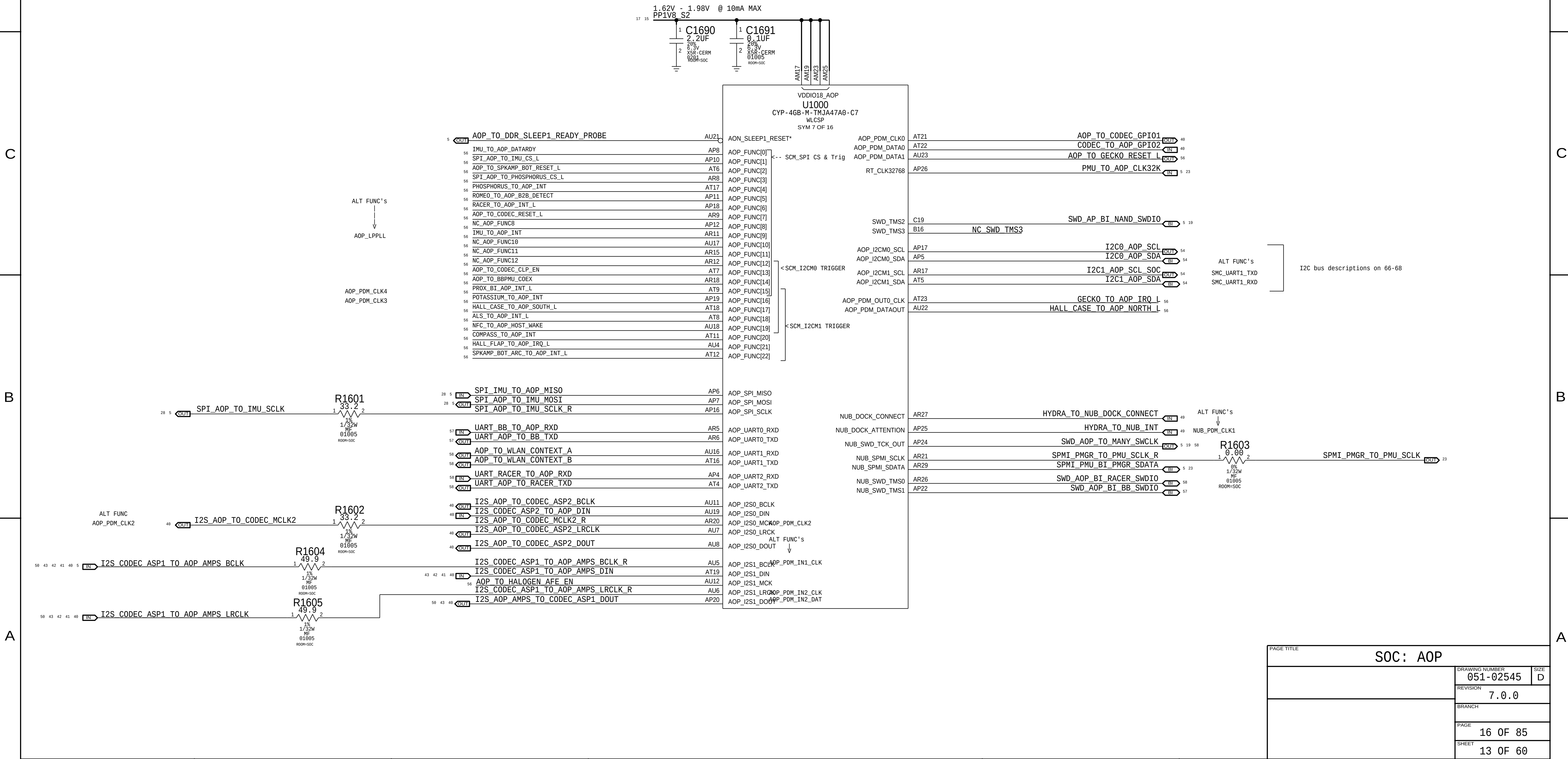
A

D

C

B

A



PAGE TITLE		
SOC: AOP		
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D

17 PP_CPU_PCORE

D

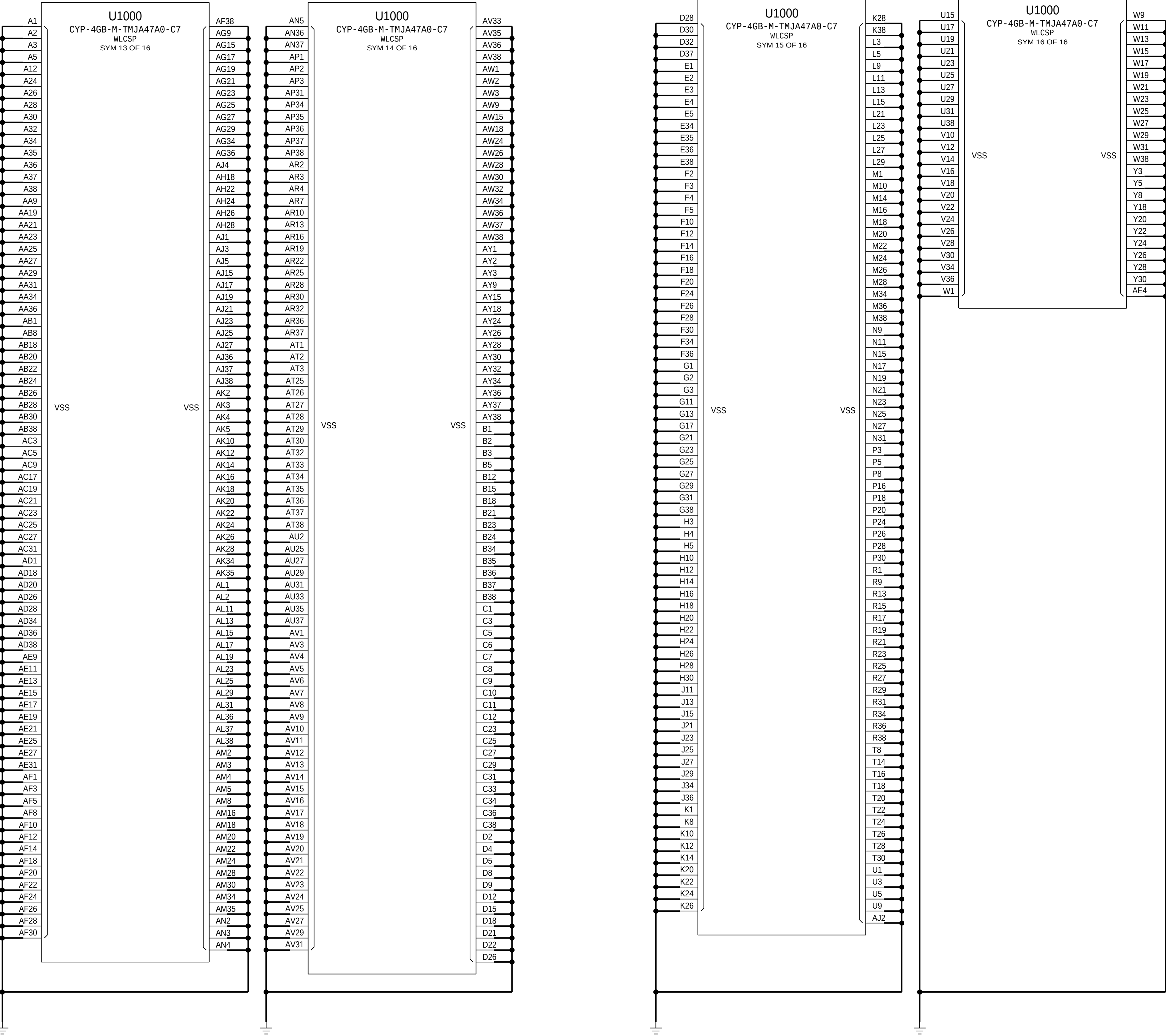
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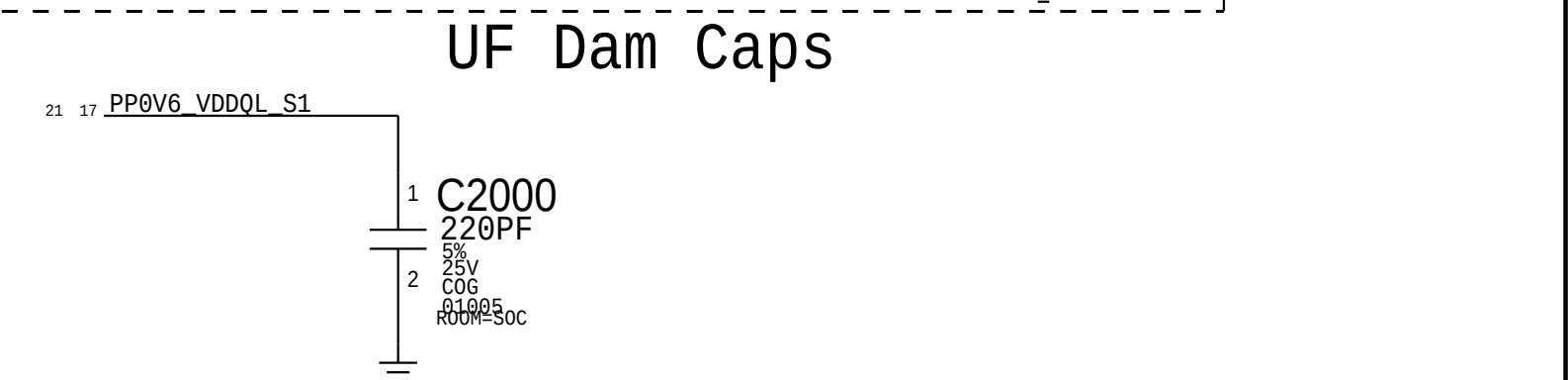
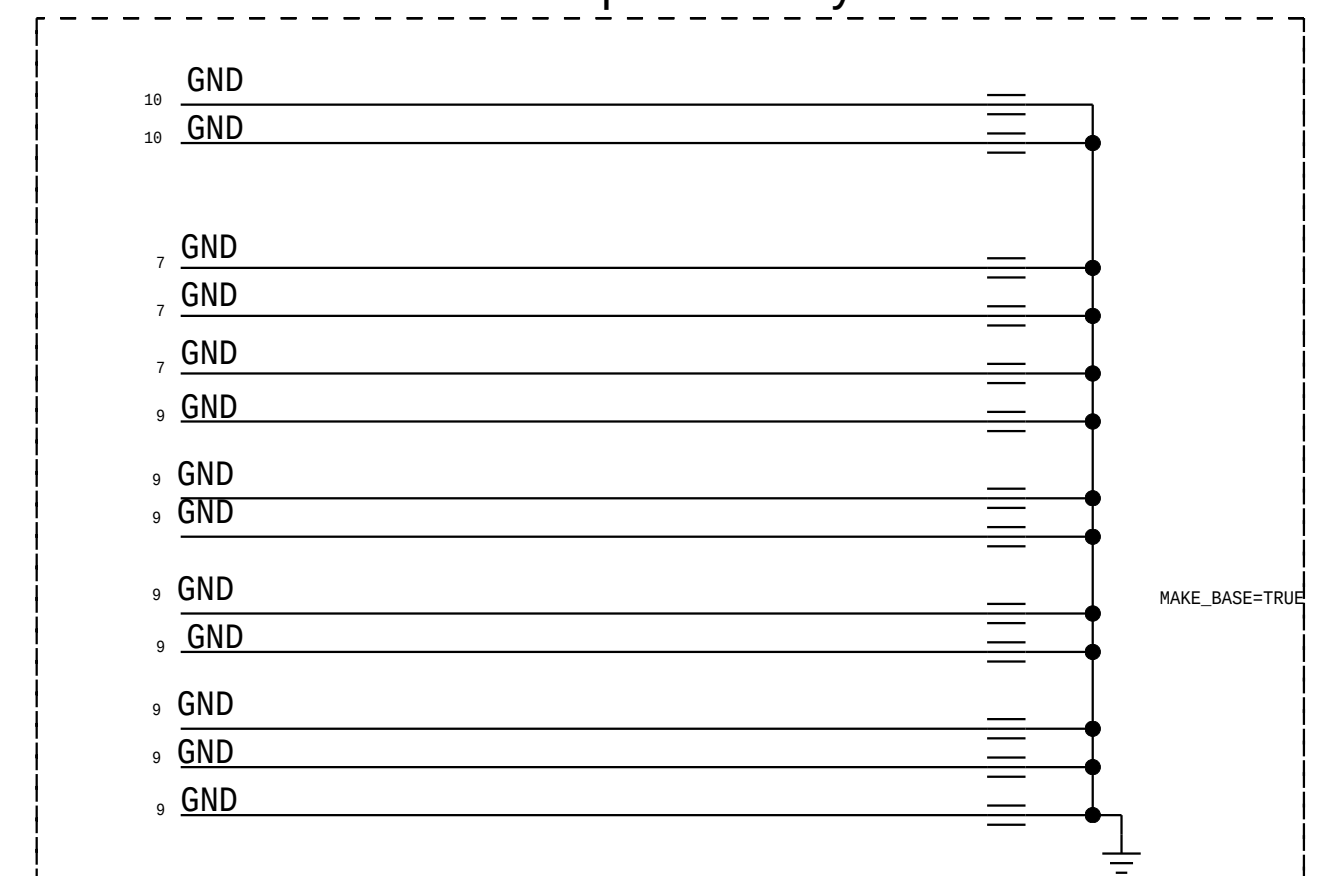
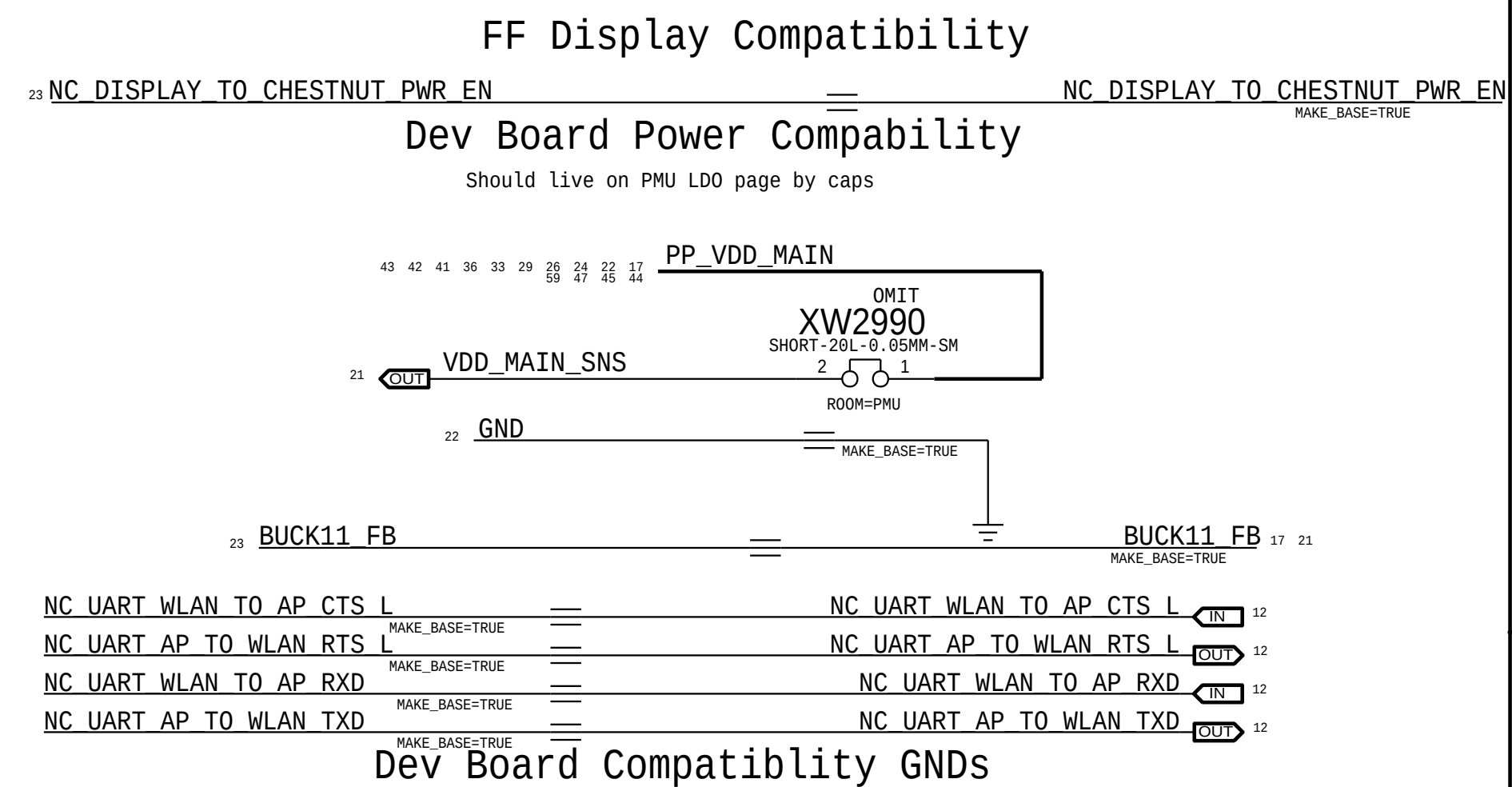
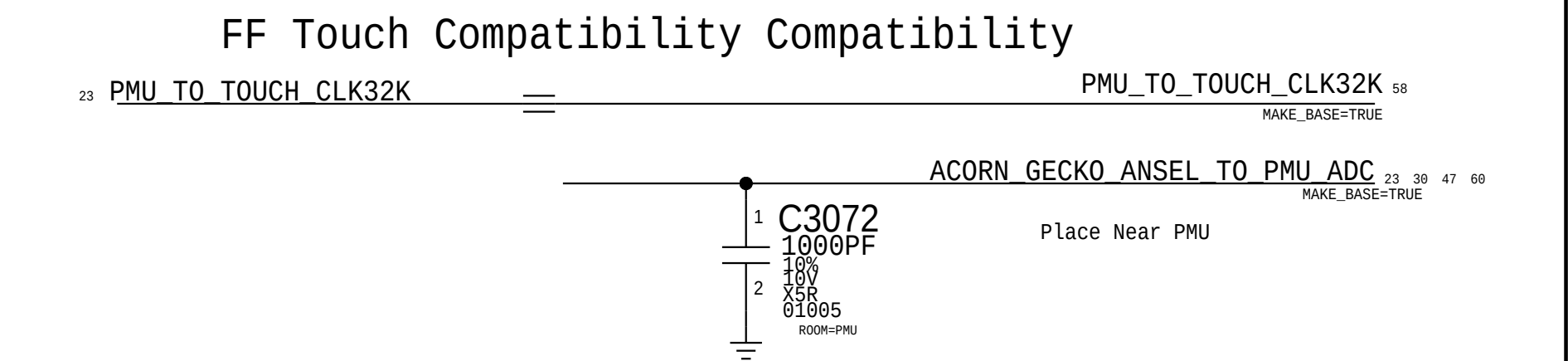
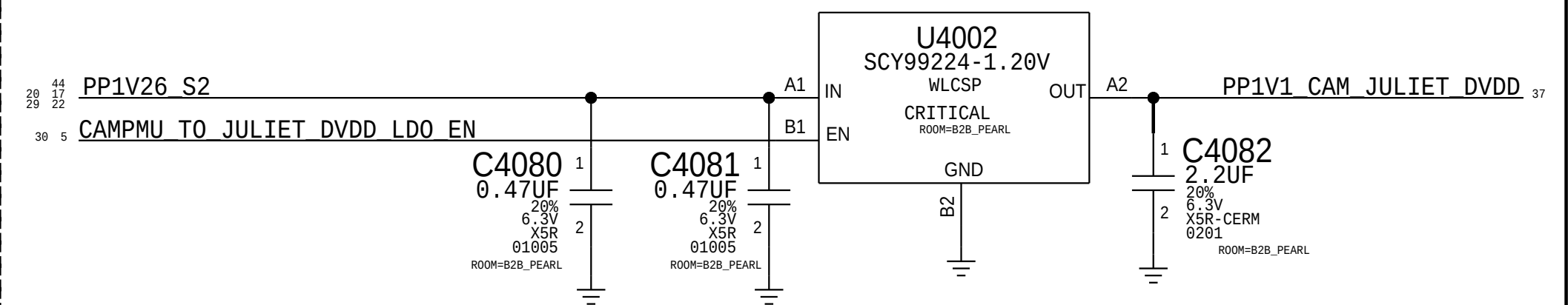
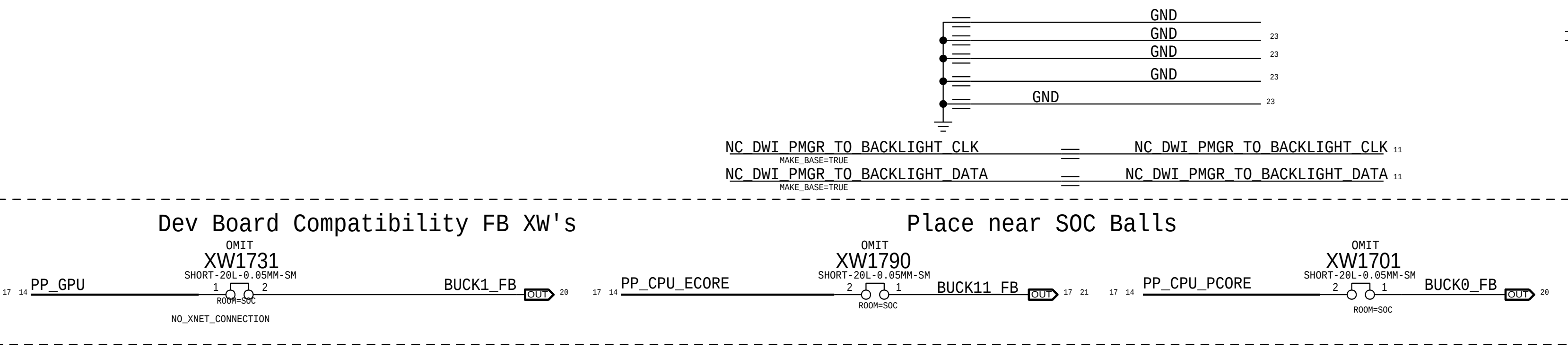
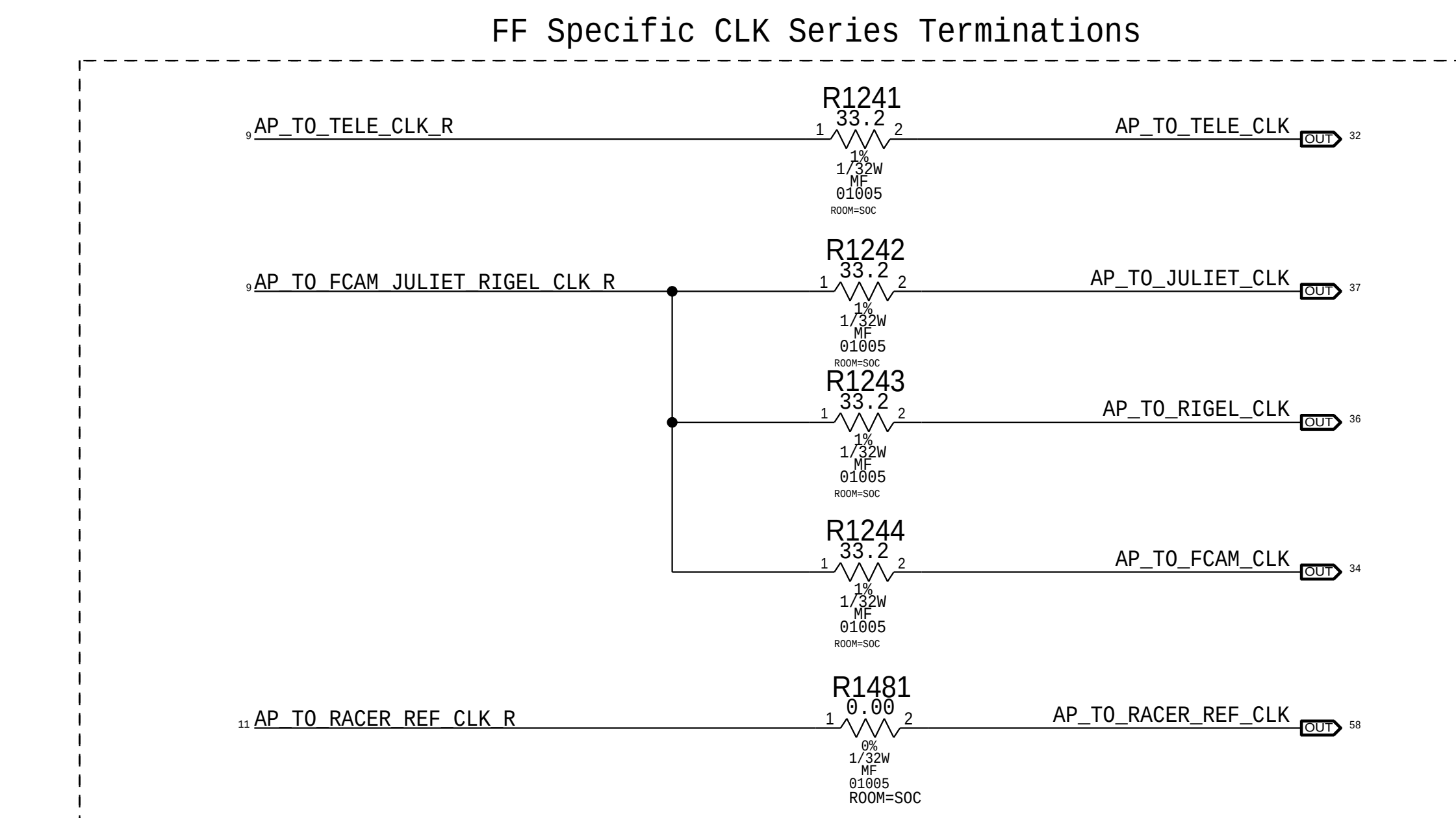
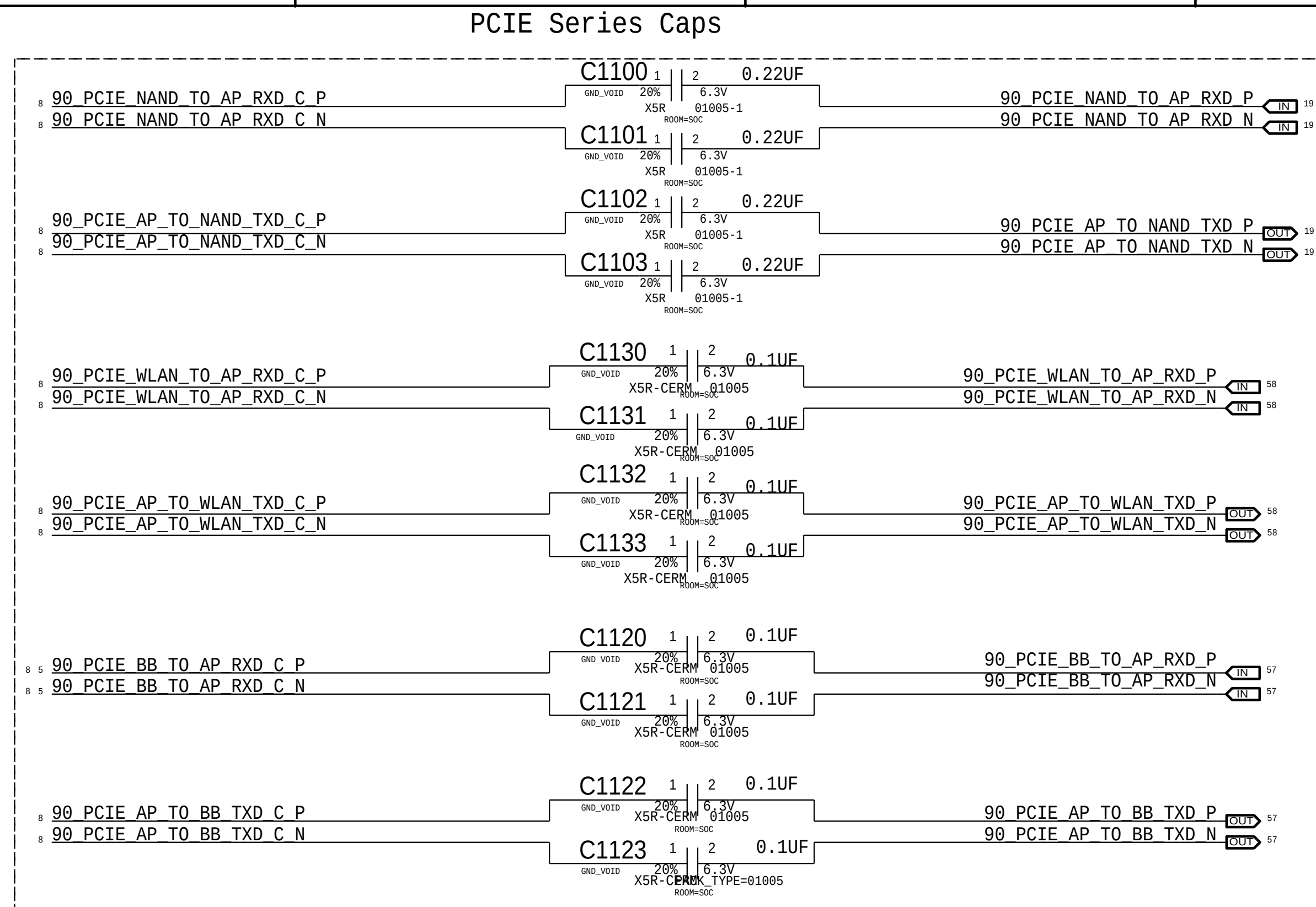
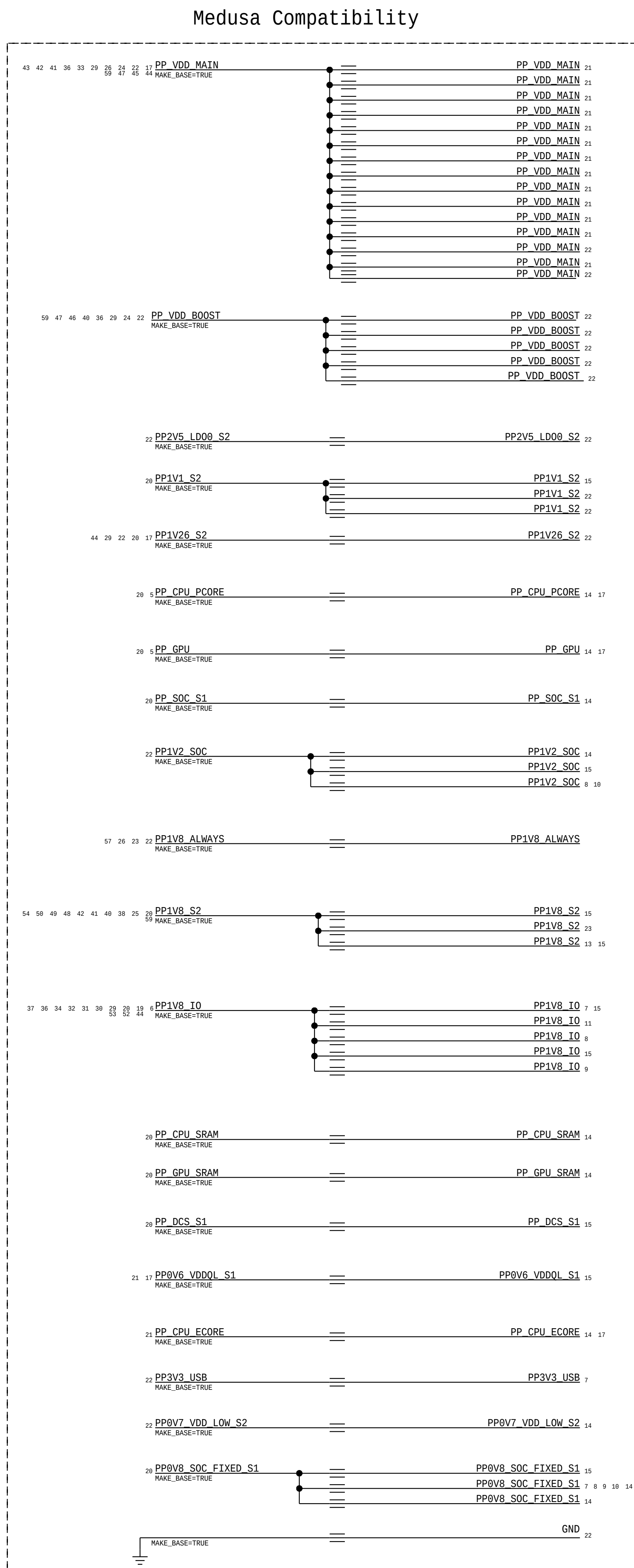
Δ

SOC - CPU, GPU & SOC RAILS

SOC - POWER SUPPLIES



PAGE TITLE		
SOC: POWER (3/3)		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	19 OF 85
	SHEET	16 OF 60
SIZE		
D		



PAGE TITLE		DRAWING NUMBER		SIZE
SOC: DEV BOARD ALIASES		051-02545		D
		REVISION		
		7.0.0		
		BRANCH		
		PAGE		
		20 OF 85		
		SHEET		
		17 OF 60		

8		7		6		5		4		3		2		1	
D														D	
C														C	
B														B	
A														A	

31

ITV

90_LPDP_WIDE_TO_AP_D0_P

31

ITV

90_LPDP_WIDE_TO_AP_D0_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_WIDE_TO_AP_D0_P

90_LPDP_WIDE_TO_AP_D0_N

10

10

31

ITV

90_LPDP_WIDE_TO_AP_D1_P

31

ITV

90_LPDP_WIDE_TO_AP_D1_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_WIDE_TO_AP_D1_P

90_LPDP_WIDE_TO_AP_D1_N

10

10

31

ITV

90_LPDP_WIDE_TO_AP_D2_P

31

ITV

90_LPDP_WIDE_TO_AP_D2_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_WIDE_TO_AP_D2_P

90_LPDP_WIDE_TO_AP_D2_N

10

10

32

ITV

90_LPDP_TELE_TO_AP_D0_P

32

ITV

90_LPDP_TELE_TO_AP_D0_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_TELE_TO_AP_D0_P

90_LPDP_TELE_TO_AP_D0_N

10

10

32

ITV

90_LPDP_TELE_TO_AP_D1_P

32

ITV

90_LPDP_TELE_TO_AP_D1_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_TELE_TO_AP_D1_P

90_LPDP_TELE_TO_AP_D1_N

10

10

32

ITV

90_LPDP_TELE_TO_AP_D2_P

32

ITV

90_LPDP_TELE_TO_AP_D2_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_TELE_TO_AP_D2_P

90_LPDP_TELE_TO_AP_D2_N

10

10

34

ITV

90_LPDP_FCAM_TO_AP_D0_P

34

ITV

90_LPDP_FCAM_TO_AP_D0_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_FCAM_TO_AP_D0_P

90_LPDP_FCAM_TO_AP_D0_N

10

10

34

ITV

90_LPDP_FCAM_TO_AP_D1_P

34

ITV

90_LPDP_FCAM_TO_AP_D1_N

MAKE_BASE=TRUE

MAKE_BASE=TRUE

90_LPDP_FCAM_TO_AP_D1_P

90_LPDP_FCAM_TO_AP_D1_N

10

10

PAGE TITLE

SOC: LPDP ALIASES

DRAWING NUMBER

051-02545

REVISION

7.0.0

BRANCH

PAGE

21 OF 85

SHEET

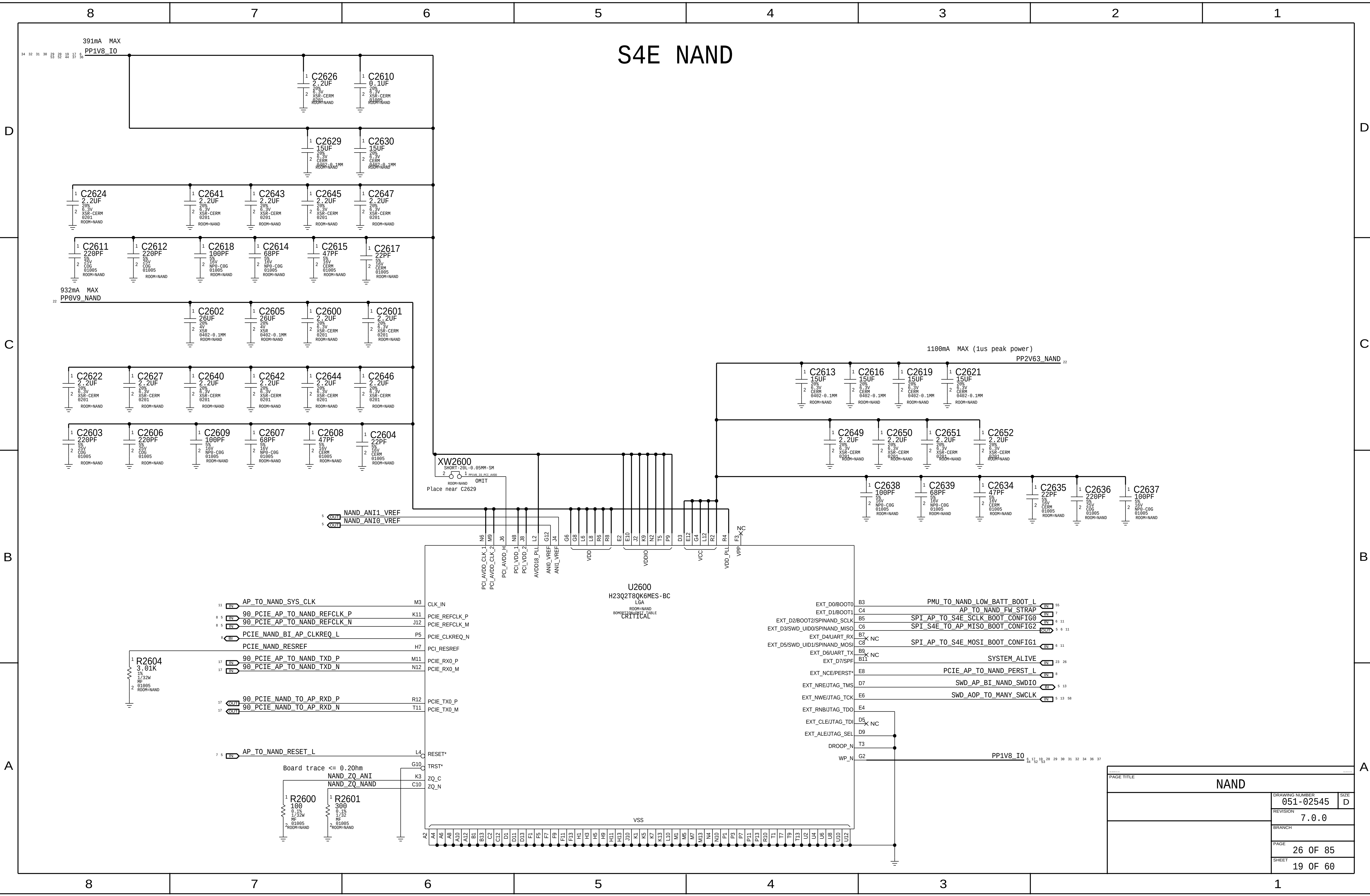
18 OF 60

SIZE

D

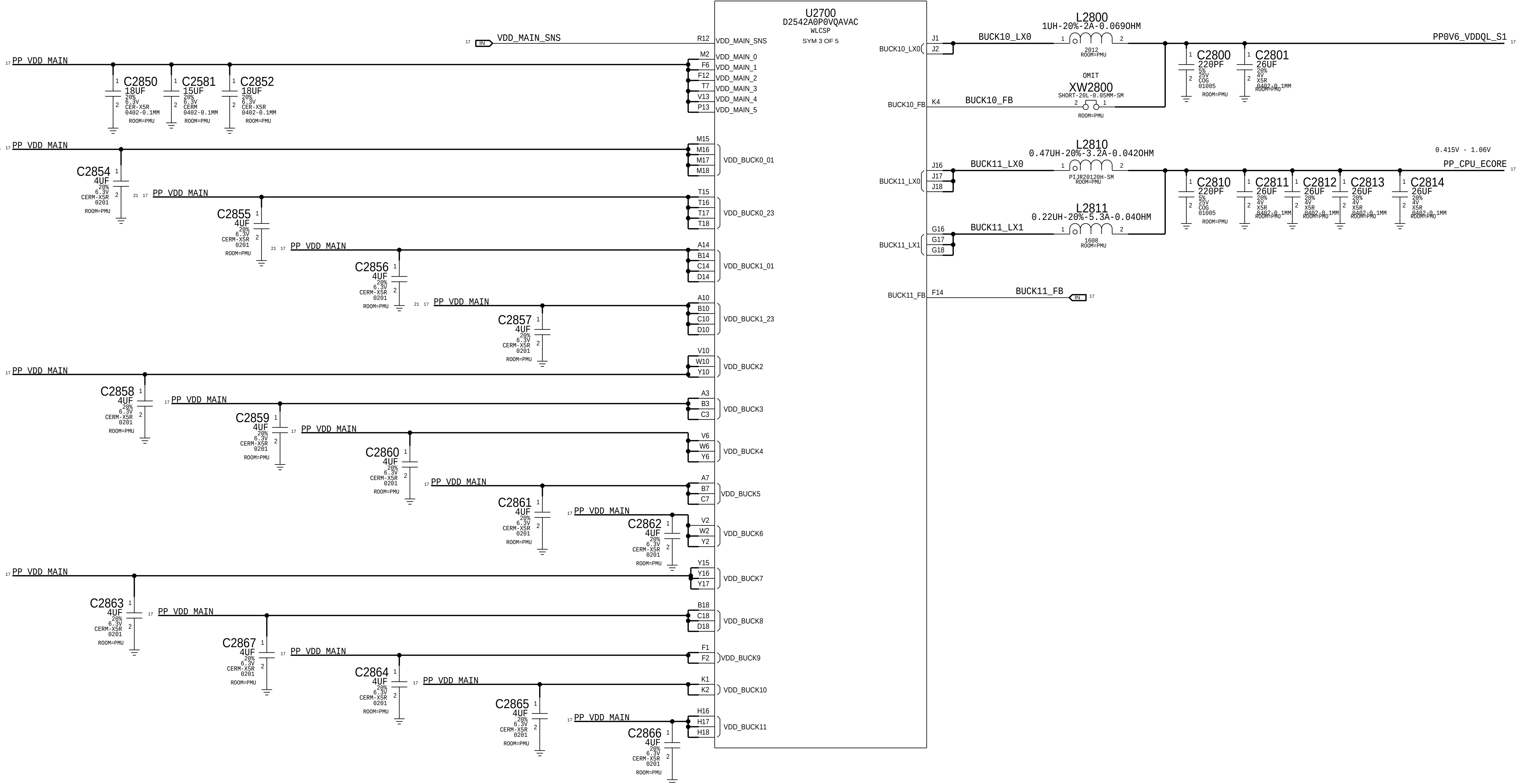
SYNC DATE=08/17/2017

S4E NAND

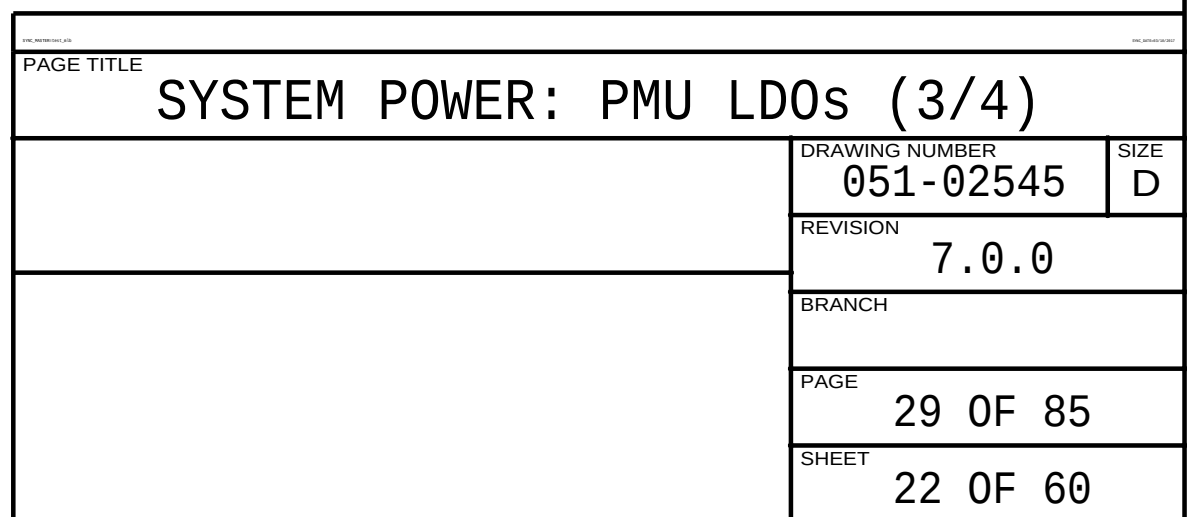


PAGE TITLE		
NAND		
DRAWING NUMBER	051-02545	SIZE D
REVISION	7.0.0	
BRANCH		
PAGE	26 OF 85	
SHEET	19 OF 60	

PMU - BUCKS



SYSTEM POWER: PMU Bucks (2/4)		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	28 OF 85
	SHEET	21 OF 60



PMU - GPIOs

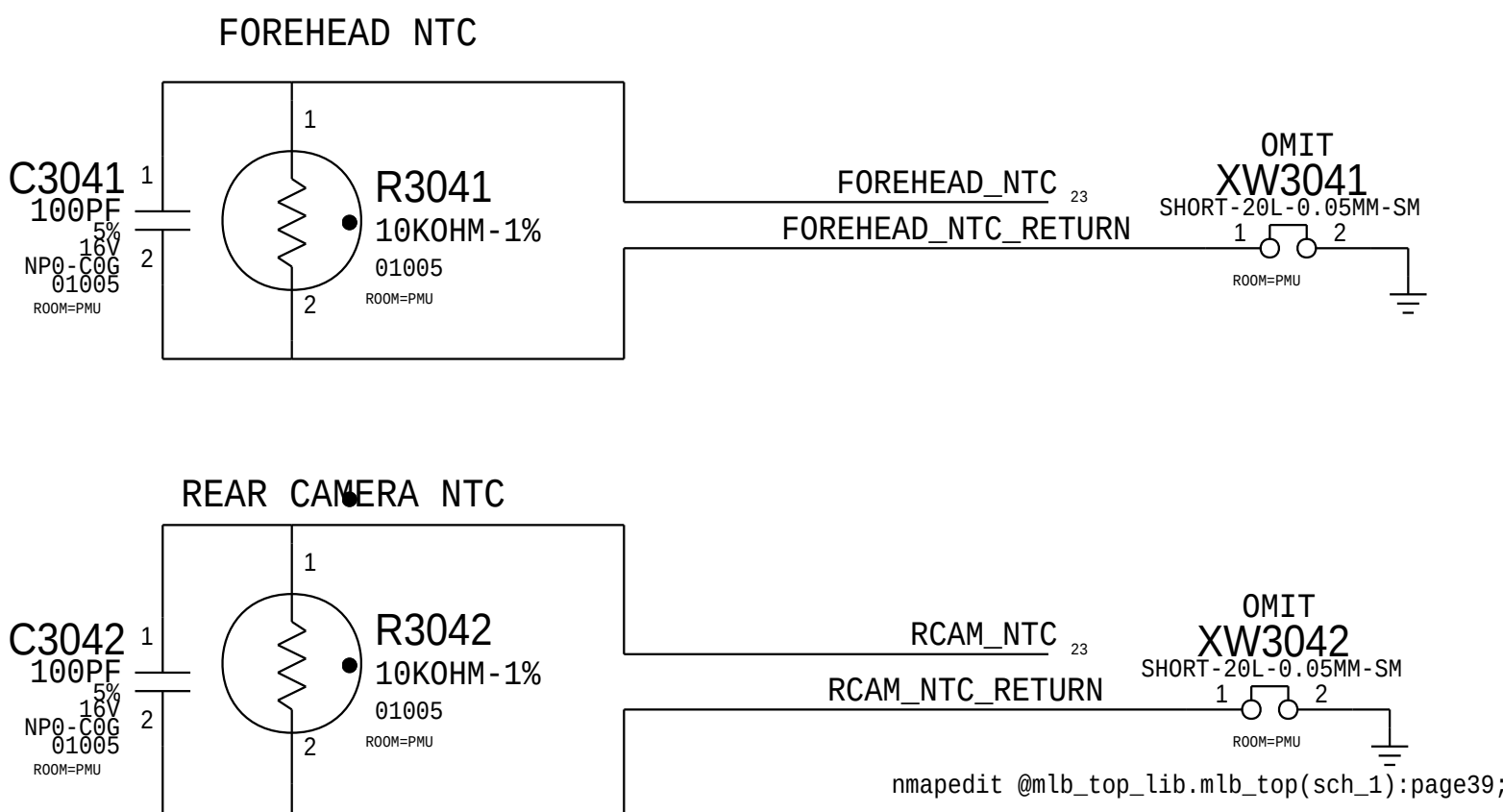
TODO: Update
CONTROL PIN NOTES:

- NOTE (1):INPUT PULL-DOWN 100-300k
NOTE (2):INPUT PULL-DOWN 1M
NOTE (3):INPUT PULL-UP OR DOWN 100k-300k
NOTE (4):OUTPUT OPEN-DRAIN, REQUIRES PULL-UP

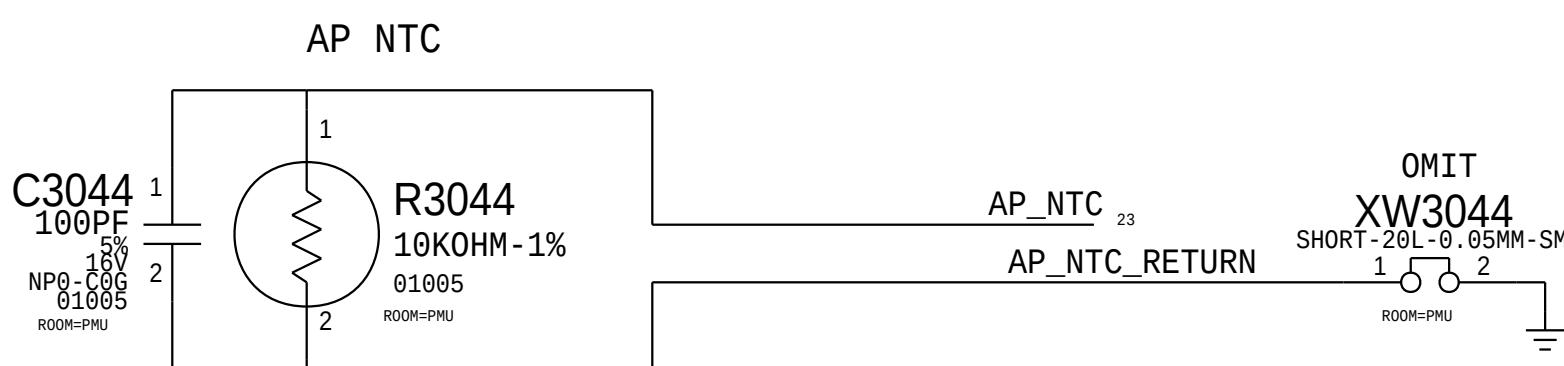
COLD_RESET & SYSTEM_ALIVE



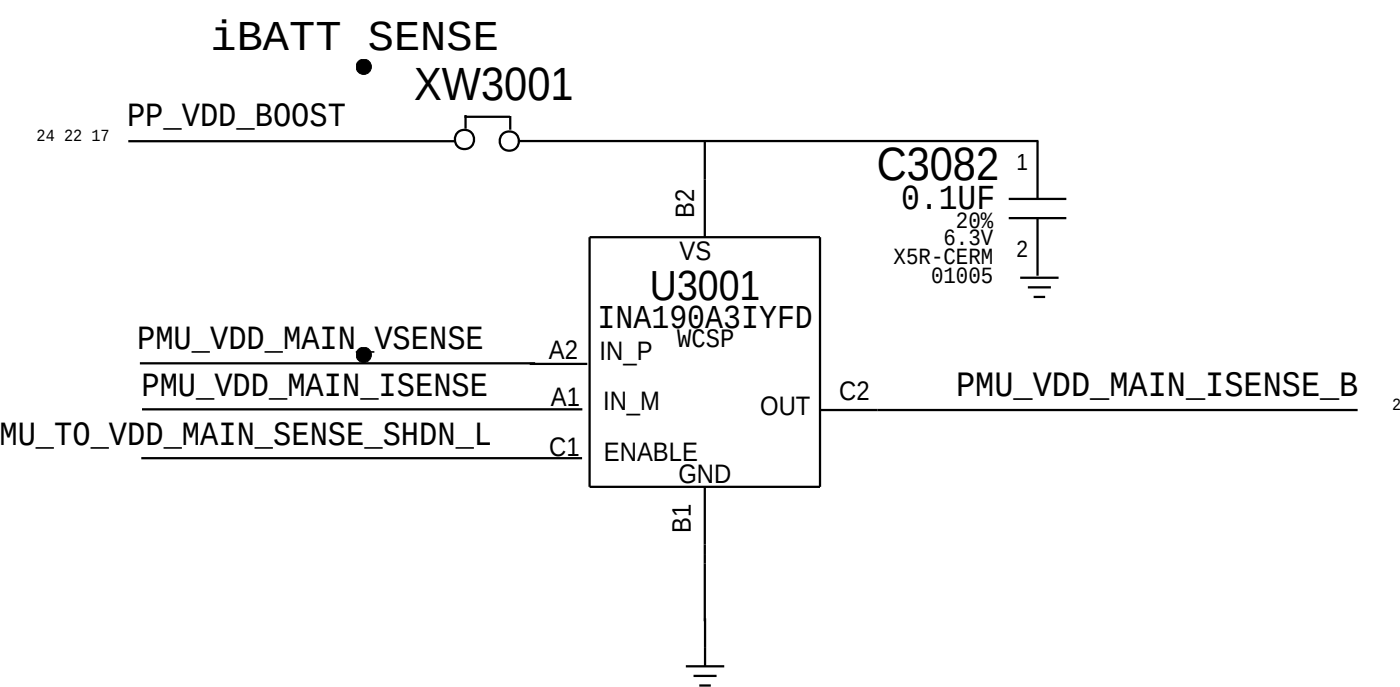
NTCs



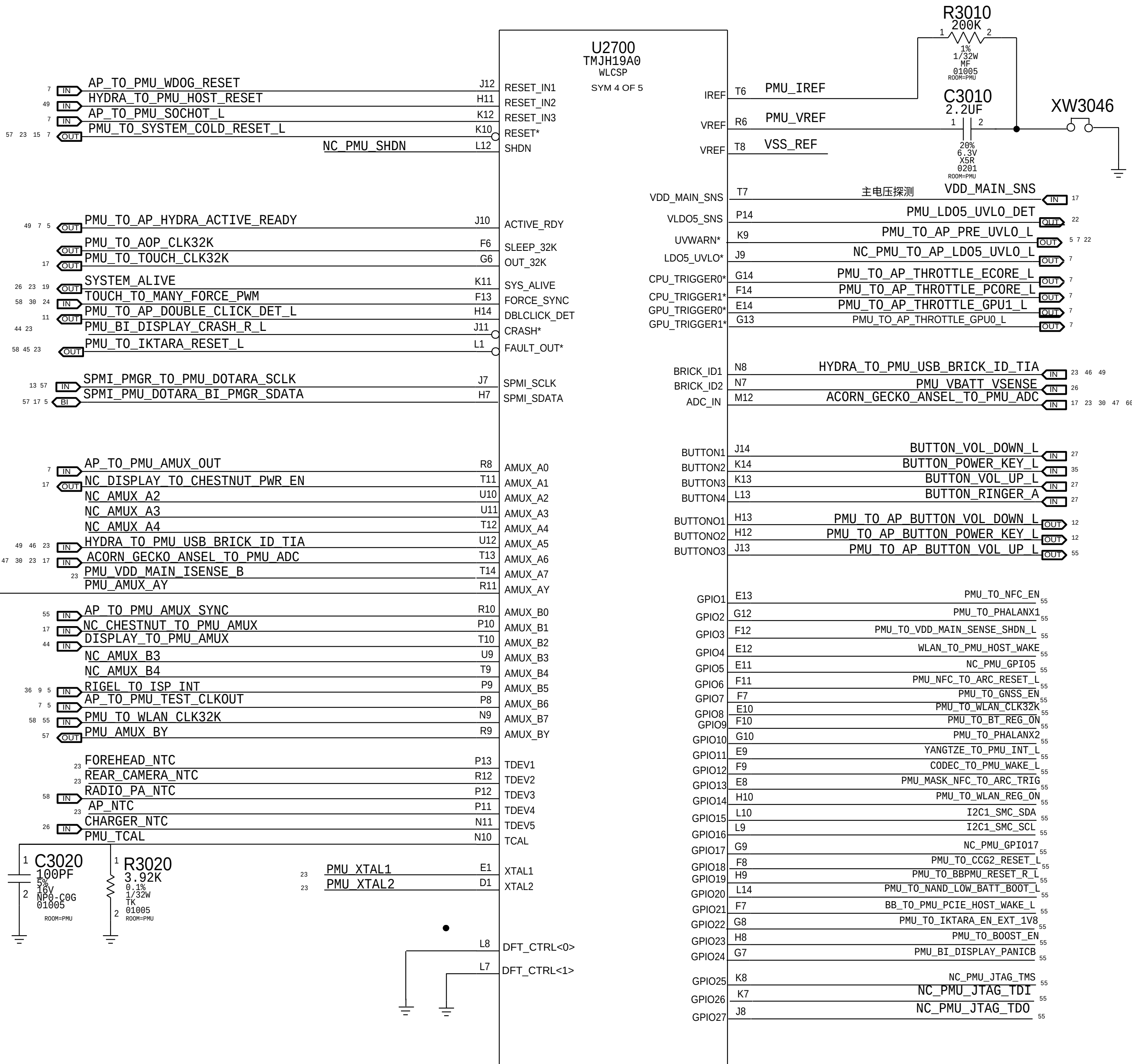
RADIO PA NTC on MLB Bottom



CHARGER NTC on Chrager Page

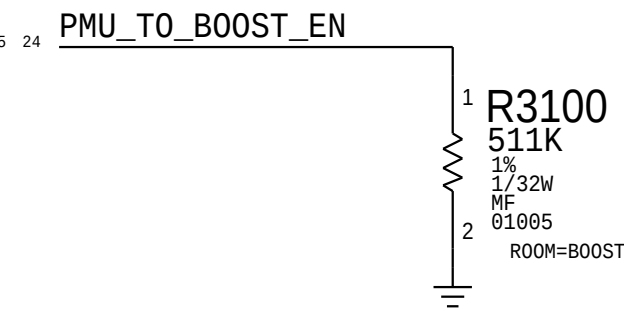


NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC



PAGE TITLE			
SYSTEM POWER: PMU (4/4)			
DRAWING NUMBER		051-02545	SIZE
REVISION		7.0.0	
BRANCH			
PAGE		30 OF 85	
SHEET		23 OF 60	

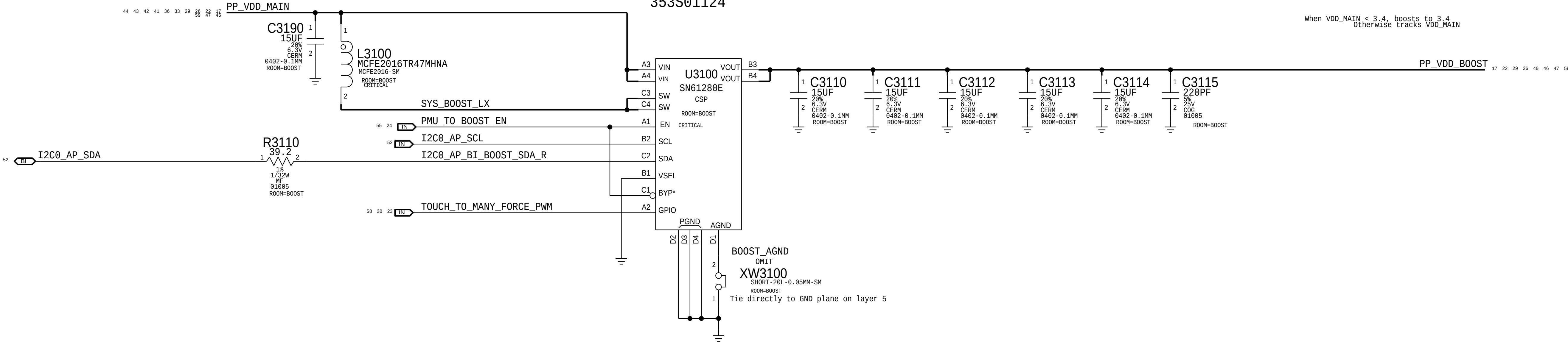
Boost Enable Pull



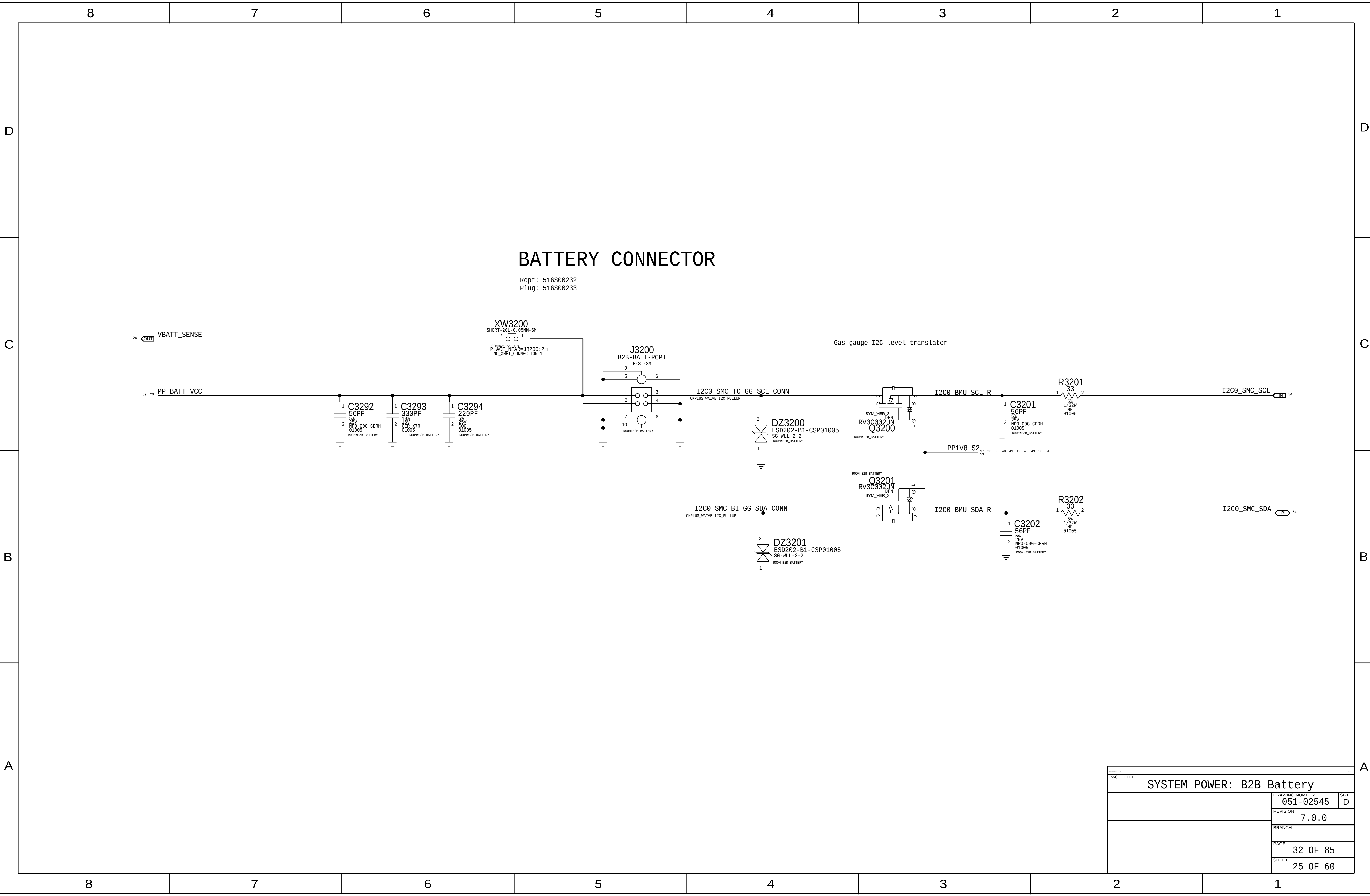
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00871	152S00869	ALT_PARTS	L3100	BOOST IND ALT, CYN
152S00873	152S00869	ALT_PARTS	L3100	BOOST IND ALT, YDK

BOOST

353S01124



SYSTEM POWER: Boost		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	31 OF 85
	SHEET	24 OF 60



BATTERY CONNECTOR

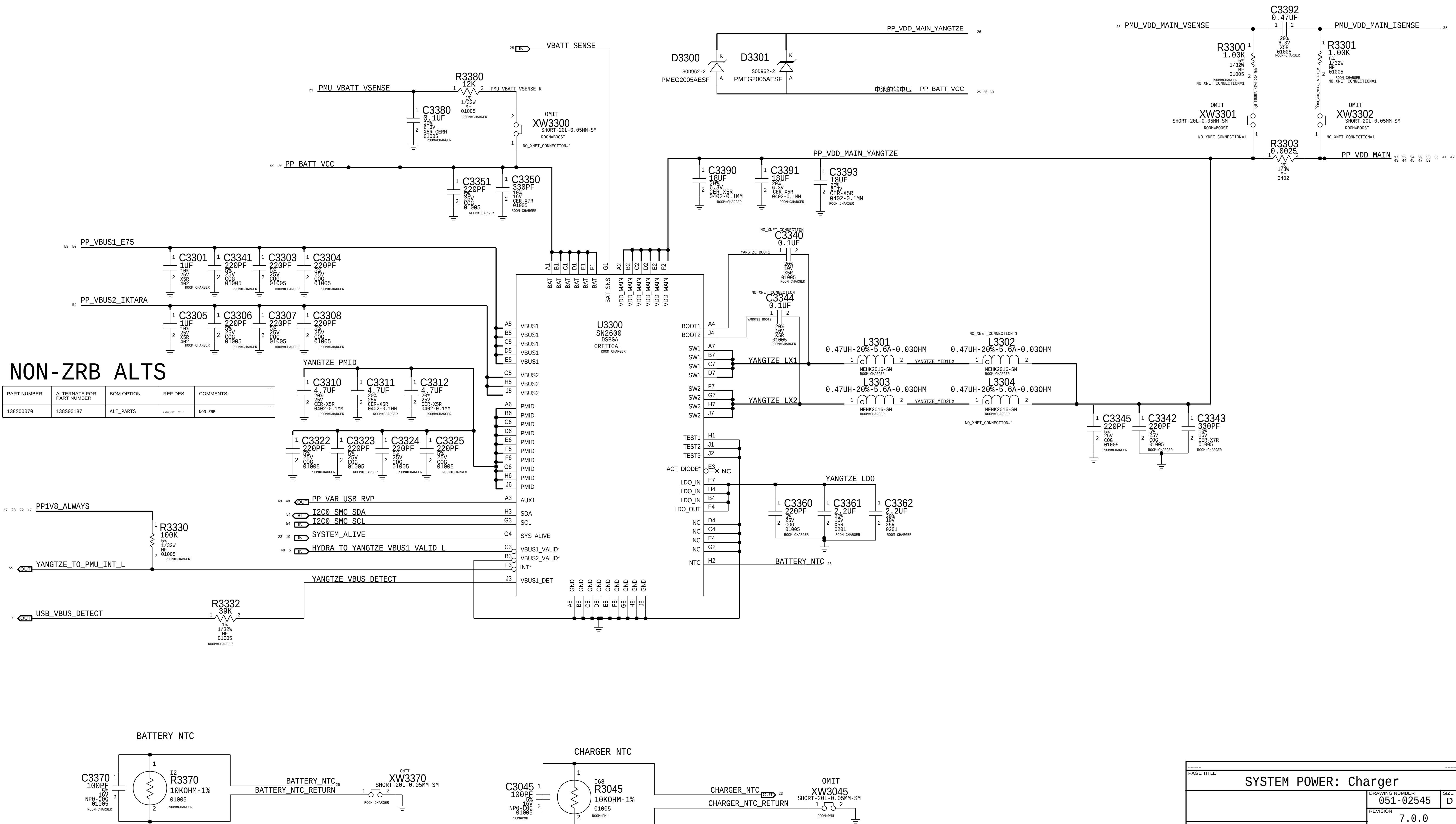
Rcpt: 516S00232
Plug: 516S00233

PAGE TITLE		
SYSTEM POWER: B2B Battery		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	32 OF 85
	SHEET	25 OF 60

YANGTZE CHARGER

NON-ZRB ALTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00070	138S00187	ALT_PARTS	CHRG CHRG1,CHRG2	NON-ZRB



PAGE TITLE		
SYSTEM POWER: Charger		
DRAWING NUMBER	051-02545	SIZE D
REVISION	7.0.0	
BRANCH		
PAGE	33 OF 85	
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D



B

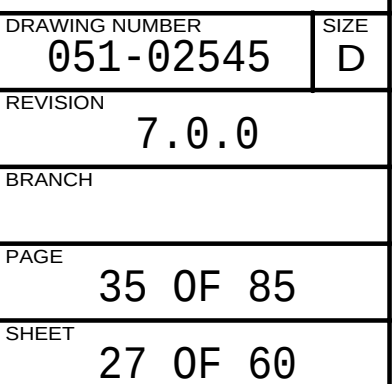


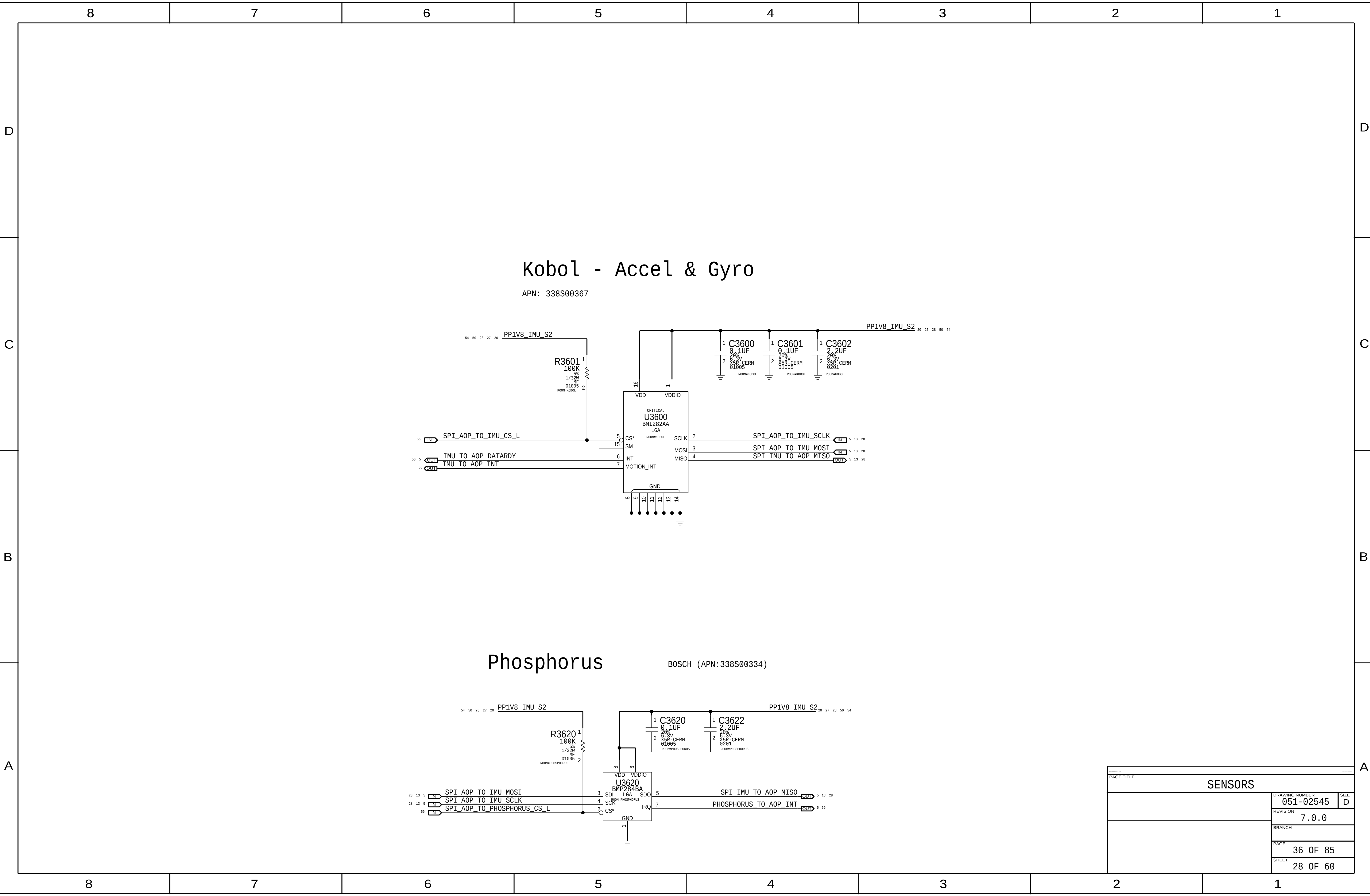
D

C

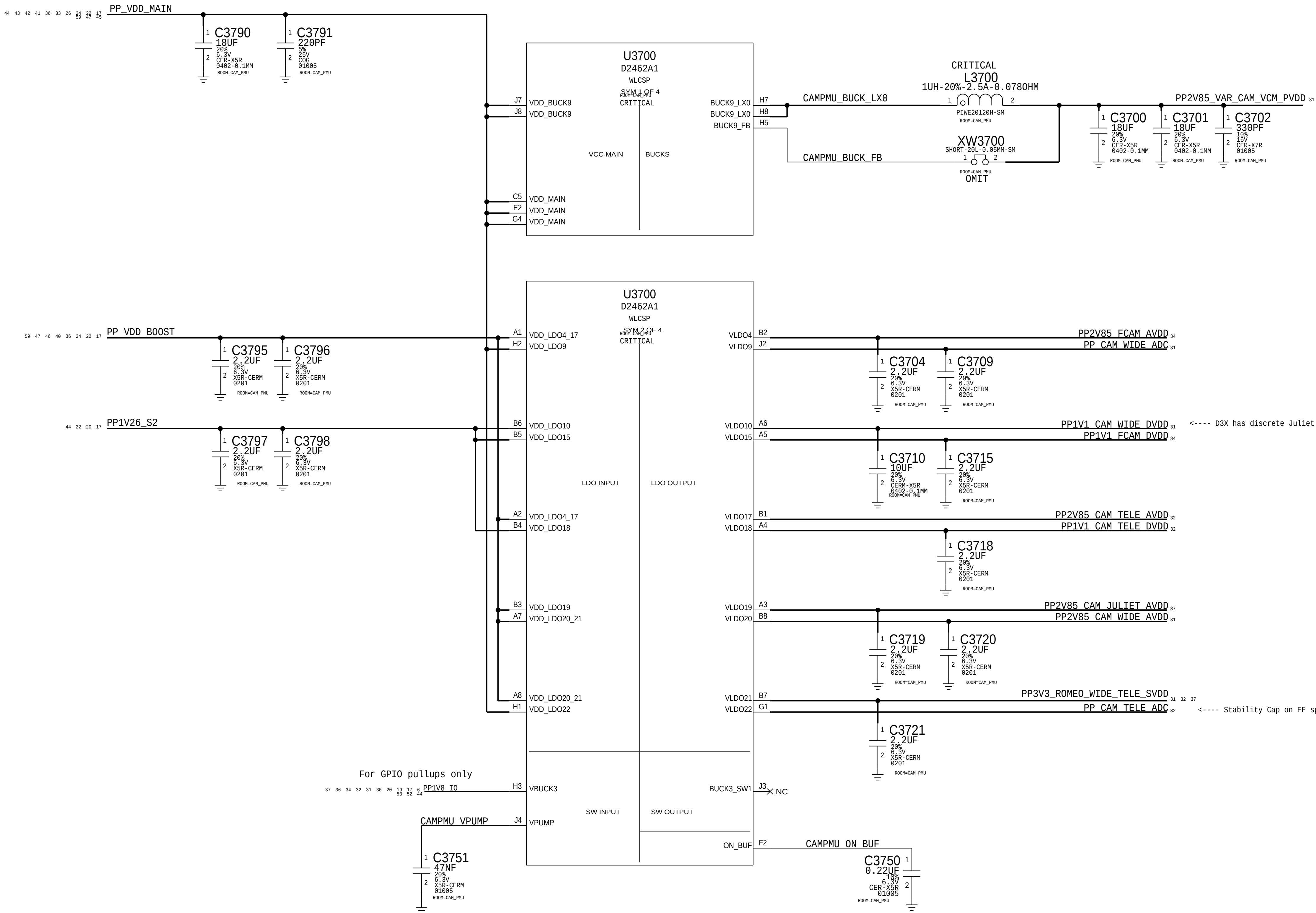


A





Camera PMU



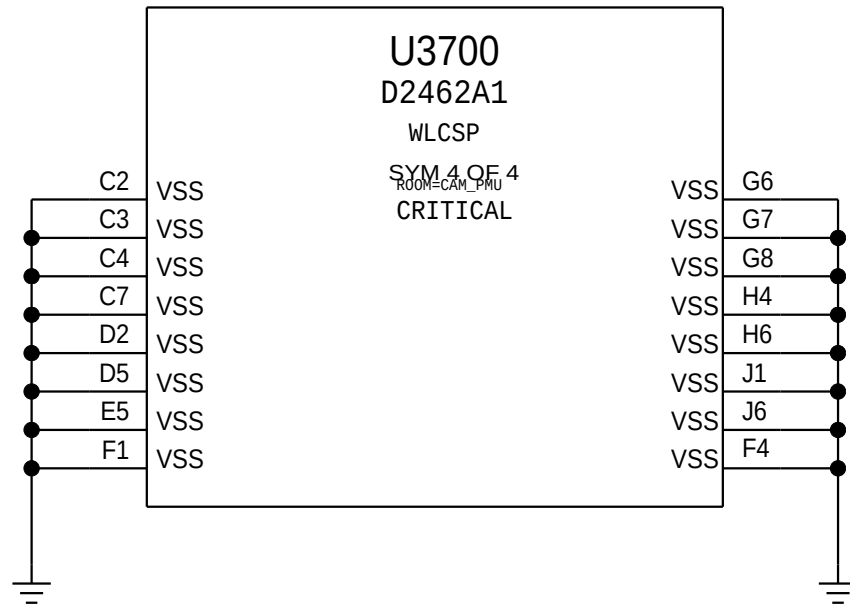
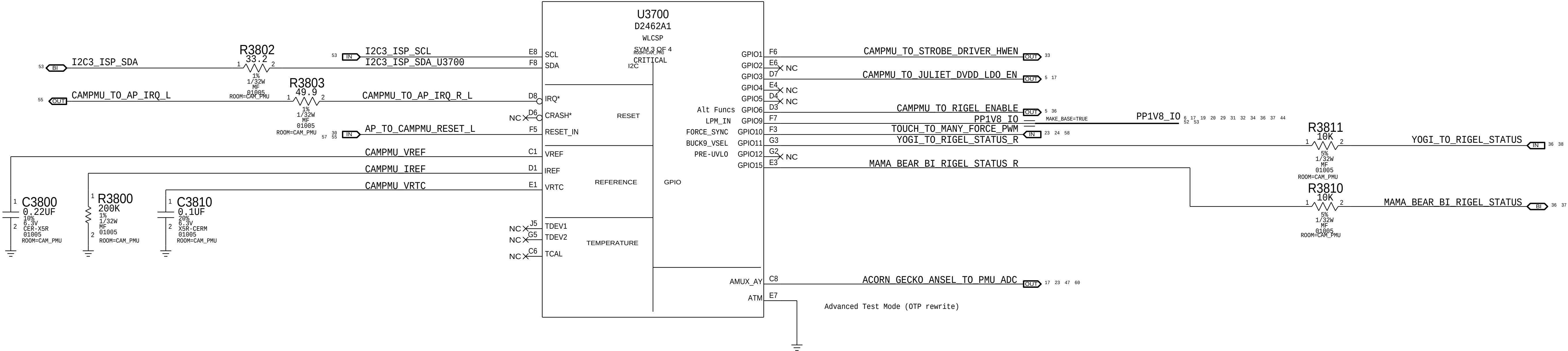
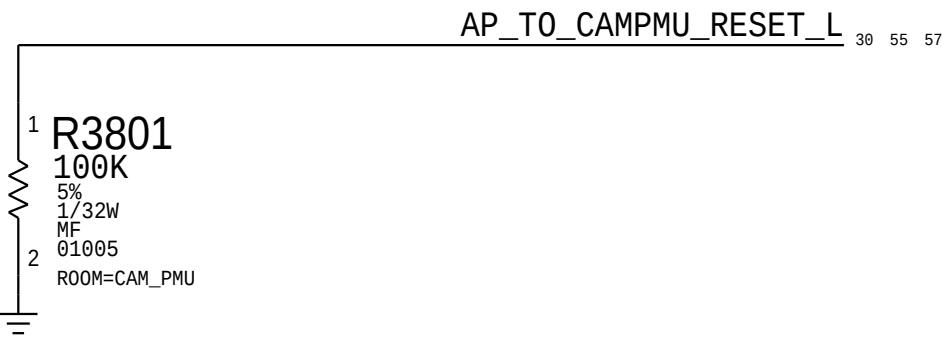
AVDD: Analog Supply (Pixels)
ADC: ADC Supply
DVDD: Digital Supply
SVDD: AF Sensor Supply
PVDD: AF Driver Supply

<---- D3X has discrete Juliet DVDD LDO

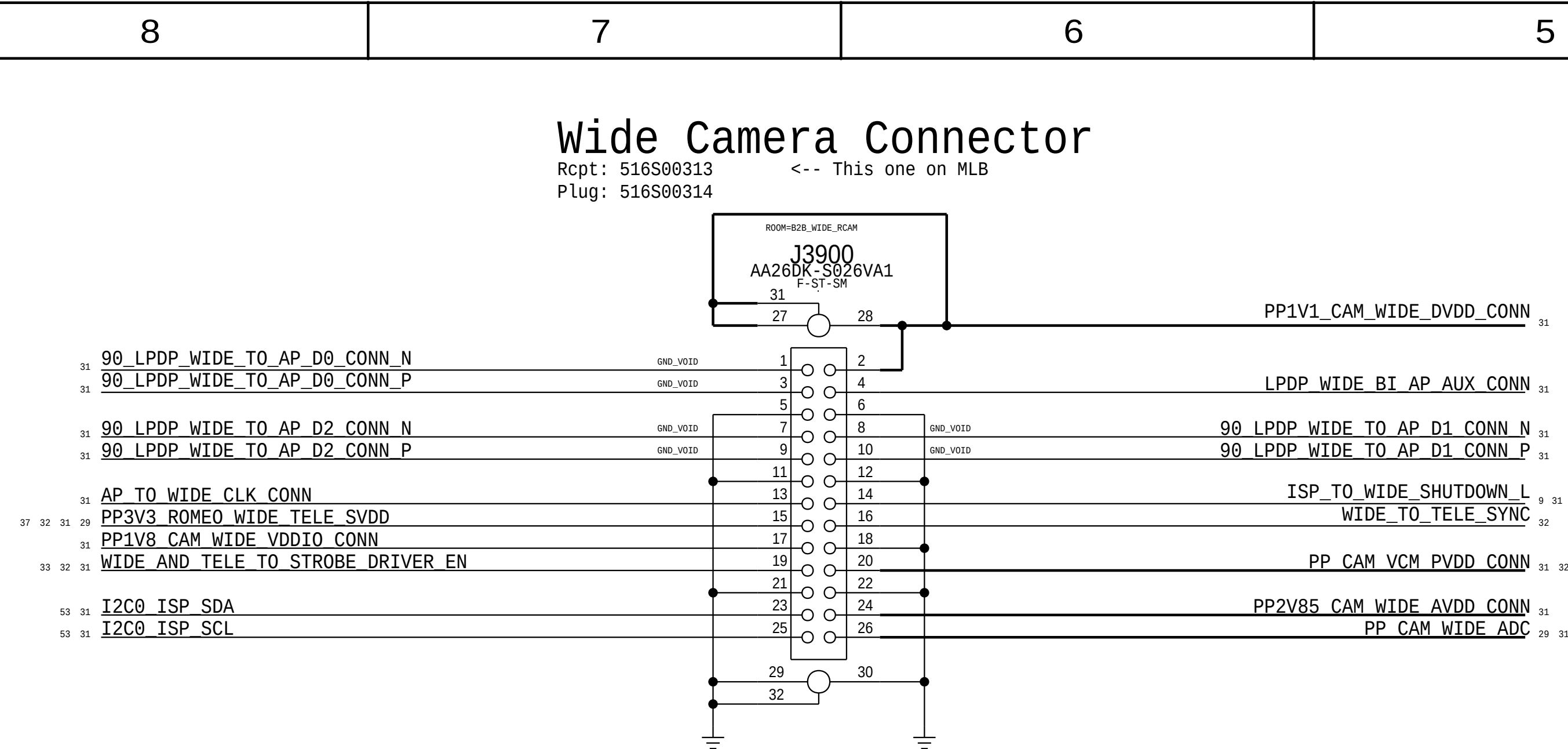
<---- Stability Cap on FF specific page

PAGE TITLE			CAMERA: PMU (1/2)		
	DRAWING NUMBER		051-02545		SIZE
	REVISION		7.0.0		D
	BRANCH				
	PAGE		37 OF 85		
	SHEET		29 OF 60		

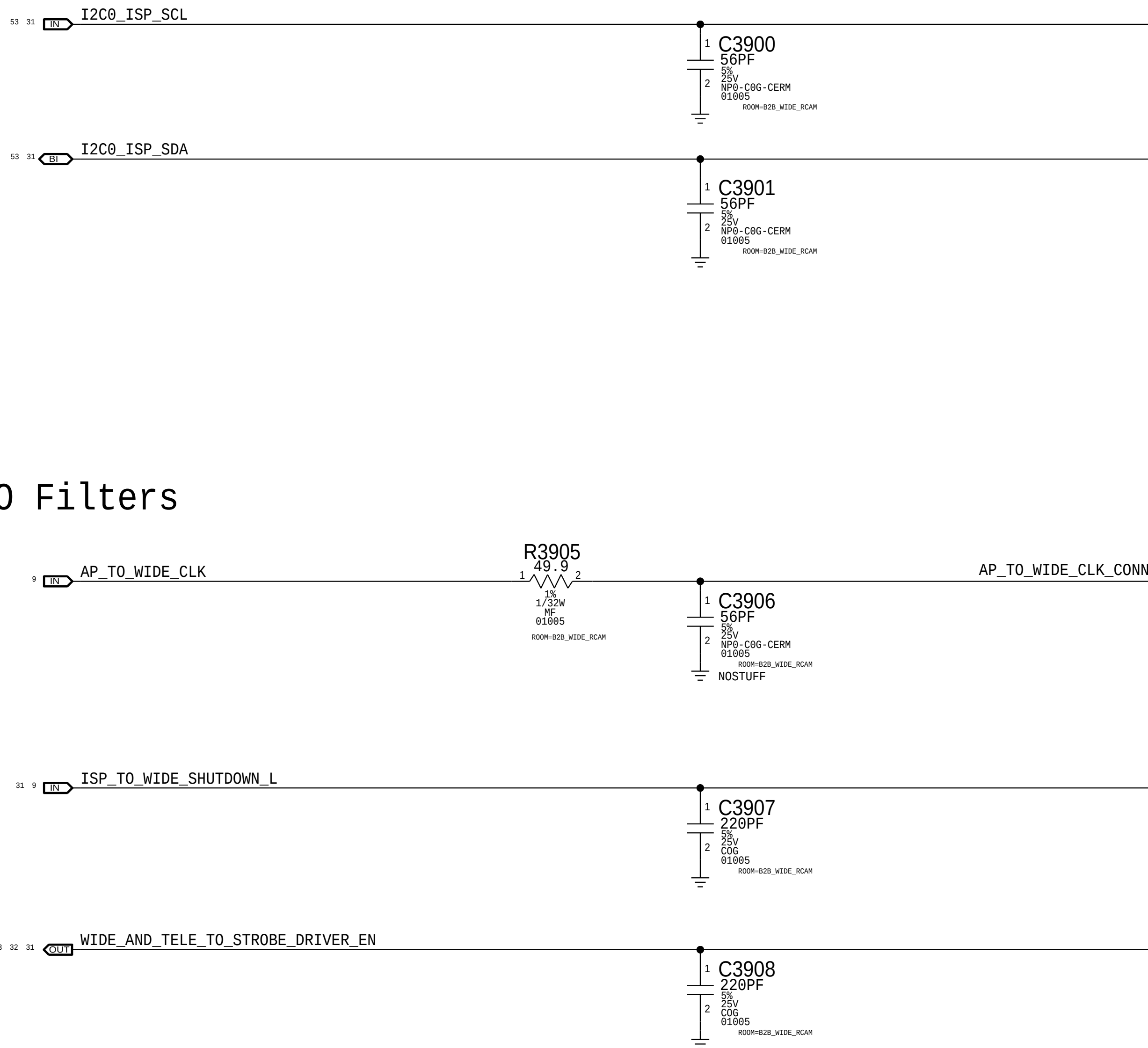
Pull Downs



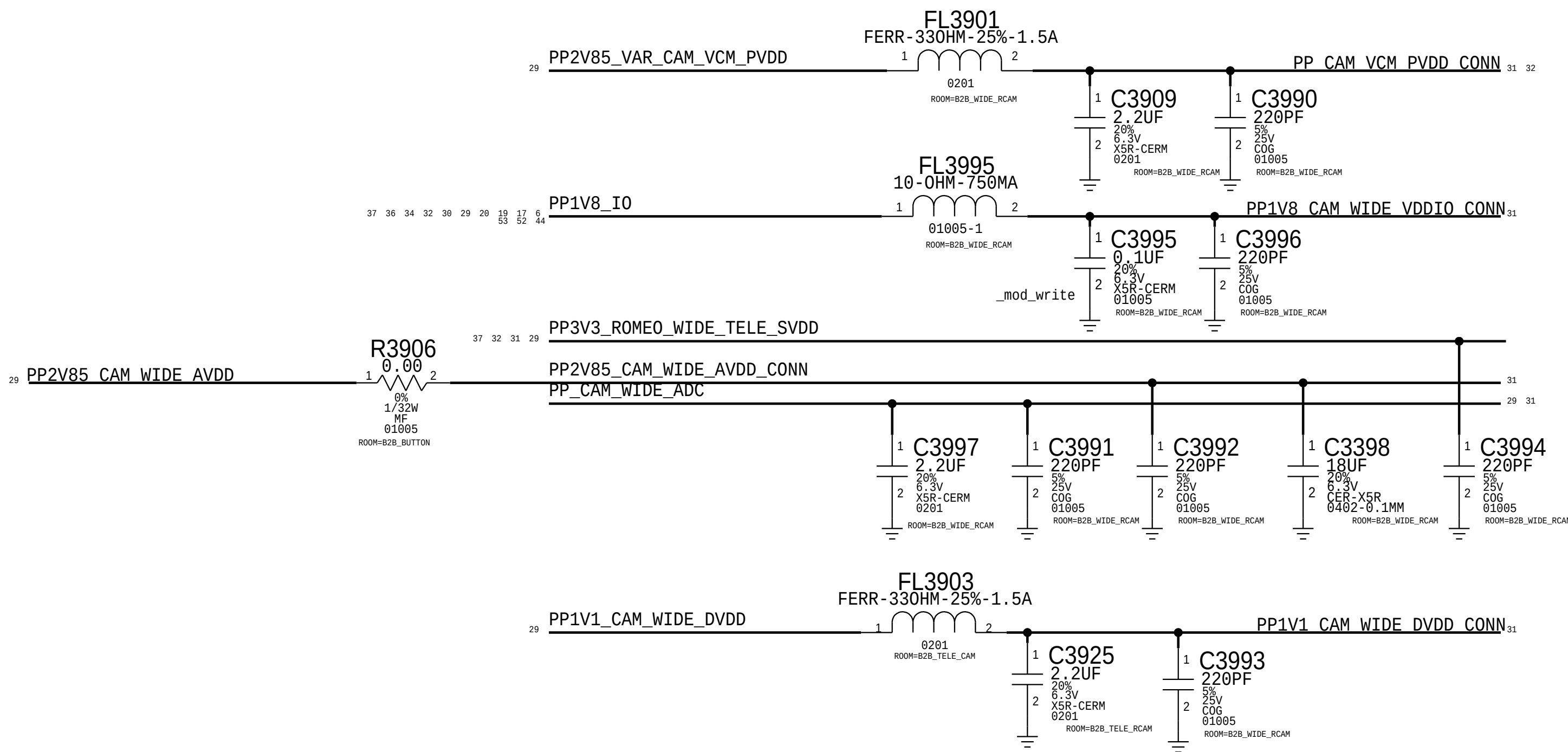
PAGE TITLE		
CAMERA: PMU (2/2)		
	DRAWING NUMBER	051-02545
	SIZE	D
	REVISION	7.0.0
	BRANCH	
	PAGE	38 OF 85
	SHEET	30 OF 60



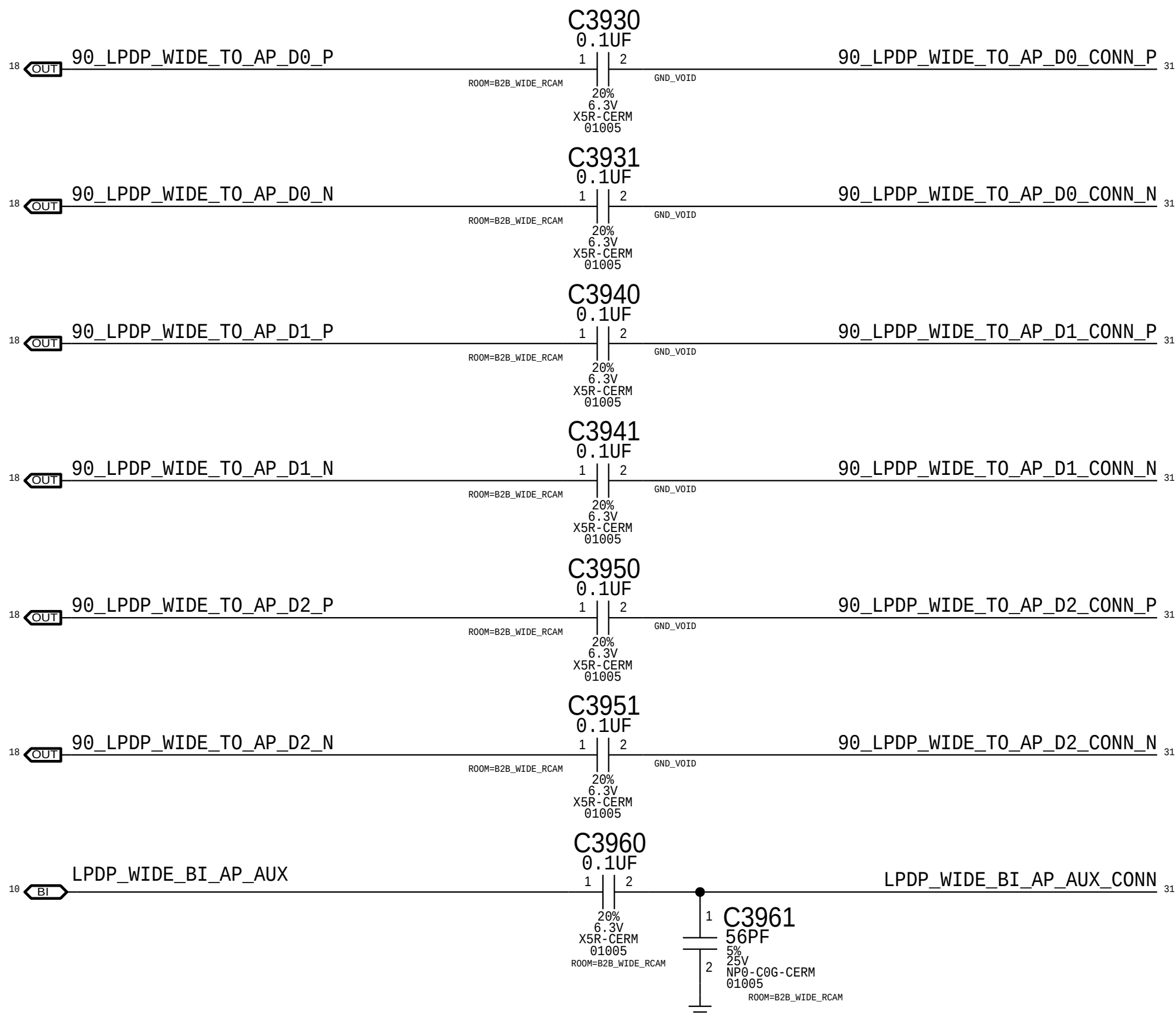
ISP I2C



Power Filtering



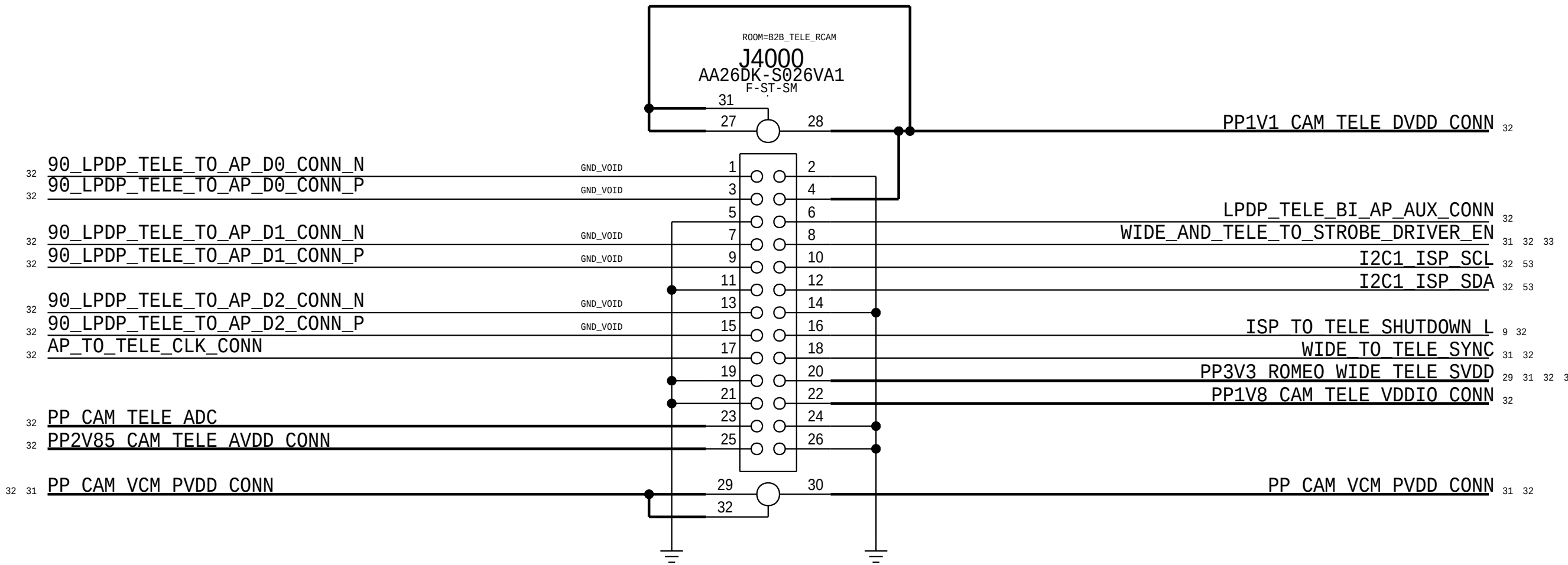
LPDP Filters



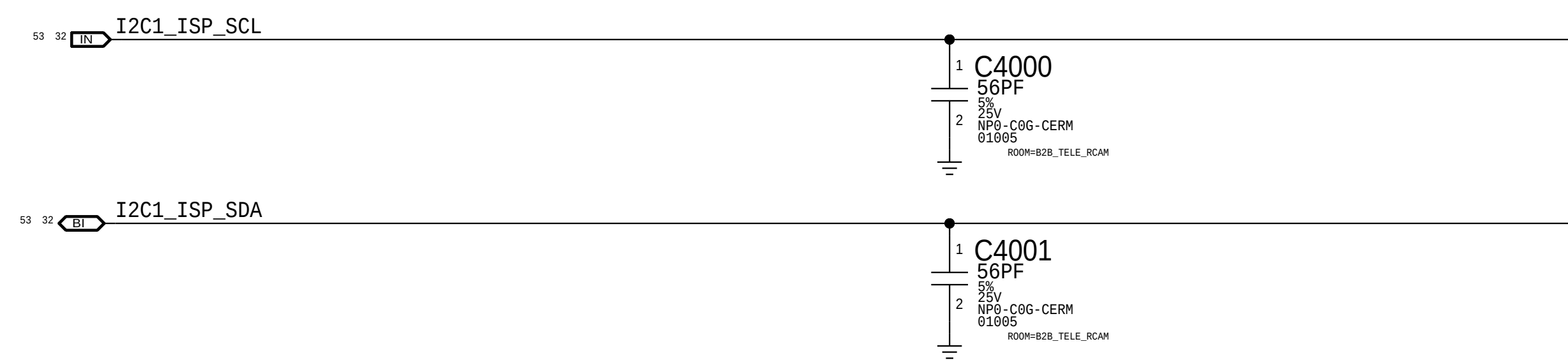
PAGE TITLE			
CAMERA: B2B Wide (TX)			
	DRAWING NUMBER		SIZE
	051-02545		D
	REVISION		
	7.0.0		
	BRANCH		
	PAGE		
	39 OF 85		
	SHEET		
	31 OF 60		

Tele Camera Connector

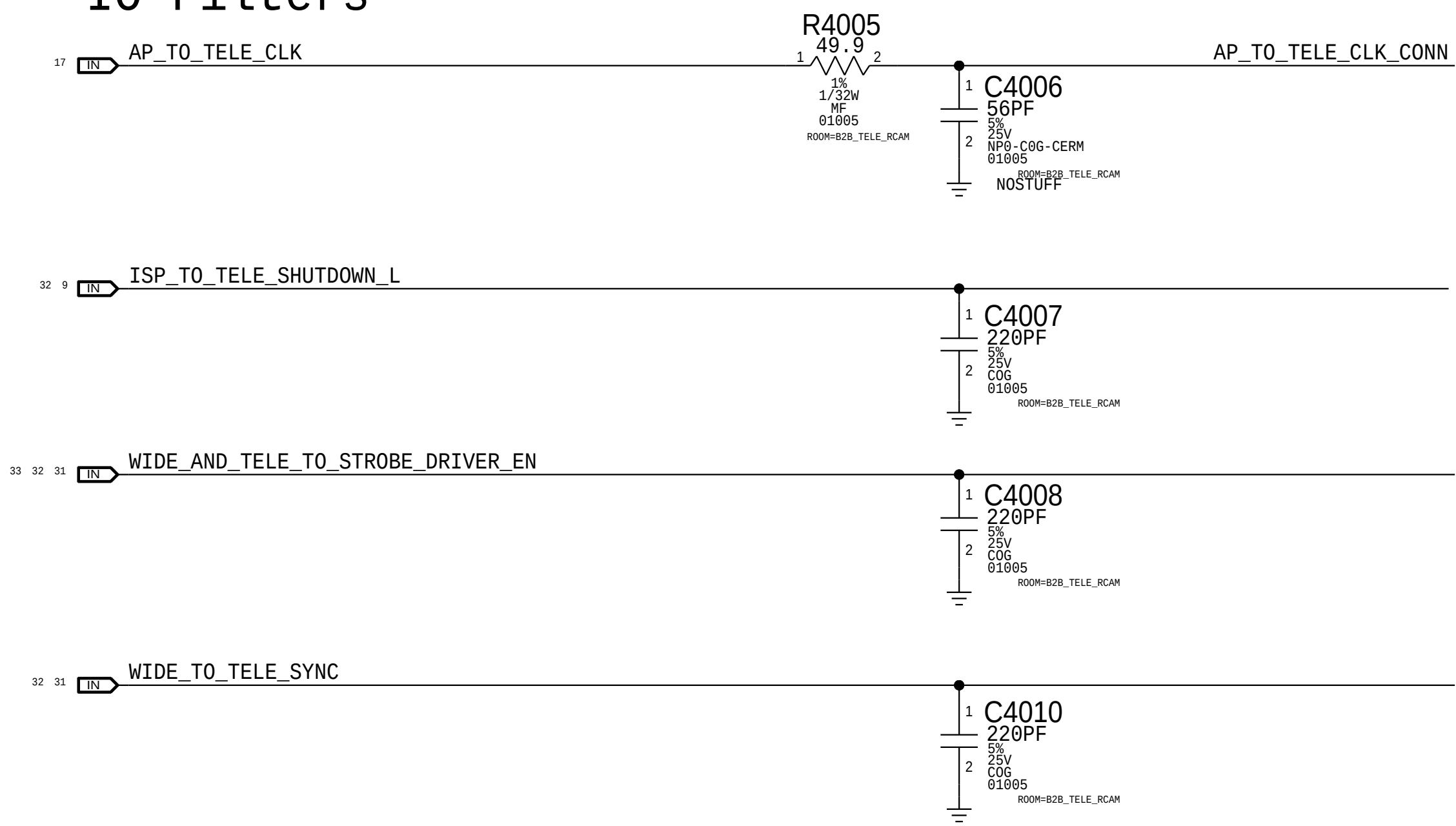
Rcpt: 516S00313 <-- This one on MLB
Plug: 516S00314



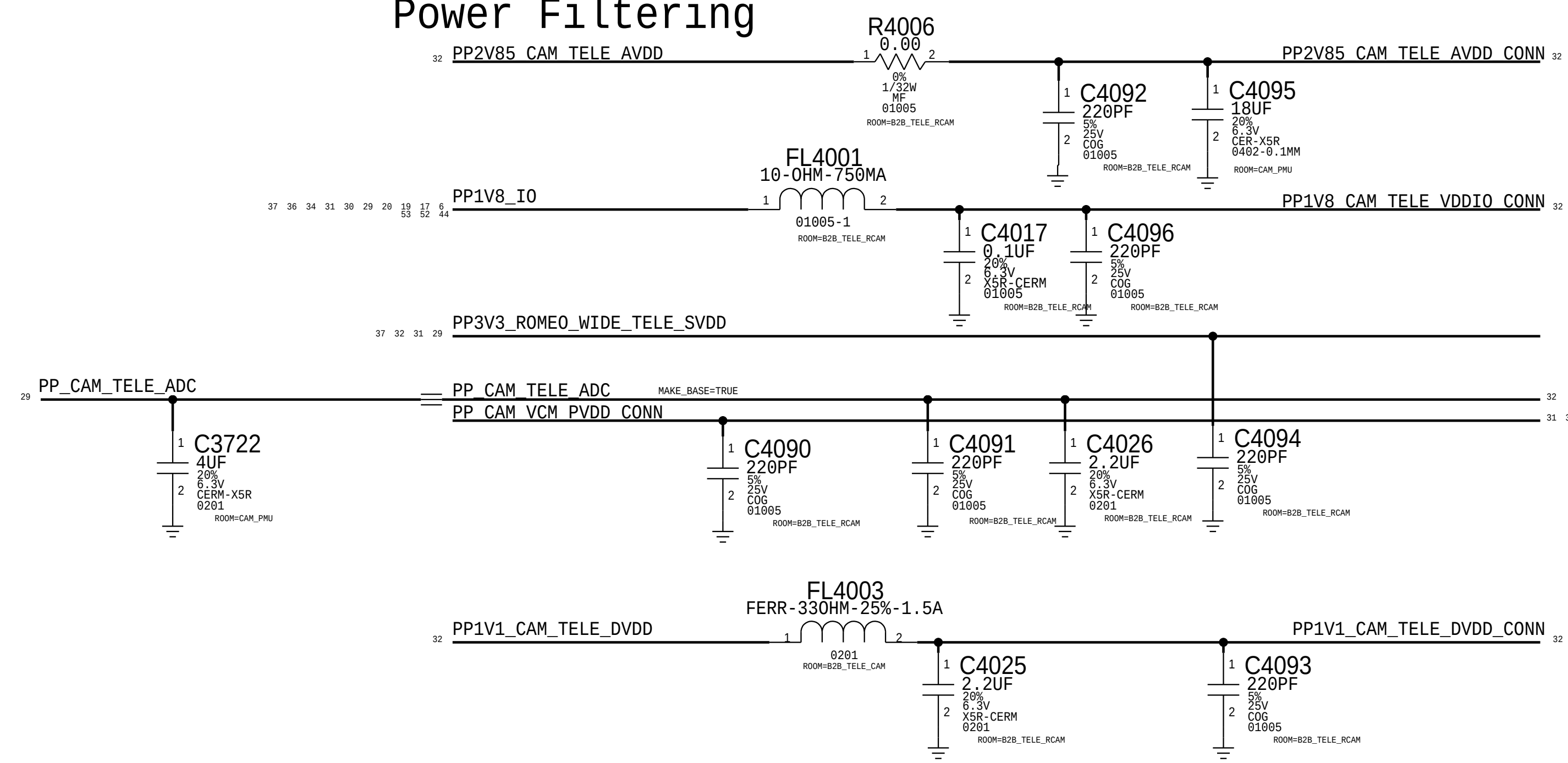
ISP I2C



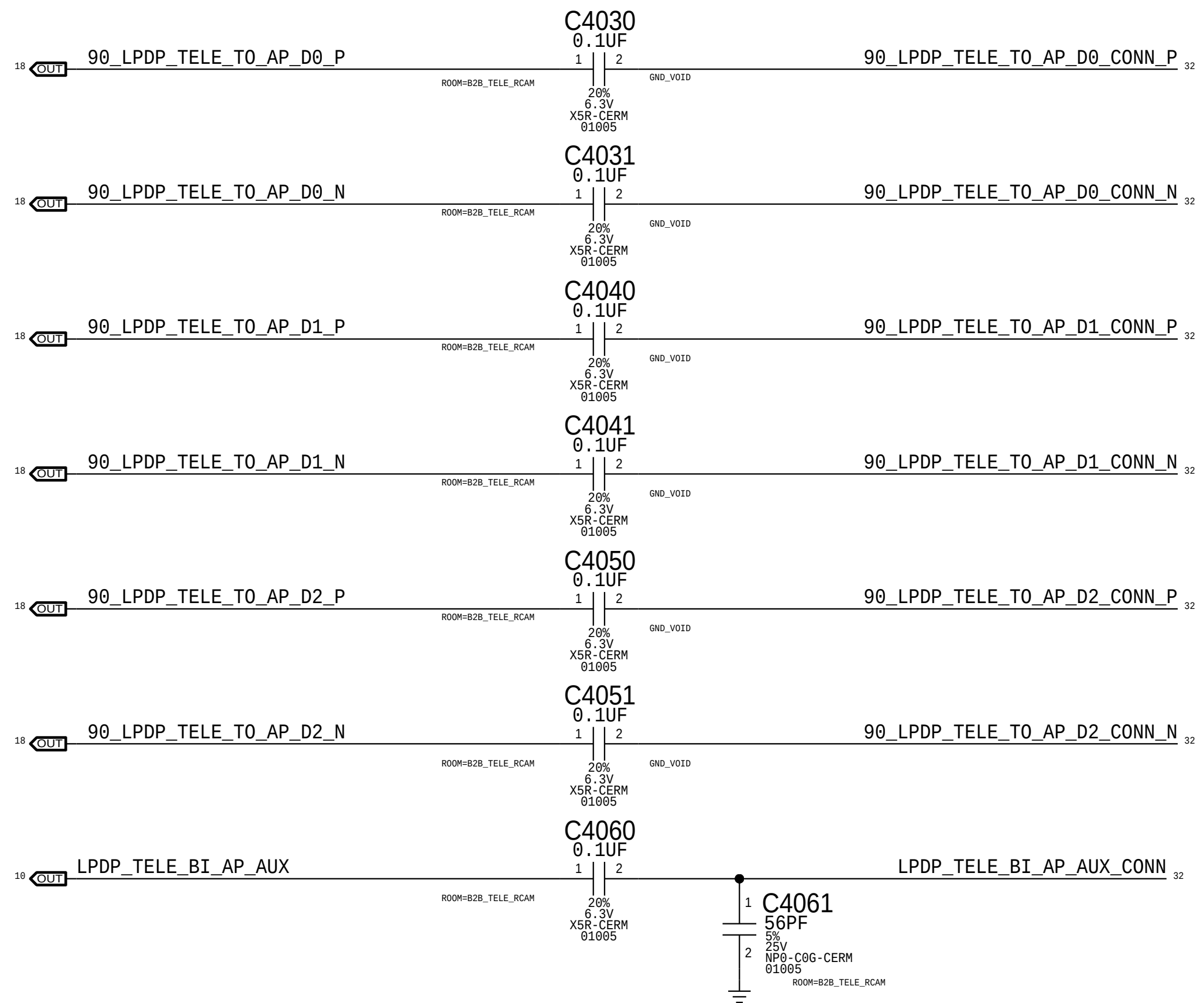
IO Filters



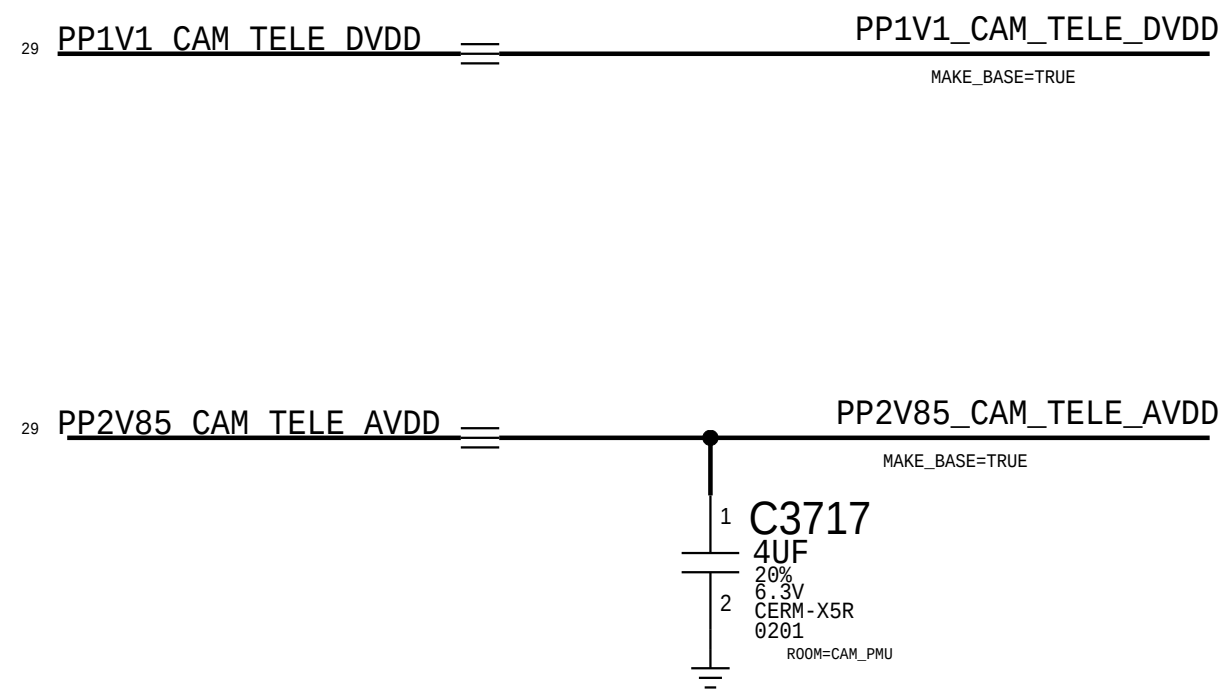
Power Filtering



LPDP



Lives Here For synccng purposes

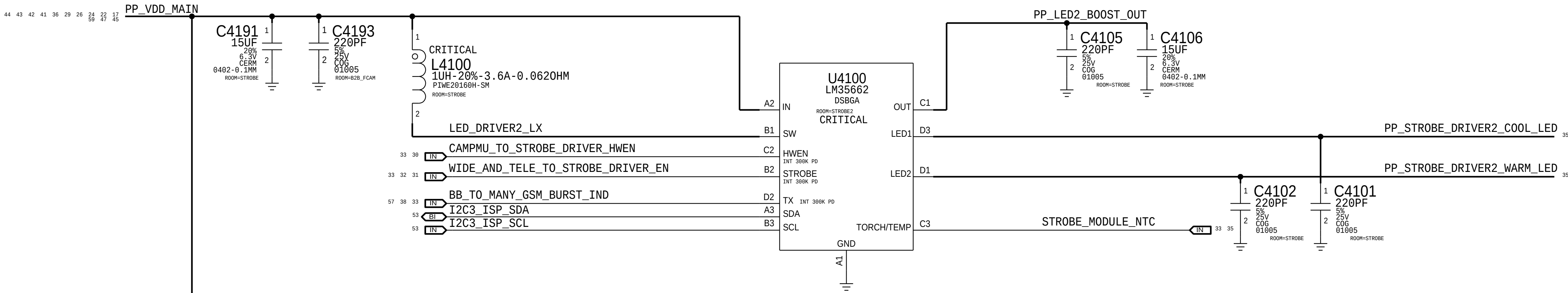


CAMERA: B2B Tele [MT]

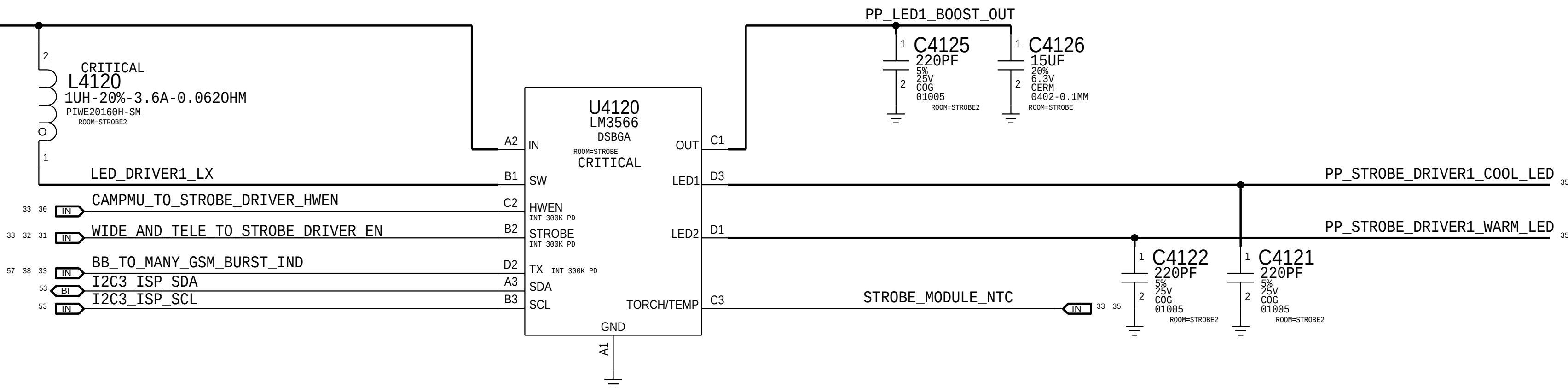
DRAWING NUMBER	051-02545	SIZE	D
REVISION	7.0.0	BRANCH	
PAGE	40 OF 85	SHEET	32 OF 60

LED STROBE DRIVERS (NEON)

APN:353S00868
I2C Address (7-bit): 0x67

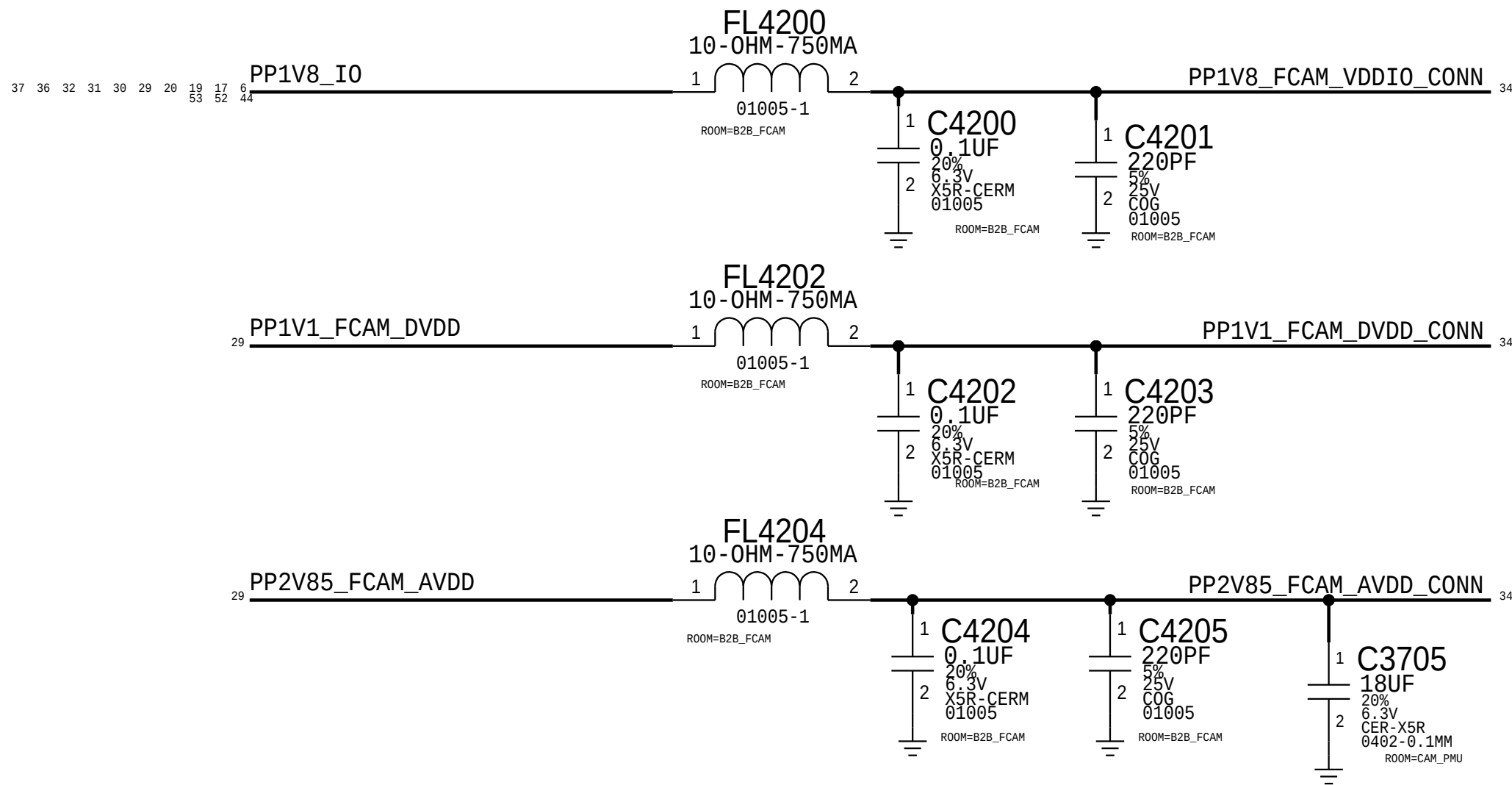


APN:353S00558
I2C Address (7-bit): 0x63



PAGE TITLE		
CAMERA: Strobe Drivers		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	41 OF 85
	SHEET	33 OF 60

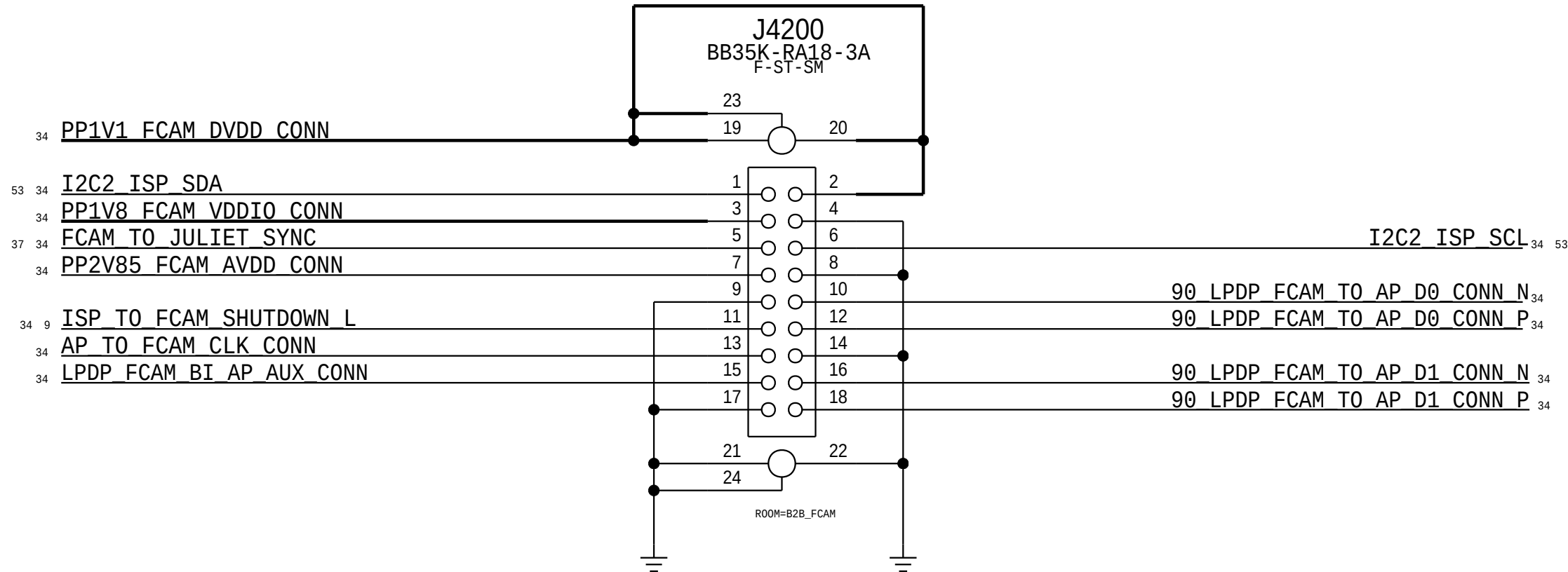
LONG ISLAND POWER



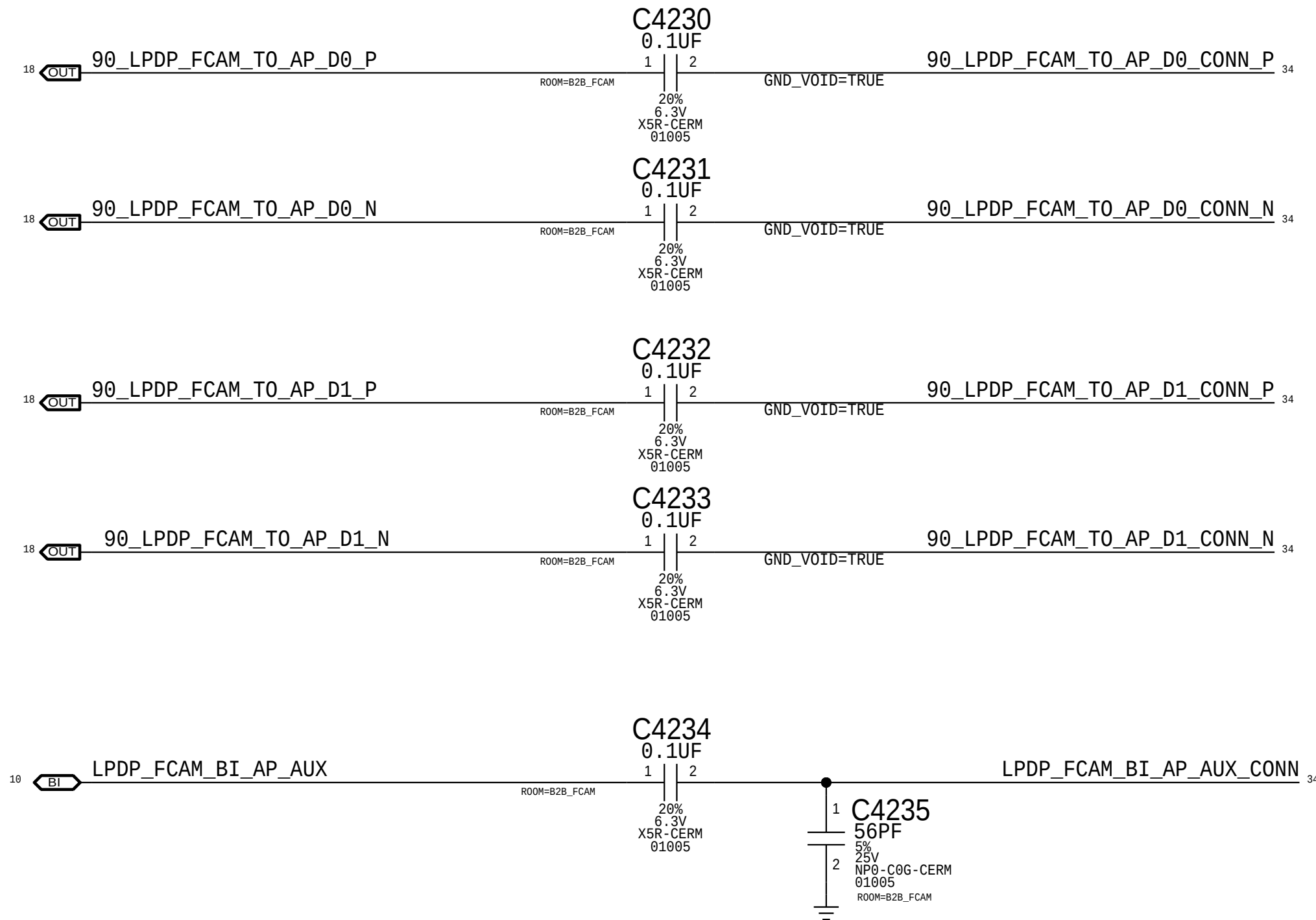
FCAM Connector

Rcpt: 516S00244
Plug: 516S00245

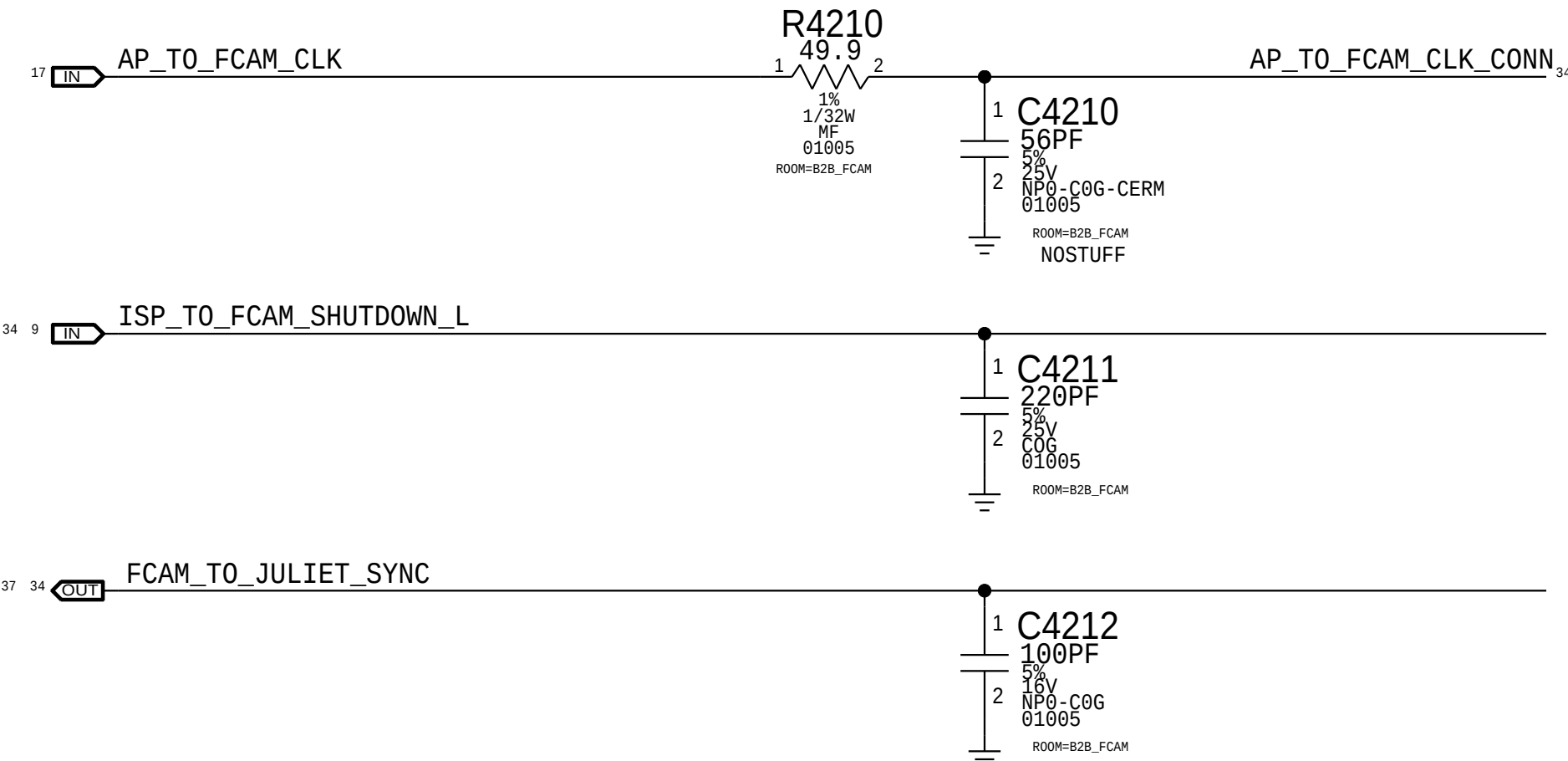
<-- This one on MLB



LPDP FILTERS

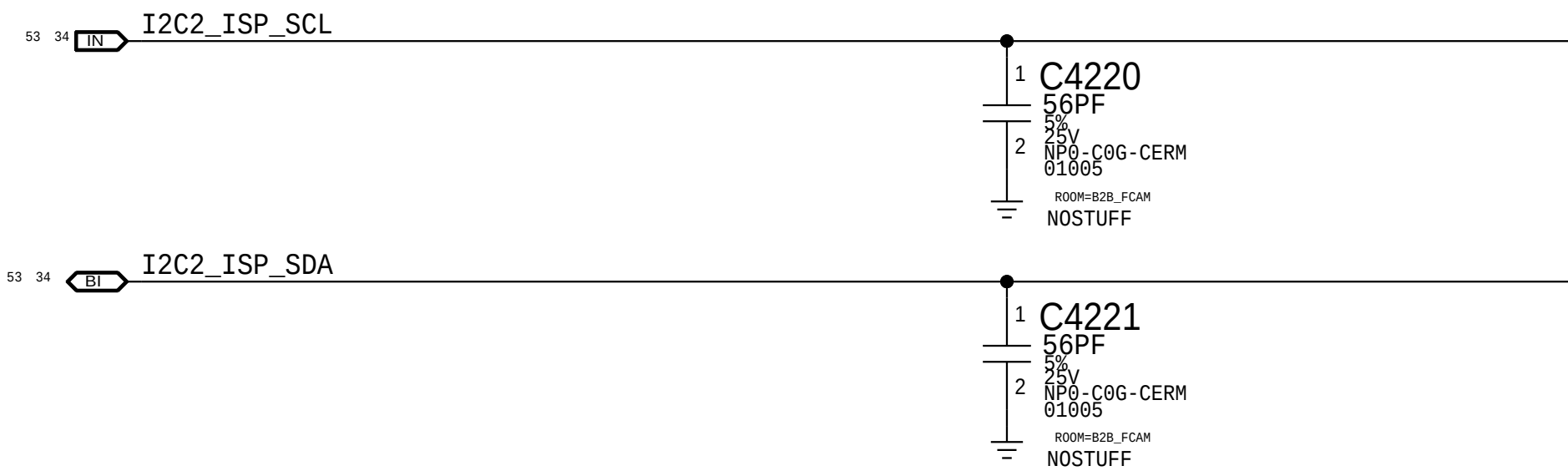


FCAM I/O



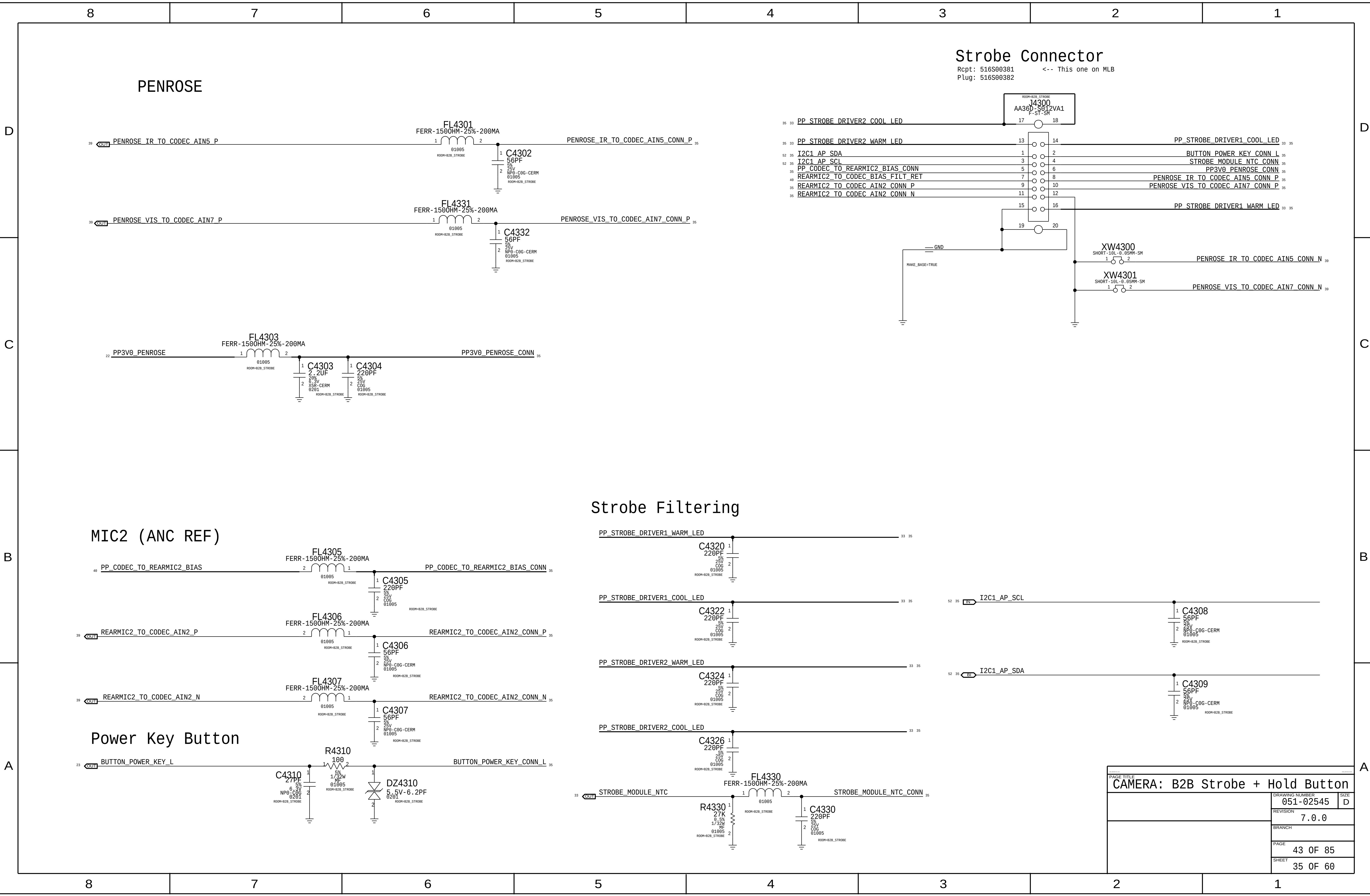
ISP I2C2

NOTE: SAME I2C as FCAM



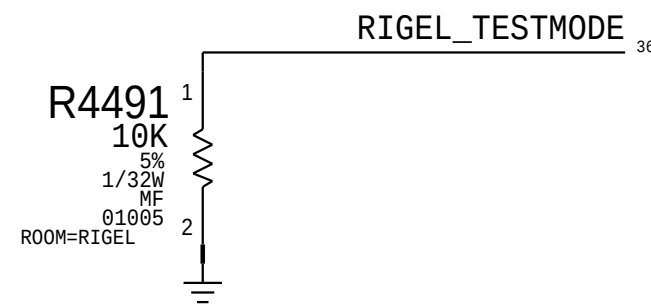
CAMERA: B2B Fcam

DRAWING NUMBER	051-02545	SIZE	D
REVISION	7.0.0	BRANCH	
PAGE	42 OF 85	SHEET	34 OF 60



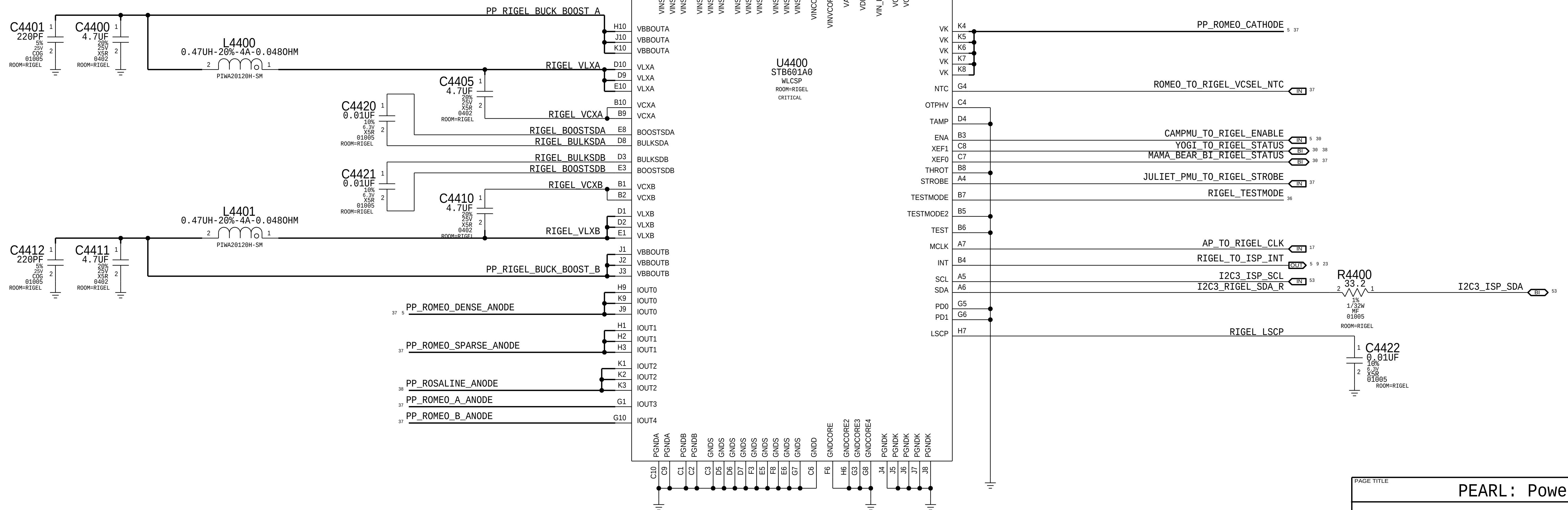
Rigel Driver

Test Mode Debugging



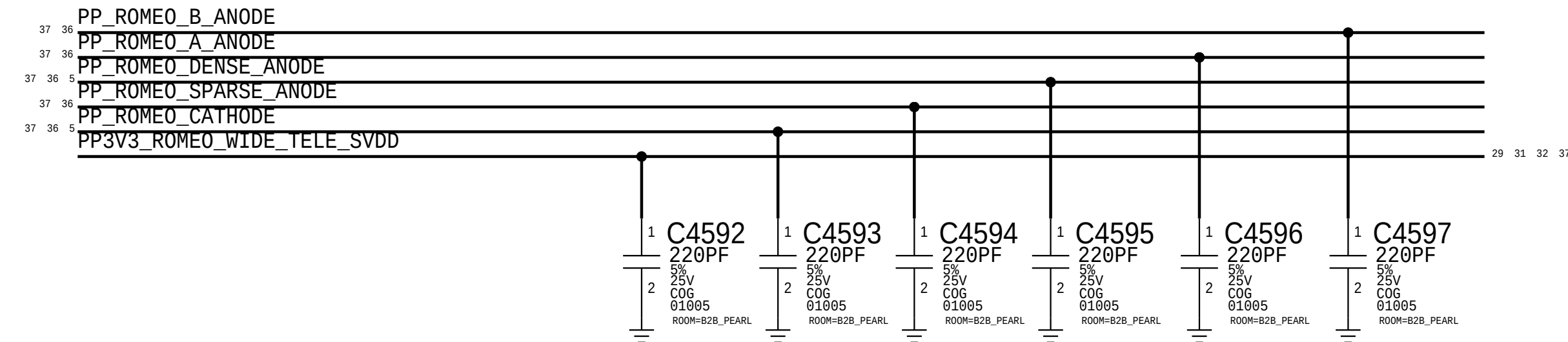
Rigel ALTs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00720	152S00640	ALT_PARTS	L4400,L4401	RIGEL Inductors

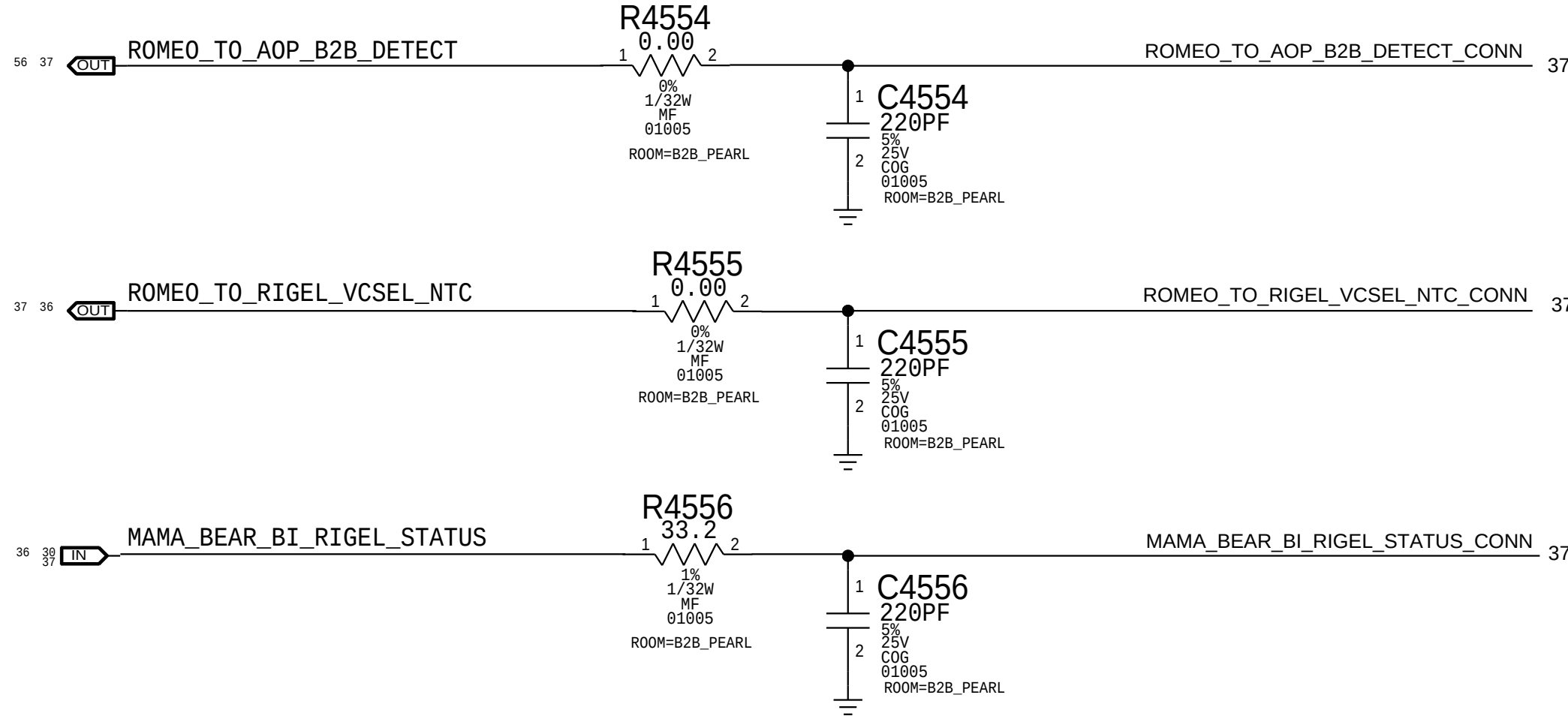


PAGE TITLE		PEARL: Power	
	DRAWING NUMBER	051-02545	SIZE D
	REVISION 7.0.0		
	BRANCH		
	PAGE 44 OF 85		
	SHEET 36 OF 60		

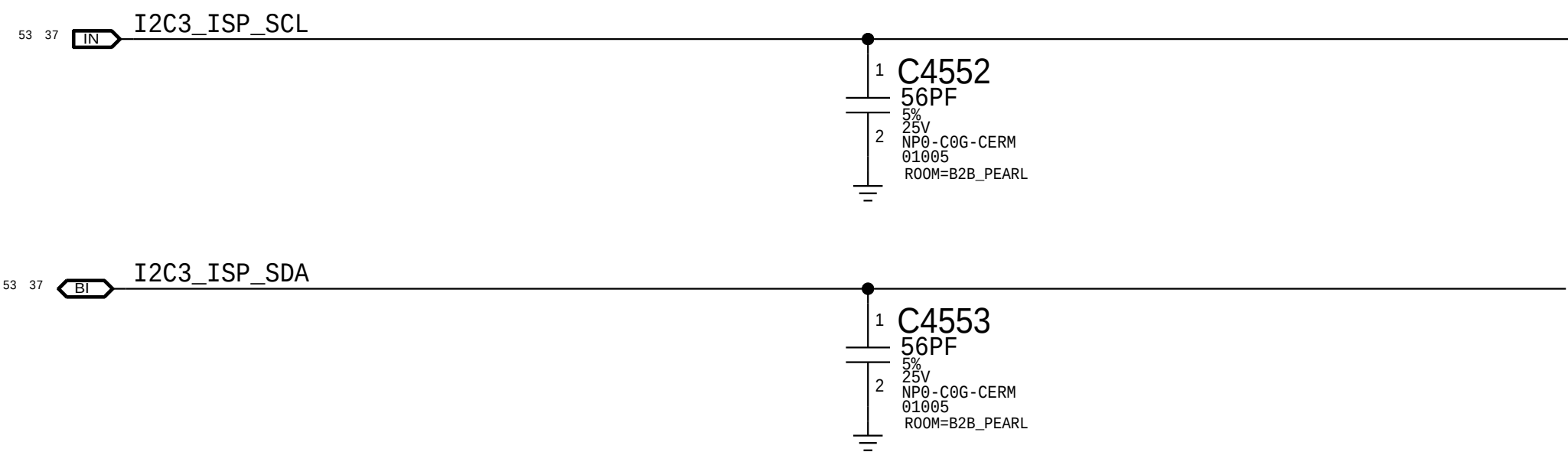
Romeo Power Filtering



Romeo I/O

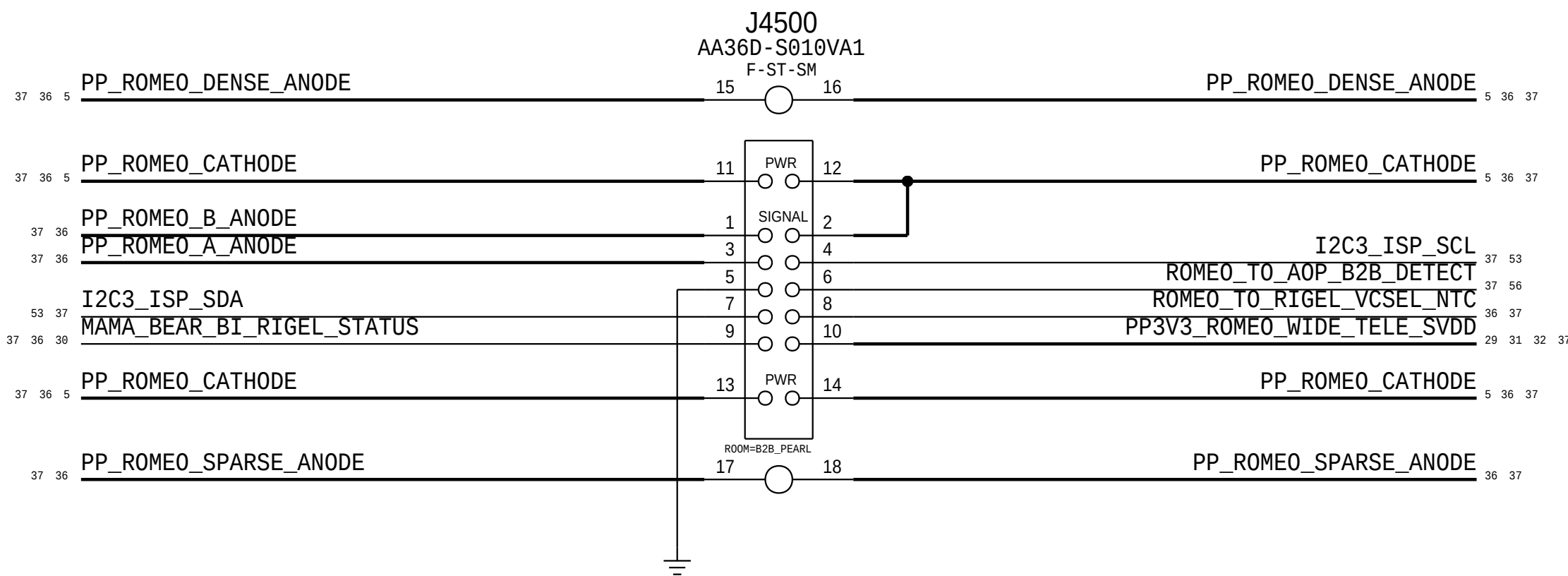


ISP I2C3



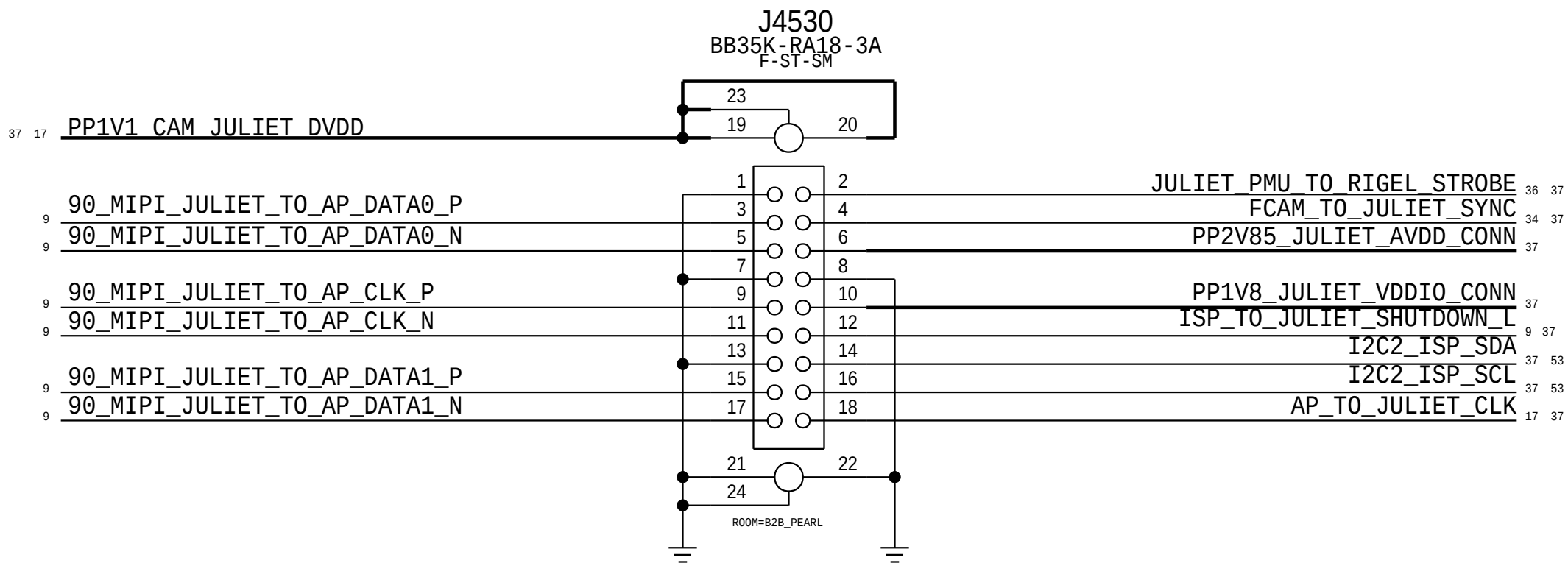
Romeo Connector

Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268

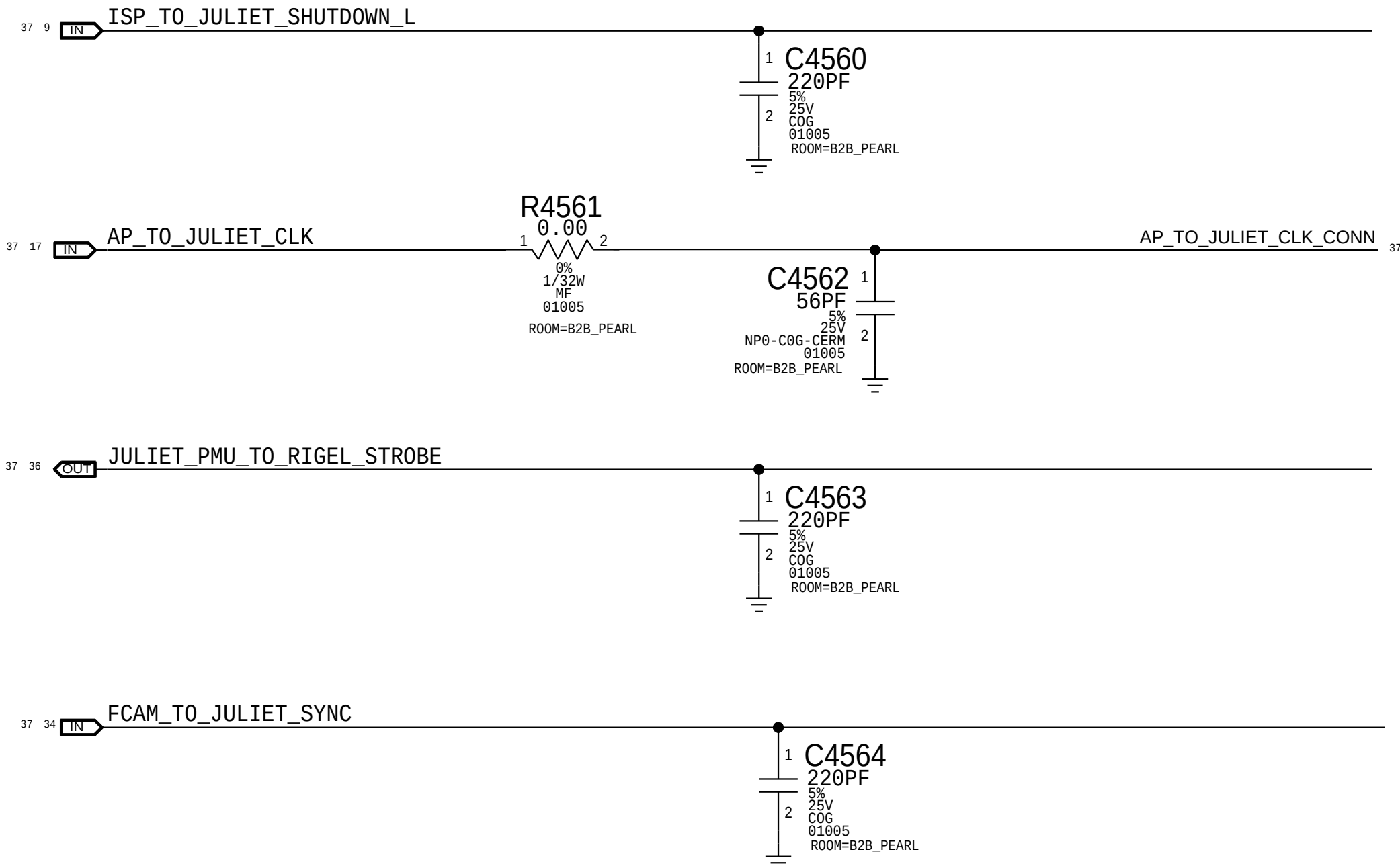
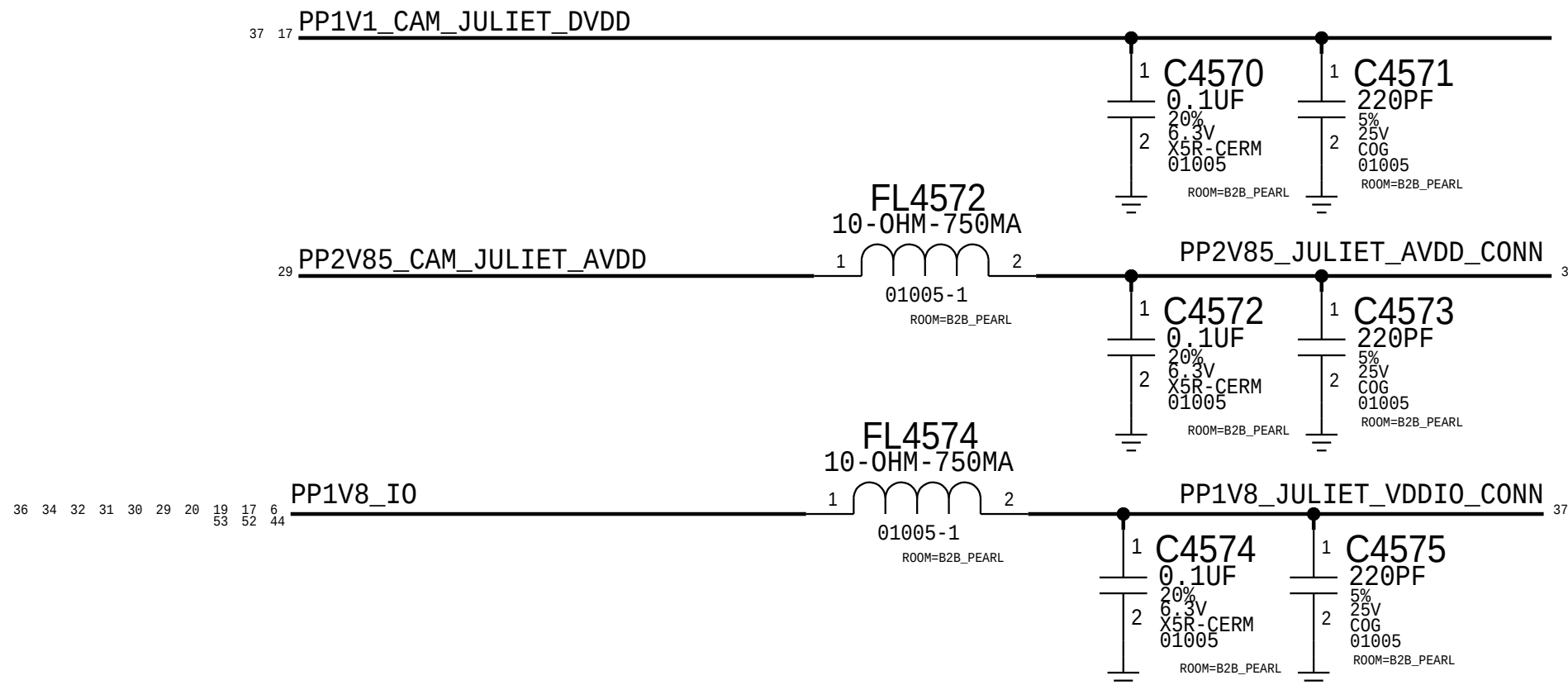


Juliet Connector

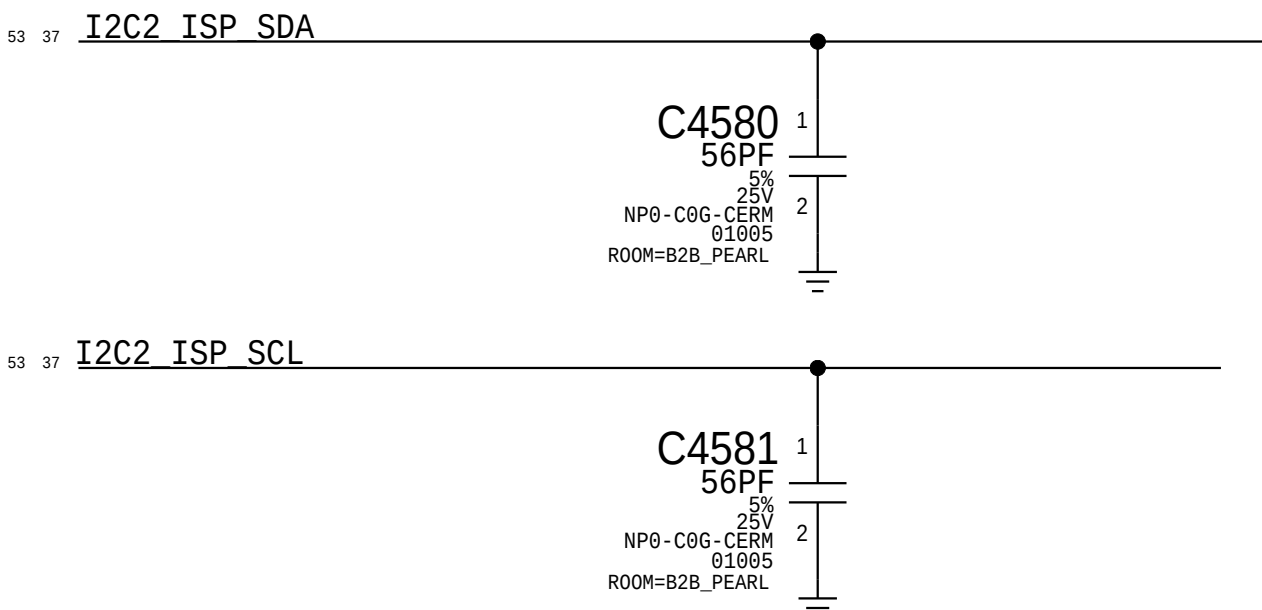
Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245



Juliet Power and I/O

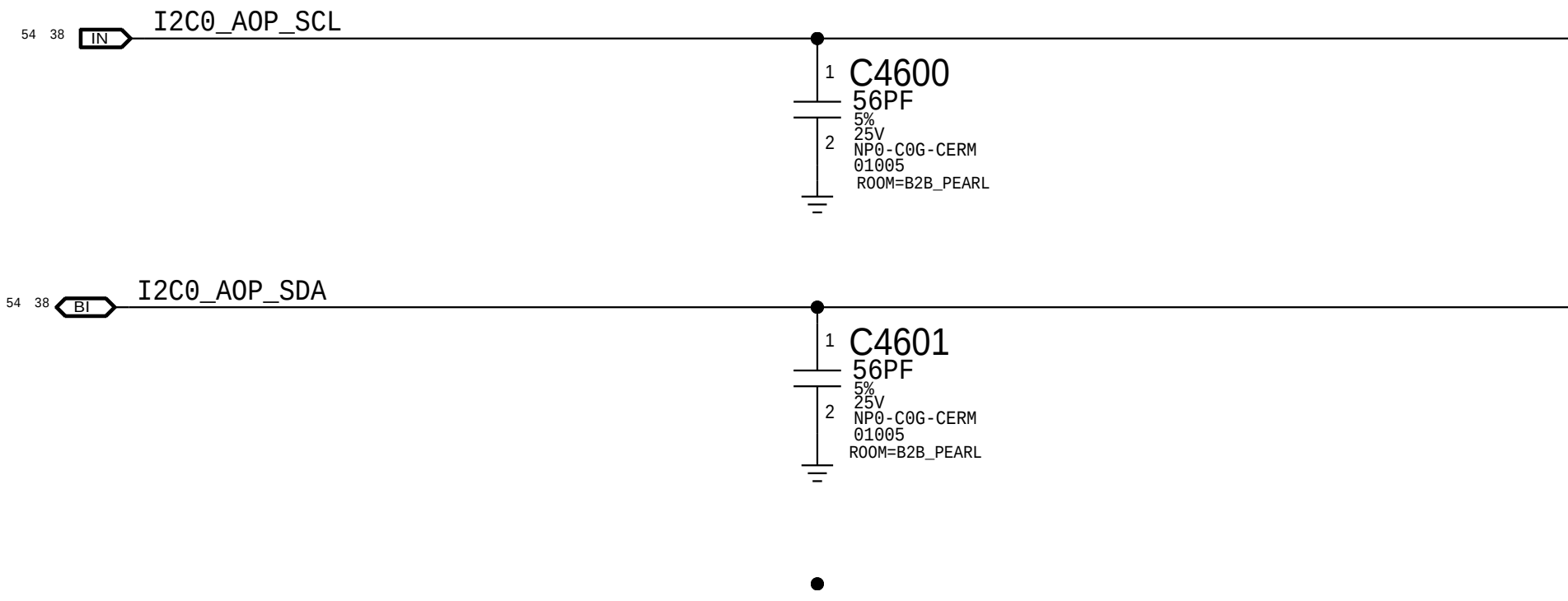


NOTE: SAME I2C as FCAM

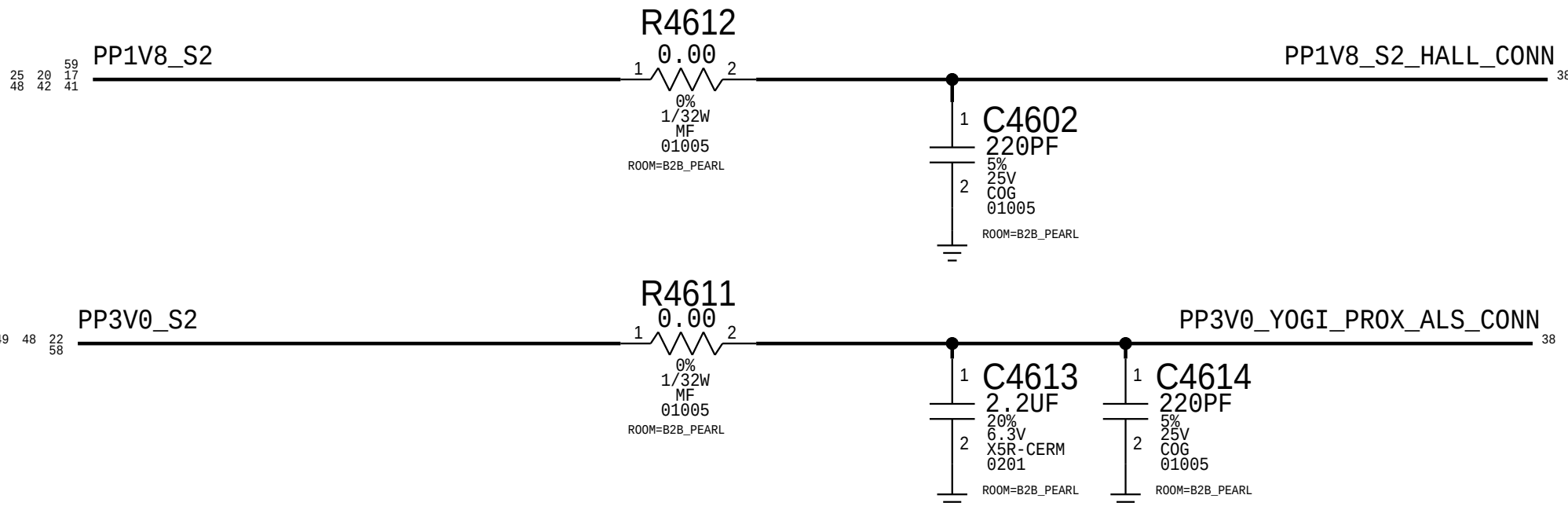


PAGE TITLE		
PEARL: B2B Romeo + Juliet		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	45 OF 85	
SHEET	37 OF 60	

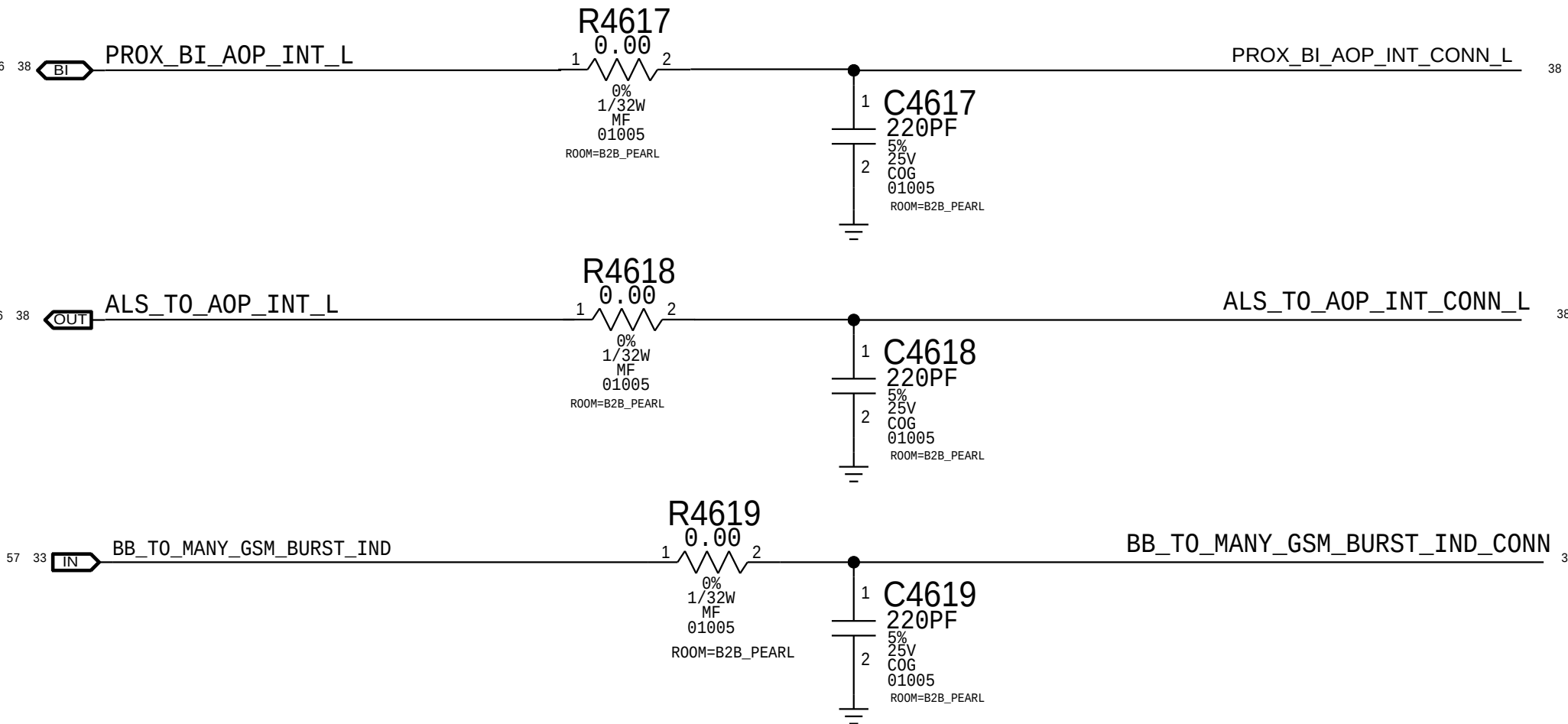
AOP I2C



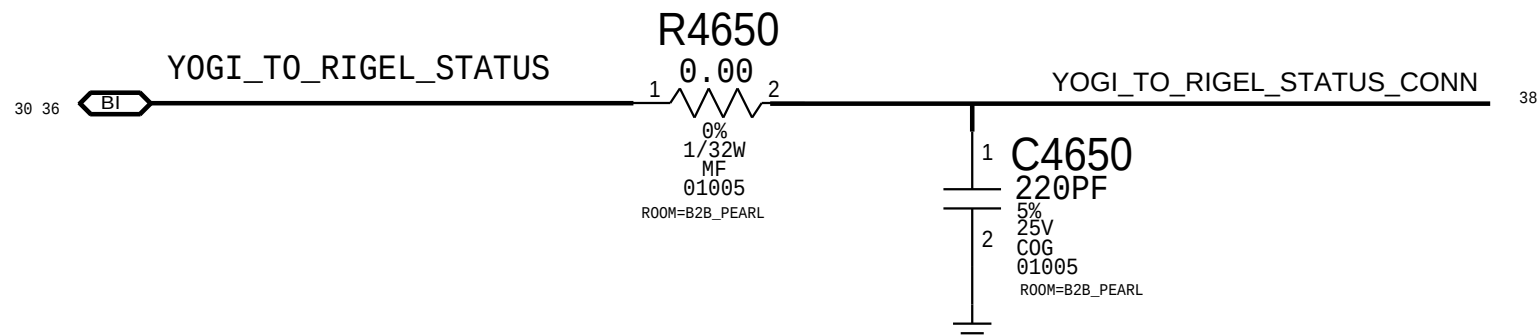
PROX & HALL POWER



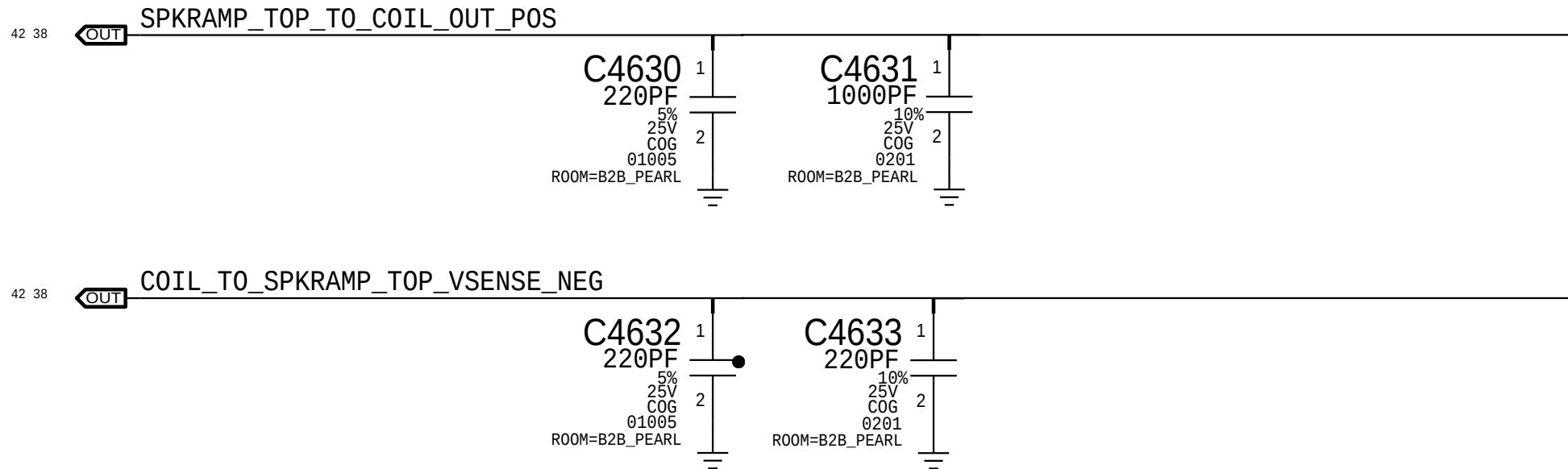
PROX/ALS/YOGI I/O



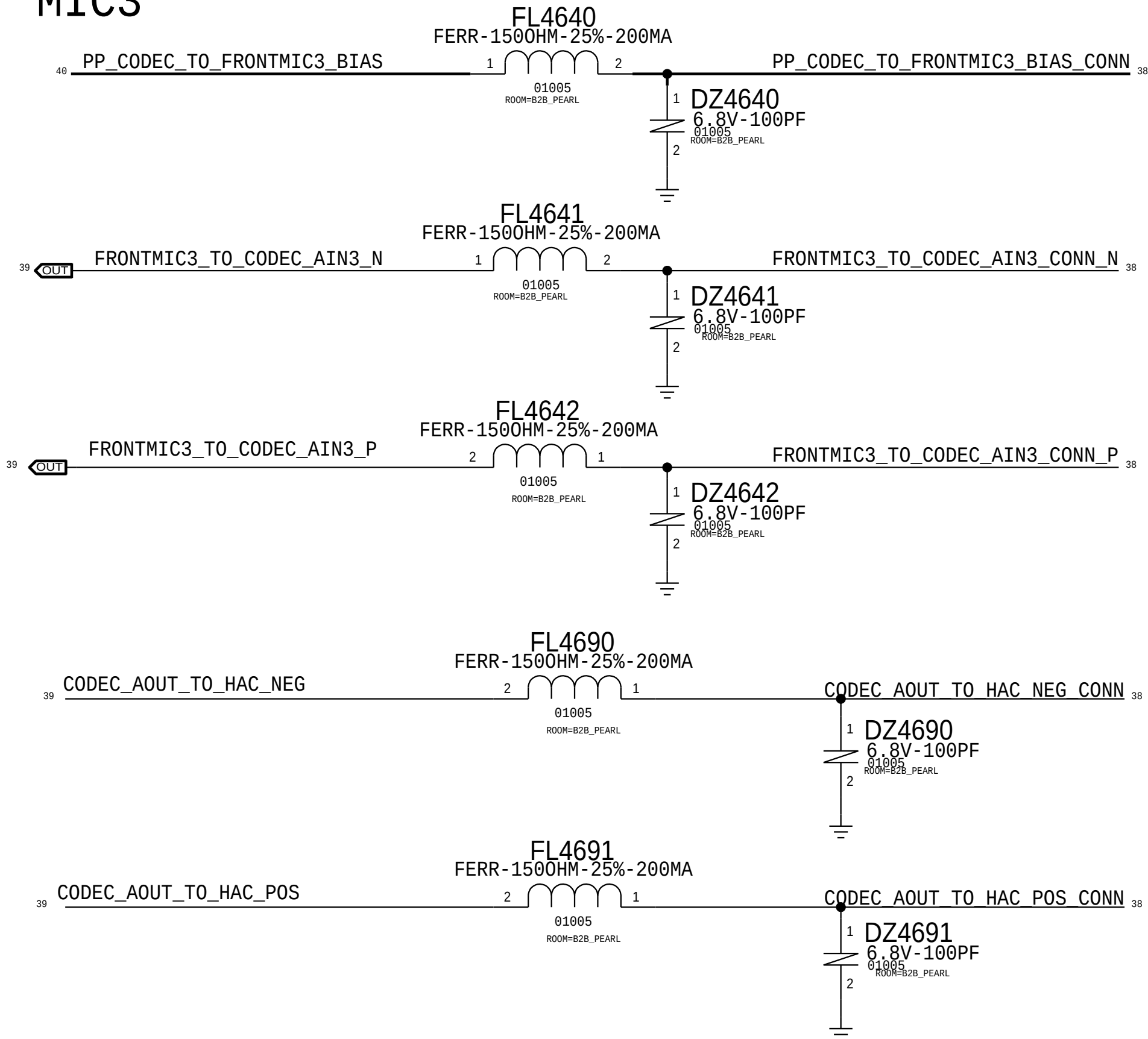
Yogi Signals



SPEAKER2

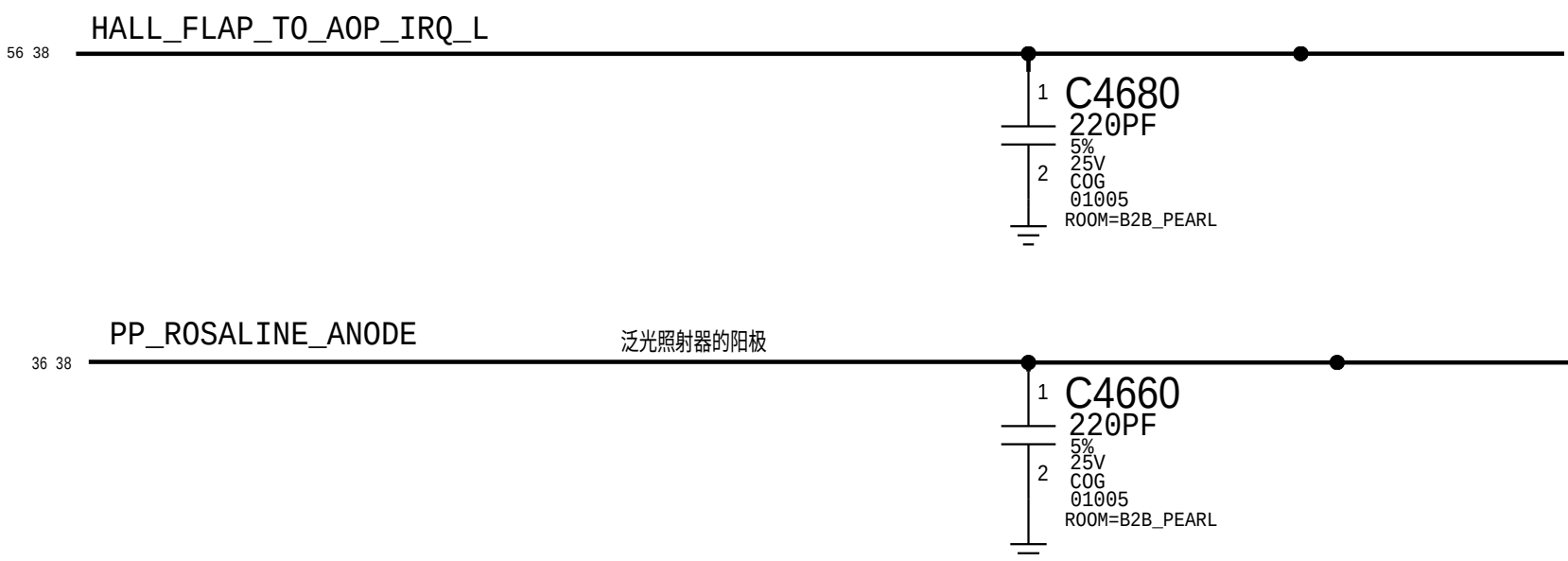
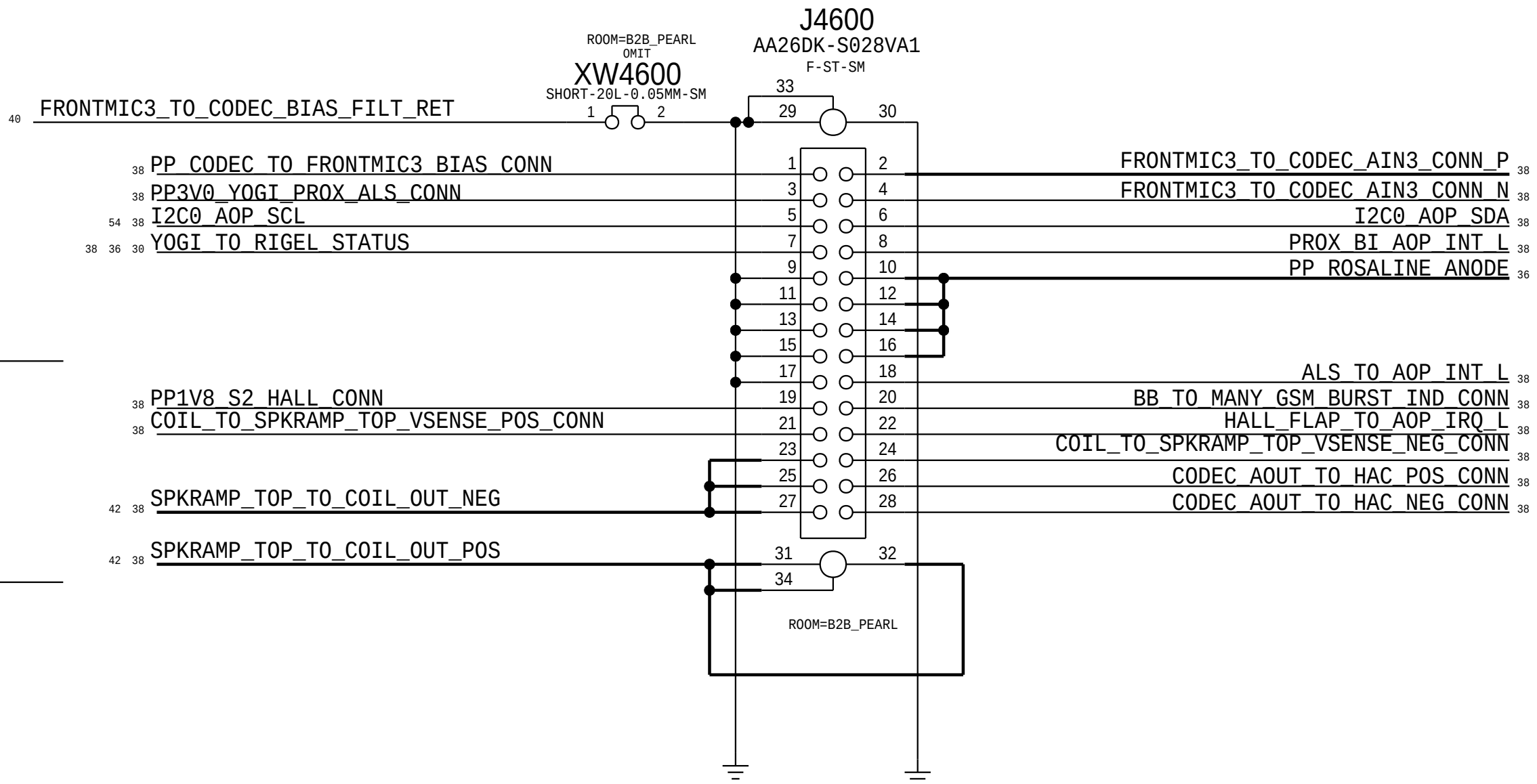


MIC3



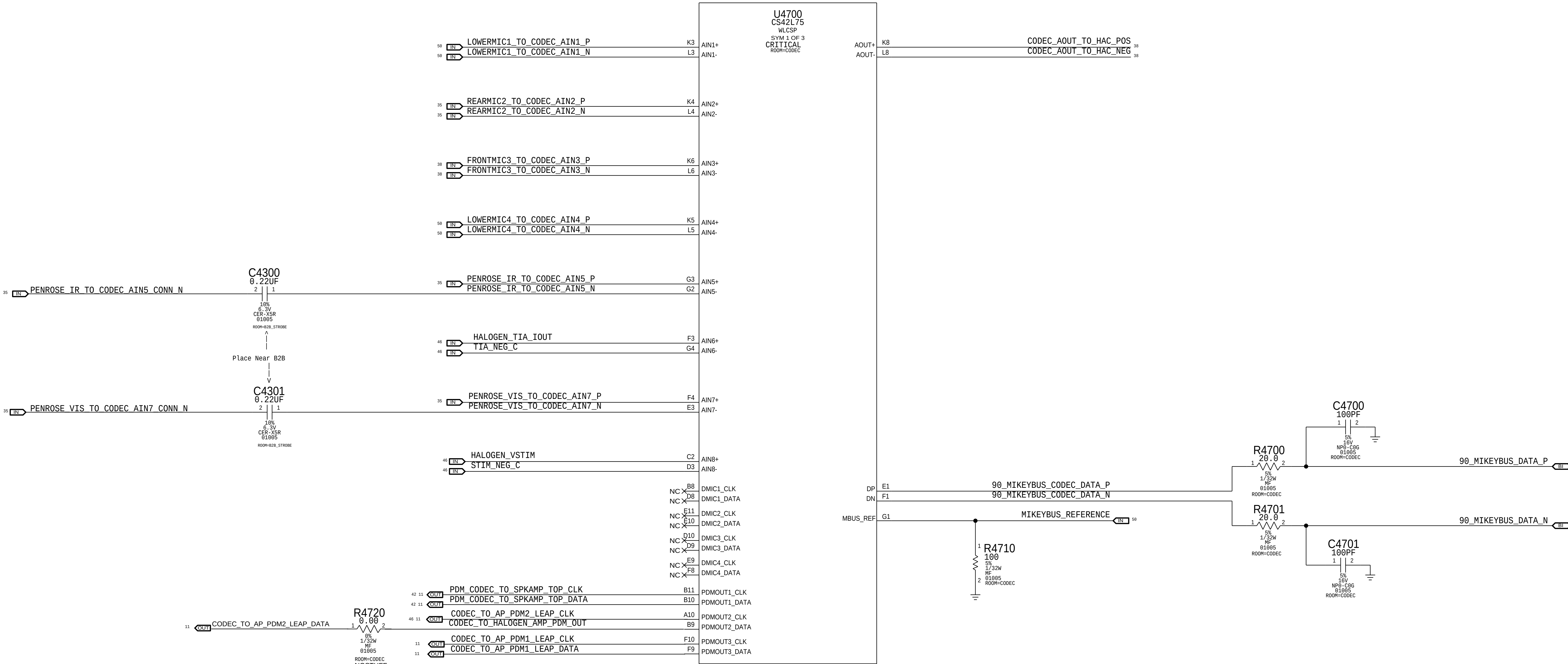
Rosaline + Sensor Connector

Rcpt: 516S00325 <-- This one on MLB
Plug: 516S00326



PAGE TITLE		
PEARL: B2B Rosaline + Sensor		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	46 OF 85	
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CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



PAGE TITLE		
AUDIO: CODEC (1/2)		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE		
47 OF 85		
SHEET		
39 OF 60		

CALLAN AUDIO CODEC (POWER & I/O)

D

C

B

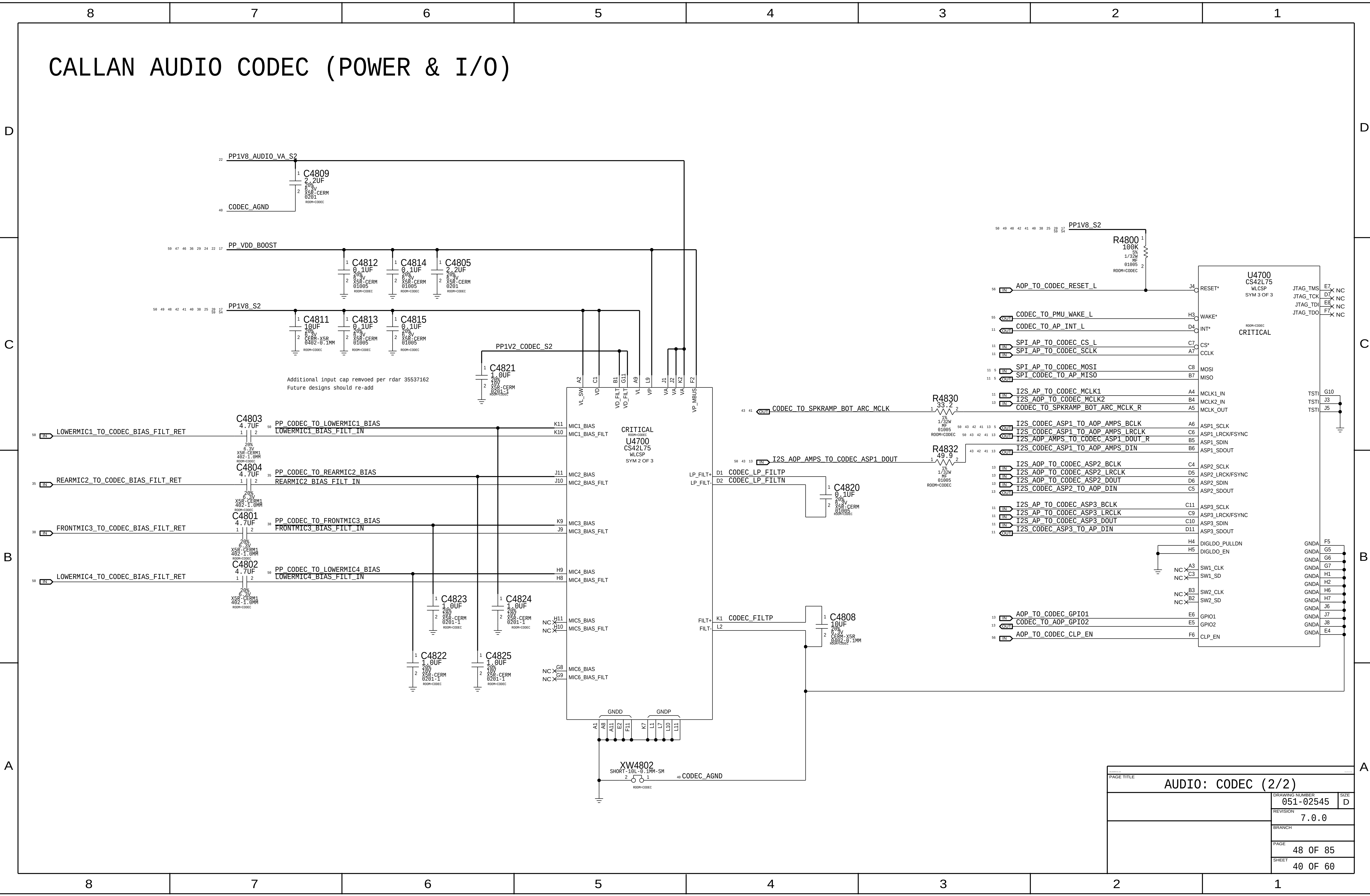
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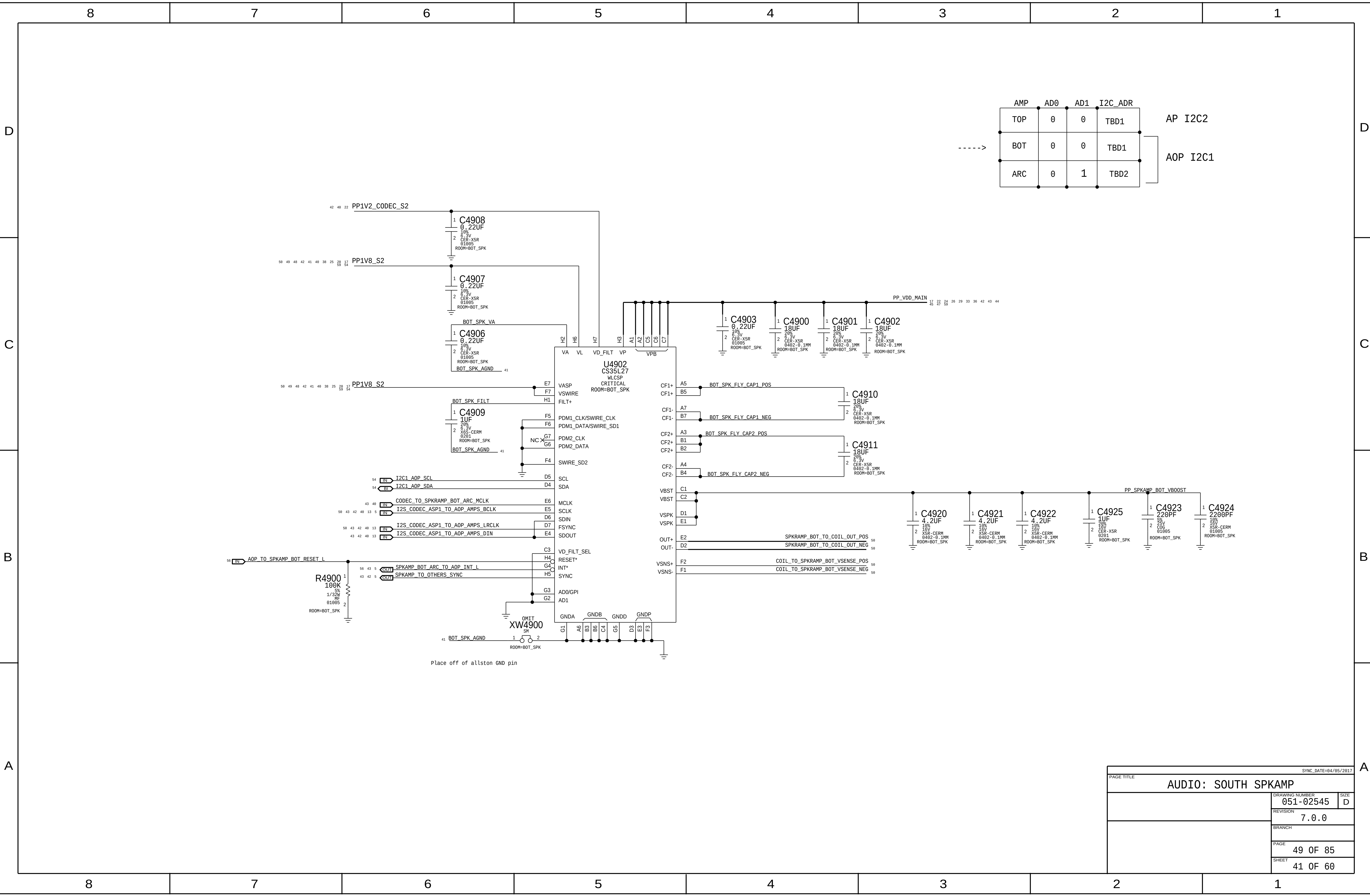
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B

A



PAGE TITLE		
AUDIO: CODEC (2/2)		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	48 OF 85	
SHEET	40 OF 60	



AMP	AD0	AD1	I2C_ADR
TOP	0	0	TBD1
BOT	0	0	TBD1
ARC	0	1	TBD2

AP I2C2

AOP I2C1

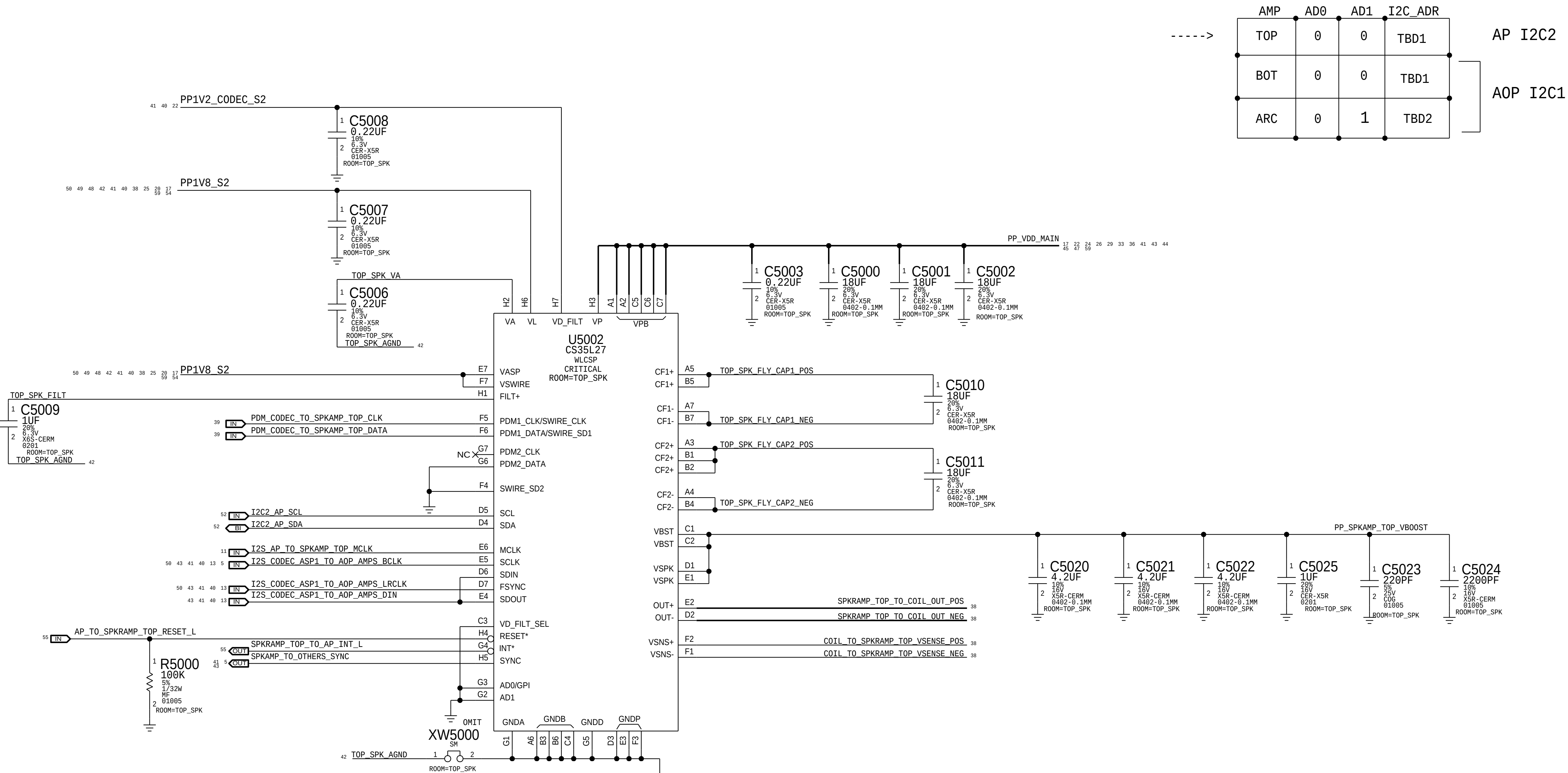
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AUDIO: SOUTH SPKAMP		
DRAWING NUMBER	051-02545	SIZE
		D
REVISION		7.0.0
BRANCH		
PAGE		49 OF 85
SHEET		41 OF 60

D

C

B

A



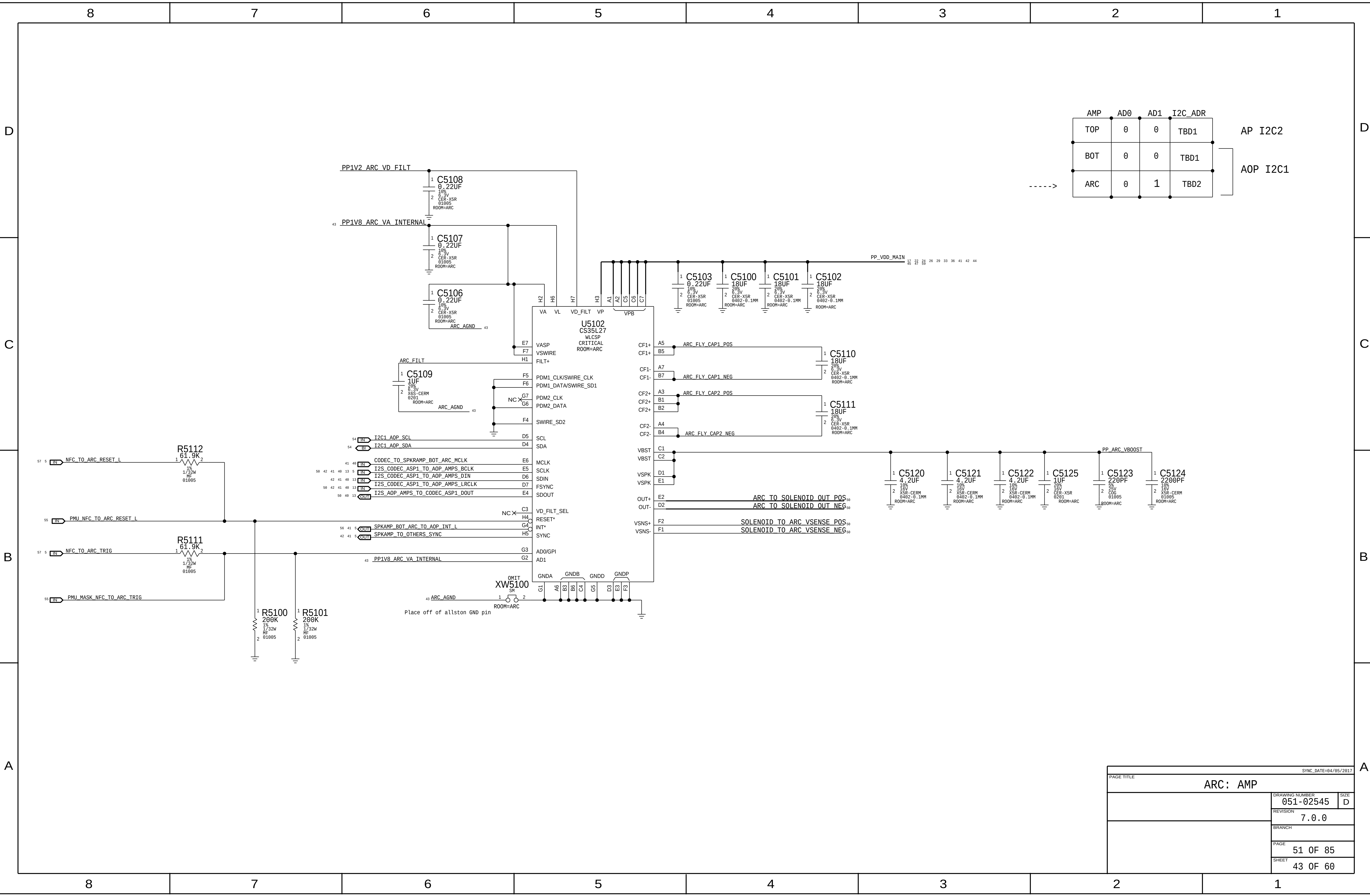
D

C

B

A

PAGE TITLE			SYNC_DATE=04/05/2017	
AUDIO: NORTH SPKAMP			DRAWING NUMBER	SIZE
			051-02545	D
			REVISION	7.0.0
			BRANCH	
			PAGE	50 OF 85
			SHEET	42 OF 60



----->

AMP	AD0	AD1	I2C_ADR	
TOP	0	0	TBD1	AP I2C2
BOT	0	0	TBD1	
ARC	0	1	TBD2	AOP I2C1

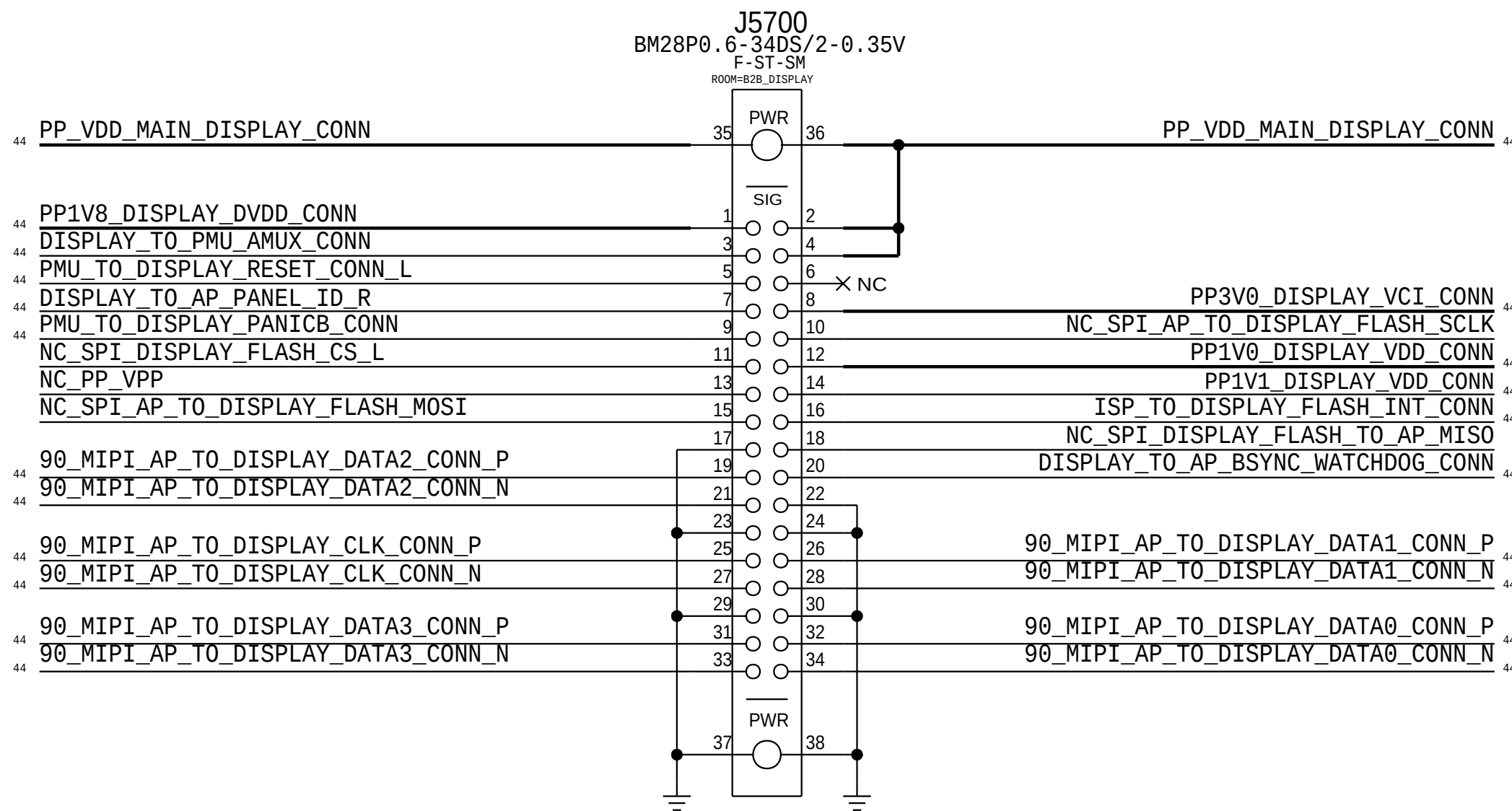
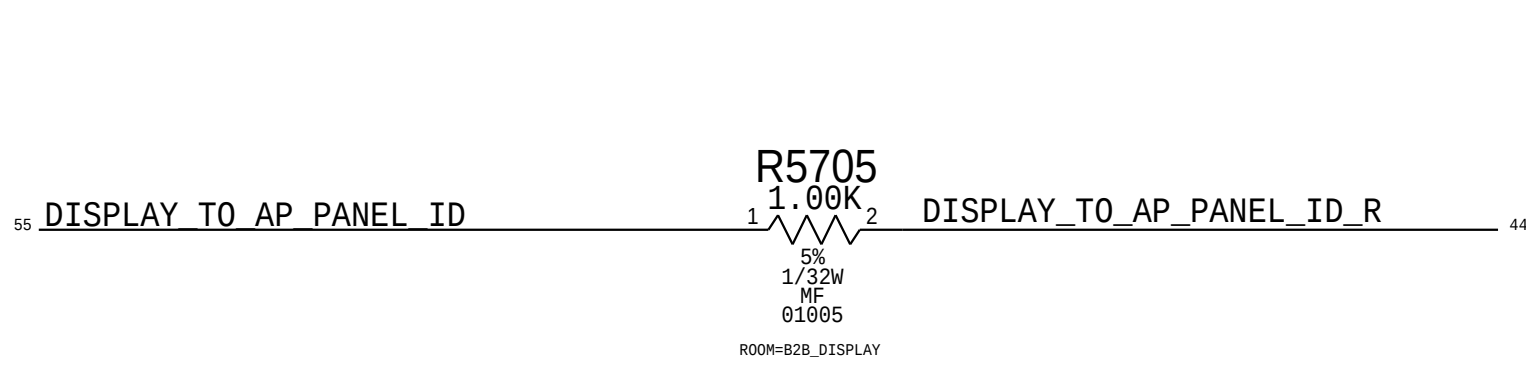
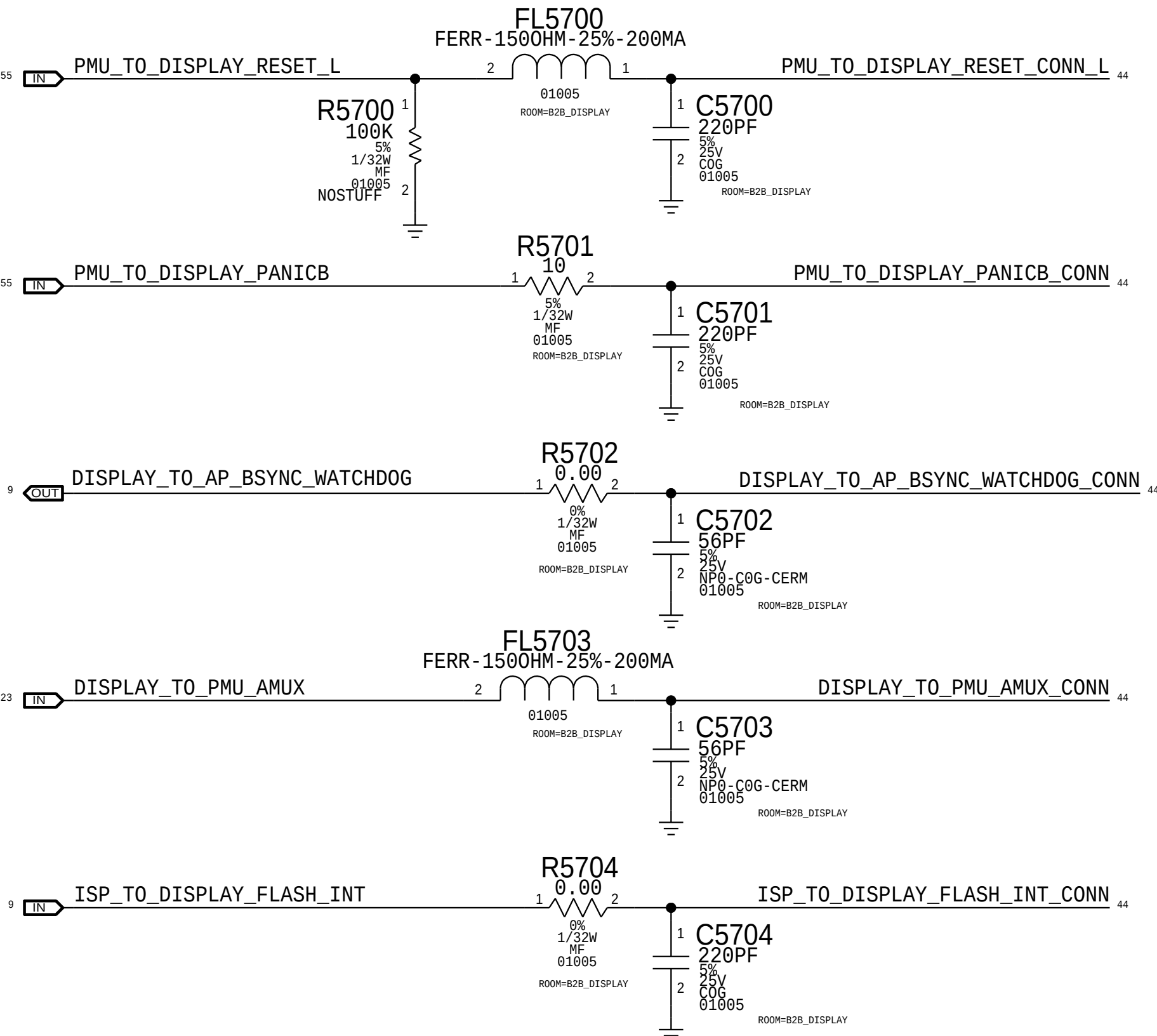
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					REVISION		7.0.0					
					BRANCH							
					PAGE		51 OF 85					
					SHEET		43 OF 60					

Display Control Signals

Display 1V0 LD0 for D33 second display vendor
rdar: #29872369

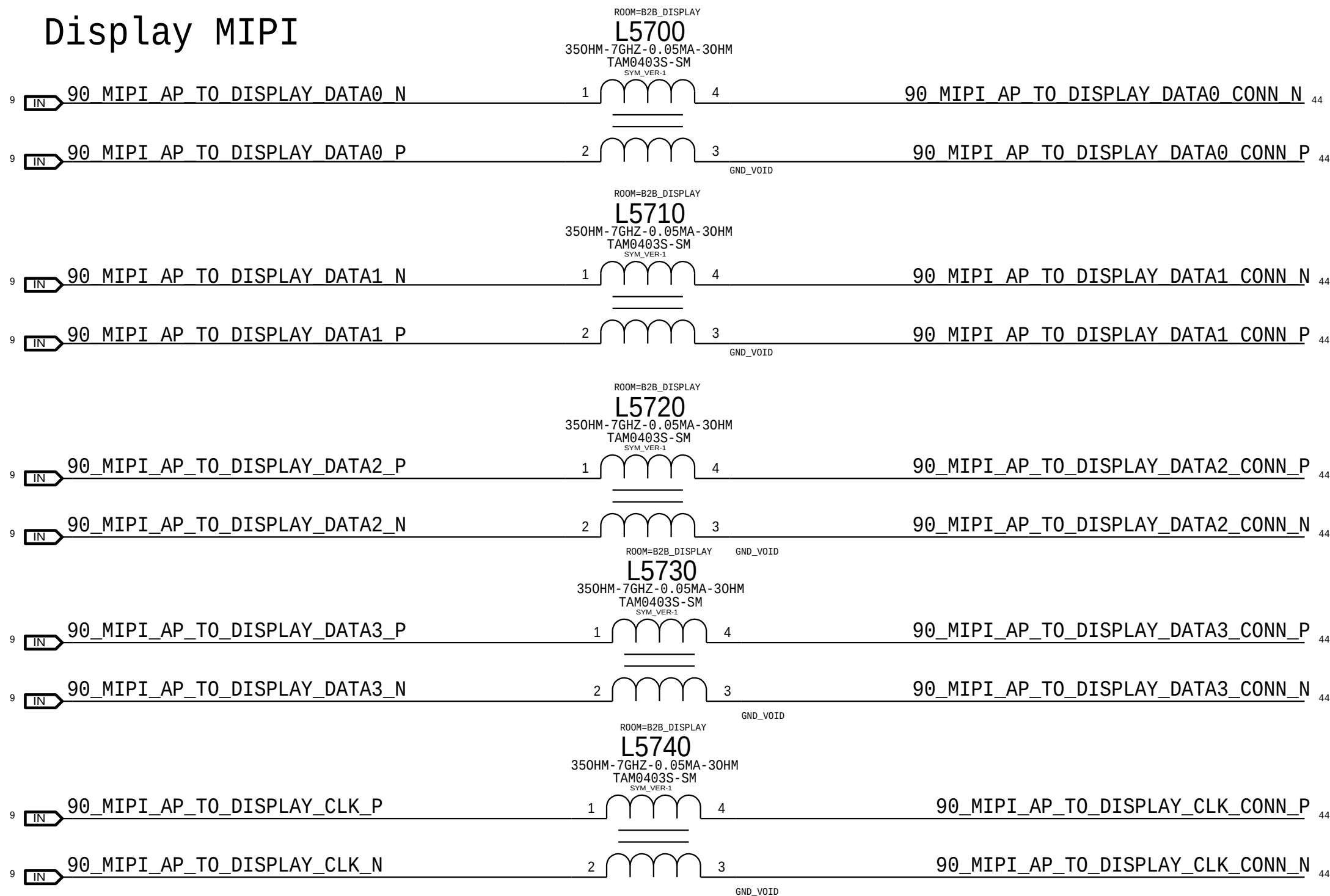
Display Flex Connector

Rcpt: 516S00210 <-- This one on MLB
Plug: 516S00211

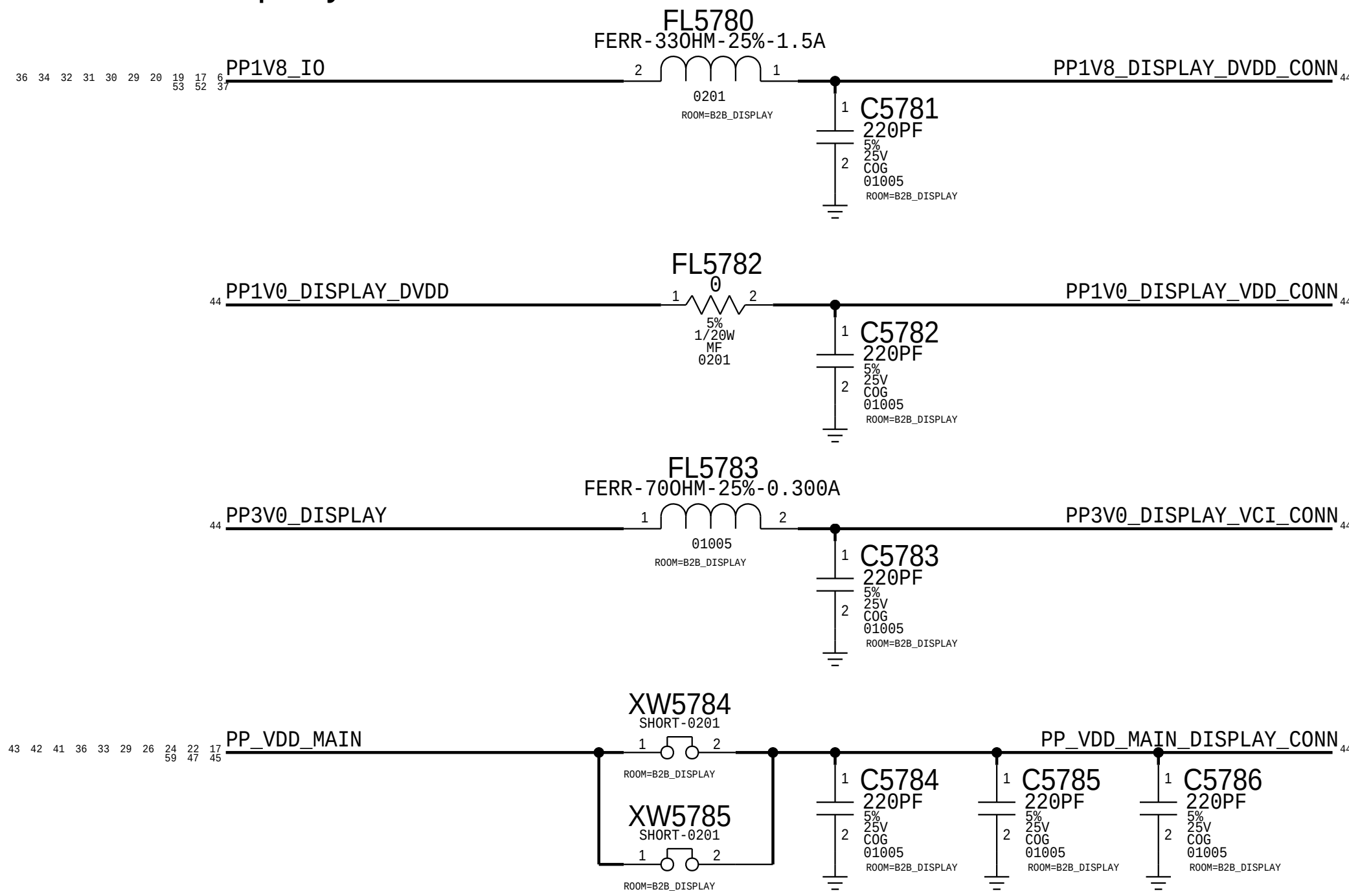


Display MIPI

Display MIPI

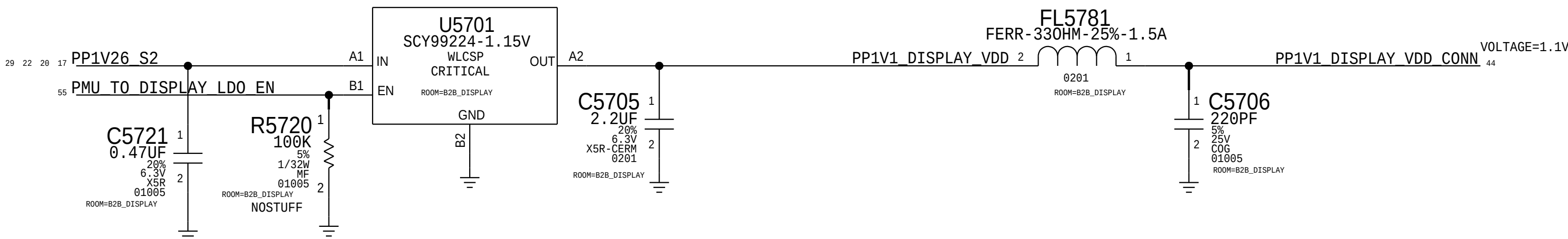


Display Power



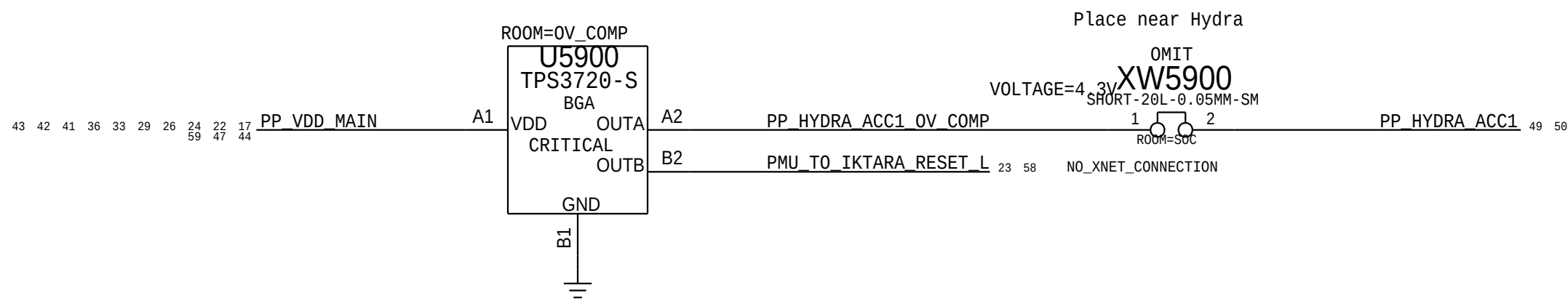
1.2V LD0 is for LGC test chip

Once normal panel is available switch to 1.1V



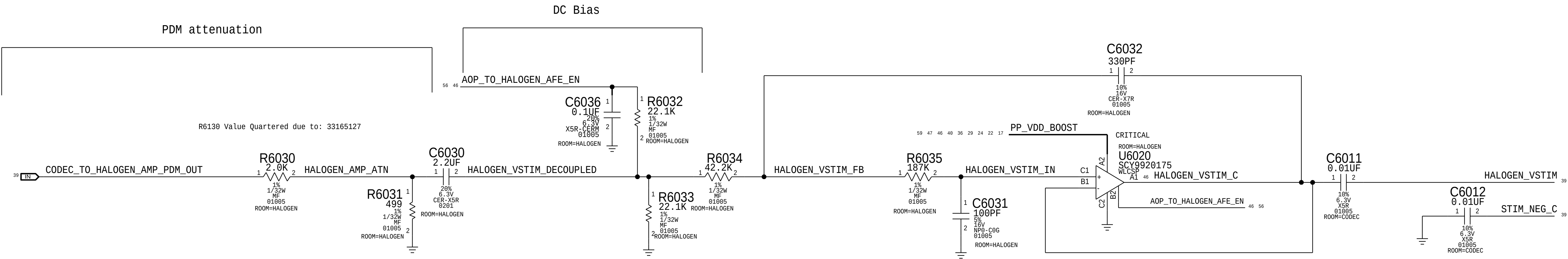
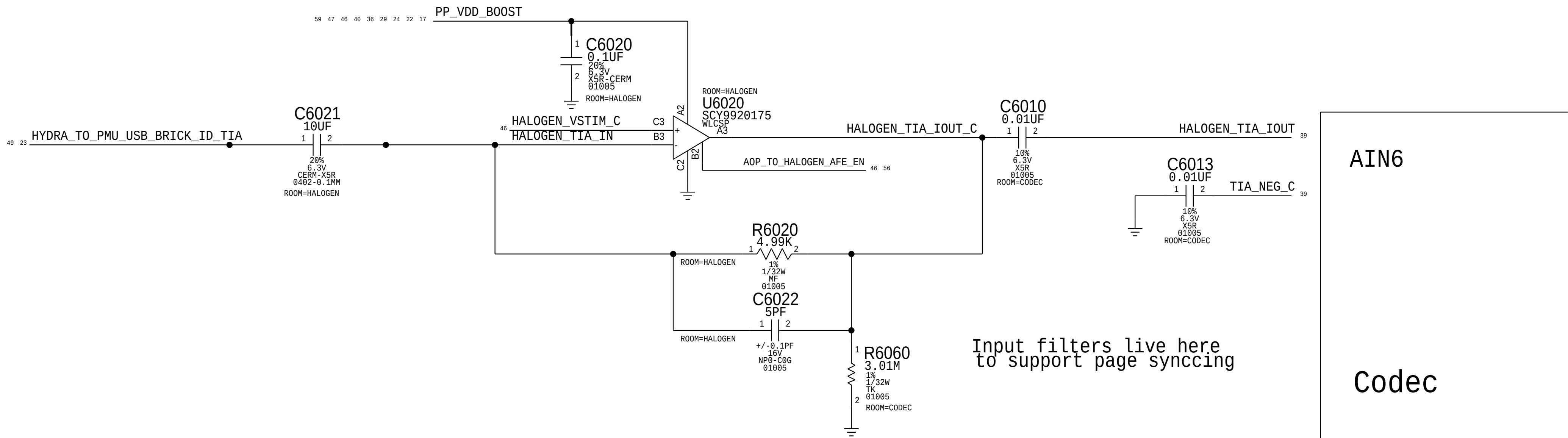
PAGE TITLE		
CG: B2B Display		
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VDD_MAIN OV CUT-OFF CIRCUIT



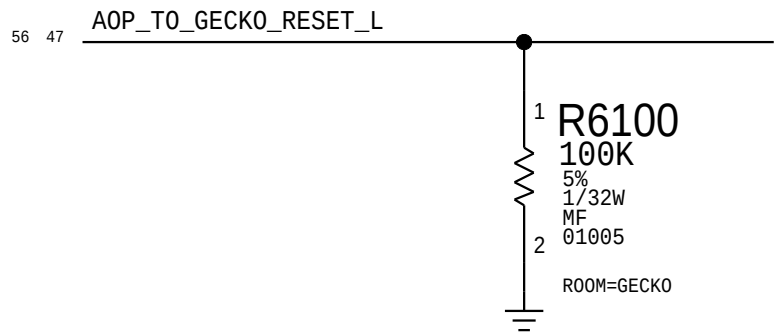
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I/O: Overvoltage Cut-Off Circuit		
	DRAWING NUMBER	SIZE
	051-02545	D
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	BRANCH	
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LDCM



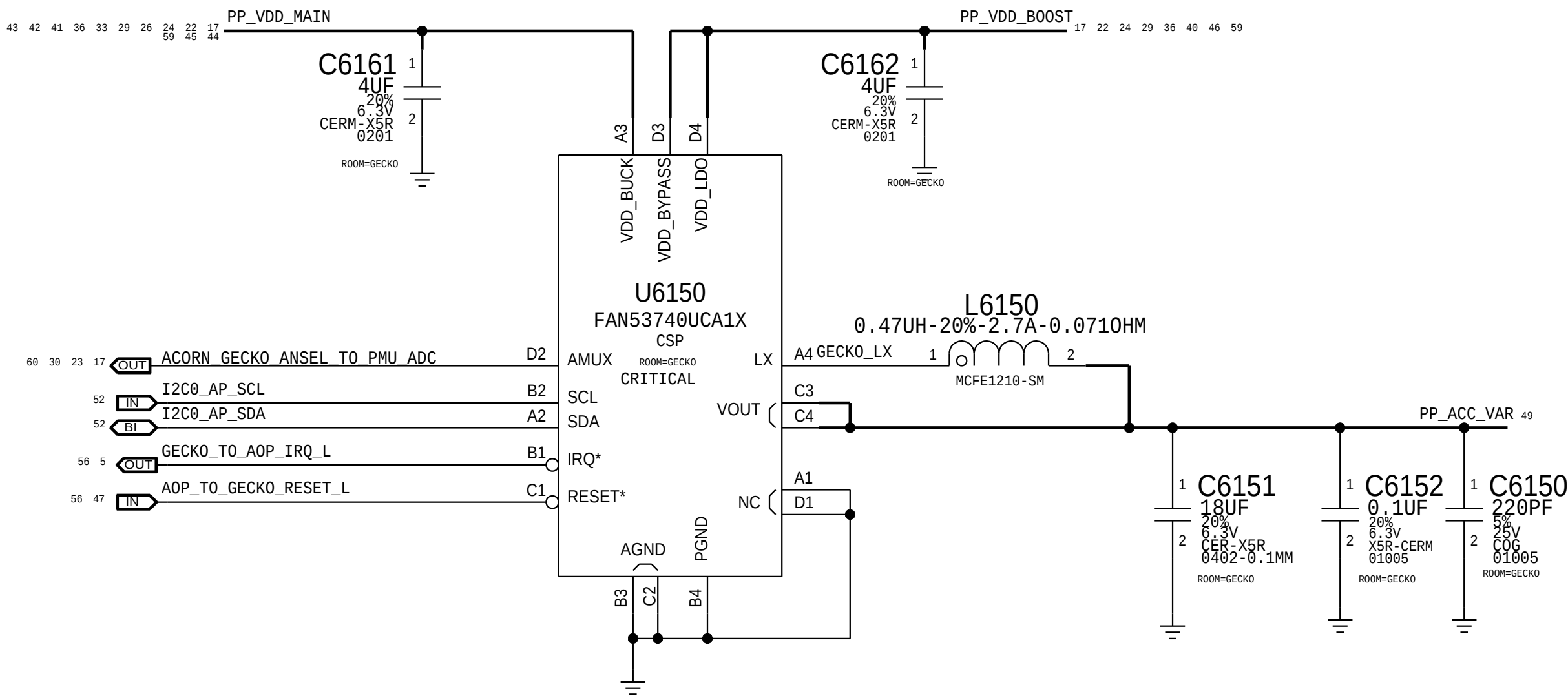
PAGE TITLE		
I/O: LDCM		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	60 OF 85
	SHEET	46 OF 60

GECK0 Reset Pull Down



Gecko

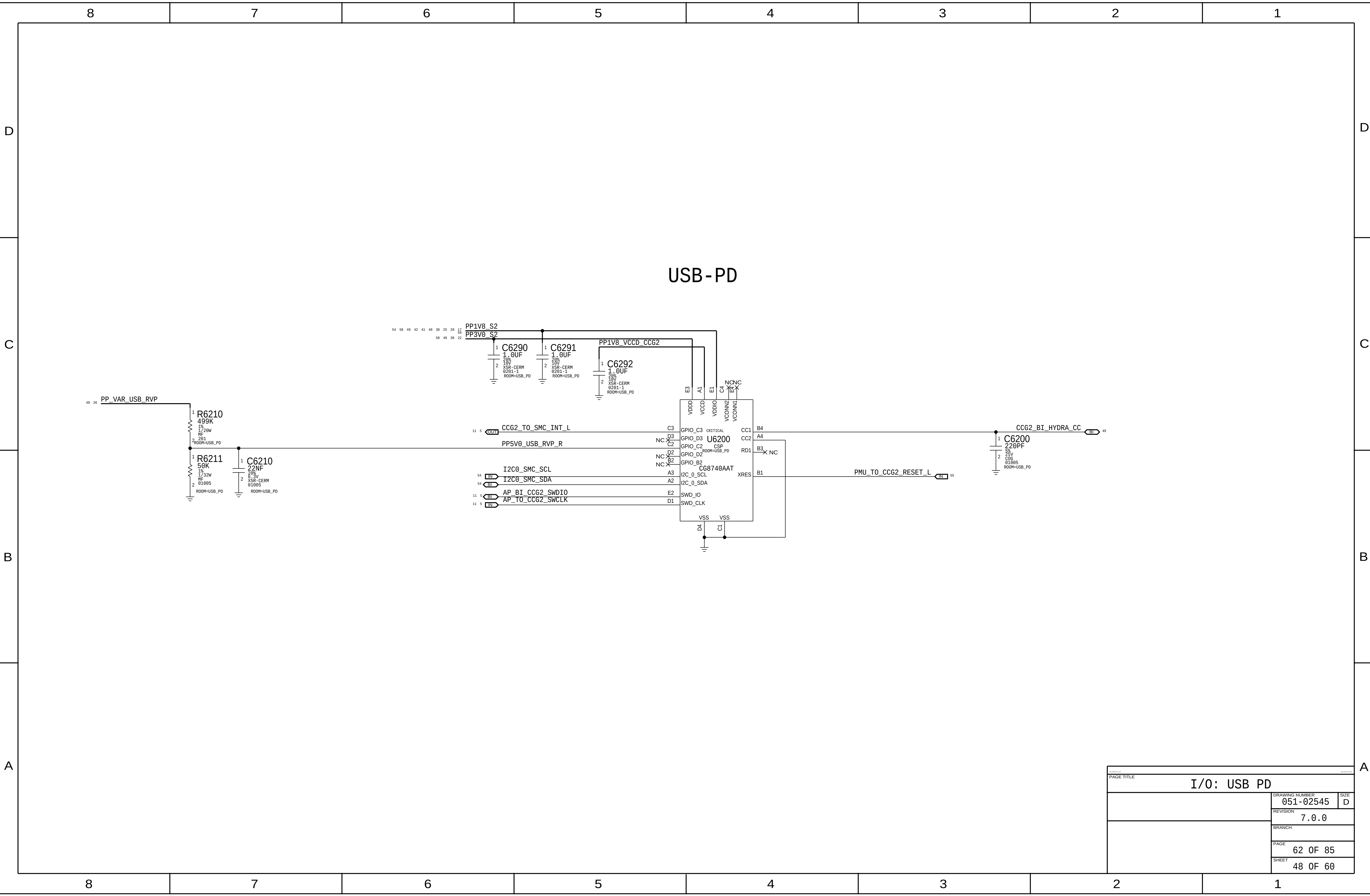
I2C ADDRESS: 0X52



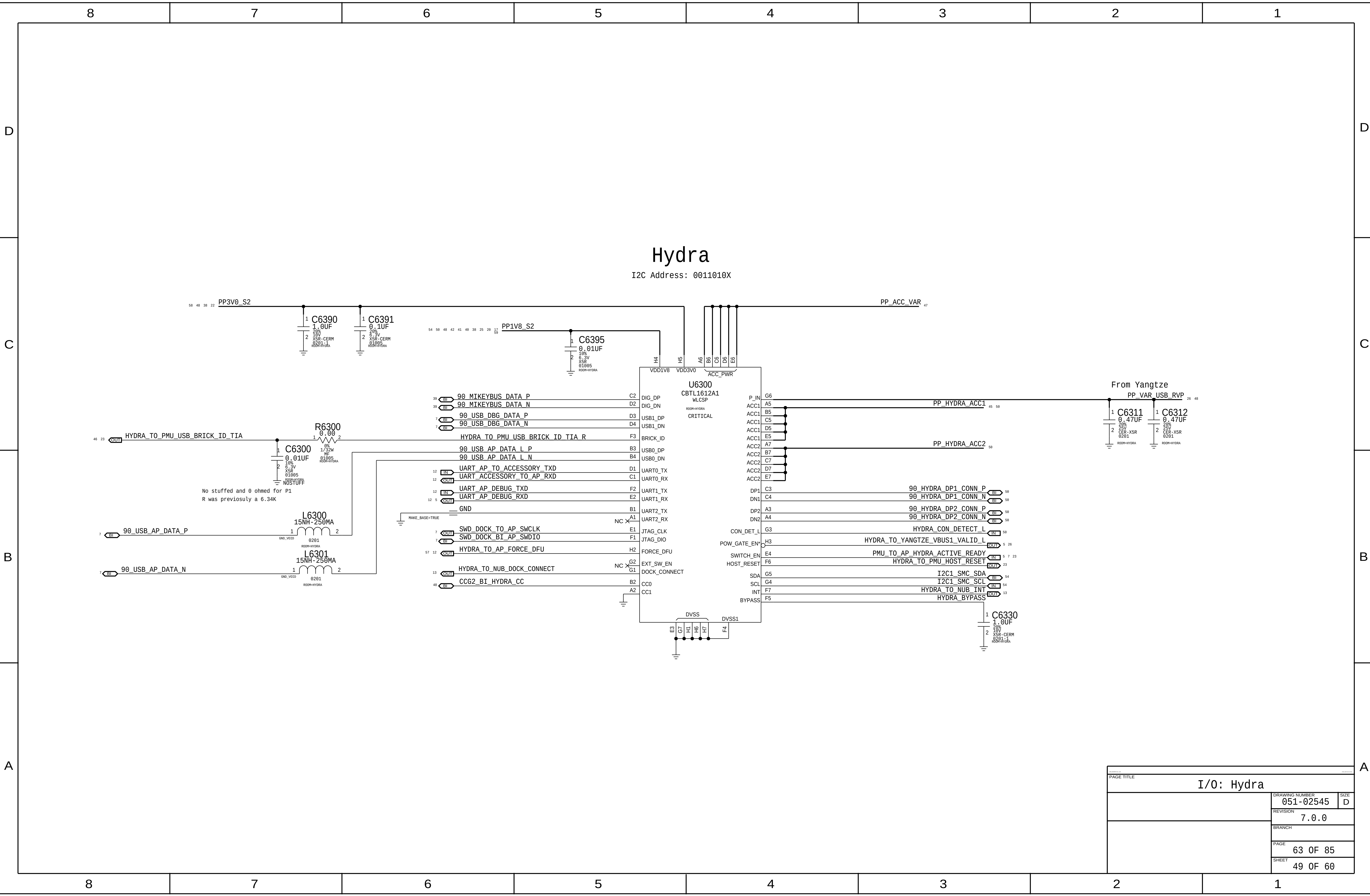
IND Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00854	152S00853	ALT_PARTS	L6150	IND, PMR, 0.47uH, 20%, 2.5A, CV
152S00855	152S00853	ALT_PARTS	L6150	IND, PMR, 0.47uH, 20%, 2.1A, Murata

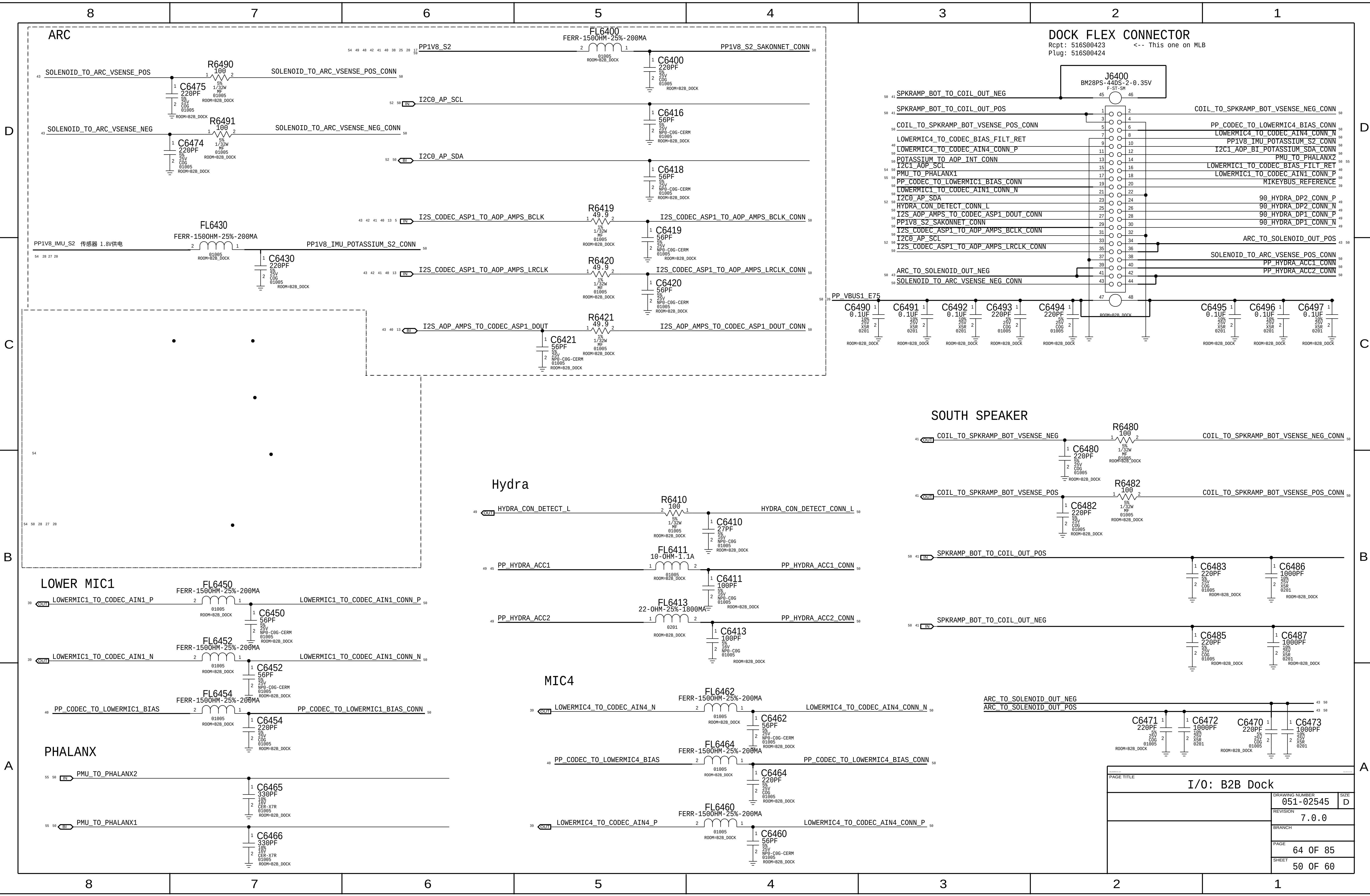
PAGE TITLE		
I/O: Gecko		
	DRAWING NUMBER	051-02545
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PAGE TITLE		
I/O: USB PD		
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PAGE TITLE		
I/O: Hydra		
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PAGE TITLE		
I/O: B2B Dock		
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BRANCH		
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AP I2C

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C0	PP1V8_I0	400 kHz	GECK0	0x52	1010 010X	0xA4, 0xA5	-	1 MHz	TOP MLB
			SAKONNET	0x08	0001 000X	0x10, 0x11	-	1 MHz	Dock Flex
			BOOST	0x75	1110 101X	0xEA, 0xEB	-	400 KHz	TOP MLB
			ARC EEPROM	0x50	1010 000X	0xA0, 0xA1	-	400 KHz	Dock Flex

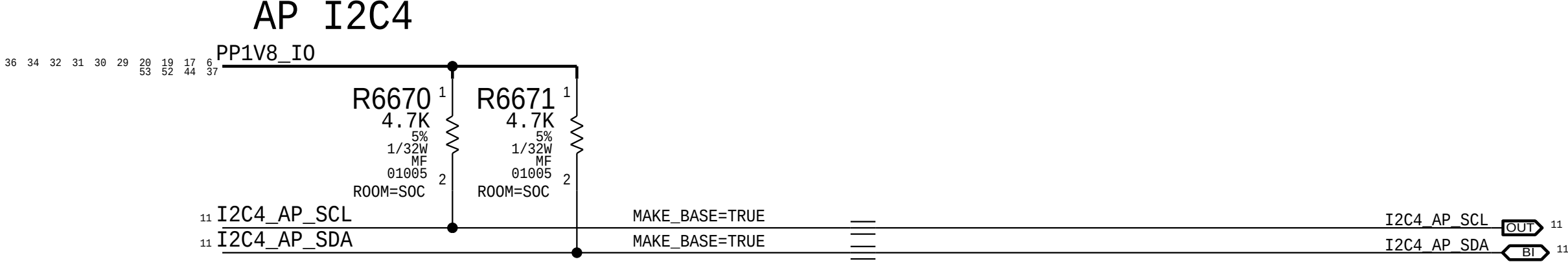
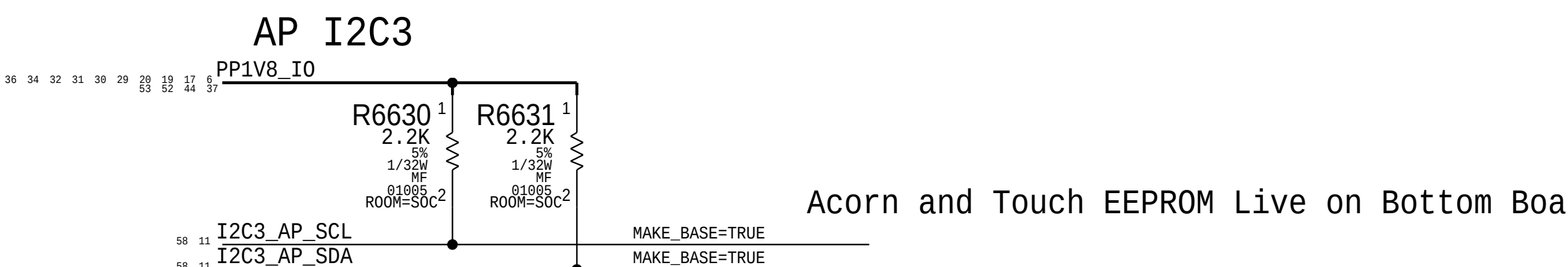
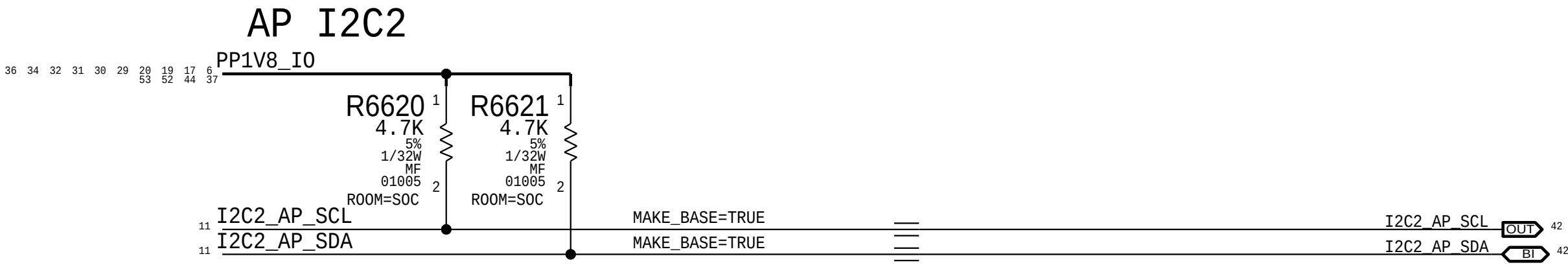
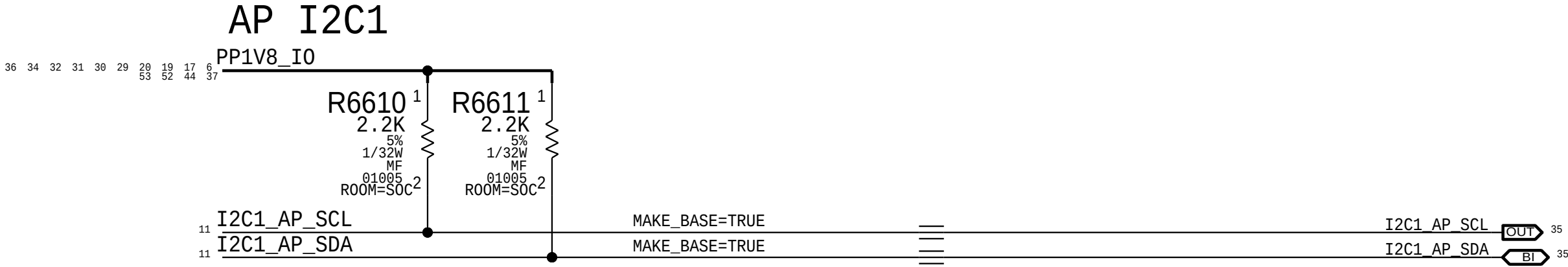
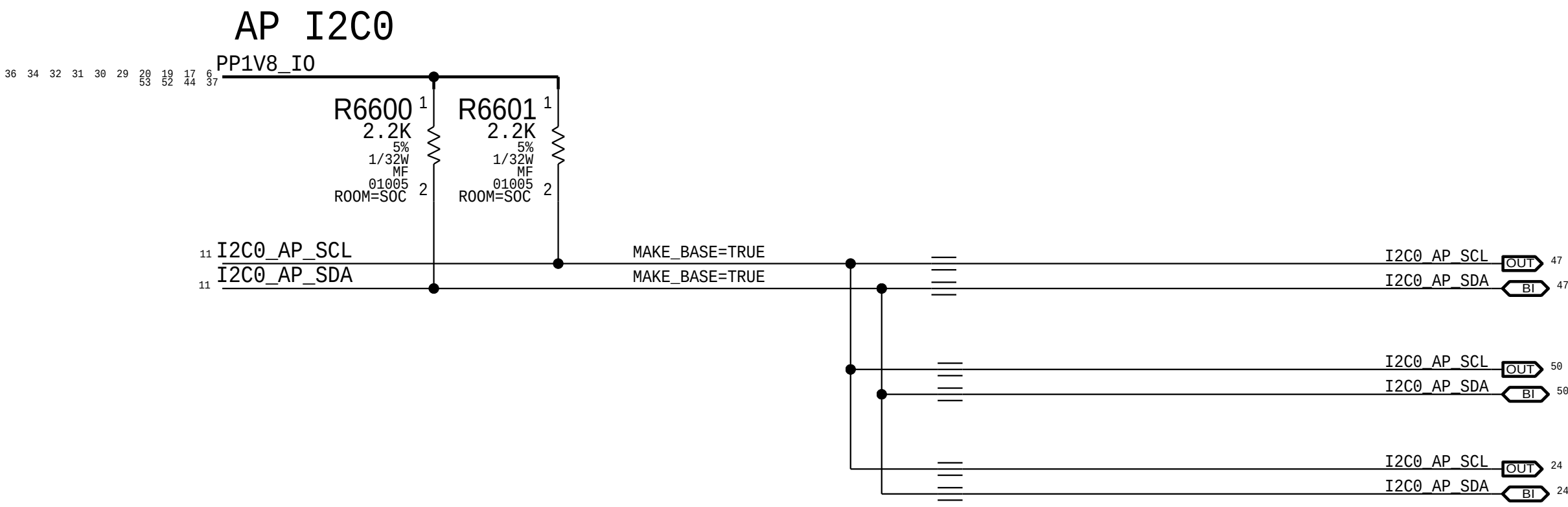
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C1	PP1V8_I0	100 kHz	MIC2	0x56	1010 100X	0xA8, 0xA9	-	1 MHz	Strobe Flex

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C2	PP1V8_I0	1 MHz	Top Speaker Amp	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top MLB

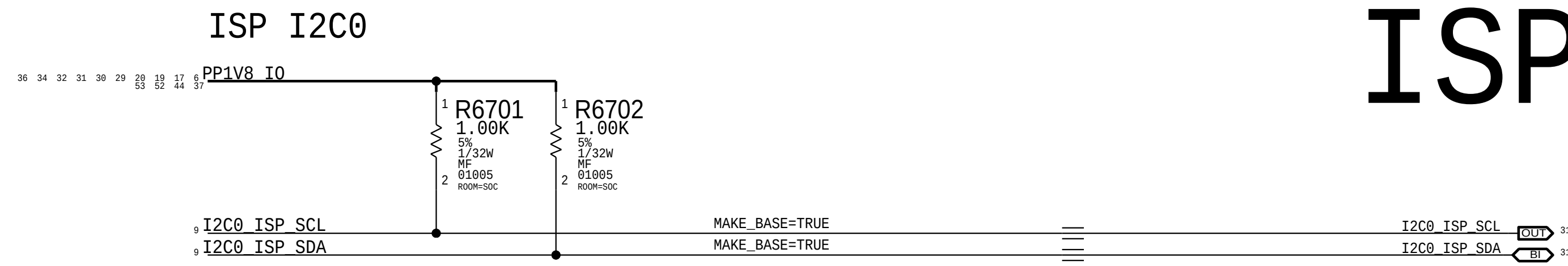
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C3	PP1V8_I0	400 kHz	ACORN	0X2A	0101 010X	0x54, 0x55	-	1 MHz	Bot MLB
			TOUCH EEPROM	0x51	1010 001X	0xA2, 0xA3	-	1 MHz	Touch Flex

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Location
AP I2C4	PP1V8_I0	400 kHz	LYNX	0X71	Top MLB

PAGE TITLE		
SYSTEM: AP I2C		
	DRAWING NUMBER	051-02545
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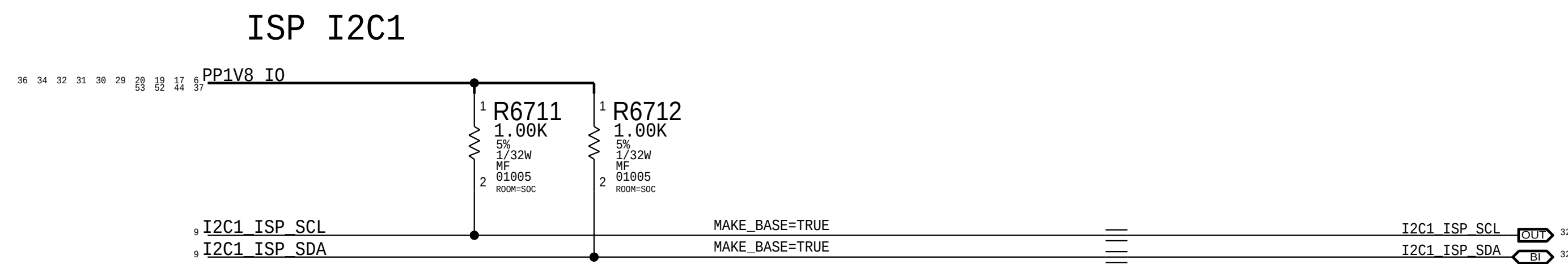


Acorn and Touch EEPROM Live on Bottom Board

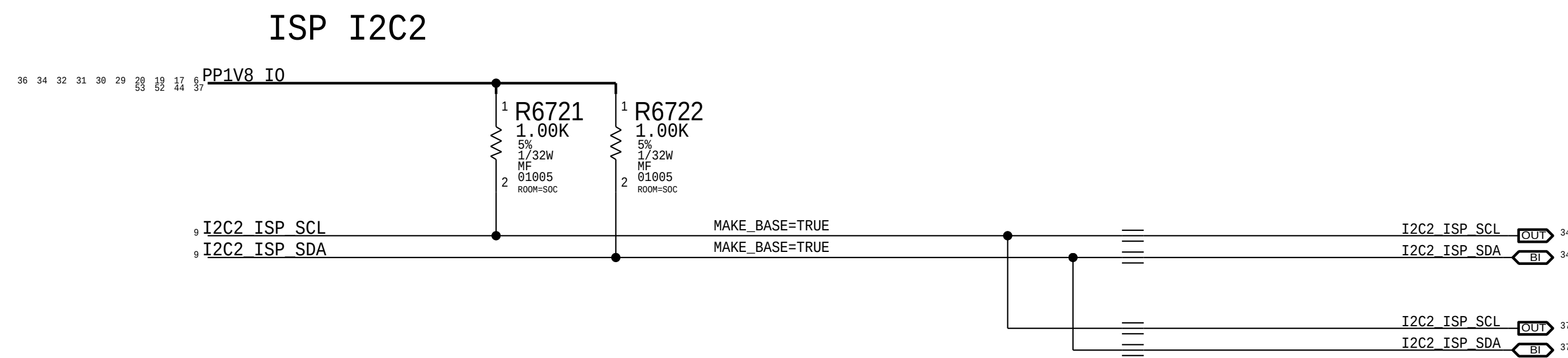


ISP I2C

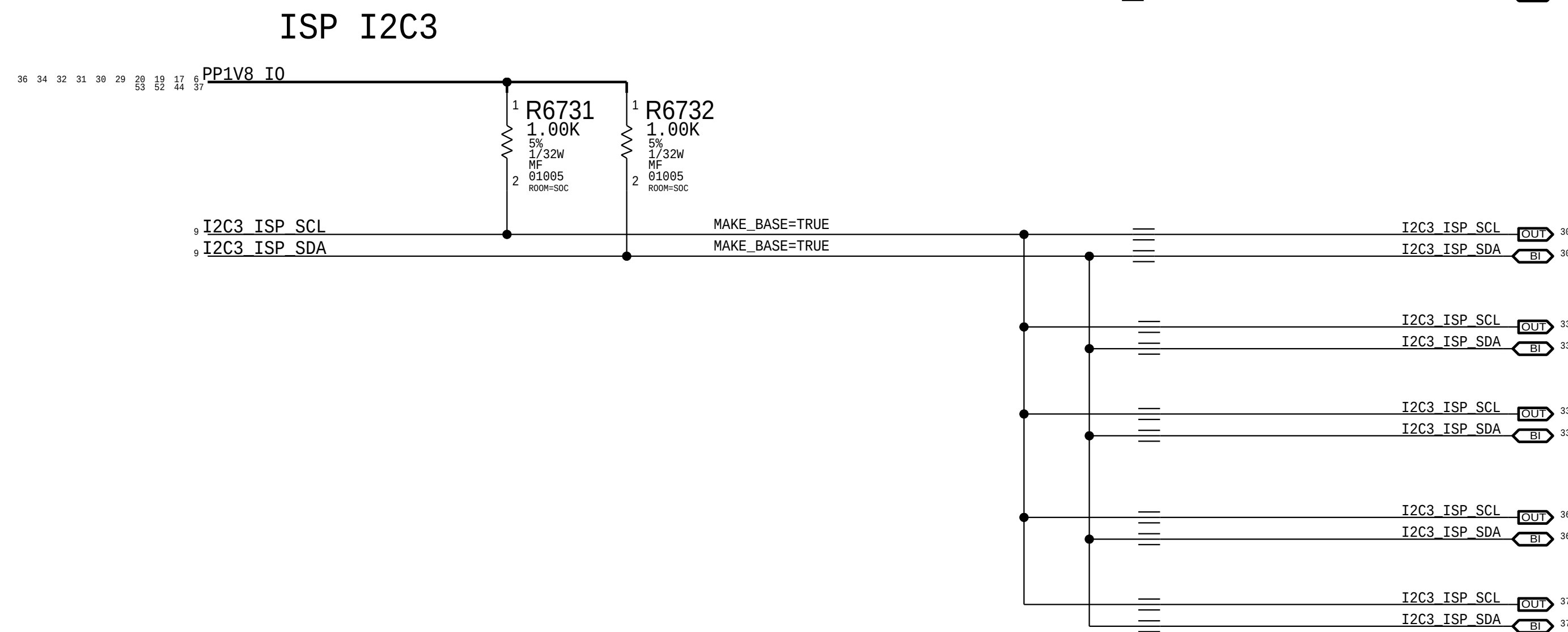
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C0	PP1V8_IO	1 MHz	Austin	0x10	0010 000X	0x20, 0x21	-	1 MHz	Wide Cam
			Raman	0x3C	0111 100X	0x78, 0x79	-	1 MHz	Wide Cam



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C1	PP1V8_I0	1 MHz	Billings	0x20	0100 000X	0x40, 0x41	-	1 MHz	Tele Cam
			Grunberg+	0x1C	0011 100X	0x38, 0x39	-	1 MHz	Tele Cam



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C2	PP1V8_I0	1 MHz	Yonkers	0x10	0010 000X	0x20, 0x21	-	1 MHz	Fcam
			Flatiron	0x70	1110 000X	0xE0, 0xE1	-	1 MHz	Fcam
			Savage	0x18	0011 000X	0x30, 0x31	-	1 MHz	Juliet Flex

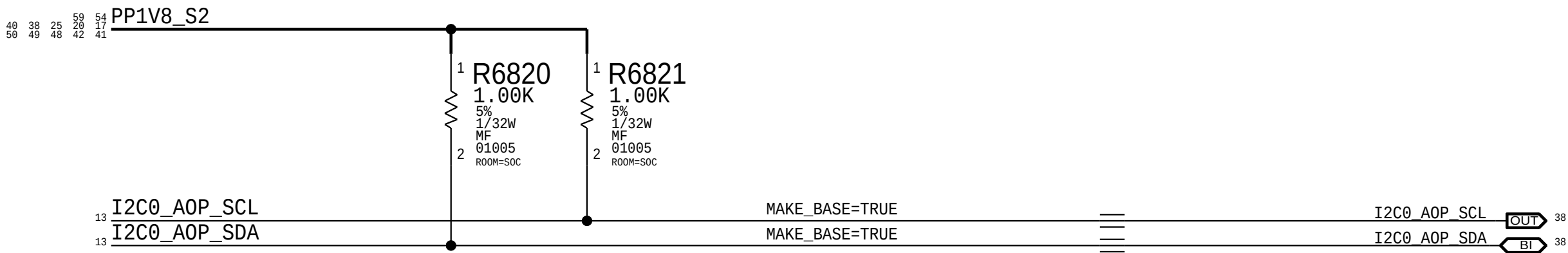


Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C3	PP1V8_I0	1 MHz	Ansel	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top Board
			Neon	0x63	1100 011X	0xC6, 0xC7	-	1 MHz	Top Board
			Neon	0x67	1100 111X	0xCE, 0xCF	-	1 MHz	Top Board
			Rigel	0x55	1100 011X	0xAA, 0xAB	-	1 MHz	Top Board
			Mama Bear	0x50	1010 000X	0xA0, 0xA1	-	1 MHz	Romeo Flex

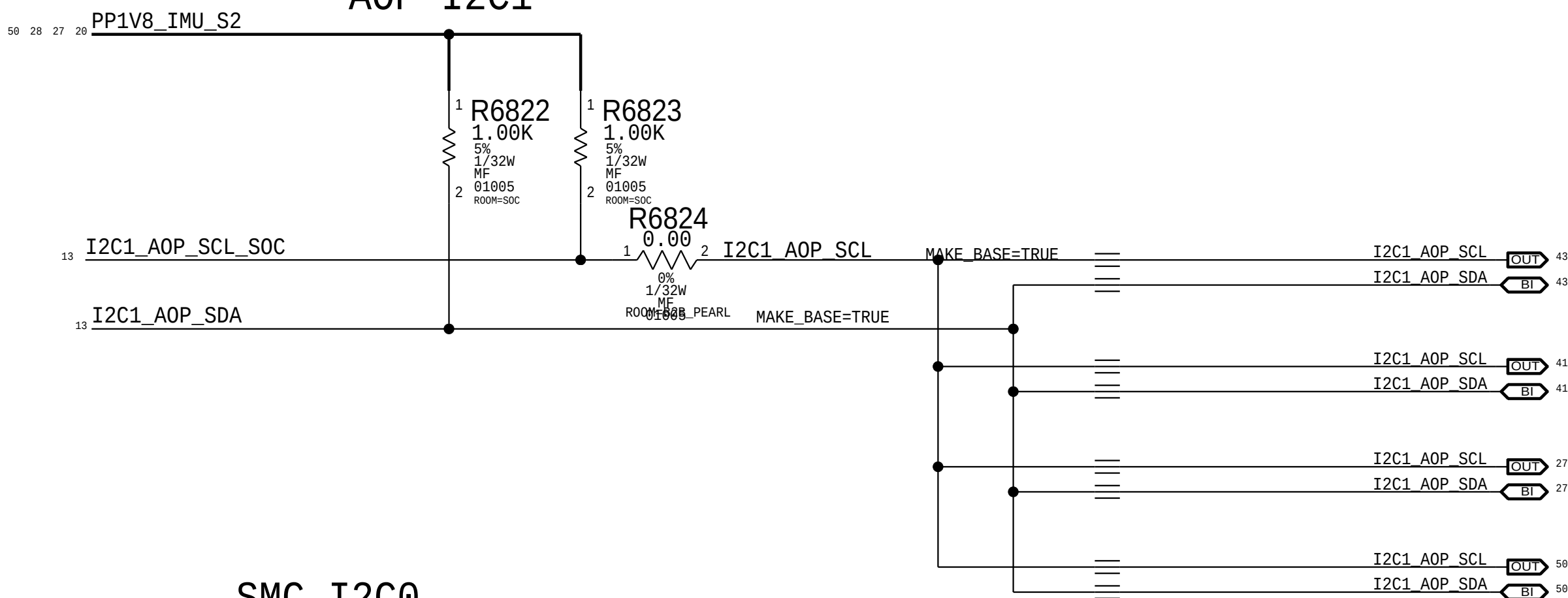
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	DRAWING NUMBER	051-02545	SIZE D
	REVISION 7.0.0		
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AOP/SMC I2C

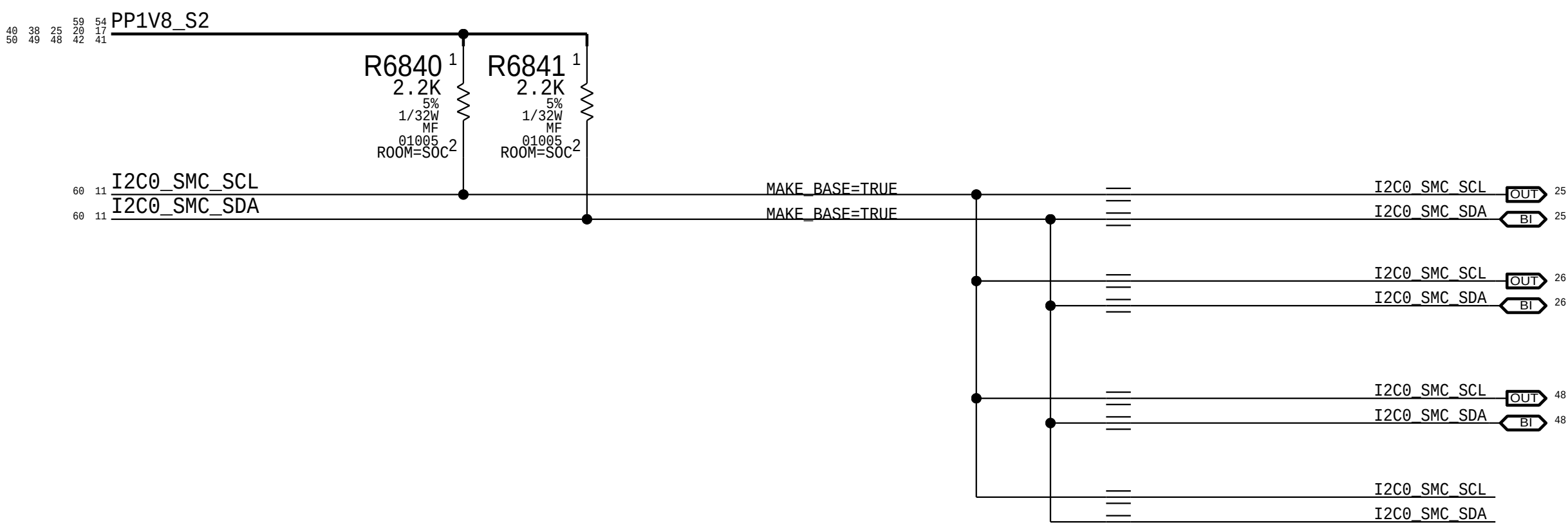
AOP I2C0



AOP I2C1

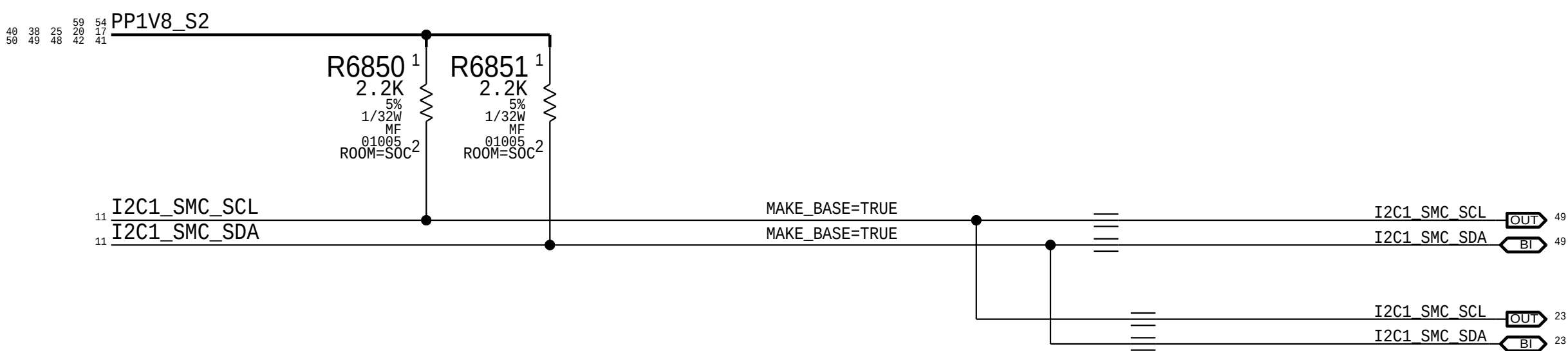


SMC I2C0



Lives on bottom board

SMC I2C1



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C0	PP1V8_S2	750 kHz	Doppler	0x58	1011 000X	0xB0, 0xB1	-	1 MHz	Sensor Flex
			Blackbird	0x29	0101 001X	0x52, 0x53	-	1 MHz	Sensor Flex
			Yogi	0x33	0110 011X	0x66, 0x67	-	1 MHz	Sensor Flex

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C1	PP1V8_IMU_S2	400 kHz	Arc	0x42	1000 001X	0x82, 0x83	-	1 MHz	Top Board
			Bottom Speaker	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top Board
			Moly	0x0E	0001 110X	0x1C, 0x1D	-	1 MHz	Button Cyclone
			Potassium	0x76	1110 110X	0xEC, 0xED	-	1 MHz	Dock Flex

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C0	PP1V8_S2	400 kHz	Yangtze	0x71	1110 001X	0xE2, 0xE3	-	400 KHz	Top Board
			Iktara	0x39	0111 001X	0x72, 0x73	-	400 KHz	Bot Board
			CCG2	0x12	0010 010X	0x24, 0x25	-	1 MHz	Top Board
			Gas Guage	0x55	0010 010X	0xAA, 0xAB	-	1 MHz	BMU Flex
			Roswell	0x10	0100 000X	0x20, 0x21	-	400 KHz	BMU Flex

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C1	PP1V8_S2	400 kHz	Hydra	0x1A	0011 010X	0x34, 0x35	-	400 KHz	Top Board
			Denali	0x74	1110 100X	0xE8, 0xE9	-	400 KHz	Top Board

AP/PMU GPIOs

SOC

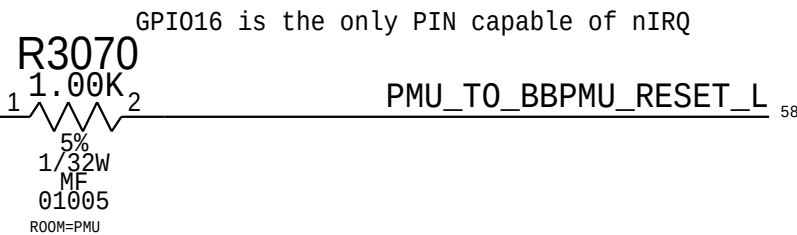
PMU

GPIO_0	12	AP_TO_BT_DEVICE_WAKE	AP_TO_BT_DEVICE_WAKE	58
GPIO_1	12	BOARD_REV0	BOARD_REV0	6
GPIO_2	12	BOARD_REV1	BOARD_REV1	6
GPIO_3	12	BOARD_REV2	BOARD_REV2	6
GPIO_4	12	AP_TO_PMU_AMUX_SYNC	AP_TO_PMU_AMUX_SYNC	23
GPIO_5	12	BOARD_REV3	BOARD_REV3	6
GPIO_6	12	AP_CANARY1	AP_CANARY1	68
GPIO_7	12	AP_BUTTON_VOL_UP_L	PMU_TO_AP_BUTTON_VOL_UP_L	23
GPIO_8	12	NC_AP_GPIO8	NC_AP_GPIO8	
GPIO_9	12	AP_TO_BBPMU_RADIO_ON_L	AP_TO_BBPMU_RADIO_ON_L	57
GPIO_10	12	AP_TO_SPKRAMP_TOP_RESET_L	AP_TO_SPKRAMP_TOP_RESET_L	42
GPIO_11	12	AP_TO_NFC_FW_DWLD_REQ	AP_TO_NFC_FW_DWLD_REQ	57
GPIO_12	15	AP_TO_BB_PEAK_POWER_INDICATOR	AP_TO_BB_PEAK_POWER_INDICATOR	57
GPIO_13	12	AP_TO_NFC_DEV_WAKE	AP_TO_NFC_DEV_WAKE	57
GPIO_14	12	CAMPMU_TO_AP_IRQ_L	CAMPMU_TO_AP_IRQ_L	38
GPIO_15	12	AP_TO_GNSS_TIME_MARK	AP_TO_GNSS_TIME_MARK	57
GPIO_16	15	SPKRAMP_TOP_TO_AP_INT_L	SPKRAMP_TOP_TO_AP_INT_L	42
GPIO_17	12	BB_TO_AP_COEX	BB_TO_AP_COEX	57
GPIO_18	12	BT_TO_AP_TIME_SYNC	BT_TO_AP_TIME_SYNC	58
GPIO_19	12	AP_TO_BB_RESET_L	AP_TO_BB_RESET_L	57
GPIO_20	12	BB_TO_AP_PEAK_POWER_INDICATOR	BB_TO_AP_PEAK_POWER_INDICATOR	57
GPIO_21	12	BB_TO_AP_RESET_DETECT_L	BB_TO_AP_RESET_DETECT_L	57
GPIO_22	12	AP_TO_BB_COREDUMP_TRIG	AP_TO_BB_COREDUMP_TRIG	57
GPIO_23	12	AP_TO_CAMPMU_RESET_L	AP_TO_CAMPMU_RESET_L	38
GPIO_24	12	AP_TO_BB_COEX	AP_TO_BB_COEX	57
GPIO_25	12	DISPLAY_TO_AP_PANEL_ID	DISPLAY_TO_AP_PANEL_ID	44
GPIO_26	12	AP_CANARY2	AP_CANARY2	58
GPIO_27	12	NC_AP_GPIO27	NC_AP_GPIO27	
GPIO_28	12	NC_AP_GPIO28	NC_AP_GPIO28	
GPIO_29	12	AP_TO_RACER_RESET_L	AP_TO_RACER_RESET_L	57
GPIO_30	15	GNSS_TO_AP_LOW_PWR_IND	GNSS_TO_AP_LOW_PWR_IND	57

GPIO_1	23	PMU_TO_CCG2_RESET_L	PMU_TO_CCG2_RESET_L	48
GPIO_2	23	PMU_TO_AP_THROTTLE_GPU1_L	PMU_TO_AP_THROTTLE_GPU1_L	7
GPIO_3	23	NC_BT_TO_PMU_HOST_WAKE	NC_BT_TO_PMU_HOST_WAKE	
GPIO_4	23	WLAN_TO_PMU_HOST_WAKE	WLAN_TO_PMU_HOST_WAKE	58
GPIO_5	23	BB_TO_PMU_PCIE_HOST_WAKE_L	BB_TO_PMU_PCIE_HOST_WAKE_L	58
GPIO_6	23	PMU_NFC_TO_ARC_RESET_L	PMU_NFC_TO_ARC_RESET_L	43
GPIO_7	23	PMU_TO_GNSS_EN	PMU_TO_GNSS_EN	58
GPIO_8	23	PMU_TO_WLAN_CLK32K	PMU_TO_WLAN_CLK32K	23
GPIO_9	23	PMU_TO_BT_REG_ON	PMU_TO_BT_REG_ON	58
GPIO_10	23	PMU_TO_PHALANX2	PMU_TO_PHALANX2	50
GPIO_11	23	YANGTZE_TO_PMU_INT_L	YANGTZE_TO_PMU_INT_L	26
GPIO_12	23	CODEC_TO_PMU_WAKE_L	CODEC_TO_PMU_WAKE_L	46
GPIO_13	23	PMU_MASK_NFC_TO_ARC_TRIG	PMU_MASK_NFC_TO_ARC_TRIG	43
GPIO_14	23	PMU_TO_WLAN_REG_ON	PMU_TO_WLAN_REG_ON	58
GPIO_15	23	PMU_TO_NFC_VDD_MAIN_EN	PMU_TO_NFC_VDD_MAIN_EN	57
GPIO_16	23	PMU_TO_NAND_LOW_BATT_BOOT_L	PMU_TO_NAND_LOW_BATT_BOOT_L	19
GPIO_17	23	PMU_TO_PHALANX1	PMU_TO_PHALANX1	50
GPIO_18	23	PMU_TO_DISPLAY_RESET_L	PMU_TO_DISPLAY_RESET_L	44
GPIO_19	23	PMU_TO_BBPMU_RESET_R_L	PMU_TO_BBPMU_RESET_R_L	
GPIO_20	23	PMU_TO_NFC_EN	PMU_TO_NFC_EN	58
GPIO_21	23	NC_PMU_GPIO21	NC_PMU_GPIO21	
GPIO_22	23	PMU_TO_IKTARA_EN_EXT_1V8	PMU_TO_IKTARA_EN_EXT_1V8	68
GPIO_23	23	PMU_TO_BOOST_EN	PMU_TO_BOOST_EN	24
GPIO_24	23	PMU_TO_DISPLAY_PANICB	PMU_TO_DISPLAY_PANICB	44
GPIO_25	23	PMU_TO_DISPLAY_LDO_EN	PMU_TO_DISPLAY_LDO_EN	44

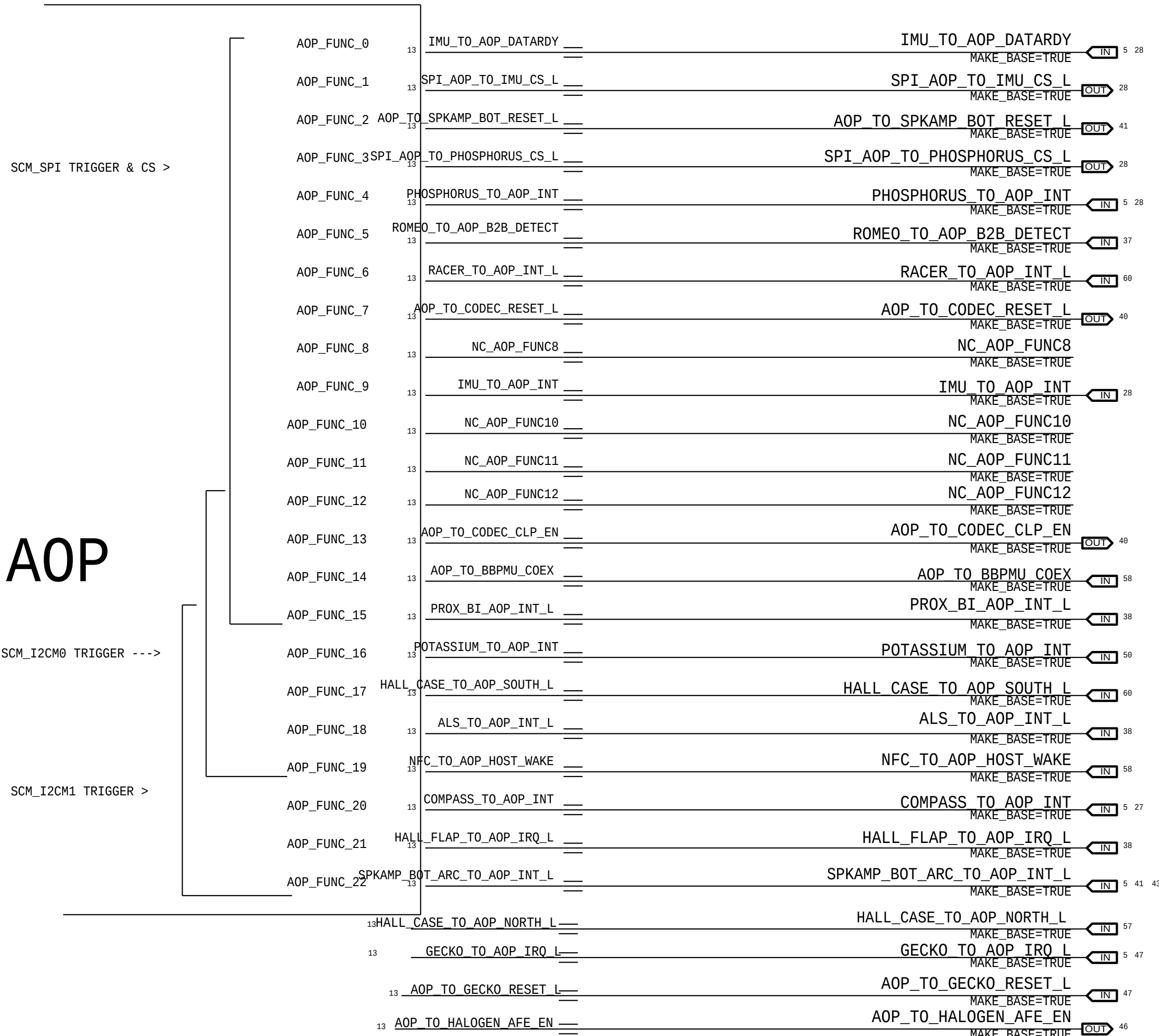
Held Through 1 Reset

Sequenced GPIOs



PAGE TITLE			SYNC_DATE=05/09/2017		
SYSTEM: SOC/PMU GPIOs			DRAWING NUMBER	051-02545	SIZE
			REVISION	7.0.0	D
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AOP GPIOs



PAGE TITLE			SYSTEM: AOP GPIOs			DRAWING NUMBER			051-02545			SIZE			D		
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Interposer Aliases: Pins 1-144

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	PMU_TO_SYSTEM_COLD_RESET_L	=====	MAKE_BASE=TRUE	PMU_TO_SYSTEM_COLD_RESET_L	7 15 23
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	AP_TO_CAMPMU_RESET_L	=====	MAKE_BASE=TRUE	AP_TO_CAMPMU_RESET_L	30 55
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	BB_TO_MANY_GSM_BURST_IND	=====	MAKE_BASE=TRUE	BB_TO_MANY_GSM_BURST_IND	33 38
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	HALL_CASE_TO_AOP_NORTH_L	=====	MAKE_BASE=TRUE	HALL_CASE_TO_AOP_NORTH_L	56
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	AP_TO_TOUCH_SCAN_CLK	=====	MAKE_BASE=TRUE	AP_TO_TOUCH_SCAN_CLK	9
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	I2S_BB_TO_AP_BCLK	=====	MAKE_BASE=TRUE	I2S_BB_TO_AP_BCLK	11
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	I2S_BB_TO_AP_DIN	=====	MAKE_BASE=TRUE	I2S_BB_TO_AP_DIN	11
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	I2S_AP_TO_BB_DOUT	=====	MAKE_BASE=TRUE	I2S_AP_TO_BB_DOUT	11
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	I2S_BB_TO_AP_LRCLK	=====	MAKE_BASE=TRUE	I2S_BB_TO_AP_LRCLK	11
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	PP1V8_ALWAYS	=====	MAKE_BASE=TRUE	PP1V8_ALWAYS	17 22 23 28
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59 60
51	GND	=====	MAKE_BASE=TRUE	GND	58
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	NFC_TO_ARC_RESET_L	=====	MAKE_BASE=TRUE	NFC_TO_ARC_RESET_L	5 43
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	NFC_TO_ARC_TRIG	=====	MAKE_BASE=TRUE	NFC_TO_ARC_TRIG	5 43
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	AP_TO_BB_RESET_L	=====	MAKE_BASE=TRUE	AP_TO_BB_RESET_L	55
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	SWD_AOP_BI_BB_SWDIO	=====	MAKE_BASE=TRUE	SWD_AOP_BI_BB_SWDIO	13
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	UART_GNSS_TO_AP_CTS_L	=====	MAKE_BASE=TRUE	UART_GNSS_TO_AP_CTS_L	12
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	UART_AP_TO_GNSS_RTS_L	=====	MAKE_BASE=TRUE	UART_AP_TO_GNSS_RTS_L	12
51	GND	=====	MAKE_BASE=TRUE	GND	57 59

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

51	PCIE_AP_TO_WLAN_PERST_L	=====	MAKE_BASE=TRUE	PCIE_AP_TO_WLAN_PERST_L	8
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	AP_TO_RACER_RESET_L	=====	MAKE_BASE=TRUE	AP_TO_RACER_RESET_L	55
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	AP_TO_WLAN_TIME_SYNC	=====	MAKE_BASE=TRUE	AP_TO_WLAN_TIME_SYNC	12
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GNSS_TO_AP_LOW_PWR_IND	=====	MAKE_BASE=TRUE	GNSS_TO_AP_LOW_PWR_IND	55
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	HYDRA_TO_AP_FORCE_DFU	=====	MAKE_BASE=TRUE	HYDRA_TO_AP_FORCE_DFU	12 49
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	PP1V8_S2	=====	MAKE_BASE=TRUE	PP1V8_S2	59
51	PP1V8_S2	=====	MAKE_BASE=TRUE	PP1V8_S2	59
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59
51	INTERPOSER_PIN_90	=====	MAKE_BASE=TRUE	INTERPOSER_PIN_90	90
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	AP_TO_BB_COEX	=====	MAKE_BASE=TRUE	AP_TO_BB_COEX	55
51	BB_TO_AP_COEX	=====	MAKE_BASE=TRUE	BB_TO_AP_COEX	55
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	AP_TO_NFC_DEV_WAKE	=====	MAKE_BASE=TRUE	AP_TO_NFC_DEV_WAKE	55
51	AP_TO_NFC_FW_DWLD_REQ	=====	MAKE_BASE=TRUE	AP_TO_NFC_FW_DWLD_REQ	55
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	PMU_TO_NFC_VDD_MAIN_EN	=====	MAKE_BASE=TRUE	PMU_TO_NFC_VDD_MAIN_EN	55
51	UART_AOP_TO_BB_TXD	=====	MAKE_BASE=TRUE	UART_AOP_TO_BB_TXD	13
51	UART_AP_TO_GNSS_TXD	=====	MAKE_BASE=TRUE	UART_AP_TO_GNSS_TXD	12
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	AP_TO_BB_COREDUMP_TRIG	=====	MAKE_BASE=TRUE	AP_TO_BB_COREDUMP_TRIG	55
51	UART_AP_TO_NFC_TXD	=====	MAKE_BASE=TRUE	UART_AP_TO_NFC_TXD	12
51	UART_NFC_TO_AP_RXD	=====	MAKE_BASE=TRUE	UART_NFC_TO_AP_RXD	12
51	BB_TO_AP_RESET_DETECT_L	=====	MAKE_BASE=TRUE	BB_TO_AP_RESET_DETECT_L	55
51	GND	=====	MAKE_BASE=TRUE	GND	59
51	BOARD_ID2	=====	MAKE_BASE=TRUE	BOARD_ID2	8 12
51	AP_TO_GNSS_TIME_MARK	=====	MAKE_BASE=TRUE	AP_TO_GNSS_TIME_MARK	55
51	NC_INTERPOSER_109	=====	MAKE_BASE=TRUE	NC_INTERPOSER_109	109
51	AP_BB_PEAK_POWER_INDICATOR	=====	MAKE_BASE=TRUE	AP_BB_PEAK_POWER_INDICATOR	55
51	AP_TO_BBPMU_RADIO_ON_L	=====	MAKE_BASE=TRUE	AP_TO_BBPMU_RADIO_ON_L	55
51	PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59
51	PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59
51	PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59
51	90_PCIE_BB_TO_AP_RXD_N	=====	MAKE_BASE=TRUE	90_PCIE_BB_TO_AP_RXD_N	17
51	90_PCIE_BB_TO_AP_RXD_P	=====	MAKE_BASE=TRUE	90_PCIE_BB_TO_AP_RXD_P	17
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	90_PCIE_AP_TO_BB_TXD_N	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_BB_TXD_N	17
51	90_PCIE_AP_TO_BB_TXD_P	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_BB_TXD_P	17
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	90_PCIE_AP_TO_BB_REFCLK_P	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_BB_REFCLK_P	8
51	90_PCIE_AP_TO_BB_REFCLK_N	=====	MAKE_BASE=TRUE	90_PCIE_AP_TO_BB_REFCLK_N	8
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	UART_BB_TO_AOP_RXD	=====	MAKE_BASE=TRUE	UART_BB_TO_AOP_RXD	13
51	UART_GNSS_TO_AP_RXD	=====	MAKE_BASE=TRUE	UART_GNSS_TO_AP_RXD	12
51	PCIE_AP_TO_BB_PERST_L	=====	MAKE_BASE=TRUE	PCIE_AP_TO_BB_PERST_L	8
51	UART_NFC_TO_AP_CTS_L	=====	MAKE_BASE=TRUE	UART_NFC_TO_AP_CTS_L	12
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	UART_AP_TO_NFC_RTS_L	=====	MAKE_BASE=TRUE	UART_AP_TO_NFC_RTS_L	12
51	PMU_AMUX_BY	=====	MAKE_BASE=TRUE	PMU_AMUX_BY	23
51	PMU_AMUX_AY	=====	MAKE_BASE=TRUE	PMU_AMUX_AY	23
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	PCIE_BB_BI_AP_CLKREQ_L	=====	MAKE_BASE=TRUE	PCIE_BB_BI_AP_CLKREQ_L	8
51	NC_INT_135	=====	MAKE_BASE=TRUE	NC_INT_135	135
51	BB_BB_PEAK_POWER_INDICATOR	=====	MAKE_BASE=TRUE	BB_BB_PEAK_POWER_INDICATOR	55
51	GND	=====	MAKE_BASE=TRUE	GND	57 59
51	PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59
51	PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59
51	PP_VDD_MAIN	=====	MAKE_BASE=TRUE	PP_VDD_MAIN	59
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59
51	GND	=====	MAKE_BASE=TRUE	GND	57 58 59

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EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE (MLB_BOT_ICE,SS,ROW,639-94888)	EEEE_J99J	CRITICAL	SSROW
825-7691	1	EEEE (MLB_BOT_ICE,SS,JP,639-94881)	EEEE_J99K	CRITICAL	SSJP
825-7691	1	EEEE (MLB_BOT_ICE,SS,NA,639-95085)	EEEE_J95S	CRITICAL	SSNA
825-7691	1	EEEE (MLB_BOT_ICE,SSV,ROW,639-95086)	EEEE_J959	CRITICAL	SSVROW
825-7691	1	EEEE (MLB_BOT_ICE,SSV,JP,639-95496)	EEEE_JM18	CRITICAL	SSVJP
825-7691	1	EEEE (MLB_BOT_ICE,SSV,NA,639-95497)	EEEE_JM19	CRITICAL	SSVNA
825-7691	1	EEEE (MLB_BOT_ICE,DS,ROW,639-95648)	EEEE_JNSM	CRITICAL	DSROW

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, Kyocera
138S00150	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, SEMCO
138S00151	138S00149	BOM_TABLE_ALTS	ALL	0402-3T,10.5uF@1V, TY

CRITICAL PART#	COMMENT
138S00149	0402-3T,10.5uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00243	685-00182	BOM_TABLE_ALTS	SUBBOM_DS	COMMON,MLB_BOT_ICE00ES,00ES01,00E1

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00242	1	SUBBOM,MLB_BOT,D100ES,D100ES,X891	SUBBOM_DS	CRITICAL	COMMON
371S00133	4	D100ES,SH0TTKY D100E,30V,2A,0603	D3400,D3401,D3402,D3403	CRITICAL	D100ES_DS
371S00189	4	0NSEMT,SH0TTKY D100E,30V,2A,0603	D3400,D3401,D3402,D3403	CRITICAL	0NSEMI_DS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Kyocera
138S00163	138S00144	BOM_TABLE_ALTS	ALL	0402,16uF@1V, Taiyo

CRITICAL PART#	COMMENT
138S00144	0402,16uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Kyocera
138S00164	138S00139	BOM_TABLE_ALTS	ALL	0201,3uF@1V, Taiyo

CRITICAL PART#	COMMENT
138S00139	0201,3uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00221	138S00146	BOM_TABLE_ALTS	ALL	0402,5.1uF@3V, Kyocera

CRITICAL PART#	COMMENT
138S00146	0402,5.1uF@3V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Kyocera
138S00142	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, SEMCO
138S00166	138S00141	BOM_TABLE_ALTS	ALL	0201,1.1uF@3V, Taiyo

CRITICAL PART#	COMMENT
138S00141	0201,1.1uF@3V

Global Ferrites

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00558	152S00557	BOM_TABLE_ALTS	ALL	IND,MLD,0.47UH,20%,2.5A,80MD,1600
155S00194	155S00610	BOM_TABLE_ALTS	ALL	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005
155S00200	155S00610	BOM_TABLE_ALTS	ALL	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005
155S00338	155S00661	BOM_TABLE_ALTS	ALL	FERR BD,11 OHM,25%,1.5A,55MDMM DCR,0201

CRITICAL PART#	COMMENT
152S00557	IND,MLD,0.47UH,20%,2.5A,80MD,1600
155S00610	FERR BD,150 OHM,25%,200MA,0.7 DCR,01005

CRITICAL PART#	COMMENT
155S00661	FERR BD,33 OHM,25%,1.5A,55MDMM DCR,0201

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00648	138S00652	BOM_TABLE_ALTS	ALL	CAP,XSR,4.7UF,6.3V,0.05MM,0402,TAIYO
138S00024	138S00986	BOM_TABLE_ALTS	ALL	CAP,CER,3-TERM,1.0UF,20%,4V,0402,TAIYO/TDK
138S0706	138S0739	BOM_TABLE_ALTS	ALL	CAP,CER,1UF,20%,10V,XSR,0201,MURATA
138S0945	138S0739	BOM_TABLE_ALTS	ALL	CAP,CER,1UF,20%,10V,XSR,0201,MURATA
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,0.22UF,20%,6.3V,20%
138S00049	138S00831	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,0.22UF,20%,6.3V,01005
138S00133	138S00128	BOM_TABLE_ALTS	ALL	CAP,XSR,0.47UF,20%,6.3V,MR,01005
138S00116	138S00071	BOM_TABLE_ALTS	ALL	CAP,XSR,4UF,20%,6.3V,0201,T=0.50MM
138S00117	138S00071	BOM_TABLE_ALTS	ALL	CAP,XSR,4UF,20%,6.3V,0201,T=0.50MM
138S00048	138S00003	BOM_TABLE_ALTS	ALL	CAP,XSR,150UF,20%,6.3V,0.05MM,MRZTL,0402
131S00172	131S00164	BOM_TABLE_ALTS	ALL	CAP,CER,CBG,220PF,5%,10V,01005
131S00173	131S00164	BOM_TABLE_ALTS	ALL	CAP,CER,CBG,220PF,5%,10V,01005
131S00185	131S0316	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,0.1UF,10%,6.3V,01005
138S00048	138S00003	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,150UF,20%,6.3V,0402
132S00185	132S0316	BOM_TABLE_ALTS	ALL	CAP,CER,XSR,150UF,20%,6.3V,0402

CRITICAL PART#	COMMENT
138S00652	CAP,XSR,4.7UF,6.3V,0.05MM,0402,TAIYO
138S00986	CAP,CER,3-TERM,1.0UF,20%,4V,0402,TAIYO/TDK
138S0739	CAP,CER,1UF,20%,10V,XSR,0201,MURATA

CRITICAL PART#	COMMENT
138S0706	CAP,CER,XSR,0.22UF,20%,6.3V,20%
132S0400	CAP,CER,XSR,0.22UF,20%,6.3V,01005
138S0031	CAP,CER,XSR,2.2UF,20%,6.3V,0201
138S00128	CAP,XSR,0.47UF,20%,6.3V,KYO,01005
138S00071	CAP,XSR,4UF,20%,6.3V,0201,T=0.50MM

CRITICAL PART#	COMMENT
138S00003	CAP,XSR,15UF,20%,6.3V,0.05MM,MRZTL,0402
131S00164	CAP,CER,CBG,220PF,5%,10V,01005

Multi-Vendor Criticals

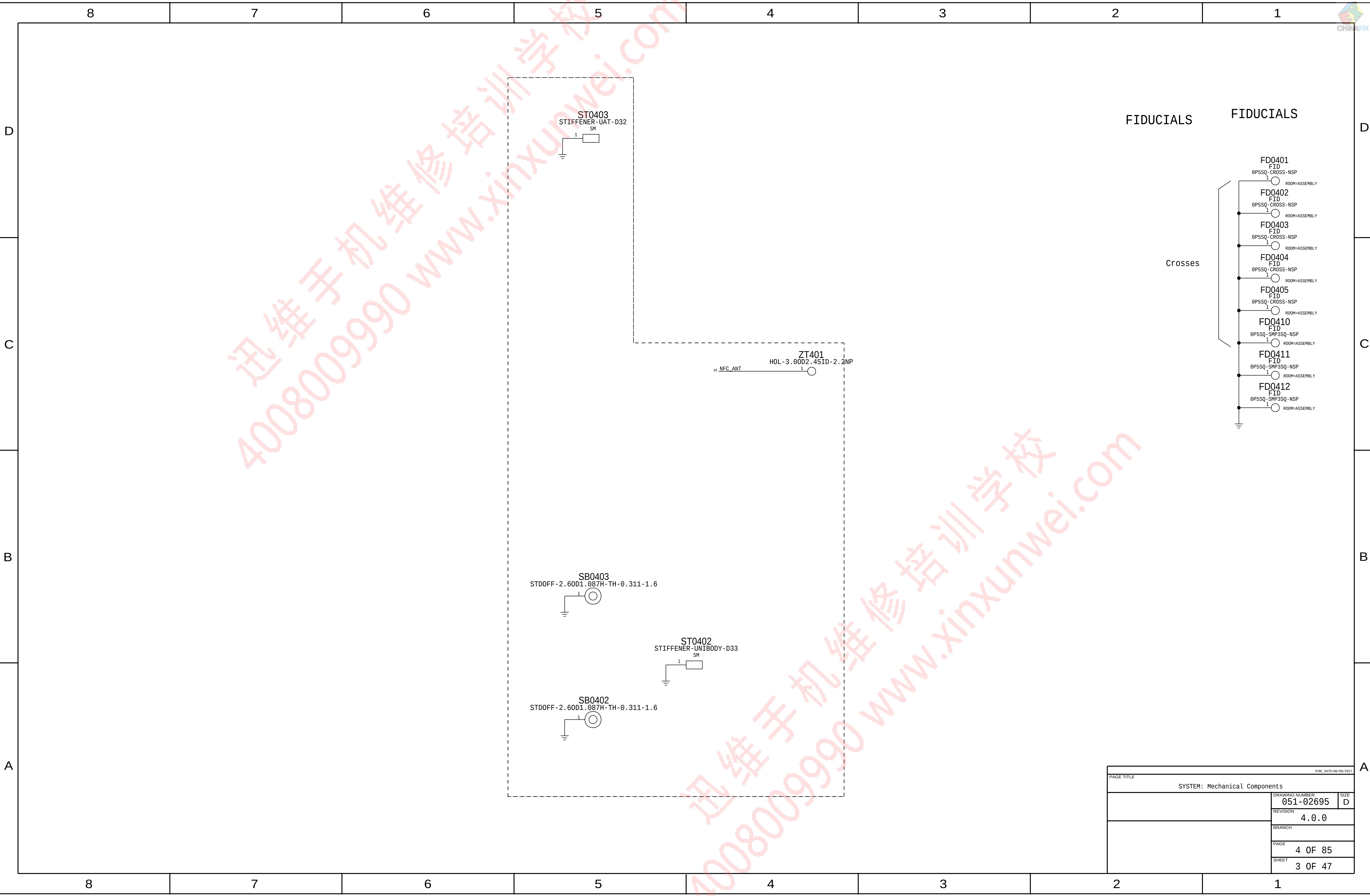
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138S00979	CAP,CER,XSR,18UF,20%,10V,0402,H=0.65MM	132S00008	CAP,CER,0.1UF,10%,50V,XTR,0402
138S00683	CAP,CER,XSR,1UF,10%,25V,0402	131S00004	CAP,CER,27PF,5%,C0G,25V,0201
132S00663	CAP,CER,XSR,1UF,10%,25V,0402	131S00307	CAP,CER,NP0/C0G,100PF,5%,10V,01005
132S0288	CAP,CER,XSR,0.1UF,10%,10V,0201	131S00053	CAP,CER,C0G,220PF,5%,10V,01005
132S0275	CAP,CER,XSR,470PF,10%,10V,01005	117S00055	RES,MF,1/20W,2M OHM,5,0201,SMD
132S0245	CAP,CER,XSR,0.01UF,10%,6.3V,01005	107S0257	THERMISTOR,NTC,10K OHM,1%,B=3435,01005
138S00692	CAP,CER,XSR,1UF,20%,6.3V,0201	131S00170	CAP,CER,C0G,220PF,5%,25V,01005
		131S00643	CAP,CER,NP0/C0G,50PF,5%,25V,01005
		131S0316	CAP,CER,XSR,0.1UF,20%,6.3V,01005

Global Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00653	152S00651	BOM_TABLE_ALTS	ALL	IND,1.2UH,3A,2016,0.65Z
152S00654	152S00652	BOM_TABLE_ALTS	ALL	IND,1.2UH,3A,2016,0.8Z

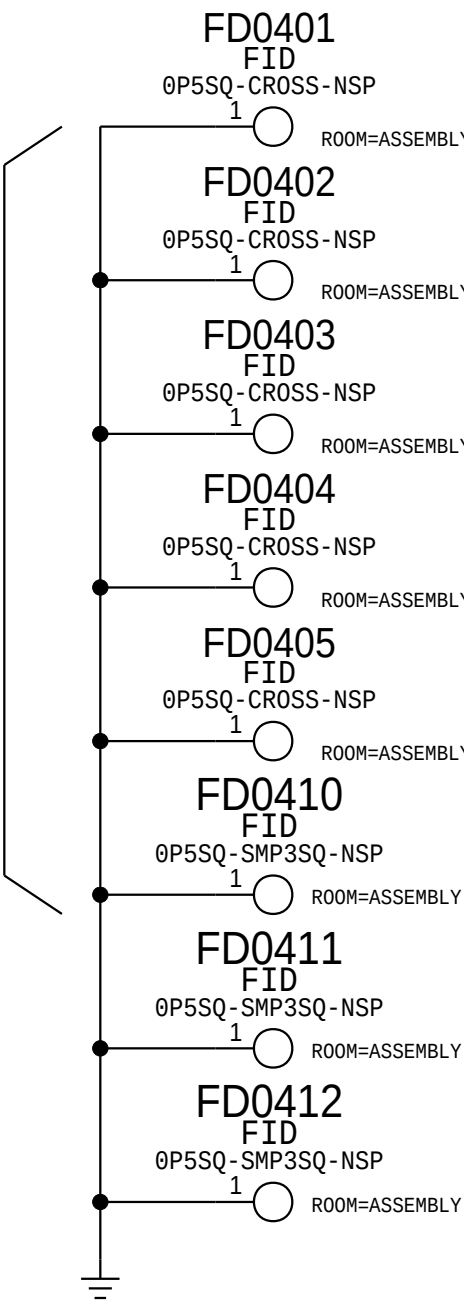
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FIDUCIALS FIDUCIALS

Crosses



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SYNMASTER

BOOTSTRAPPING:BOARD ID[3]

Board ID[4:0]					
Float = Low PU = High					
	4	3	2	1	0
	Denali = 0, Imola = 1	Mav = 0, Ice = 1	00=Open 01=D33 10=N84 11=D32	MLB = 0, Dev = 1	
D33 MLB Mav	0	0	0	1	0
D33p MLB Mav	1	0	0	1	0
D33 Dev Mav	0	0	0	1	1
D33p Dev Mav	1	0	0	1	1
D33 MLB Ice	0	1	0	1	0
D33p MLB Ice	1	1	0	1	0
D33 Dev Ice	0	1	0	1	1
D33p Dev Ice	1	1	0	1	1
D32 MLB Mav	0	0	1	1	0
D32 Dev Mav	0	0	1	1	1
D32 MLB Ice	0	1	1	1	0
D32 Dev Ice	0	1	1	1	1

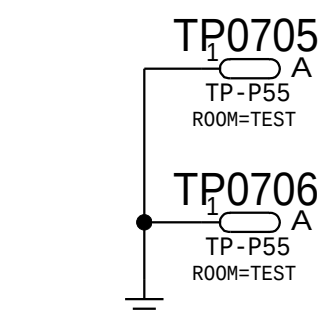
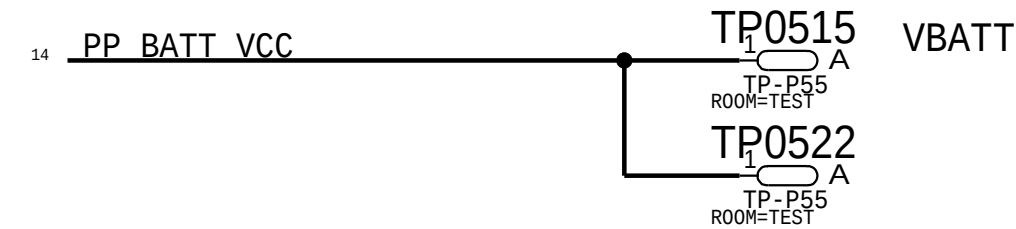
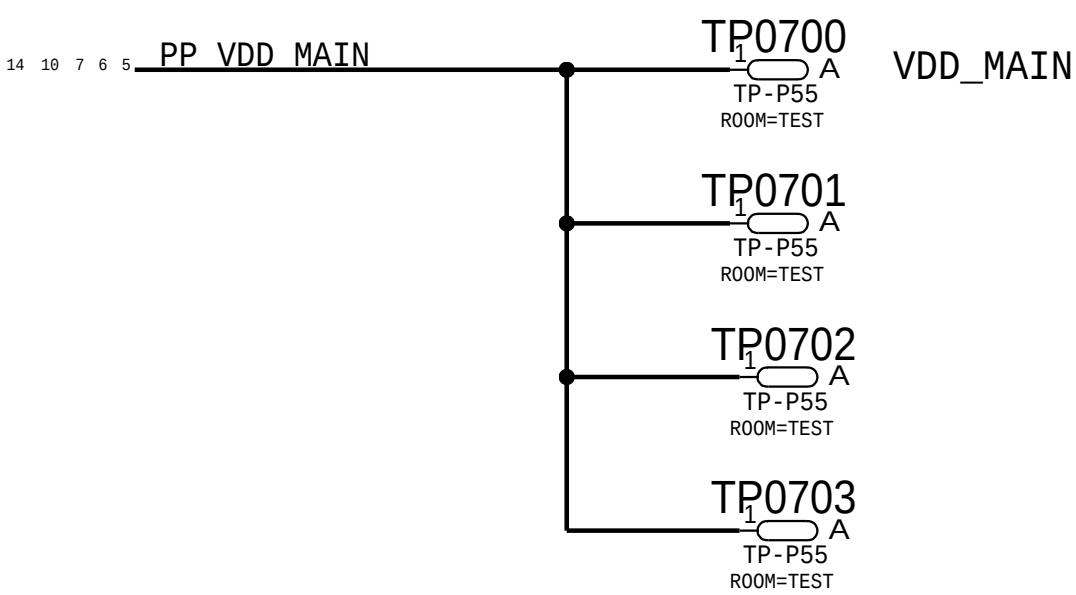
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11  BOARD_ID2
GXPLUS_WALVE-SINGLE_NODENET

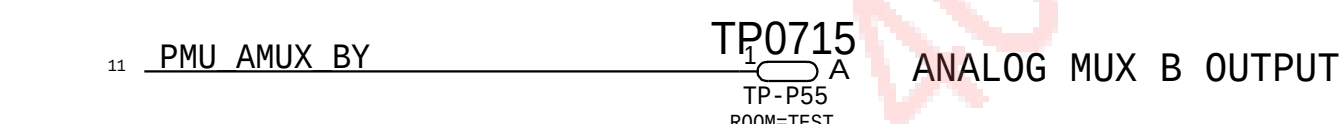
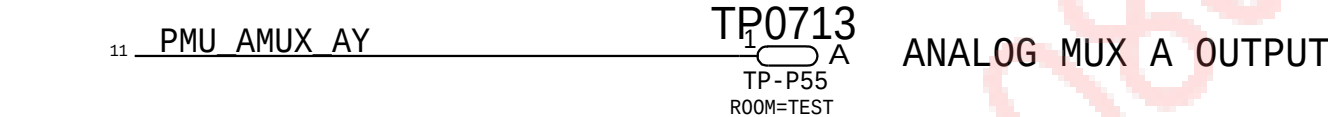
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			REVISION	4.0.0
			BRANCH	
			PAGE	
			6 OF 85	
			SHEET	
			4 OF 47	



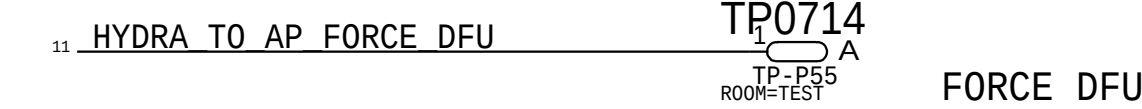
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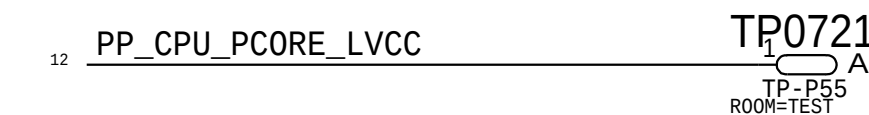
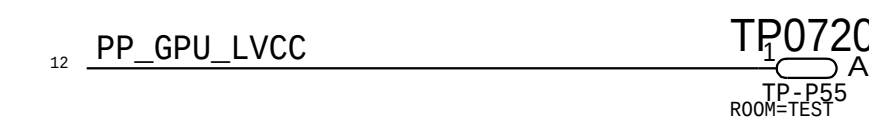
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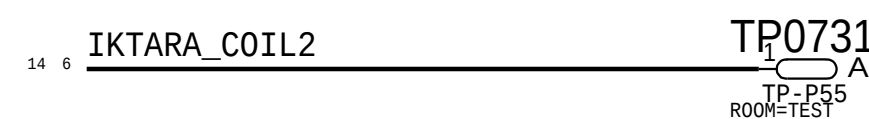
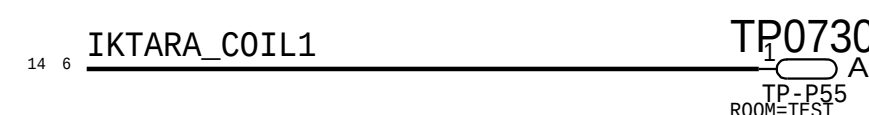
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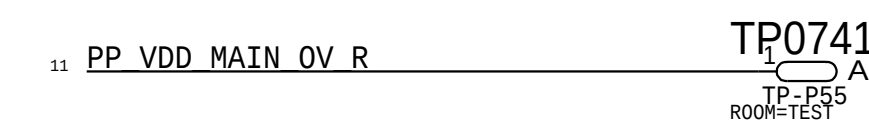
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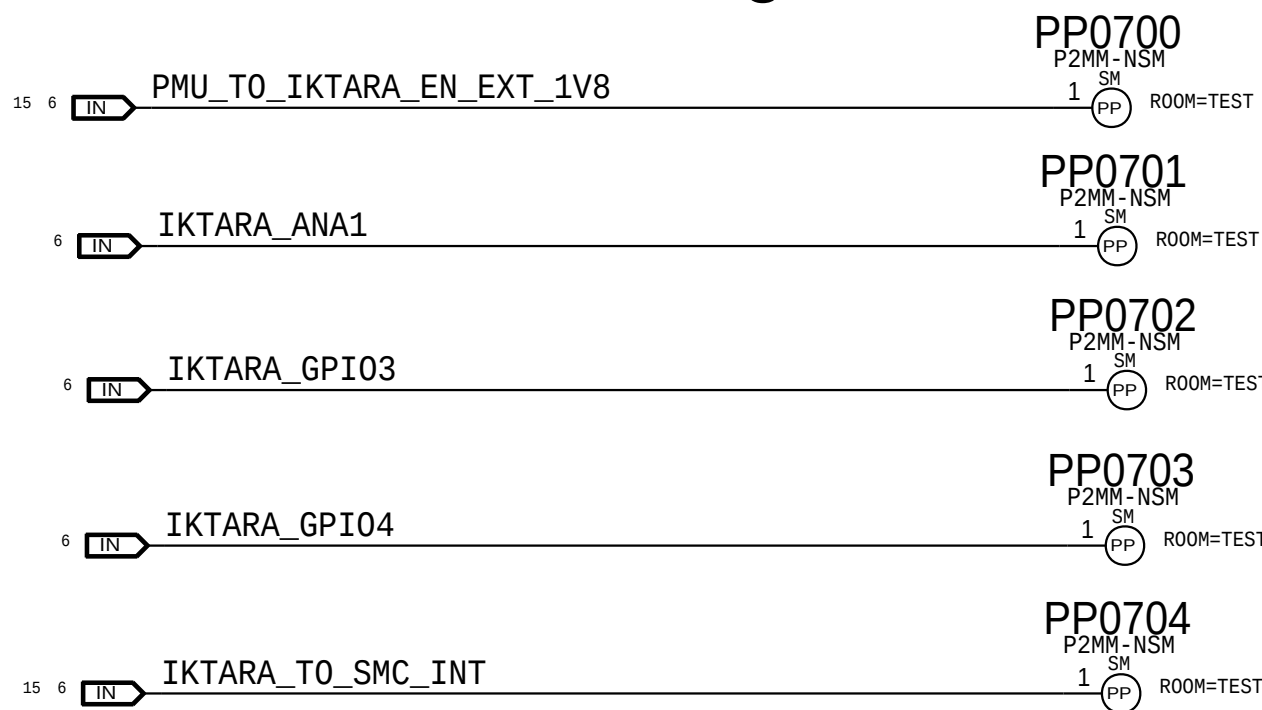


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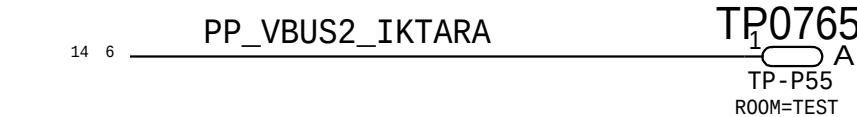
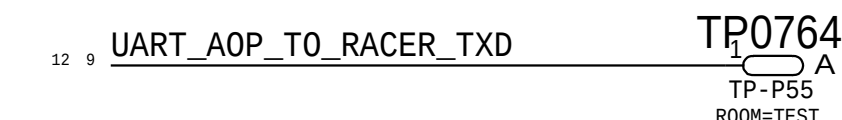
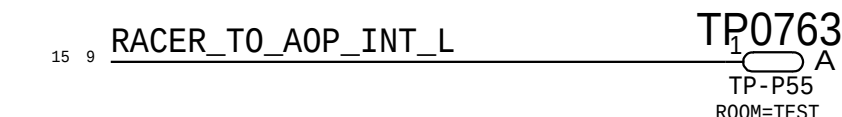
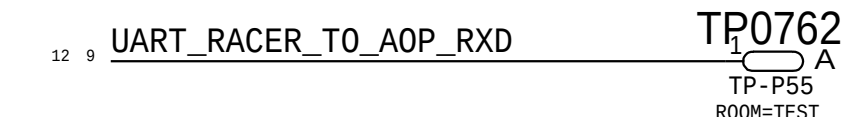
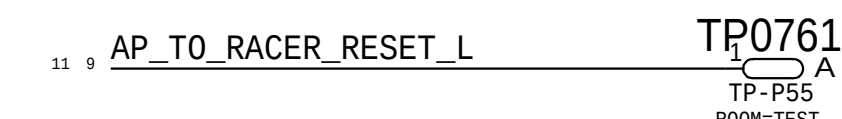
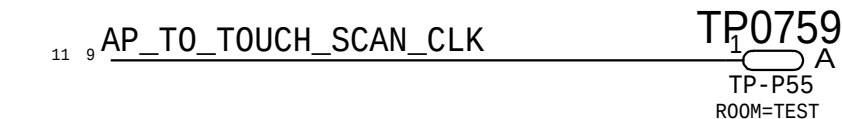
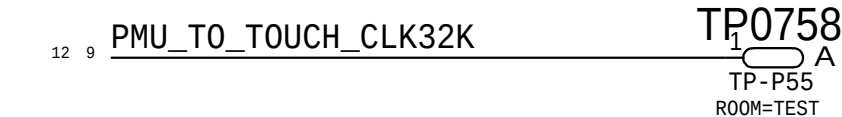
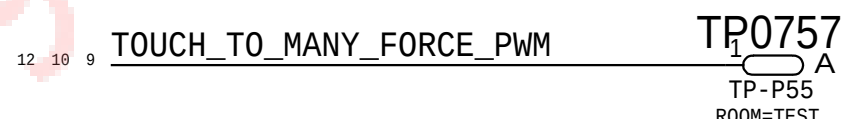
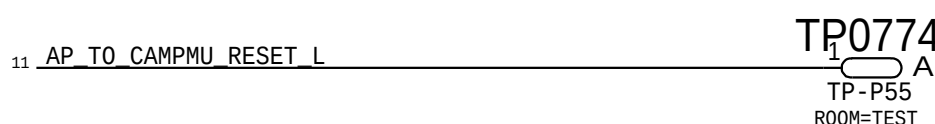
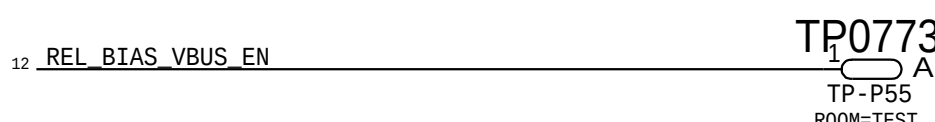
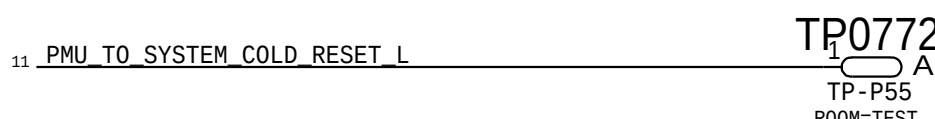


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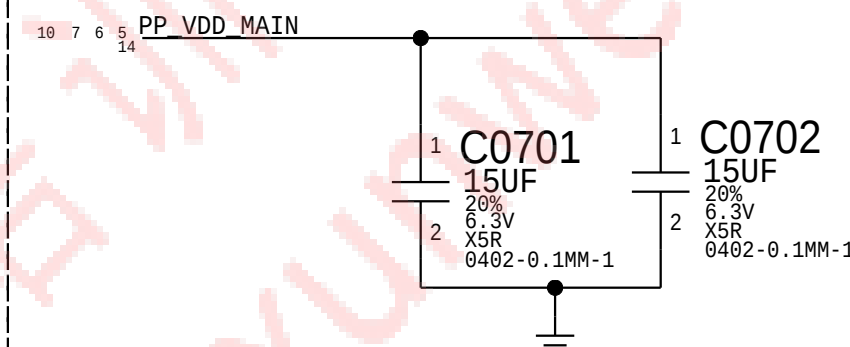
Iktara Debug



REL BIAS



VDD_MAIN TRADEOFF CAPS

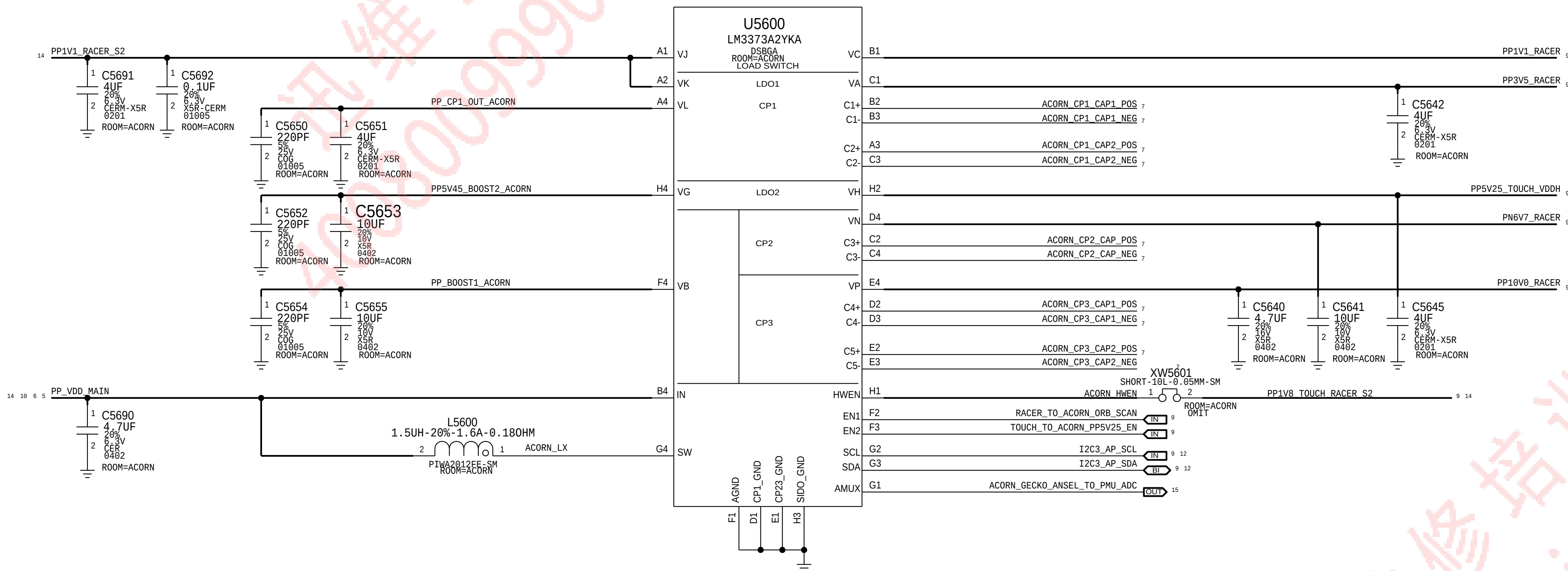
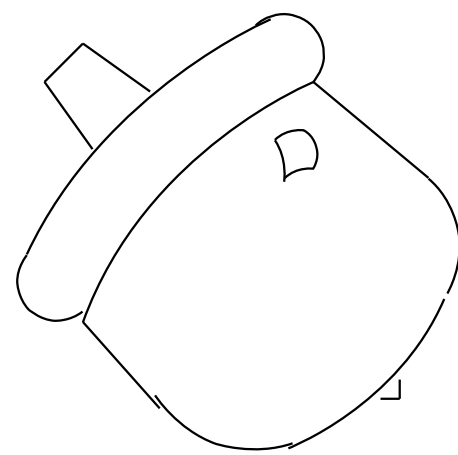


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	BRANCH	
	PAGE	7 OF 85
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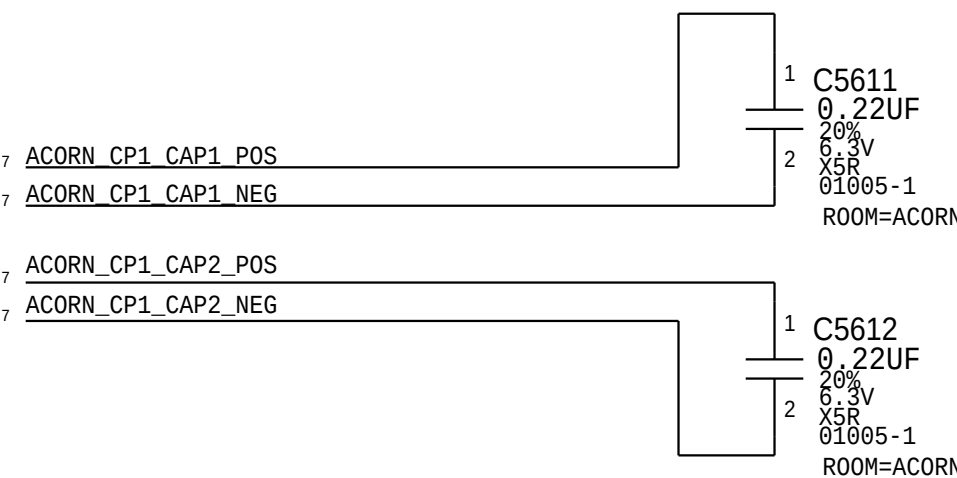


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	4.0.0	
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	SHEET	6 OF 47

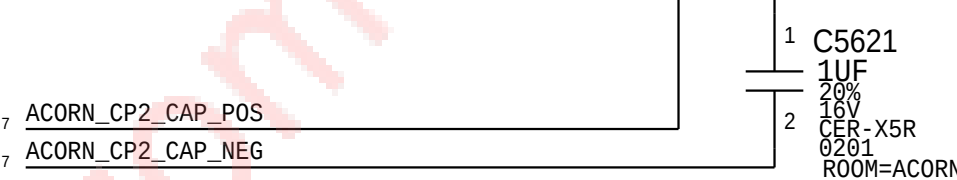
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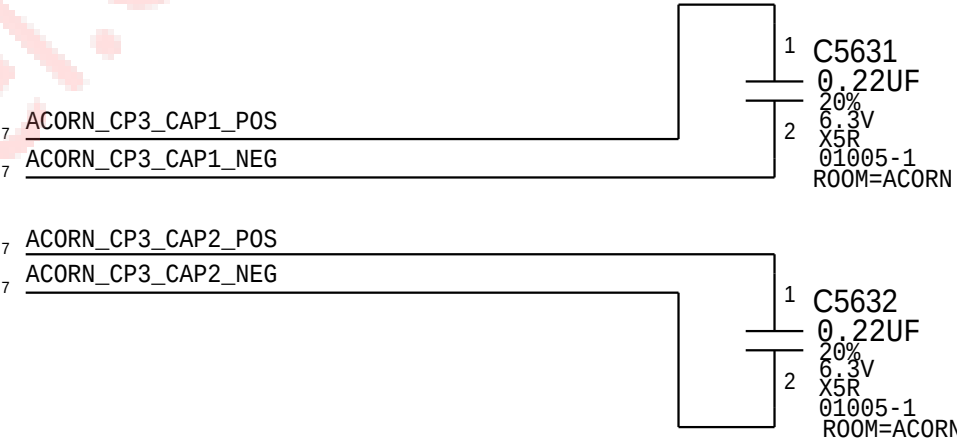
Charge Pump 1 Caps



Charge Pump 2 Caps



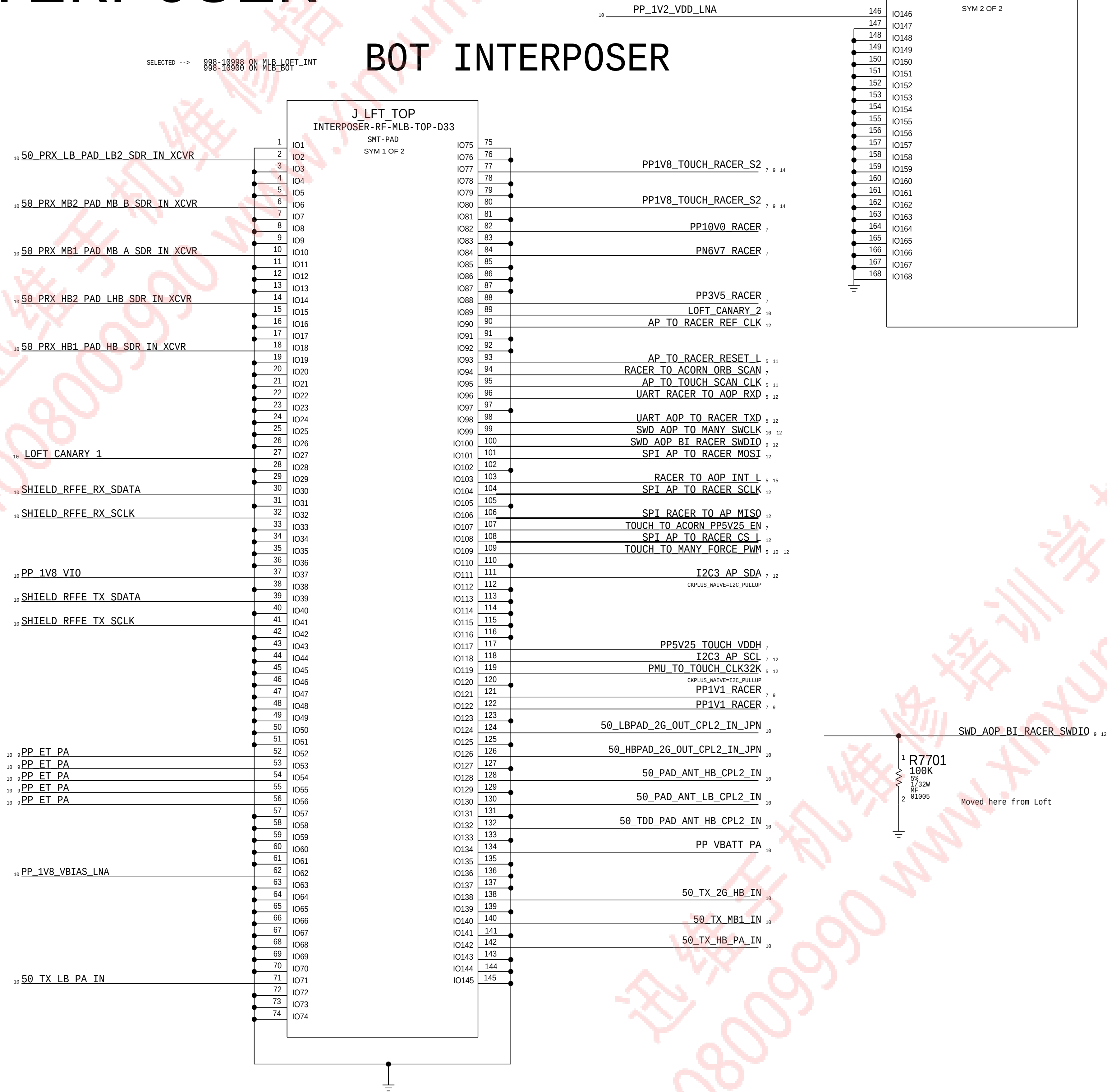
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REVISION		
4.0.0		
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LOFT INTERPOSER

BOT INTERPOSER



PAGE TITLE		
B2B: Interposer Loft		
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```

8      BB TO AP RESET
8
8      BOA
8      AP TO GNSS TIM
8      PCIE BB BI AP CI
AP TO BB PEAK POWER INC
8      AP TO BBPMU RAD
8
8      PP VC
8      PP VC
8      PP VC
8
8
8
8
8
8
8      90 PCIE BB TO AF
8      90 PCIE BB TO AF
8
8      90 PCIE AP TO BE
8      90 PCIE AP TO BE
8
8
8      90 PCIE AP TO BB RE
8      90 PCIE AP TO BB RE
8
8      UART BB TO A
8      UART GNSS TO
8      PCIE AP TO BB F
8
8
8      PMU A
8      PMU A
8
8
8      UART NFC TO AF
8      UART AP TO NFC
BB TO AP PEAK POWER INC
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8      PP VC
8      PP VC
8      PP VC
8
8
8
8

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THIS SIDE HAS ATTRIBUTE
MAKE **BASE=TRUE**

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GND	WMC_B001270F	GND	11 14
AP TO RACER RESET_L	WMC_B001270F		AP TO RACER RESET_L
GND	WMC_B001270F	GND	11 14
AP TO WLAN TIME SYNC	WMC_B001270F		AP TO WLAN TIME SYNC
GND	WMC_B001270F	GND	11 14
GNSS TO AP LOW PWR_IND	WMC_B001270F		GNSS TO AP LOW PWR_IND
GND	WMC_B001270F	GND	11 14
HYDRA TO AP FORCE_DFU	WMC_B001270F		HYDRA TO AP FORCE_DFU
GND	WMC_B001270F	GND	11 14
PP1V8_S2	WMC_B001270F		PP1V8_S2
PP1V8_S2	WMC_B001270F		PP1V8_S2
GND	WMC_B001270F	GND	11 14
GND	WMC_B001270F	GND	11 14
GND	WMC_B001270F	GND	11 14
GND	WMC_B001270F	GND	11 14
GND	WMC_B001270F	GND	11 12 14
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UART_AP_TO_NFC_TXD	WMC_B001270F		UART_AP_TO_NFC_TXD
UART_NFC_TO_AP_RXD	WMC_B001270F		UART_NFC_TO_AP_RXD
BB_TO_AP_RESET_DETECT_L	WMC_B001270F		BB_TO_AP_RESET_DETECT_L
GND	WMC_B001270F	GND	14
BOARD_ID2	WMC_B001270F		BOARD_ID2
AP_TO_GNSS_TIME_MARK	WMC_B001270F		AP_TO_GNSS_TIME_MARK
PCIE_BB_BT_AP_CLKREQ_L	WMC_B001270F		PCIE_BB_BT_AP_CLKREQ_L
AP_TO_BB_PEAK_POWER_INDICATOR	WMC_B001270F		AP_TO_BB_PEAK_POWER_INDICATOR
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PP_VDD_MAIN	WMC_B001270F		PP_VDD_MAIN
PP_VDD_MAIN	WMC_B001270F		PP_VDD_MAIN
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GND	WMC_B001270F	GND	11 14
GND	WMC_B001270F	GND	11 14
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90_PCIE_AP_TO_BB_REFCLK_P	WMC_B001270F		90_PCIE_AP_TO_BB_REFCLK_P
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UART_GNSS_TO_AP_RXD	WMC_B001270F		UART_GNSS_TO_AP_RXD
PCIE_AP_TO_BB_PERST_L	WMC_B001270F		PCIE_AP_TO_BB_PERST_L
GND	WMC_B001270F	GND	11 14
PMU_AMUX_BY	WMC_B001270F		PMU_AMUX_BY
PMU_AMUX_AY	WMC_B001270F		PMU_AMUX_AY
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UART_AP_TO_NFC_RTS_L	WMC_B001270F		UART_AP_TO_NFC_RTS_L
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PP_VDD_MAIN	WMC_B001270F		PP_VDD_MAIN
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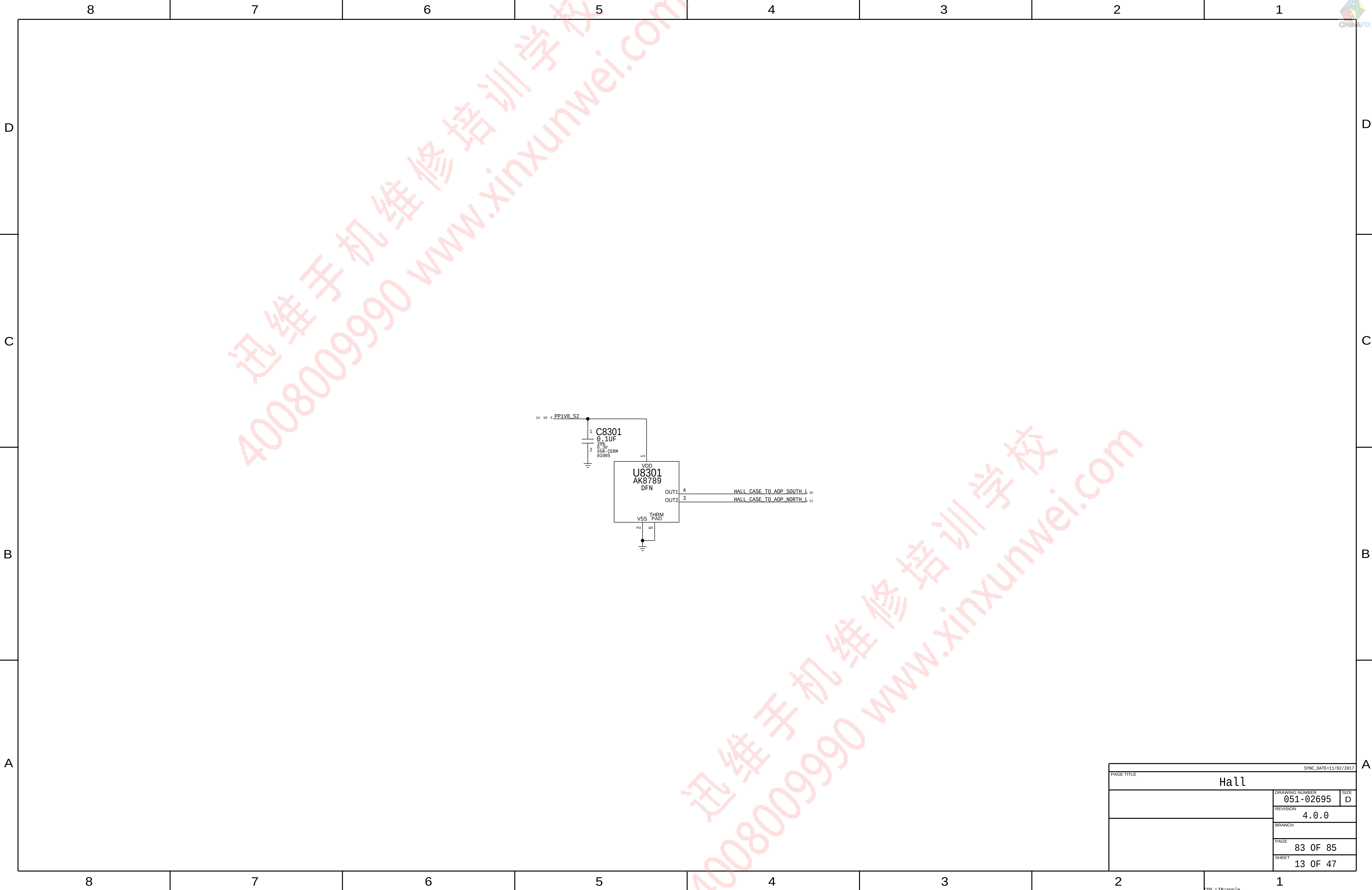
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	PAGE	81 OF 85	
	SHEET	11 OF 47	

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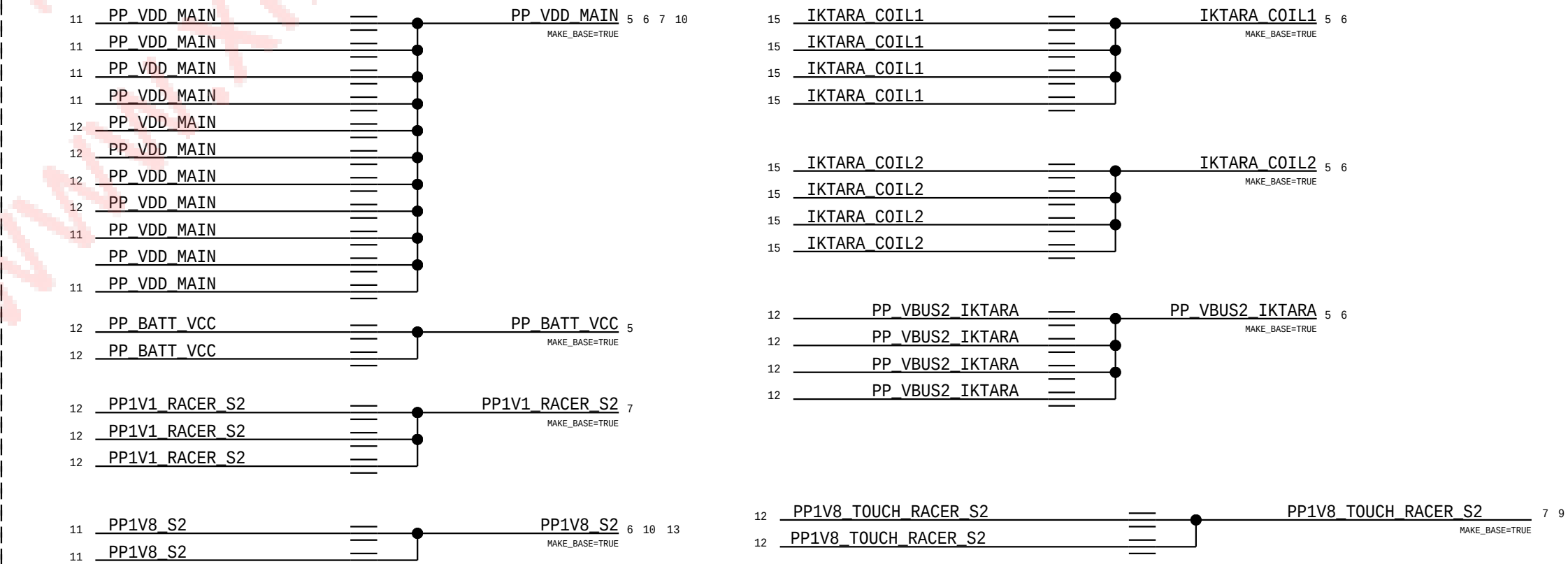
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		SHEET	
		12	OF 47



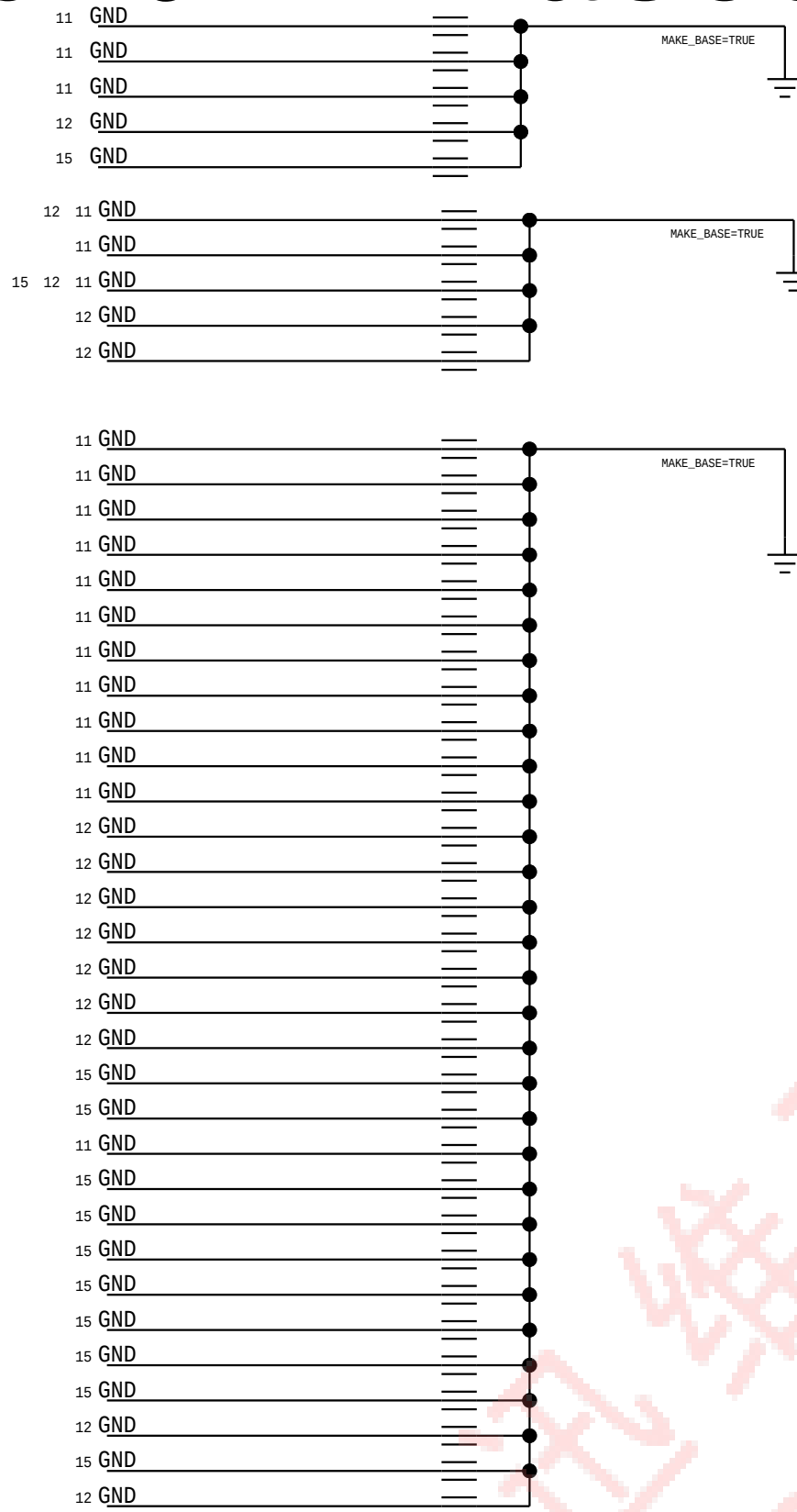
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						PAGE			83 OF 85			SHEET			13 OF 47		

Interposer Top Level Aliases

Power Aliases

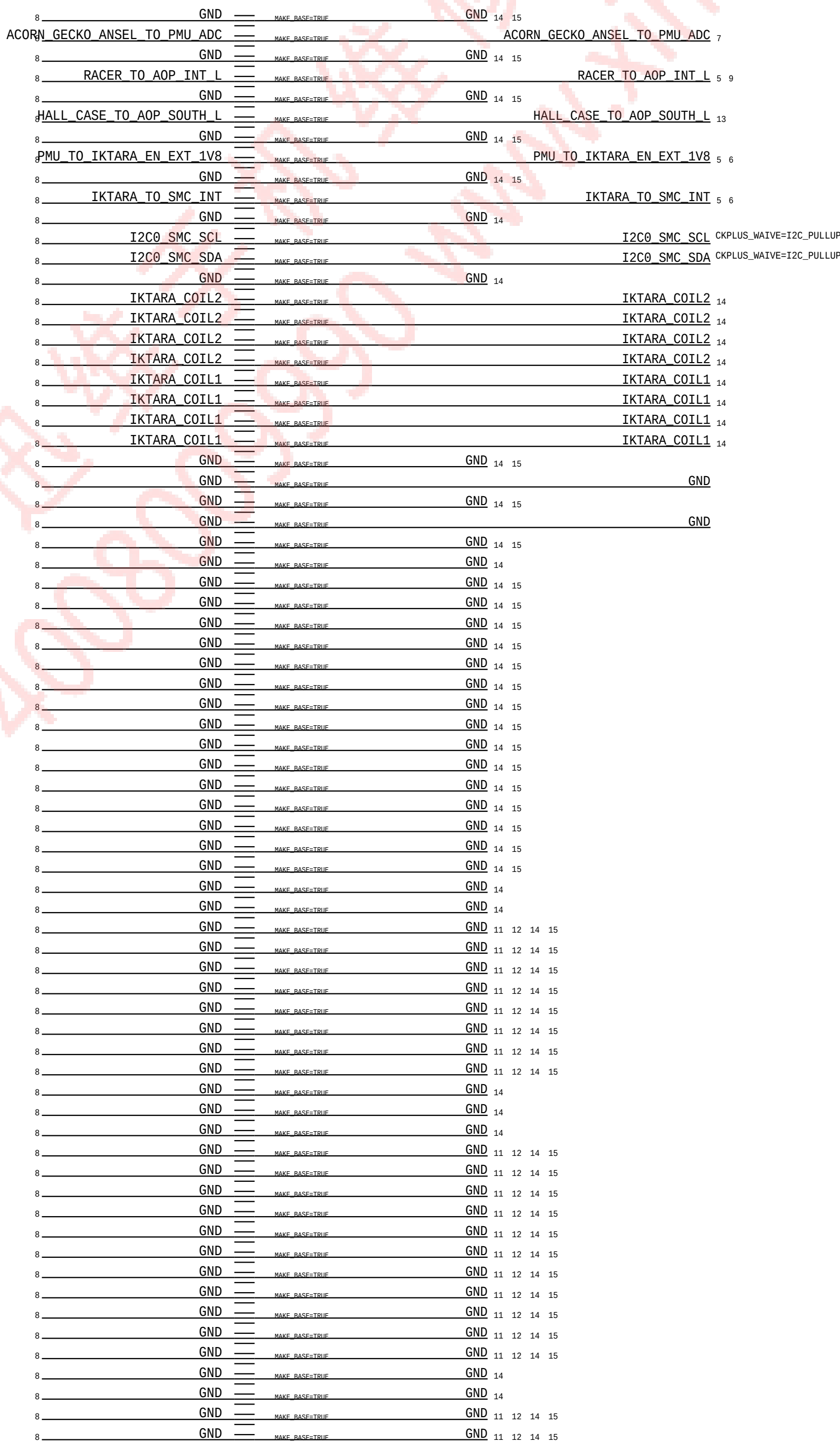


Ground Aliases



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	PAGE	84 OF 85
	SHEET	14 OF 47

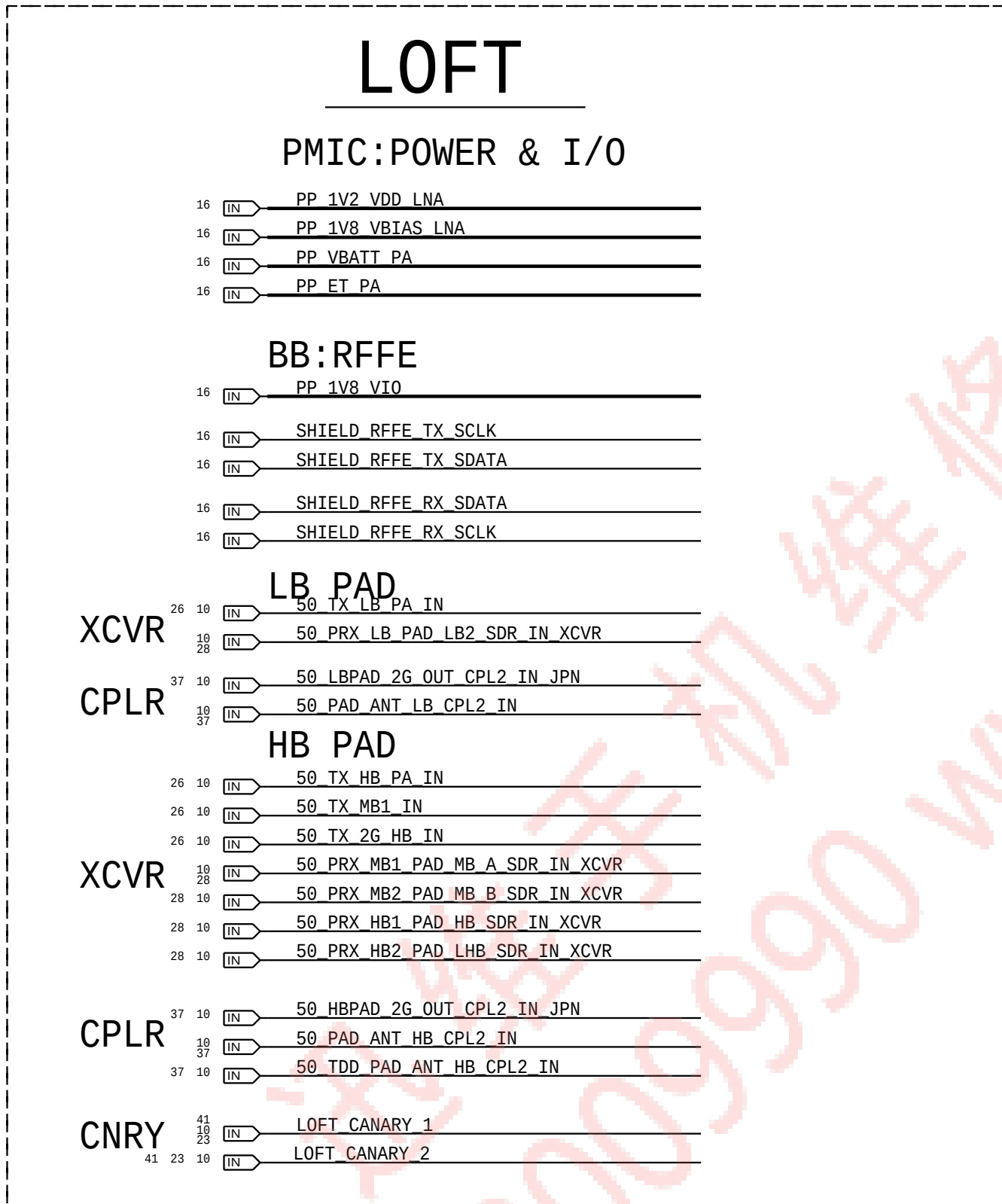
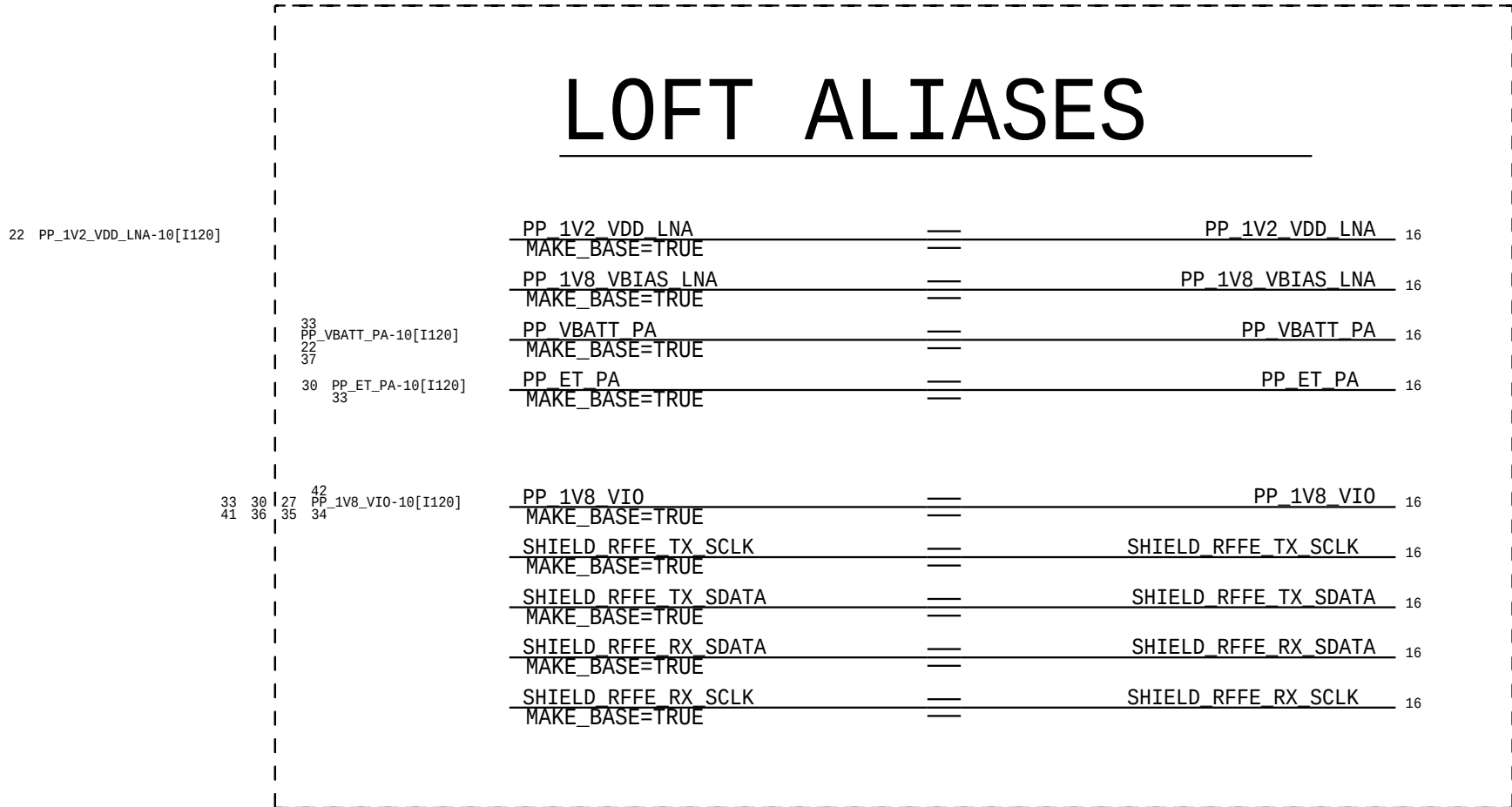
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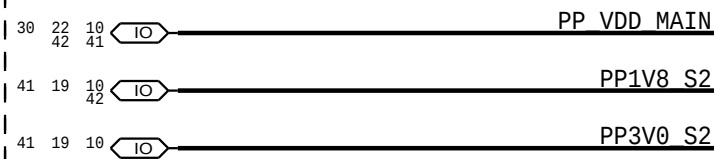
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LAST_MODIFICATION=Fri Mar 16 10:21:01 2018

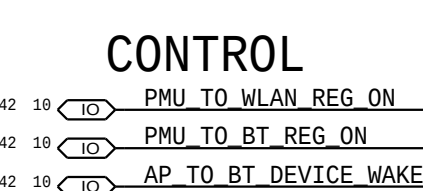
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16	1	RADIO: TABLE OF CONTENTS		
17	2	BOM TABLES		
18	3	ANTENNA DIAGRAM		
19	4	ANTENNA: B2BS		
20	5	ANTENNA: N-PLEX SHARED		
21	6	BBPMU: CONTROL		
22	7	BBPMU: RAILS		
23	8	BB: INTERFACE		
24	9	BB: DDR PWR & JTAG		
25	10	BB: DIGITAL PWR		
26	11	XCVR: TX & GNSS		
27	12	XCVR: INTERFACE & PWR		
28	13	XCVR: PRX DRX		
29	14	HW CONFIG OPTIONS		
30	15	ET		
31	16	LB SPAD		
32	17	HB SPAD		
33	18	UHB LMB SPAD		
34	19	LB DIVERSITY RECEIVE LNA		
35	20	HB DIVERSITY RECEIVE LNA		
36	21	MIMO RECEIVE LNAS		
37	22	COUPLER + LOWER ANTENNA		
38	23	UPPER ANTENNA FEEDS		
39	24	SIM: ESIM		
40	25	SIM: PSIM		
41	26	TEST POINTS		
42	27	SYMBOL: WIFI		
43	1	WIFI: TABLE OF CONTENTS		
44	2	DIETCOKE		
45	3	FEM MODULES		



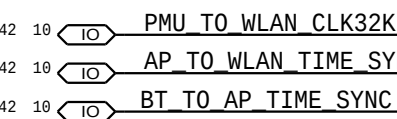
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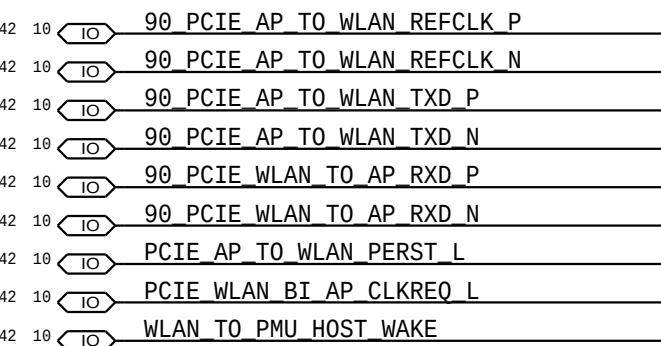
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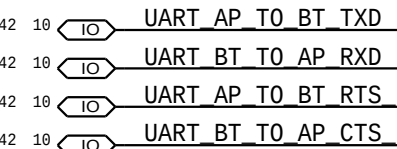
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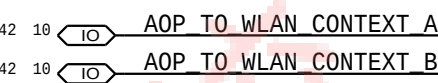
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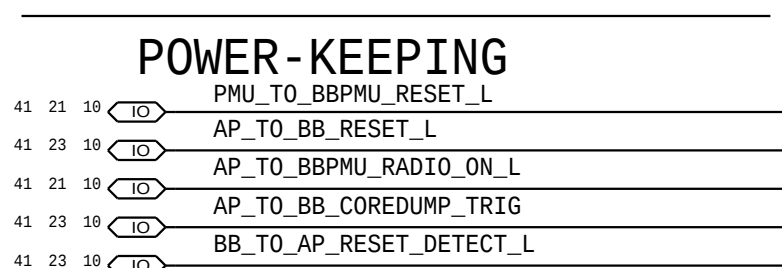
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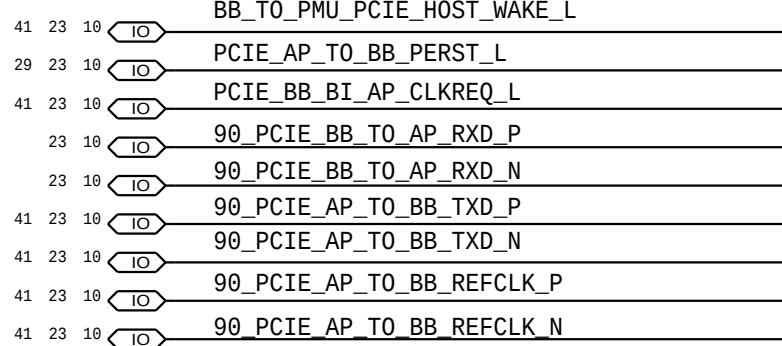
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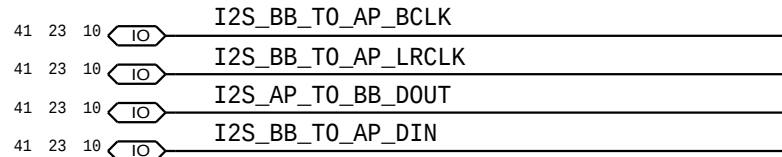
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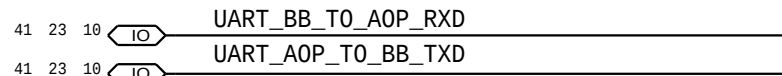
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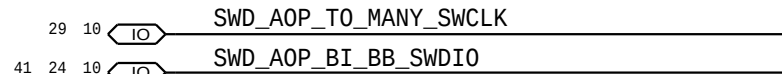
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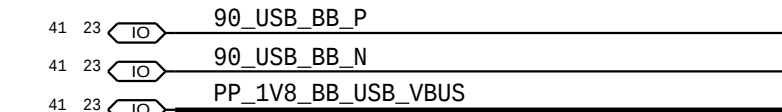
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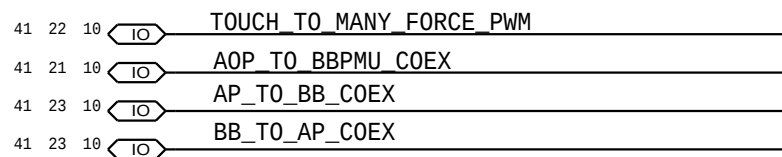
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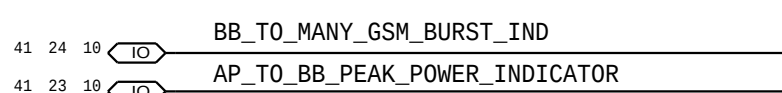
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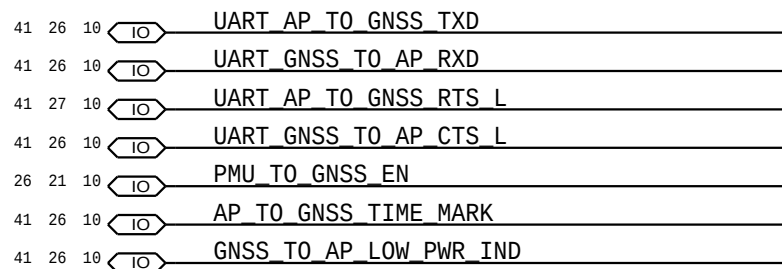
SYSTEM COEX



PEAK POWER MANAGEMENT



GNSS



RADIO: TABLE OF CONTENTS

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D32	HIER_WIFI	0.52.0	S	2018_03_07_13:05:02

SCH:951-04720
PCB:920-03588

DRAWING TITLE			
SCH,MLB,BOT,ICE,D33			
		DRAWING NUMBER	051-02695
		REVISION	4.0.0
		BRANCH	
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		SHEET	16 OF 47

BOM TABLES

RF SKU

ROW SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2003_K	ROW
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2015_K	ROW
118S00072	1	RES,MF,0.2K OHM,1/32W,01005	R312_K	ROW
197S00155	1	XTL,THERM,30.4MHZ,12PPM,6PF,1612	Y301_K	ROW
353S01647	1	20PA, SKYWORKS, 78208, REV13	PA_UHB_K	ROW
353S01301	1	SKYWORKS, LB DSM, 13765, REV16	DSM_LB_K	ROW

JP SKU

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118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R1403_K	JP
118S00120	1	RES,MF,0.0K OHM,1/32W,01005	R312_K	JP
197S00179	1	XTL,THERM,30.4MHZ,12PPM,6PF,1612	Y301_K	JP
353S01662	1	SKYWORKS, UHB PA, REV13	PA_UHB_K	JP
353S01301	1	SKYWORKS, LB DSM, 13765, REV16	DSM_LB_K	JP

NA SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2003_K	NA
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2015_K	NA
118S00073	1	RES,MF,50K OHM,1/32W,01005	R312_K	NA
197S00156	1	XTL,THERM,30.4MHZ,12PPM,6PF,1612	Y301_K	NA
353S01647	1	20PA, SKYWORKS, 78208, REV13	PA_UHB_K	NA
353S01472	1	SKYWORKS, LB DSM, 13765, REV13	DSM_LB_K	NA

SIM SKU

SINGLE 4FF SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
512S00030	1	SIM,TALL,PTH,4X3,D33	J_SIM	SINGLE
117S0158	1	RES,MF,100K OHM,1/32W,01005	R11_SIM	SINGLE

DUAL 4FF SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
512S00039	1	SIM,DUAL,BUNK,PTH,4X3,D33	J_SIM	DUAL
117S0161	1	RES,MF,0.0 OHM,1/32W,01005	R31_SIM	DUAL
117S0166	1	RES,MF,100 OHM,1/32W,01005	R27_SIM	DUAL
117S0185	1	RES,MF,47 OHM,1/32W,01005	R26_SIM	DUAL
117S0166	1	RES,MF,100 OHM,1/32W,01005	R29_SIM	DUAL

SINGLE 4FF + VINYL SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
512S00030	1	SIM,TALL,PTH,4X3,D33	J_SIM	VINYL
990-12930	1	VINYL,N10,BIRCH 4.0	U_SIM	VINYL
117S0158	1	RES,MF,100K OHM,1/32W,01005	R10_SIM	VINYL
118S00636	1	RES,MF,4.7K OHM,1/32W,01005	R12_SIM	VINYL

RULER_RULE_SET=MSAP_2017		MANUFACTURING CONFIGURATION				TABLE_REV_NUMBER=6	
		DIELECTRIC BASED SPACING RULES			DEFAULT SPACING MULTIPLES	VOID SPACE RATIO	
		MULTIPLES	SMDPIN MAX(UM)	MVIA MAX(UM)			
1		1.65, 2, 2D, 3.2D, 4D		120	120	100	2
		LAYERS		MINIMUM CU WIDTH RATIO		MINIMUM CU SPACING RATIO	MINIMUM TO DEFAULT RATIO
TOP,BOTTOM, ISL4, ISL5				1.0		1.0	1.0
ISL2, ISL3, ISL6, ISL7				1.0		1.0	1.0

POWER RULE DEFINITIONS			
LOAD CURRENT(MA)	CURRENT DENSITY(A/SQMM)	MIN NECK WIDTH(MM)	MAX NECK LENGTH(MM)
50	50	0.08	10
100	50	0.08	10
200	50	0.08	10
300	50	0.08	10
500	50	0.08	10
1000	50	0.08	10
3000	50	0.08	10

TABLE_IMPEDANCE_HYBRID=50_HYBRID	
LAYERS	IMPEDANCE RULE
TOP	50_OHM_SE
ISL3	50_OHM_SE
ISL6	50_OHM_SE_6R4R0
ISL7	50_OHM_SE
ISL5	50_OHM_SE

TABLE_IMPEDANCE_HYBRID=50_HYBRID_WIDE	
LAYERS	IMPEDANCE RULE
TOP	50_OHM_SE_1R3
ISL3	50_OHM_SE
ISL6	50_OHM_SE_6R4R0
ISL7	50_OHM_SE
ISL5	50_OHM_SE

FERRITE ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00131	155S00341	BOM_TABLE_ALTS	FL900_K	FERR BD, 2400HM
155S00414	155S00076	BOM_TABLE_ALTS	ALL	FERR BD, 100HM
155S00200	155S00400	BOM_TABLE_ALTS	ALL	FERR BD, 1500HM
155S00194	155S00400	BOM_TABLE_ALTS	ALL	FERR BD, 1500HM

ETIC CAP ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00237	138S00167	BOM_TABLE_ALTS	C1204_K	CAP, 2.2UF
138S00237	138S00167	BOM_TABLE_ALTS	C1205_K	CAP, 2.2UF
138S00237	138S00167	BOM_TABLE_ALTS	C1206_K	CAP, 2.2UF

CAP ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00086	138S00084	BOM_TABLE_ALTS	ALL	CAP, 20UF
138S0719	138S1103	BOM_TABLE_ALTS	ALL	CAP, 4.7UF
138S00128	138S00133	BOM_TABLE_ALTS	ALL	CAP, 0.47UF

VARISTOR ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S00130	377S0106	BOM_TABLE_ALTS	ALL	VARISTOR, 12V, 33PF

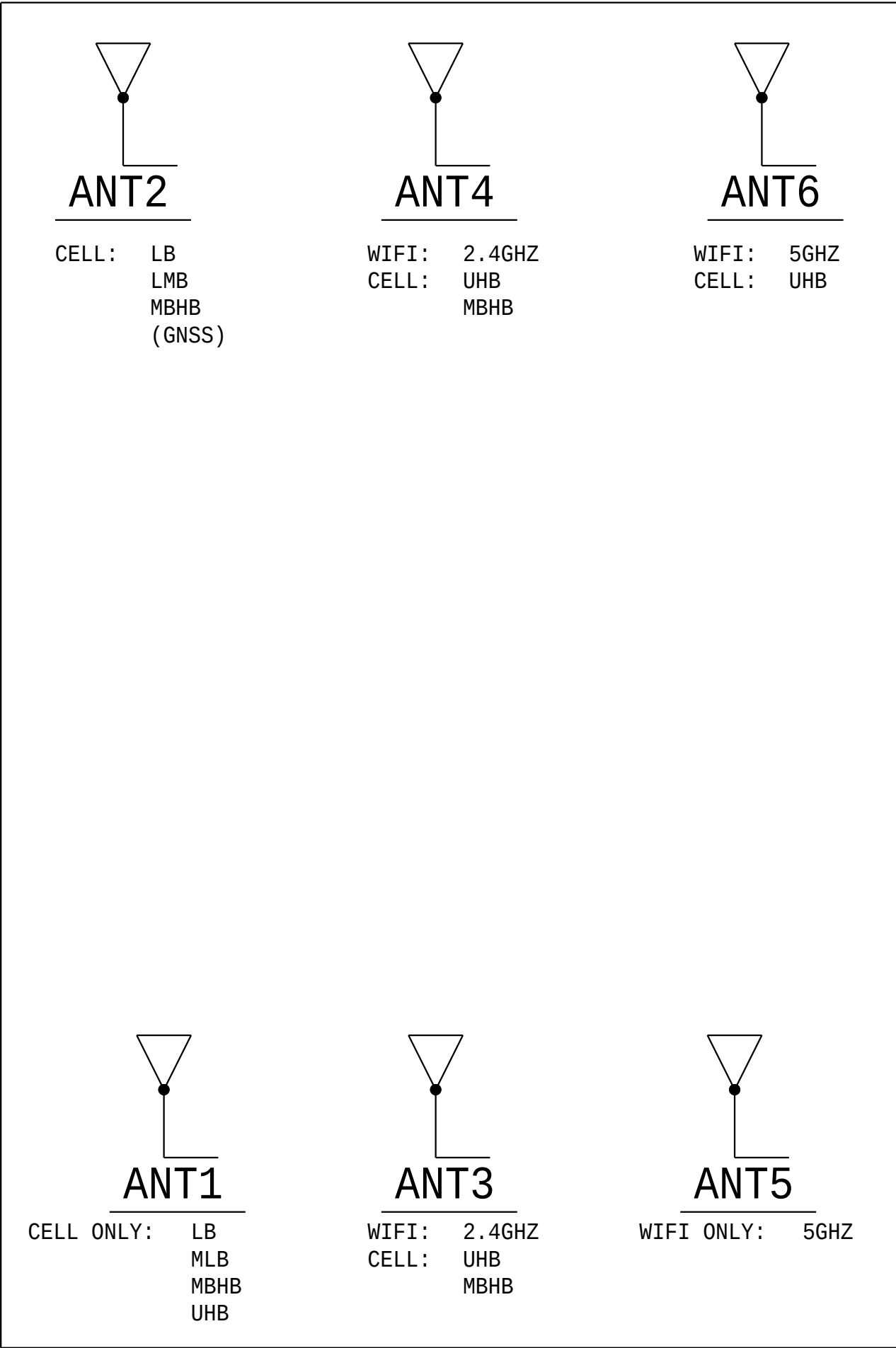
EEPROM ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00013	335S00094	BOM_TABLE_ALTS	EEPROM_K	EEPROM, 8KBIT, I2C

PAGE TITLE		
BOM TABLES		
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		SHEET
		17 OF 47

8	7	6	5	4	3	2	1
D							
C							
B							
A							
8	7	6	5	4	3	2	1

RADIO ANTENNAS



UAT

LAT

PAGE TITLE		
ANTENNA DIAGRAM		
	DRAWING NUMBER	051-02695
	REVISION	4.0.0
	BRANCH	
	PAGE	3 OF 27
	SHEET	18 OF 47

ANTENNA B2BS

UAT

UAT B2BS

GPO FILTERS

LAT

LAT B2B

RFFE FILTERS

ANTENNA CONNECTIONS

ANTENNA: B2BS

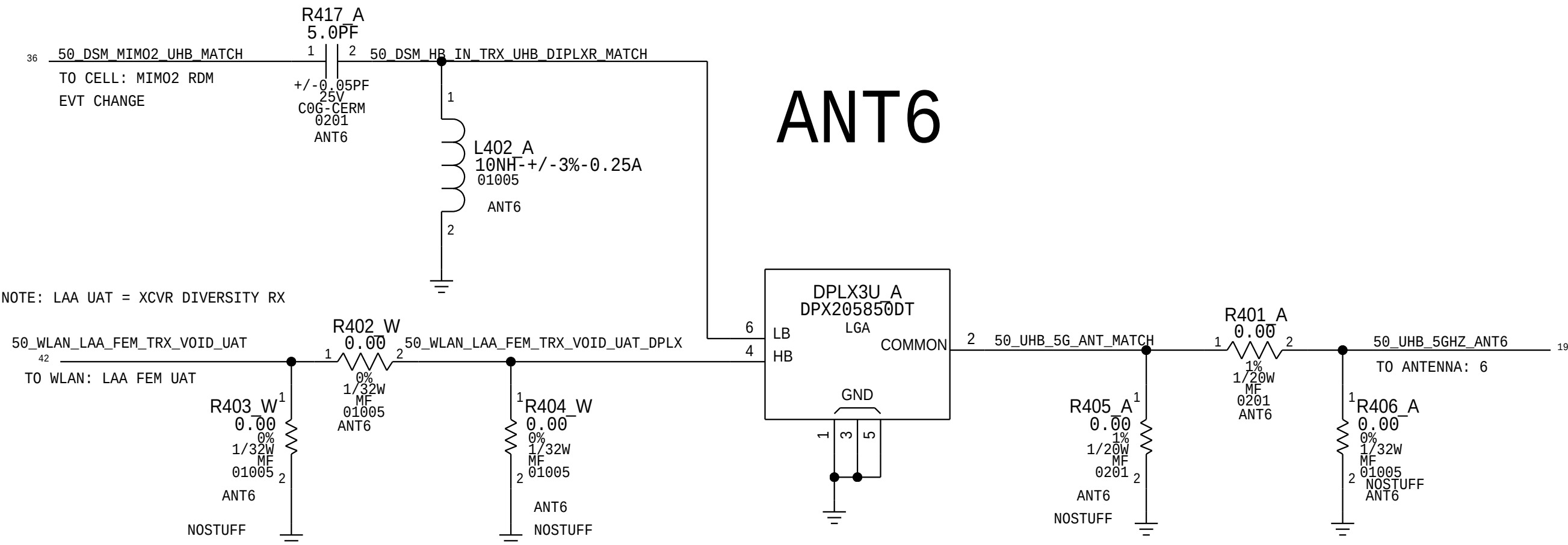
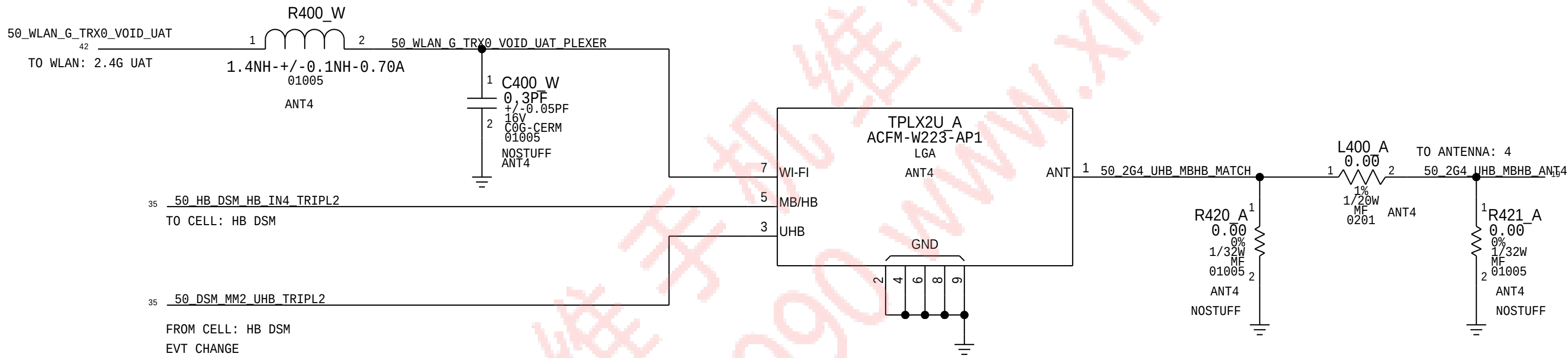
DRAWING NUMBER	051-02695	SIZE	D
REVISION	4.0.0		
BRANCH			
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SHARED N-PLEXERS

UAT

ANT4

ANT6



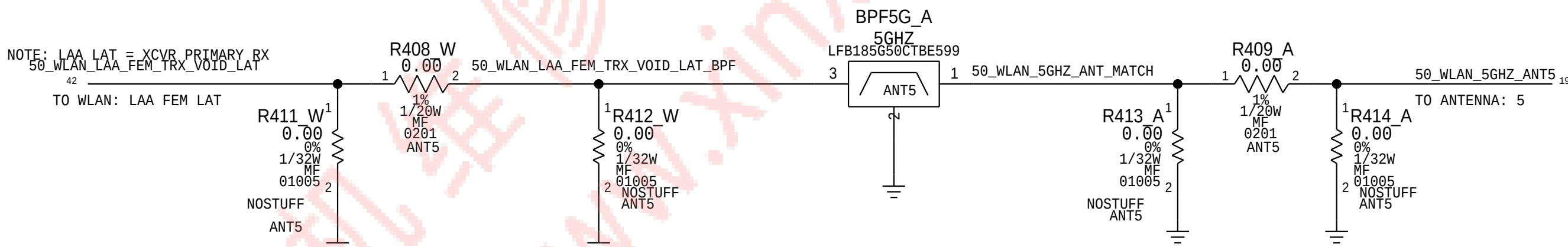
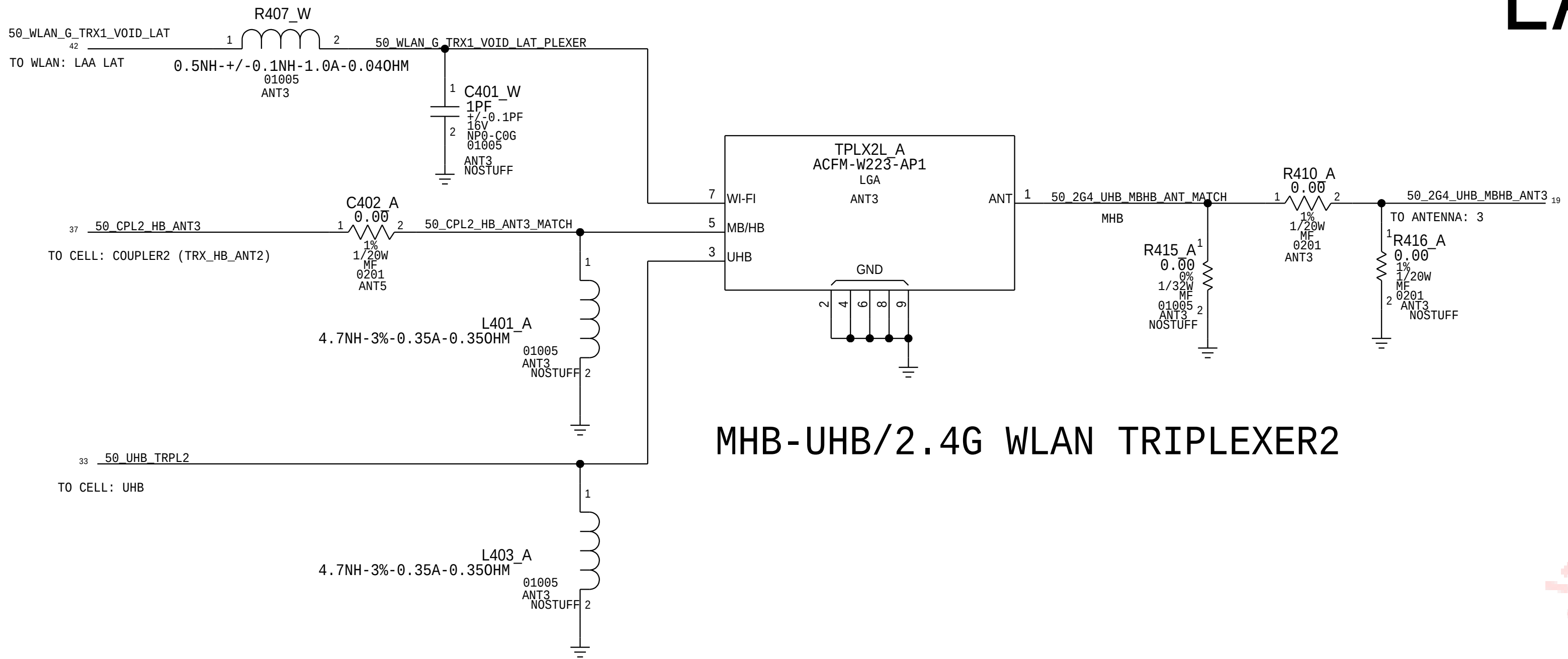
MHB-UHB/2.4G WLAN TRIPLEXER2

UHB-5G WLAN DIPLEXER3

LAT

ANT3

ANT5



MHB-UHB/2.4G WLAN TRIPLEXER2

5GHZ WLAN BPF

PAGE TITLE		
ANTENNA: N-PLEX SHARED		
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BRANCH		
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PMIC_K
3829-C1
FWLB
9 OF 14
BPMU

CT TO
VSS_PMU

HWID TABLE RDAR://32880811
ADC TABLE RDAR://27081897

BBPMU ADC TABLES

2.2K

P1

2.0

3.3K

P2

2.1

4.7K

3.0

6.8K

EVT

8.2K

EVT 1.5

4.0

10K

CARRIER

12K

DVT

5.0

15K

PVT

5.1

18K

22K

27K

6.0

33K

6.0

39K

47K

56K

DARWIN

68K

JP_SKWS

82K

100K

P1 DOE2, TXC

120K

P1 DOE1, S7 B1

150K

2.2K

ICE 18.0 ROW

(ICE 18.0)

3.3K

ICE 18.1 JP

X1049
(ICE 18.1)

4.7K

ICE 18.1 ROW

6.8K

ICE 18.2 JP

X1210
(ICE 18.2)

8.2K

ICE 18.2 ROW

10K

RFDEV ROW

12K

ICE 18.5 JP

X1170

15K

ICE 18.6 JP

X1176

18K

22K

ICE 18.5 US

X1170

27K

ICE 18.6 US

X1176

33K

ICE 18.0 NA

X1344

39K

RFDEV NA

47K

ICE 18.1 NA

X1049

56K

ICE 18.2 NA

X1210

68K

82K

KAR00 INTERNAL

100K

KAR00 INTERNAL

120K

KAR00 INTERNAL

150K

PAGE TITLE

BBPMU: CONTROL

DRAWING NUMBER

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D

REVISION

4.0.0

BRANCH

PAGE

6 OF 27

SHEET

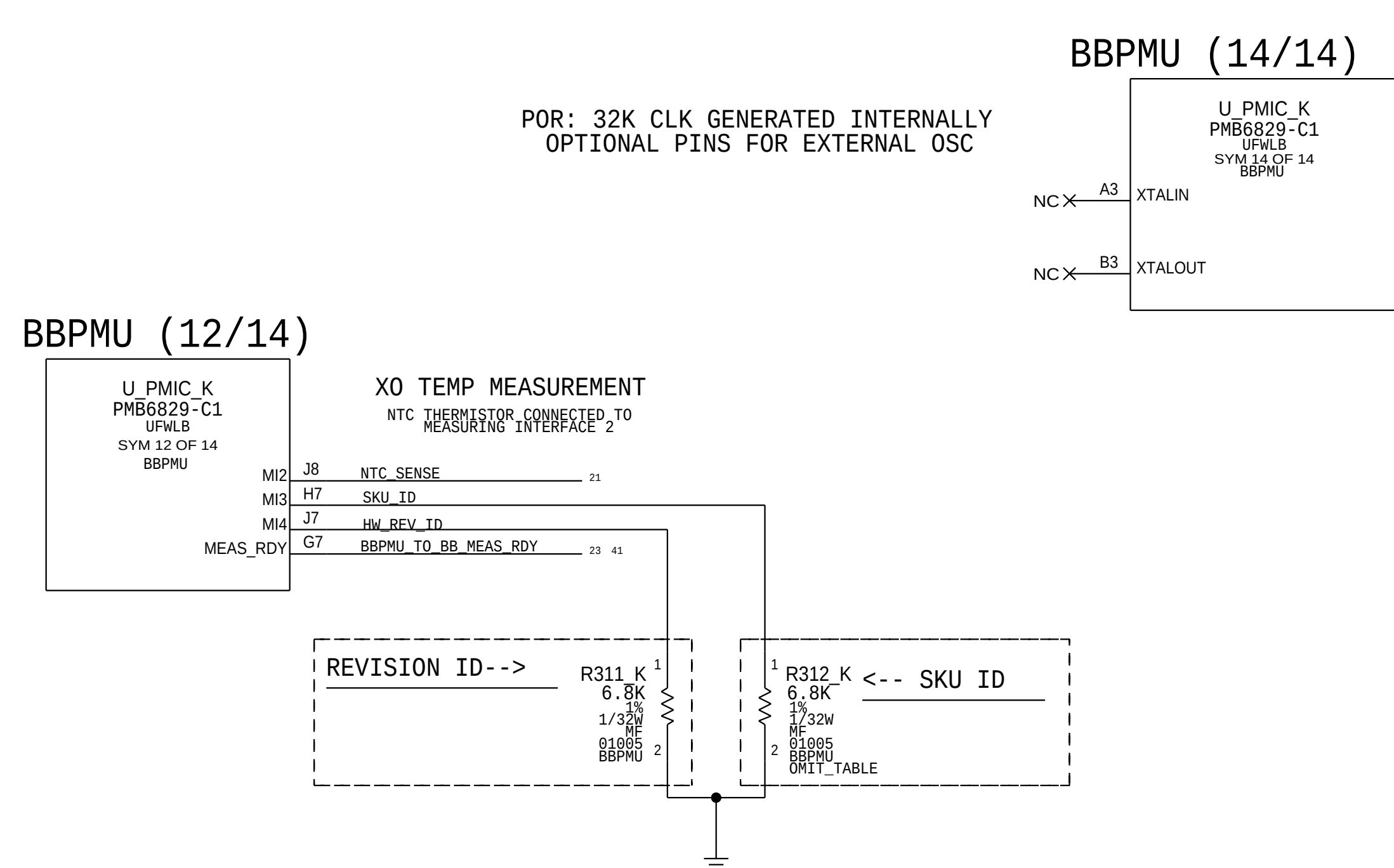
21 OF 47

4

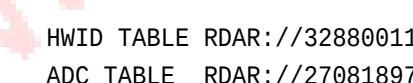
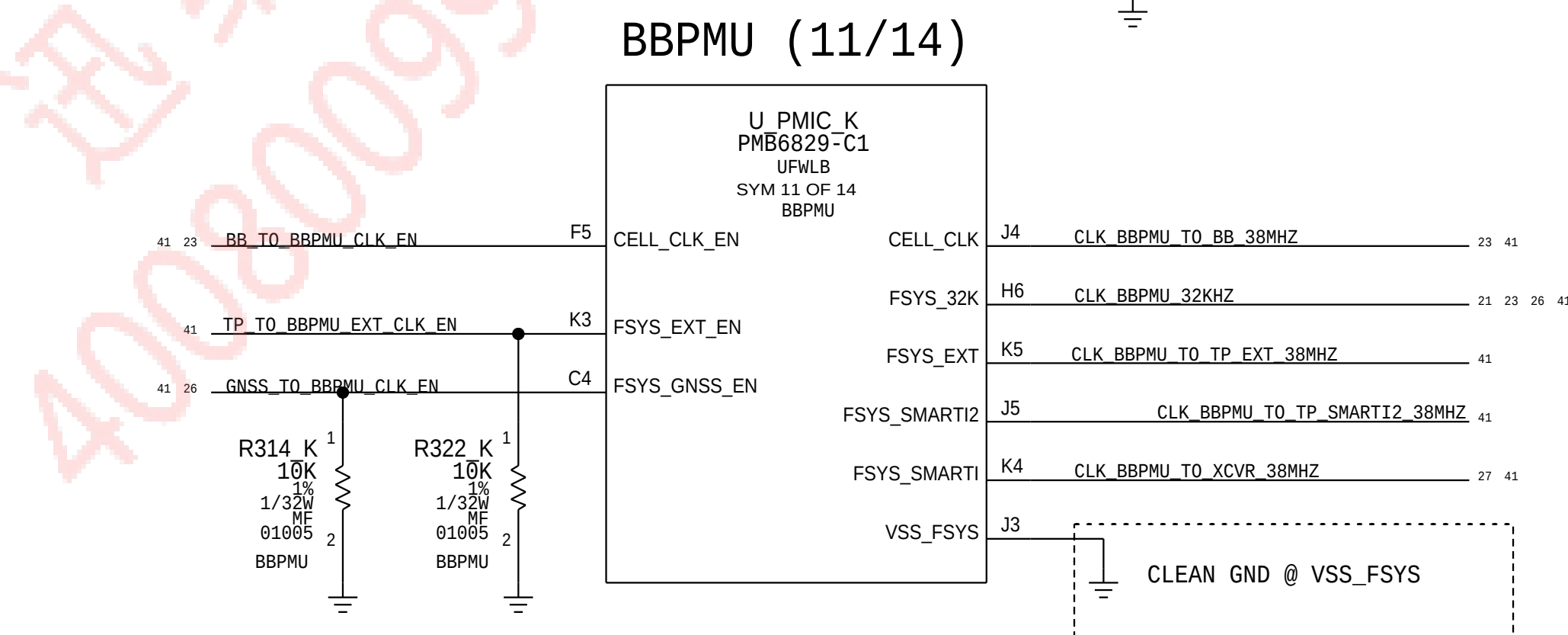
3

2

1



SKU/CATEGORY		
R312_K	HW REVISION	X-CODE
0.0	RFDEV JP	
1.2K	ICE 18.0 JP	X1344 (ICE 18.0)
2.2K	ICE 18.0 ROW	
3.3K	ICE 18.1 JP	X1049 (ICE 18.1)
4.7K	ICE 18.1 ROW	
6.8K	ICE 18.2 JP	X1210 (ICE 18.2)
8.2K	ICE 18.2 ROW	
10K	RFDEV ROW	
12K	ICE 18.5 JP	X1170
15K	ICE 18.6 JP	X1176
18K		
22K	ICE 18.5 US	X1170
27K	ICE 18.6 US	X1176
33K	ICE 18.0 NA	X1344
39K	RFDEV NA	
47K	ICE 18.1 NA	X1049
56K	ICE 18.2 NA	X1210
68K		
82K	KAR00 INTERNAL	
100K	KAR00 INTERNAL	
120K	KAR00 INTERNAL	
150K		

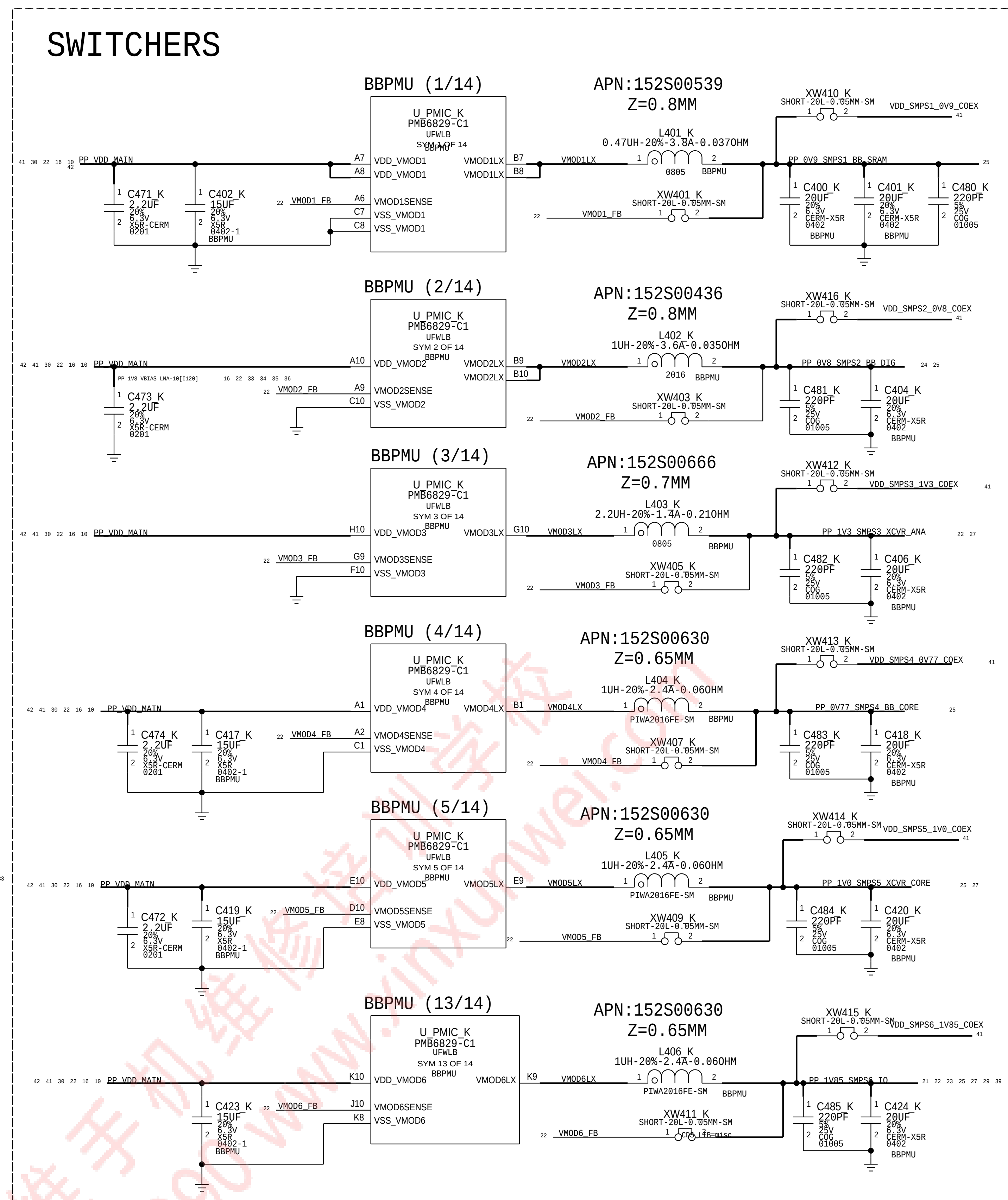


ADC TABLE RDAR://27081897

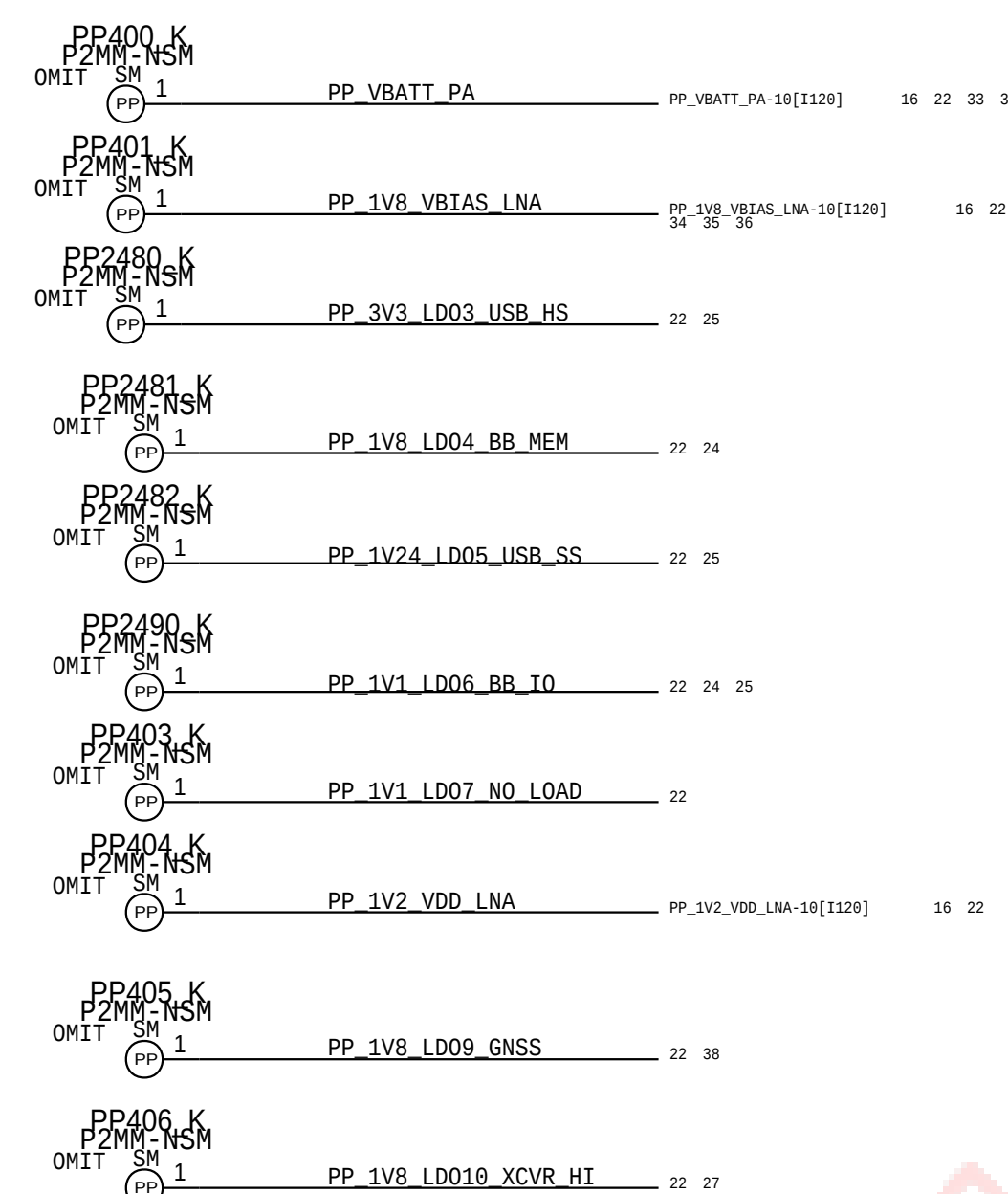
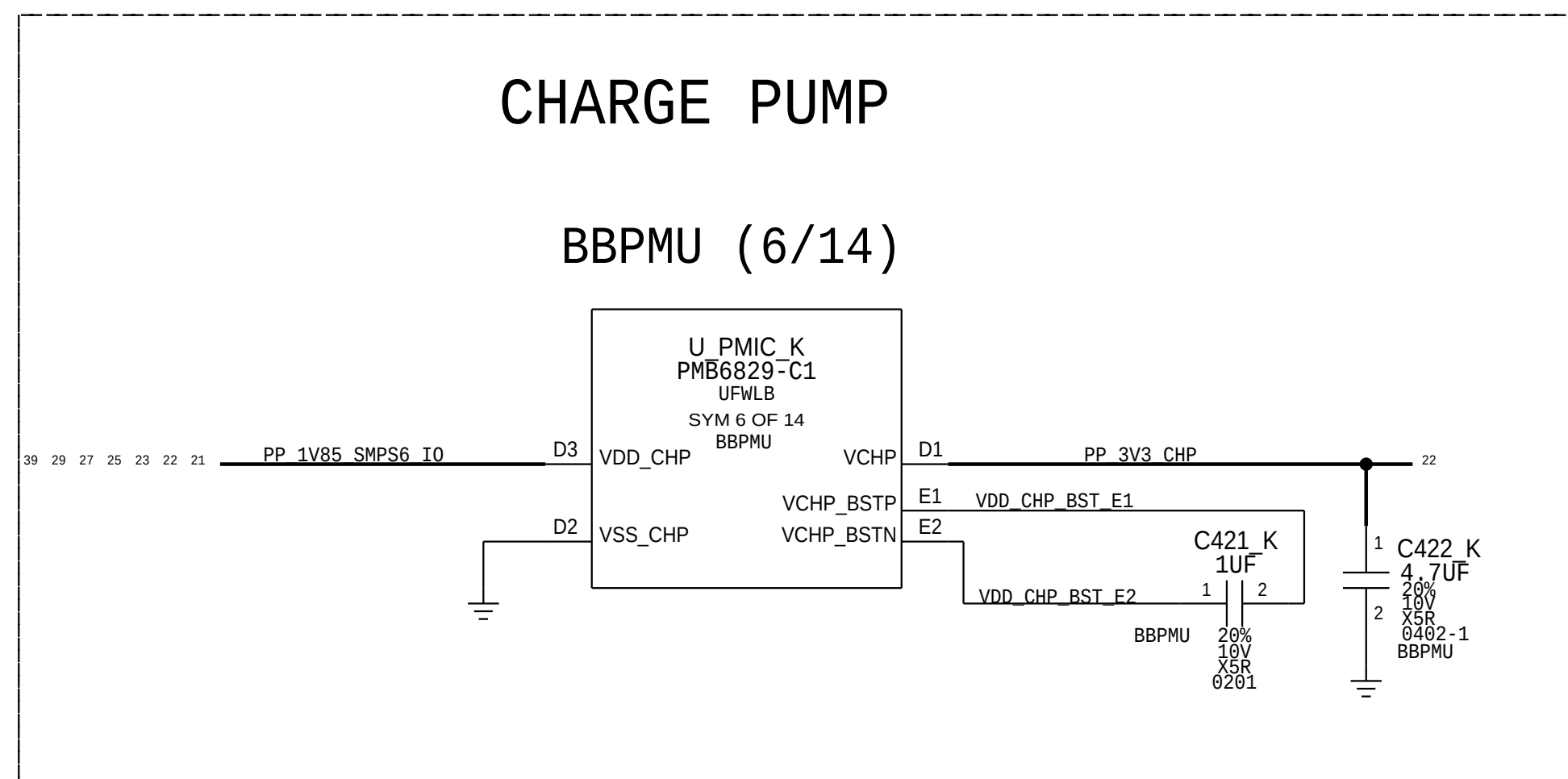
BBPMU ADC TABLES

PAGE TITLE		BBPMU: CONTROL	
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		REVISION	
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		BRANCH	
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		21 OF 47	

SWITCHERS

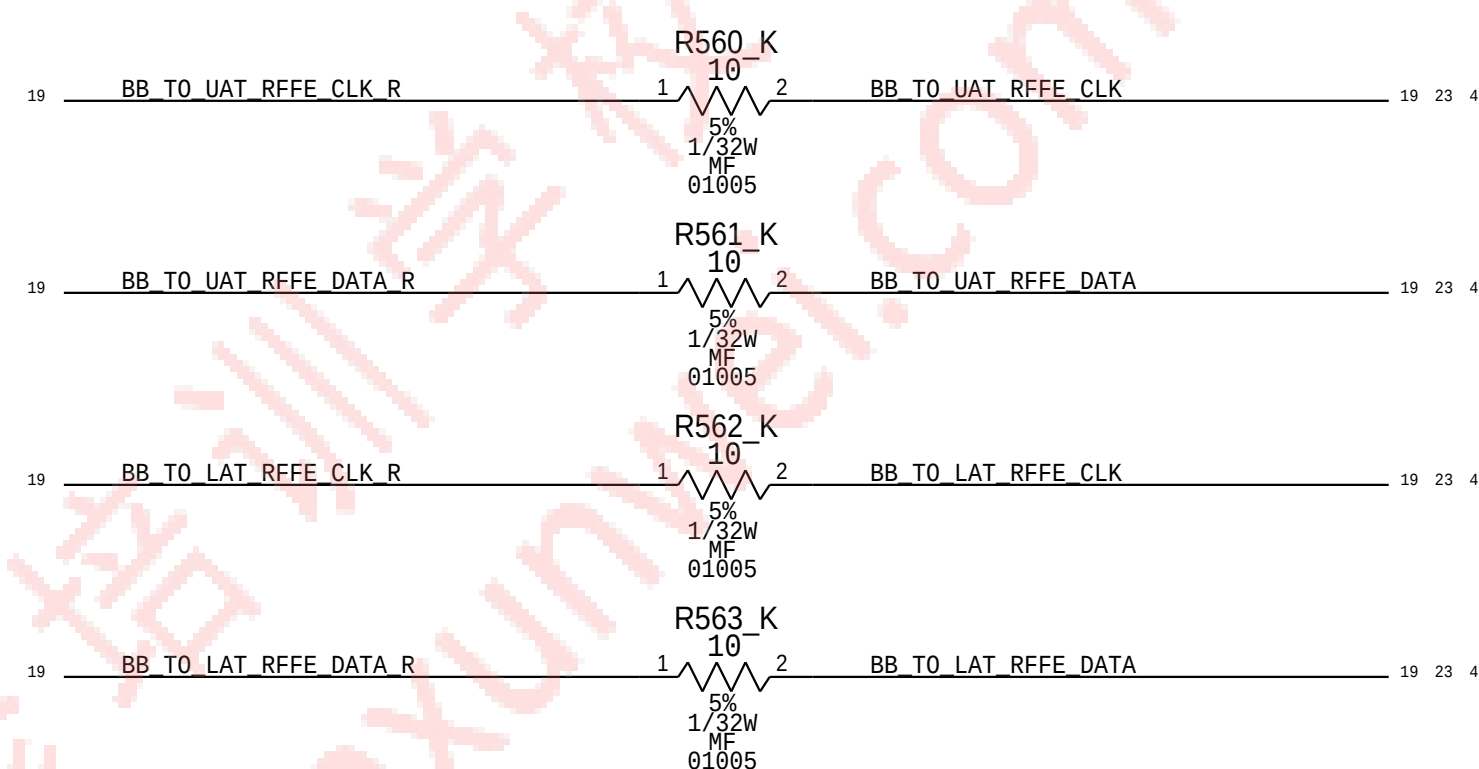
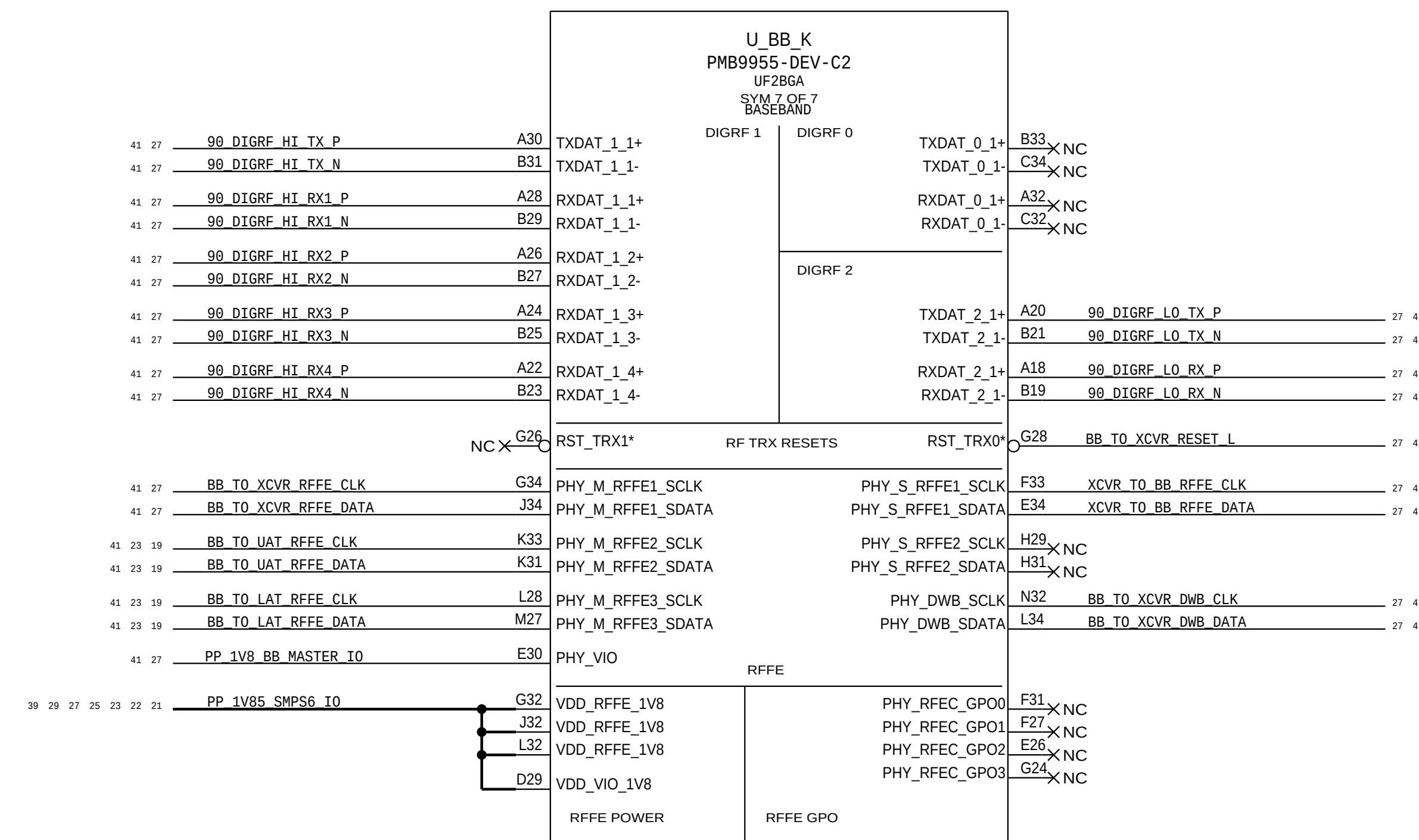


BBPMU (6/14)

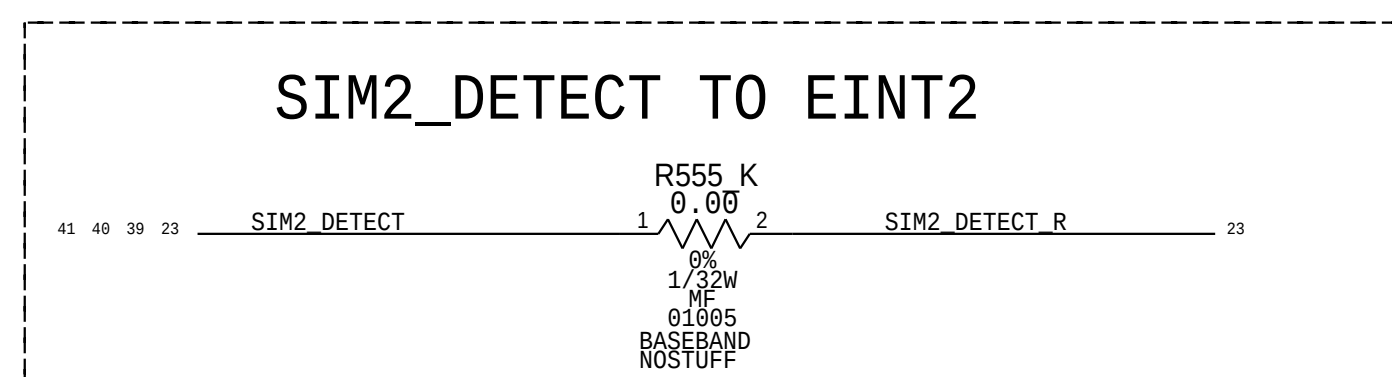
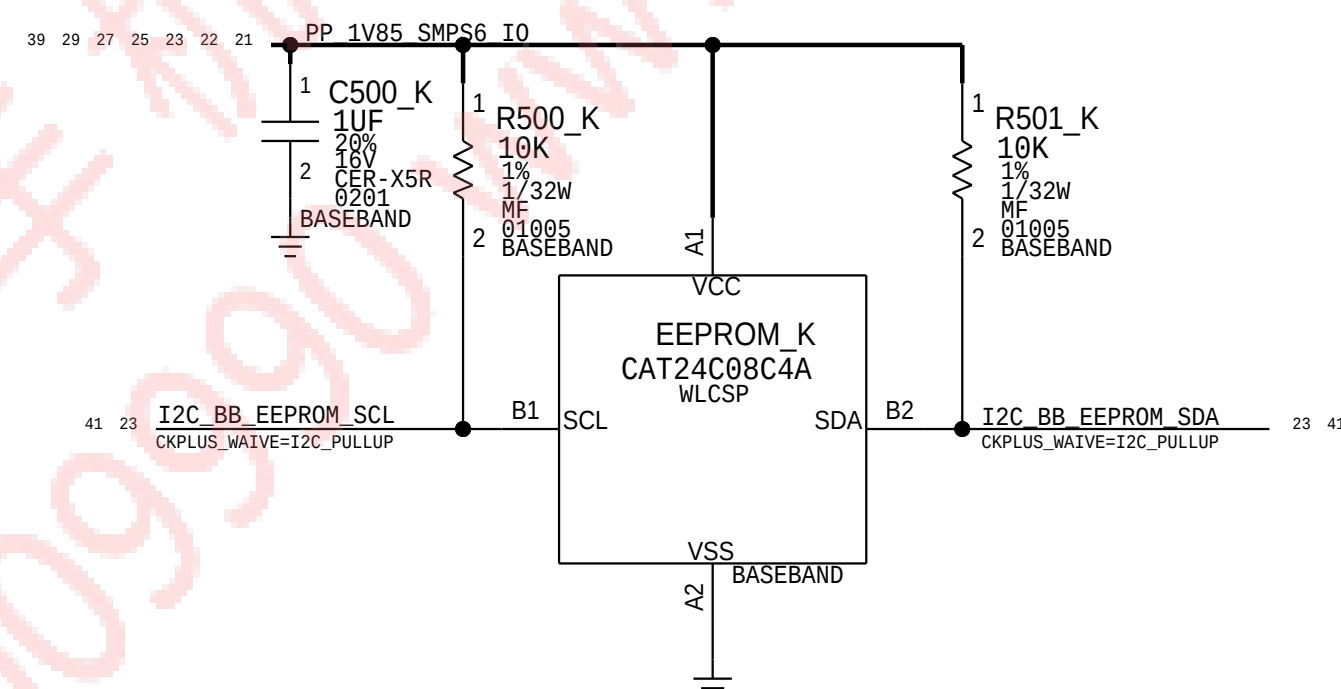


PAGE TITLE		BBPMU : RAILS	
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	REVISION 4.0.0		
	BRANCH		
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			SHEET 22 OF 47

XG756 (7/7)

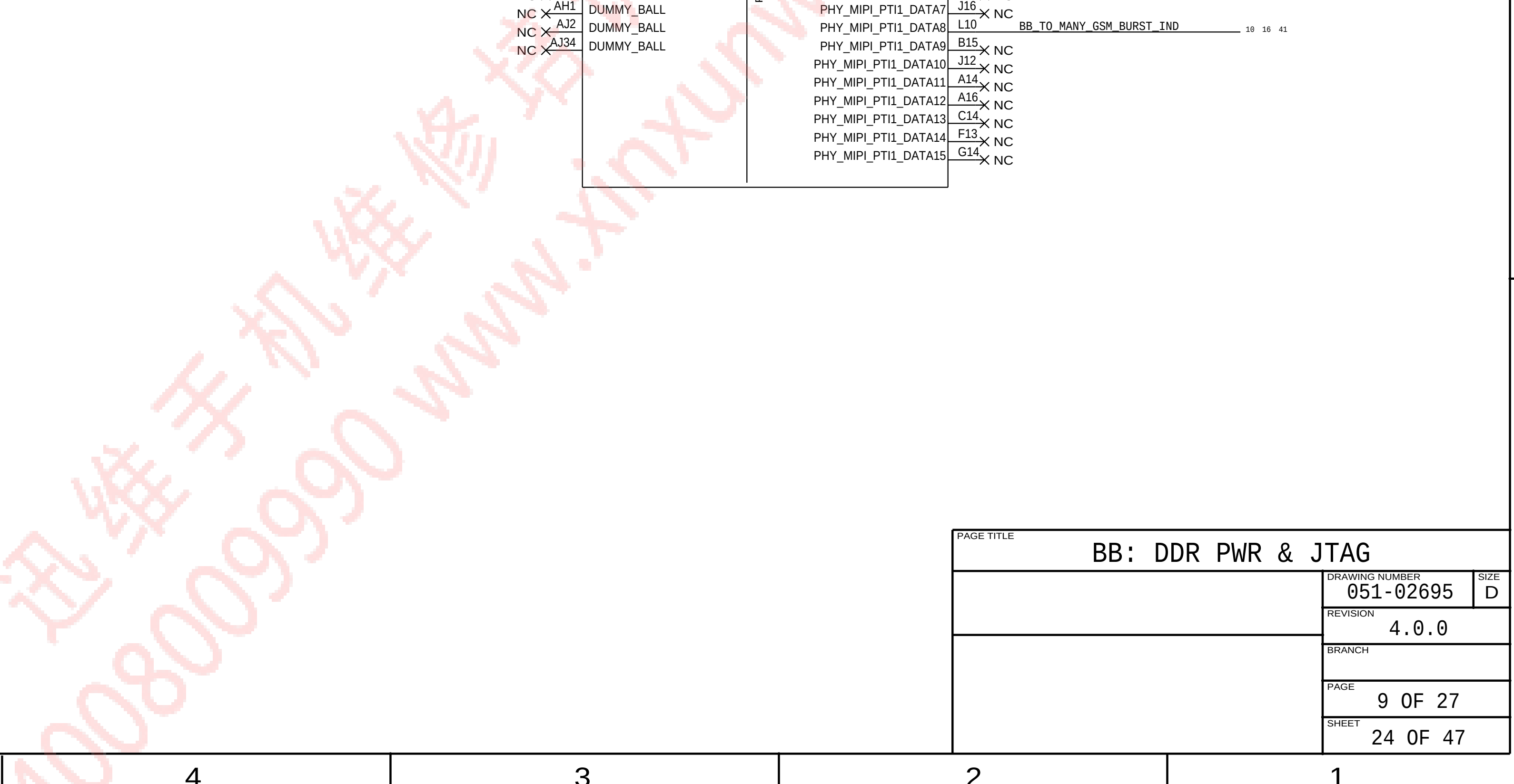
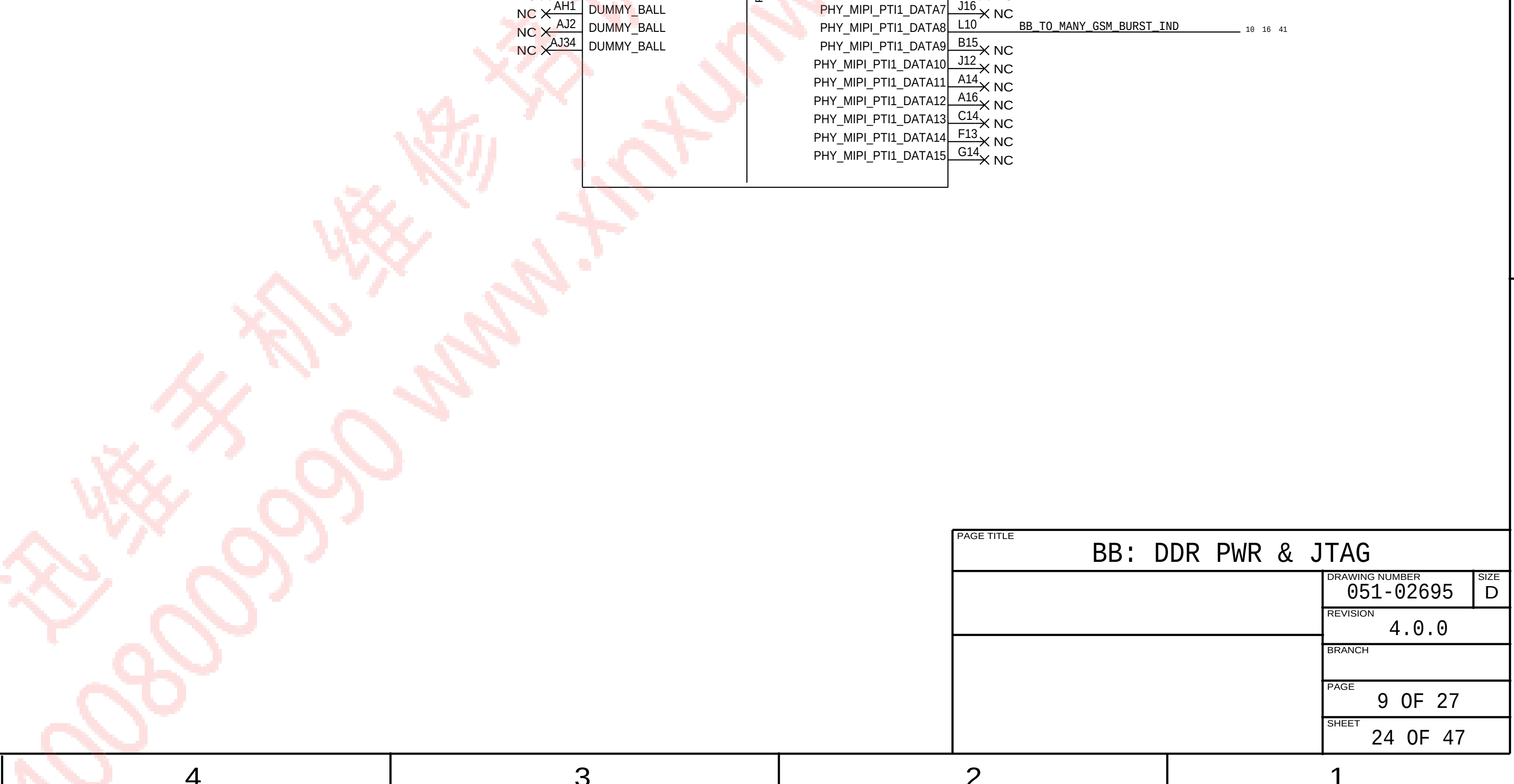


BB EEPROM



PAGE TITLE		BB: INTERFACE	
		DRAWING NUMBER	051-02695
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		REVISION	4.0.0
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		PAGE	8 OF 27
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A

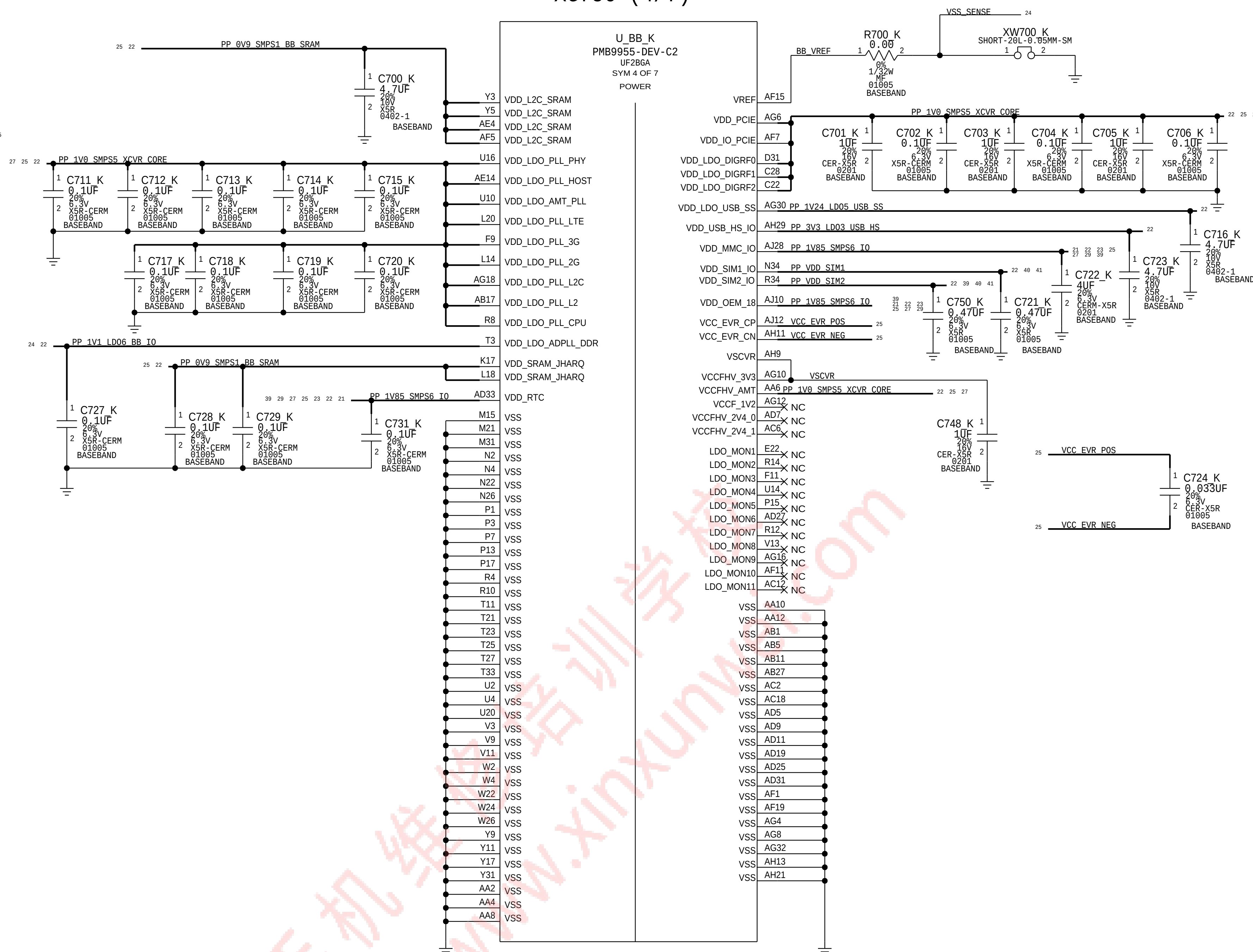


PAGE TITLE		
BB: DDR PWR & JTAG		
	DRAWING NUMBER	SIZE
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	4.0.0	
	BRANCH	
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XG756 (3/7)



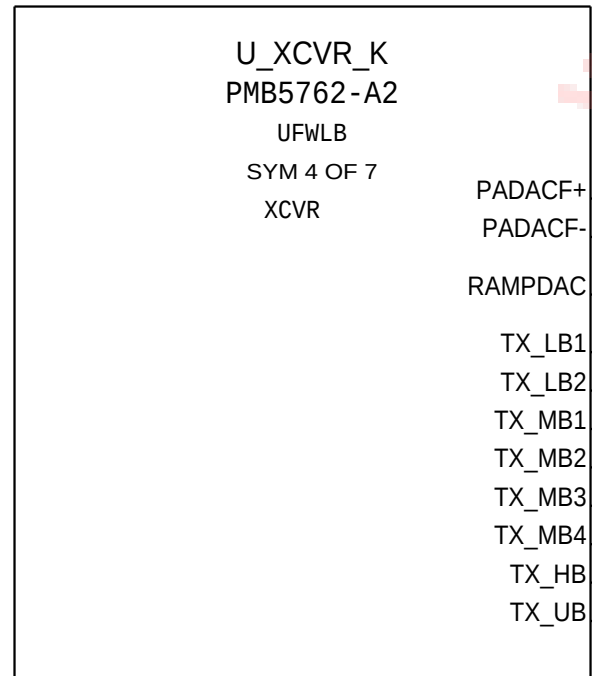
XG756 (4/7)



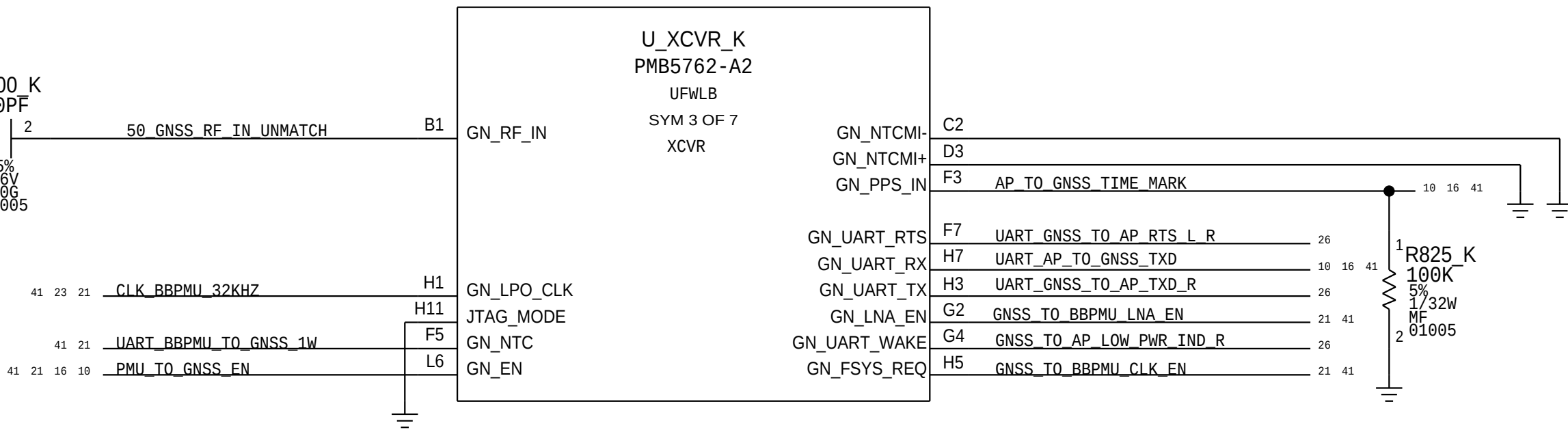
PAGE TITLE		BB: DIGITAL PWR	
	DRAWING NUMBER	051-02695	SIZE D
	REVISION 4.0.0		
	BRANCH		
	PAGE	10 OF 27	
	SHEET	25 OF 47	

XCVR: TRANSMIT & GNSS

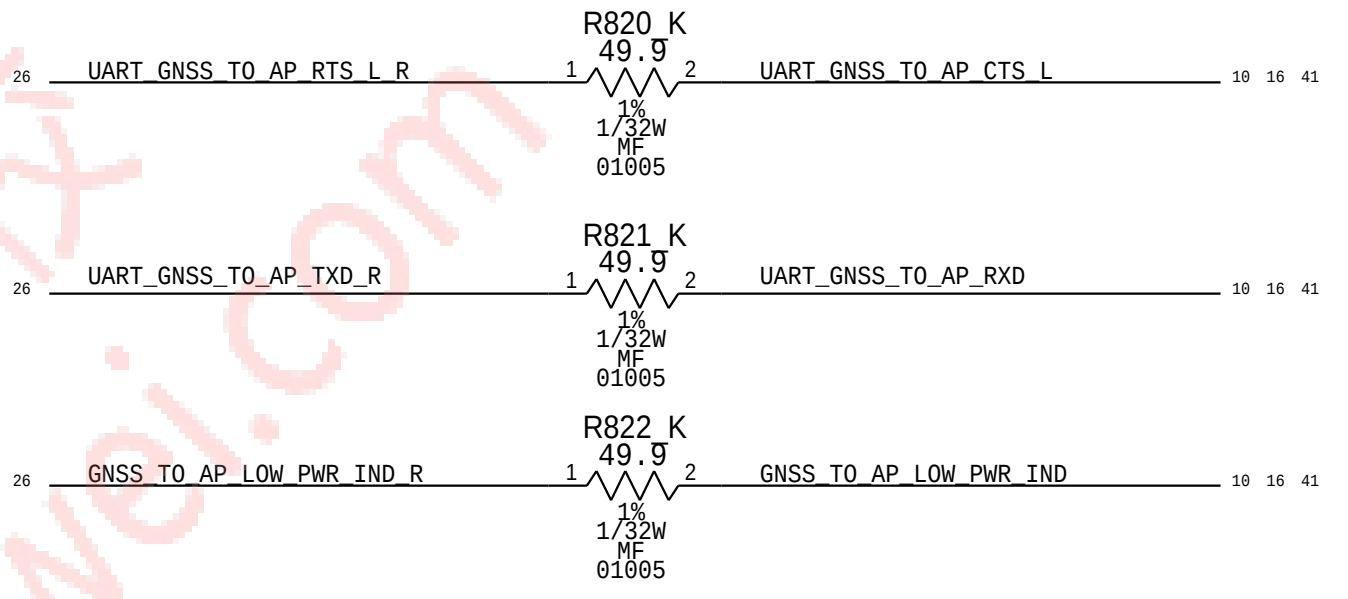
SMARTI7 (4/7)



SMARTI7 (3/7)



GNSS DAMPING RESISTORS



PAGE TITLE		
XCVR: TX & GNSS		
	DRAWING NUMBER	051-02695
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	BRANCH	
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XCVR: INTERFACE & PWR

SMARTI7 (1/7)

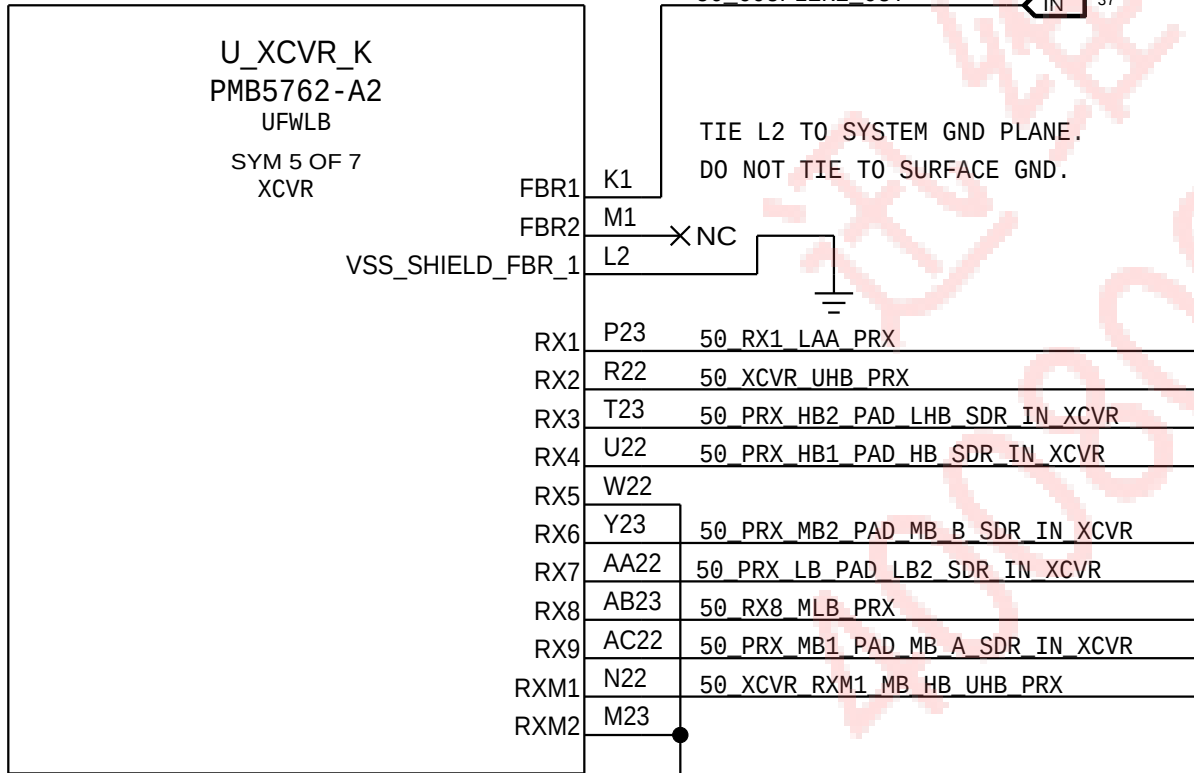
SMARTI7 (7/7)

SMARTI7 (2/7)

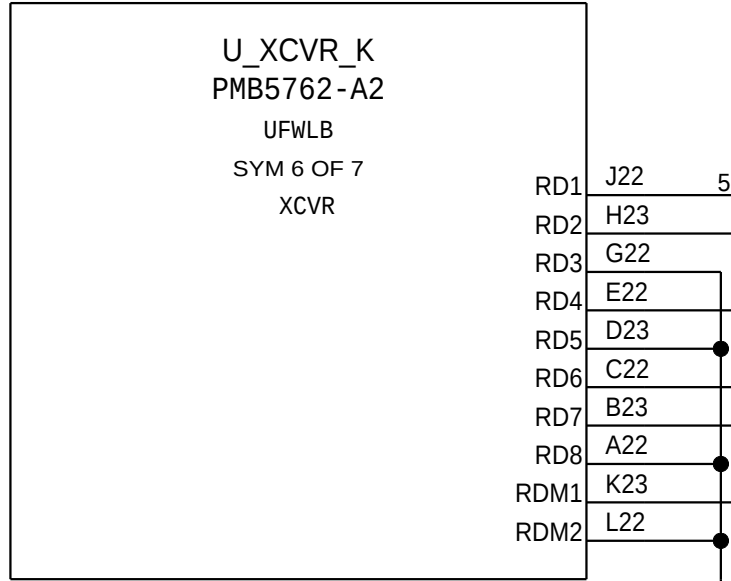
XCVR RFFE DAMPING RESISTORS

XCVR: PRIMARY/DIVERSITY RX

SMARTI7 PRX (5/7)



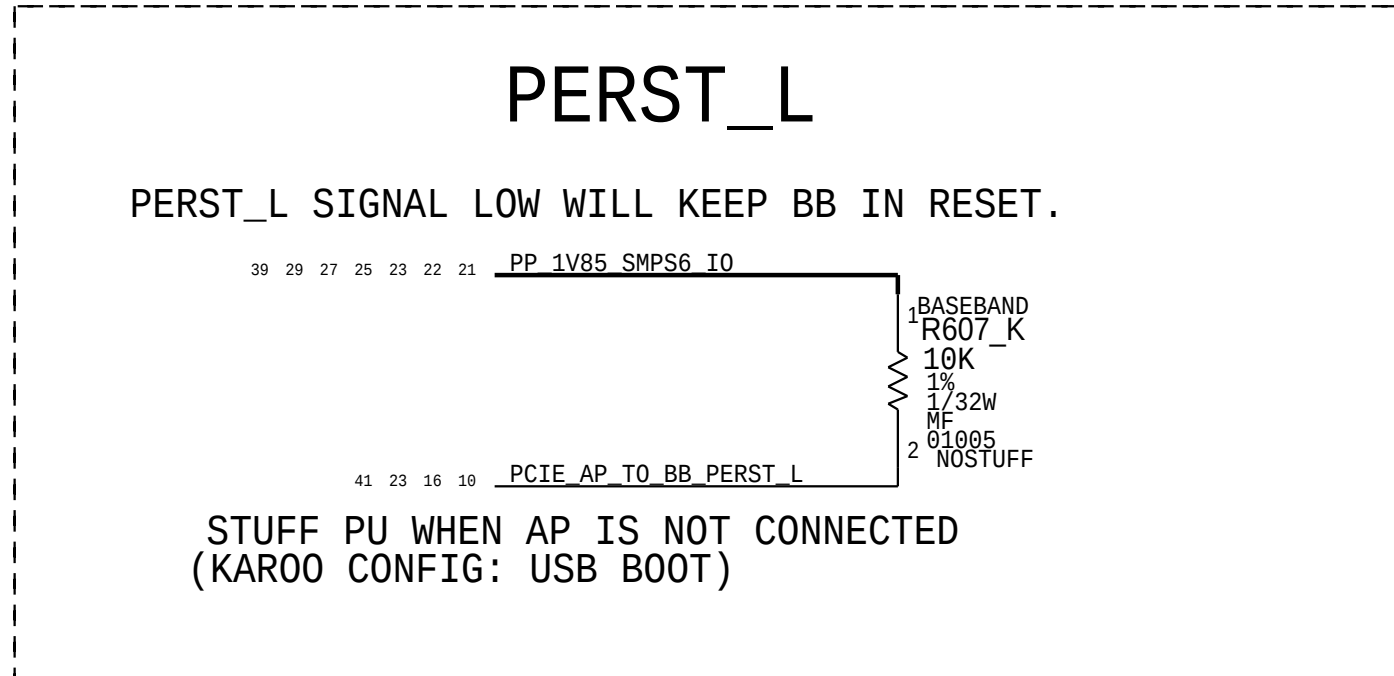
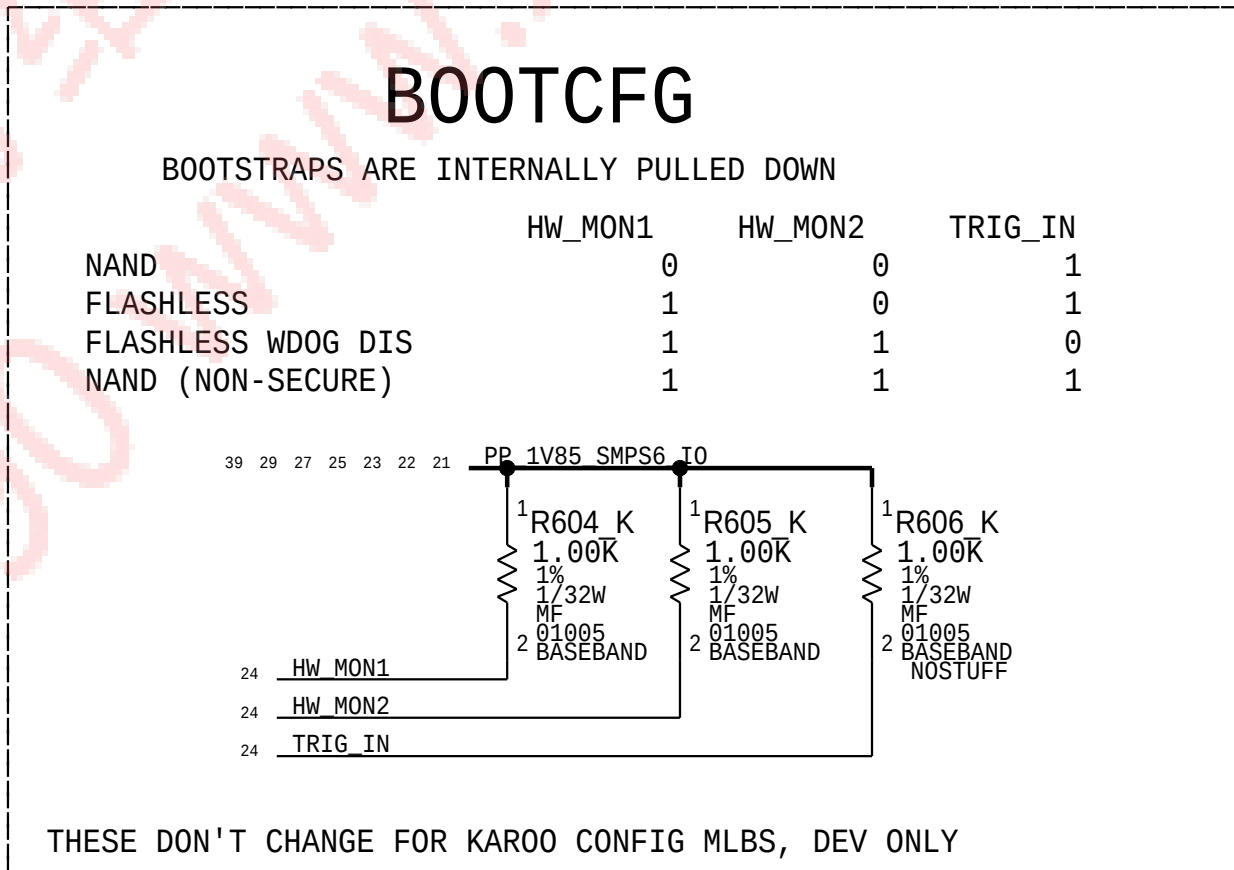
SMARTI7 DRX (6/7)



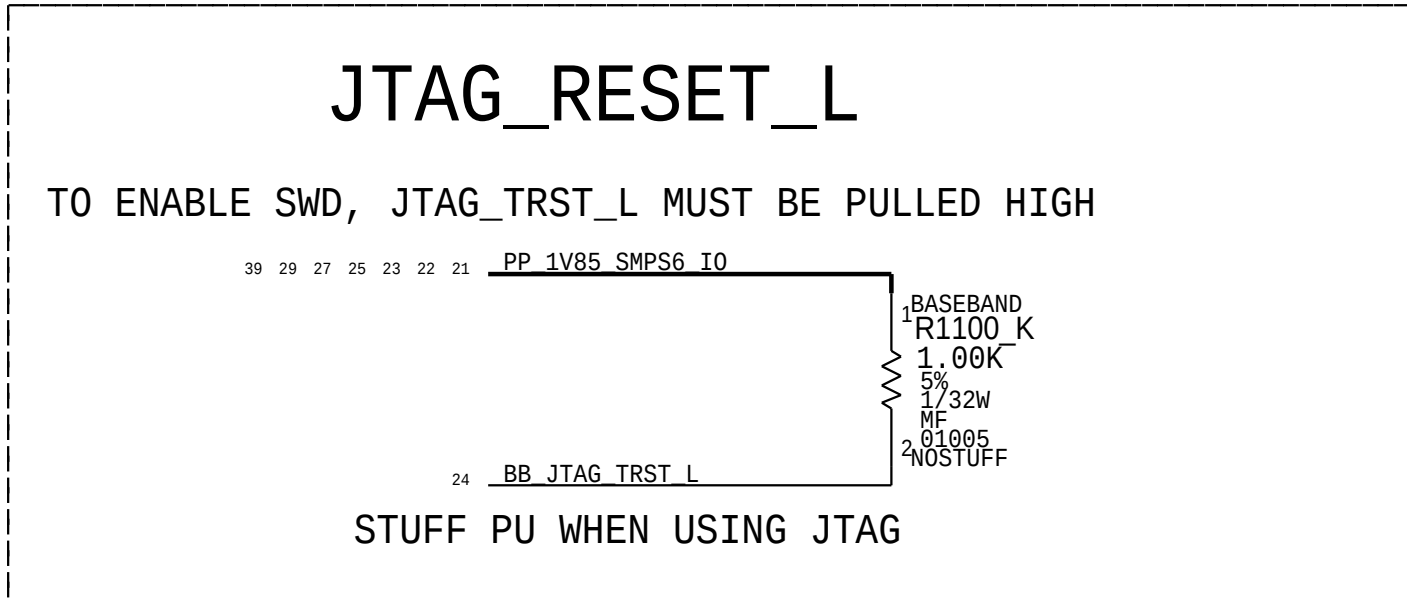
S7 PORT MAP CHANGE 10/11/2016:
USE RD6 INSTEAD OF RD3.
CONNECT ALL UNUSED RD PORTS TO GND.

PAGE TITLE		
XCVR: PRX DRX		
	DRAWING NUMBER	051-02695
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	PAGE	13 OF 27
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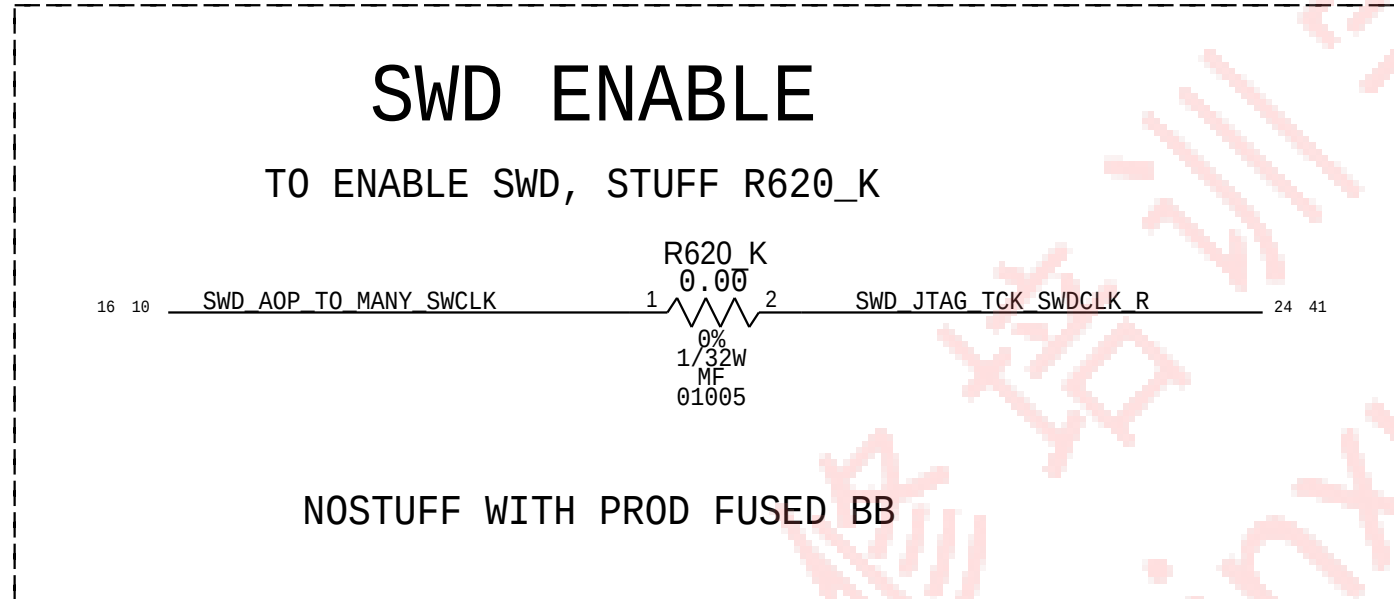
KAR00 CONFIG



DEFAULTS SET TO AP/FLASHLESS BOOT

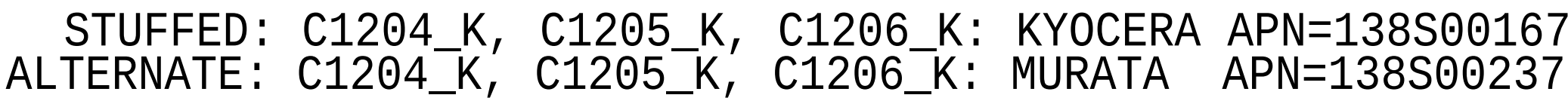


HW BUG IN BASEBAND A/B SILICON, FIXED IN C SILICON
NOSTUFF, STARTING AT EVT



PAGE TITLE		
HW CONFIG OPTIONS		
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ALPES STX QM81019 E1.0.1 MODULE



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LB SPAD

SEE MLB_LOFT

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LB SPAD		
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SIZE D		



D

C

B

A

1

2

3

4

5

6

7

8

HB SPAD

SEE MLB_LOFT

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		BRANCH			
		PAGE			
		17 OF 27			
		SHEET			
		32 OF 47			

1

2

3

4

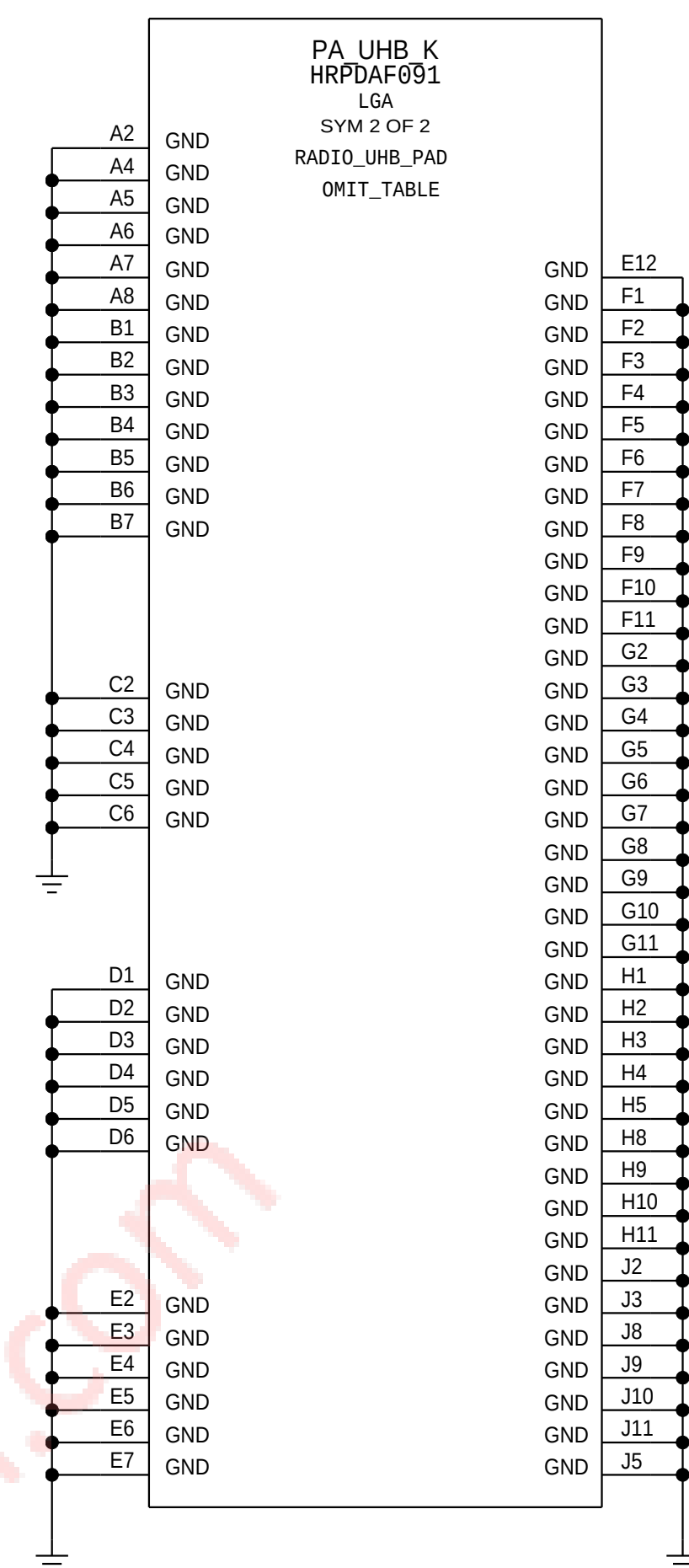
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6

7

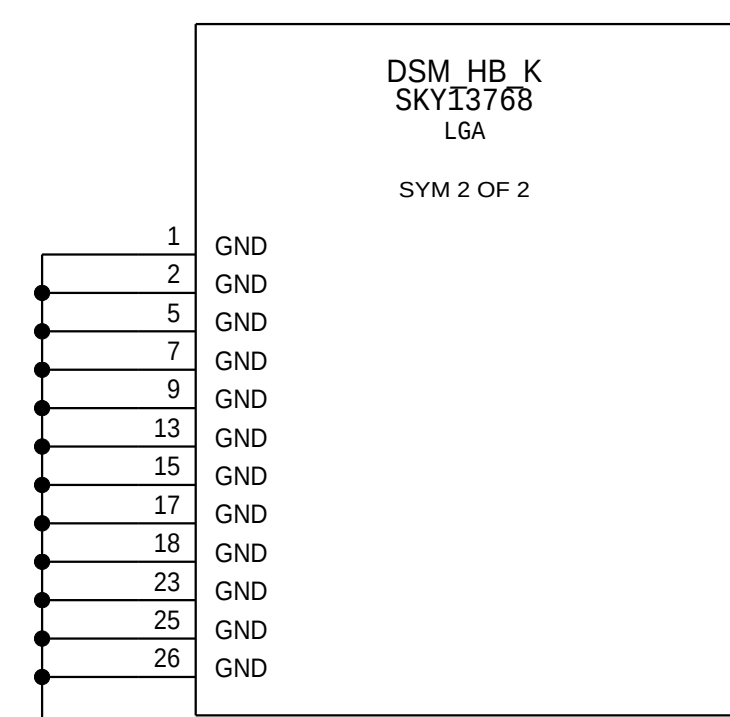
8

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PAGE TITLE		UHB LMB SPAD	
		DRAWING NUMBER 051-02695	SIZE D
		REVISION 4.0.0	
		BRANCH	
		PAGE 18	OF 27
		SHEET 33	OF 47

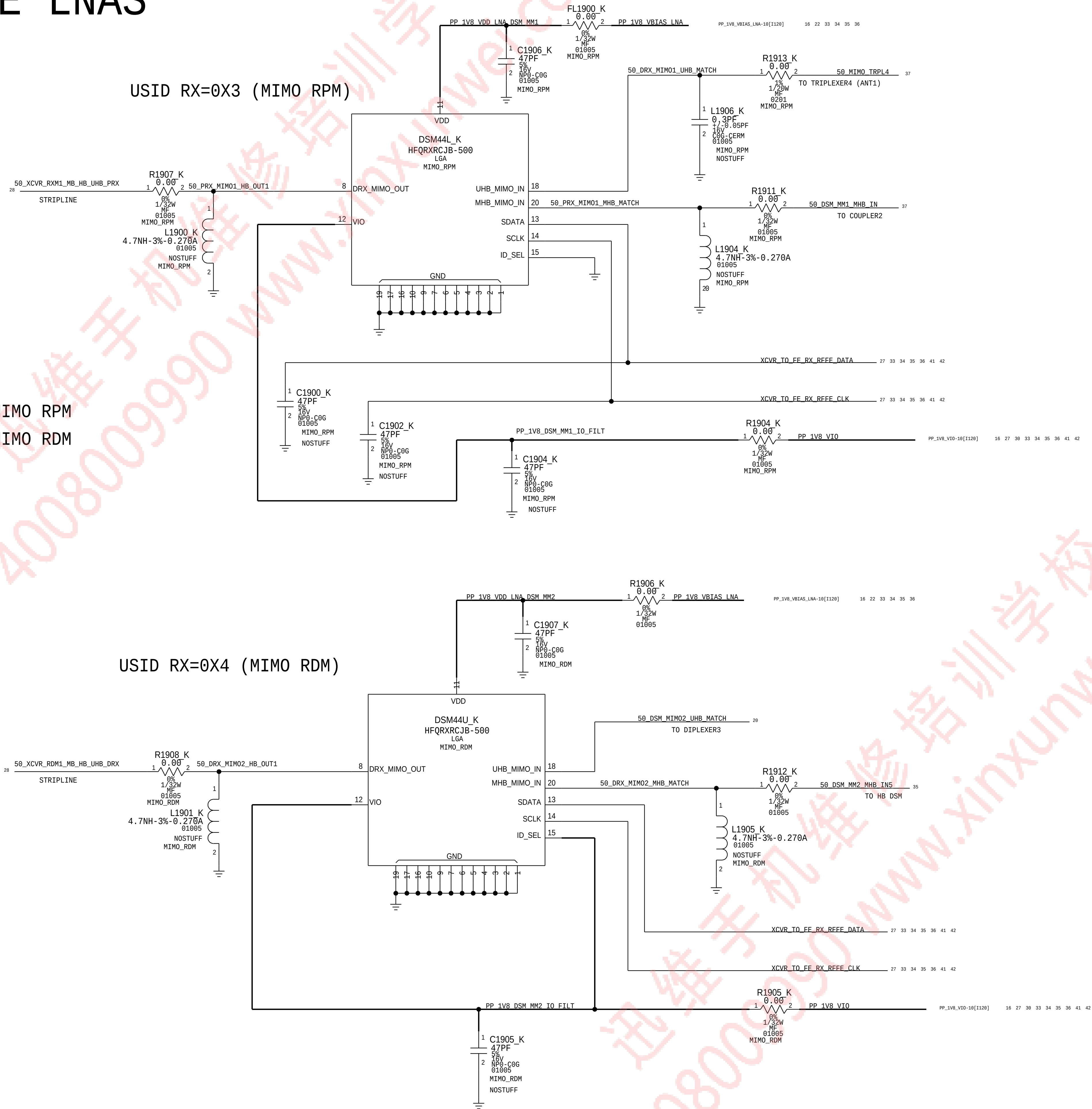
CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSE ONLY - NOT A CHANGE REQUEST



PAGE TITLE		
HB DIVERSITY RECEIVE LNA		
	DRAWING NUMBER	SIZE
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	4.0.0	
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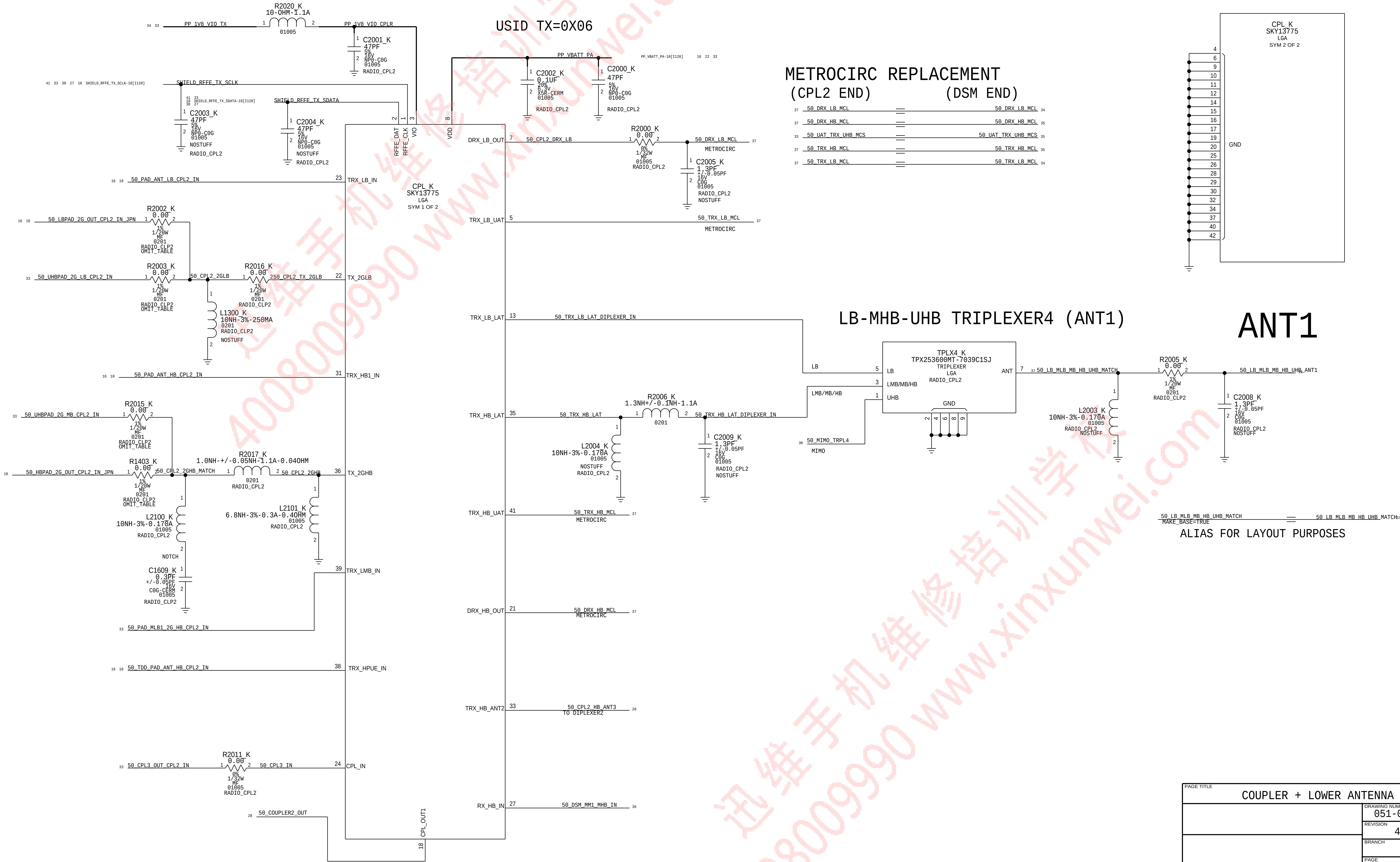
MIMO RECEIVE LNAS

ID_SEL = 0, MIMO RPM
ID_SEL = 1, MIMO RDM

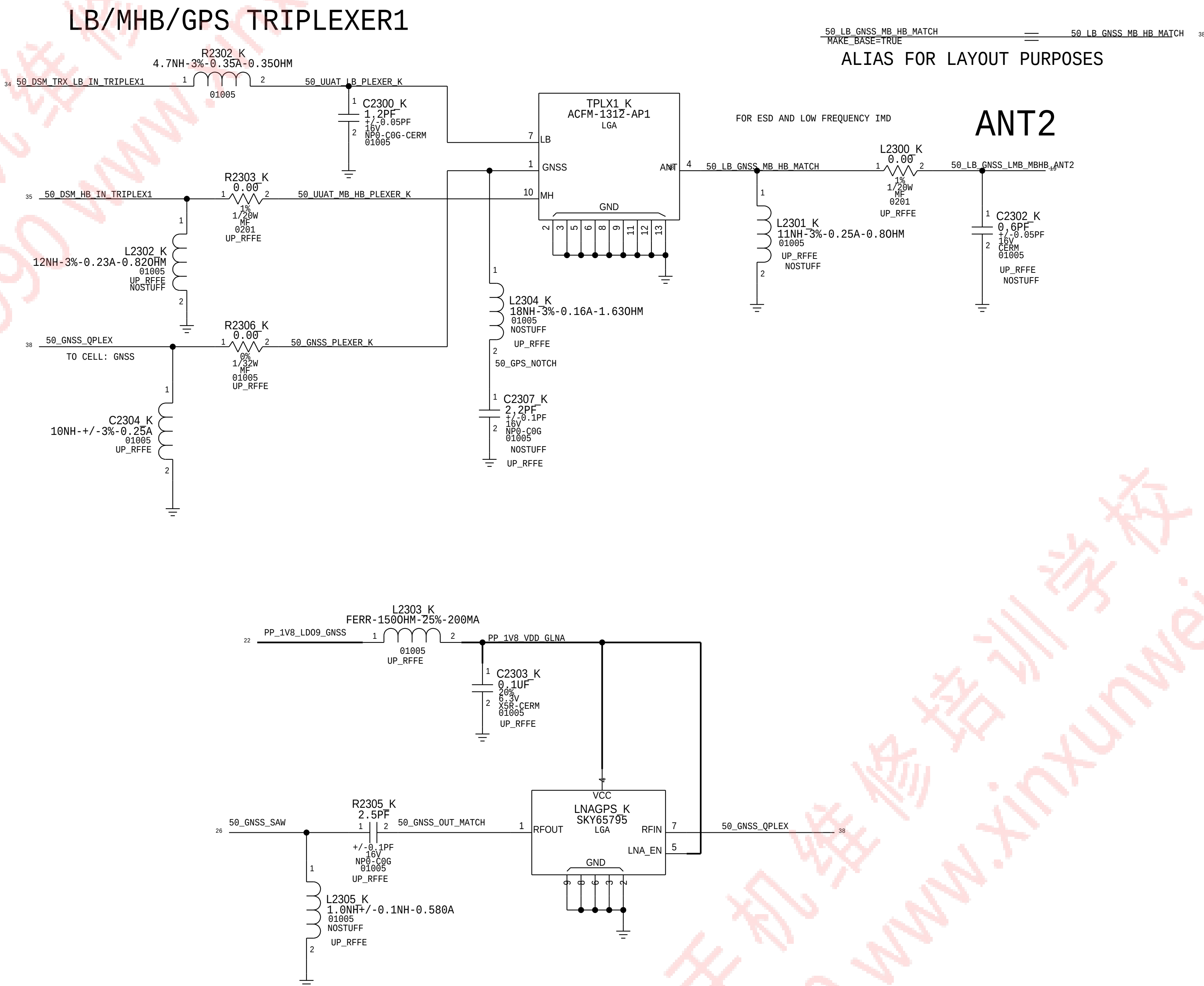


PAGE TITLE		
MIMO RECEIVE LNAS		
	DRAWING NUMBER	051-02695
	REVISION	4.0.0
	BRANCH	
	PAGE	21 OF 27
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COUPLER & LOWER ANTENNA



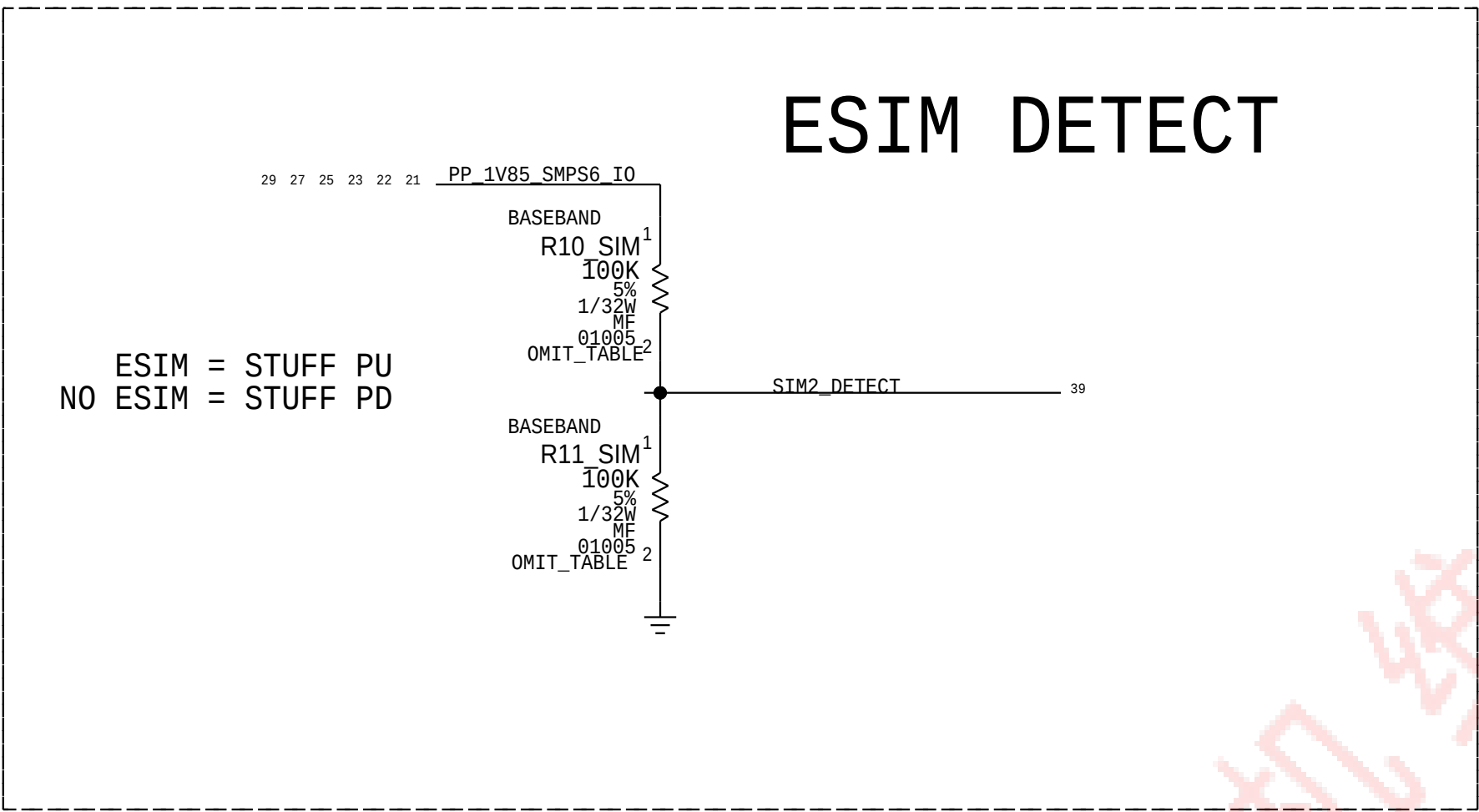
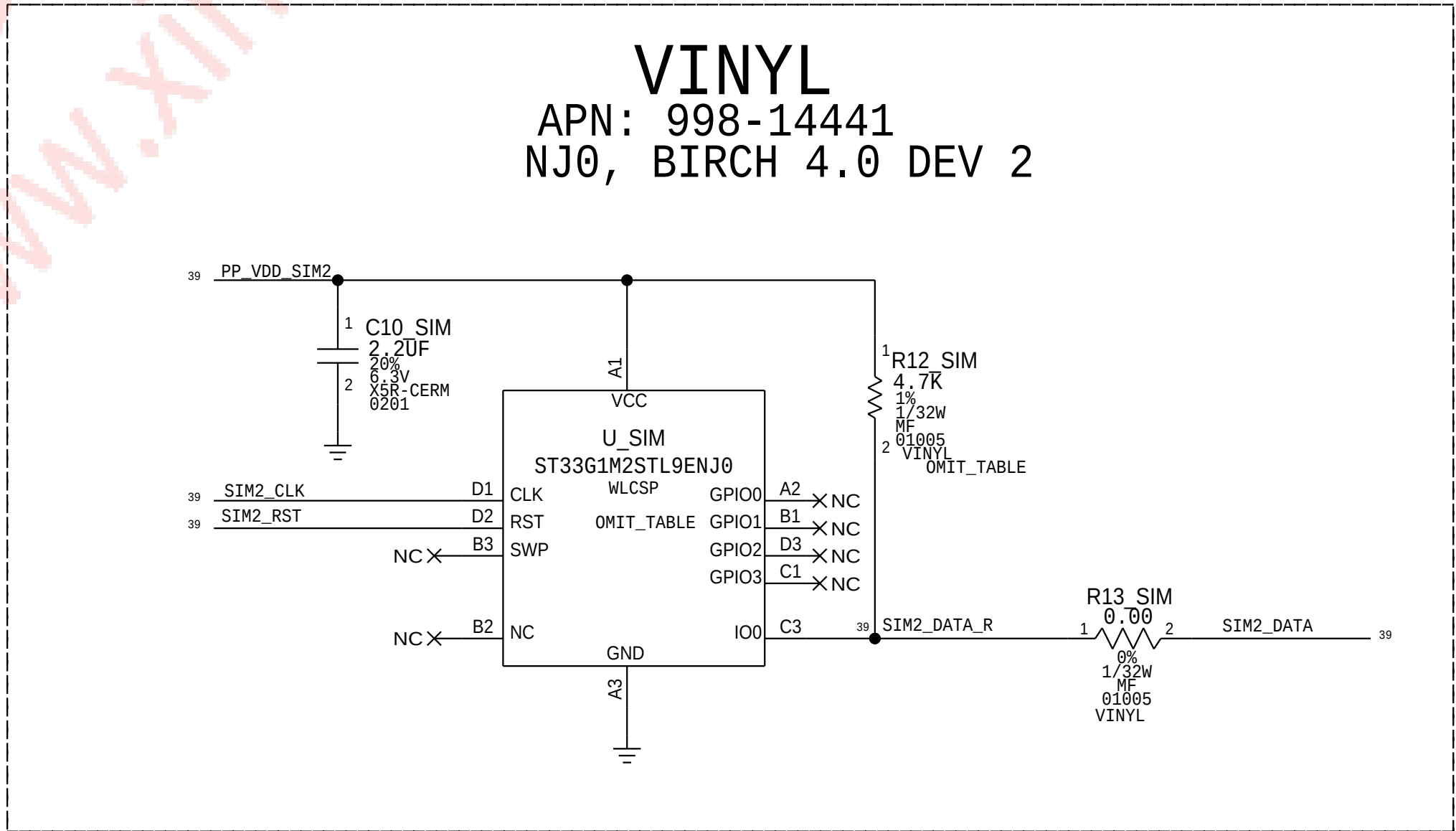
UPPER ANTENNA FEEDS



UPPER ANTENNA FEEDS		
	DRAWING NUMBER	051-02695
	REVISION	4.0.0
	BRANCH	
	PAGE	23 OF 27
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	SIZE	D

ESIM

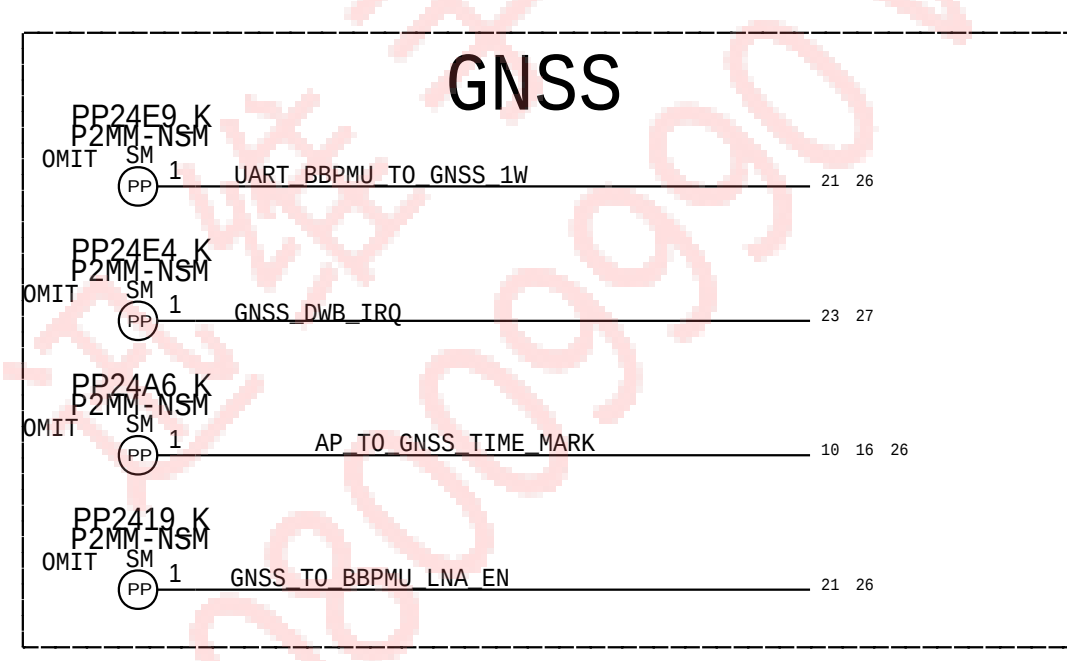
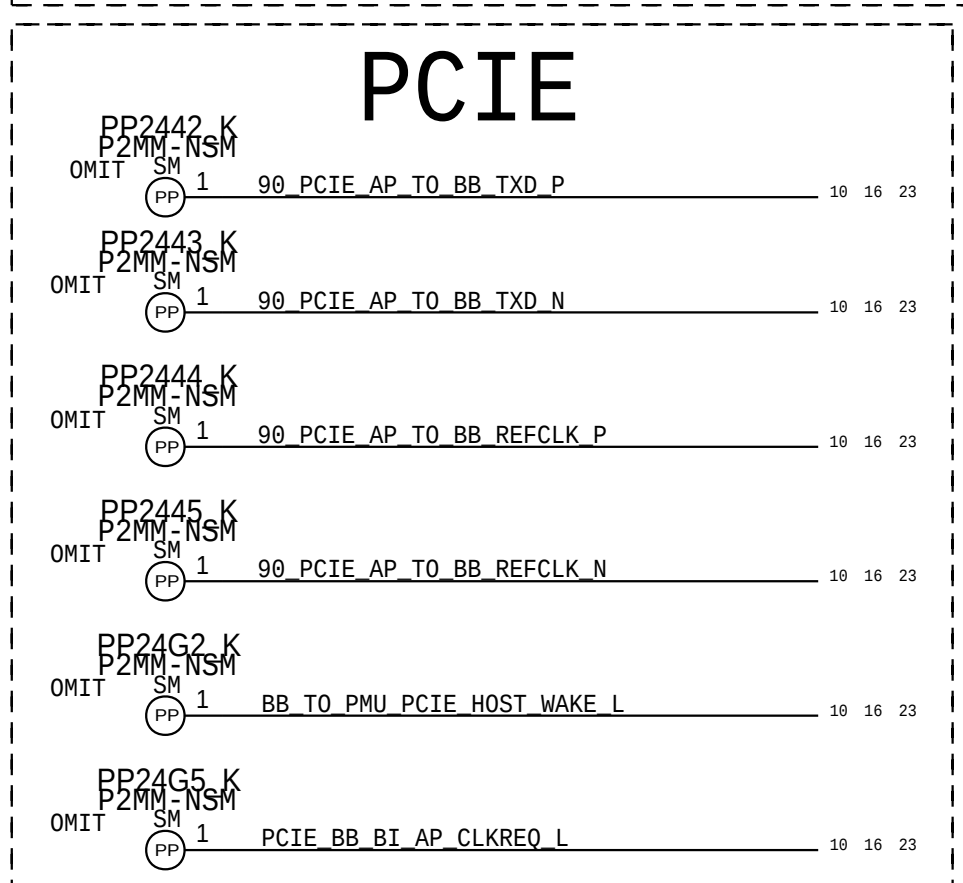
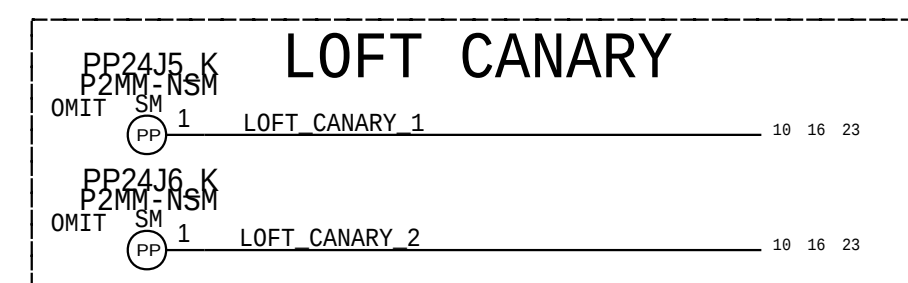
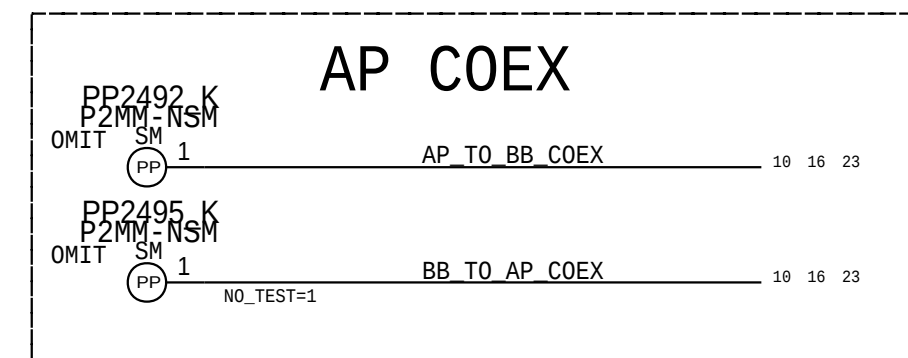
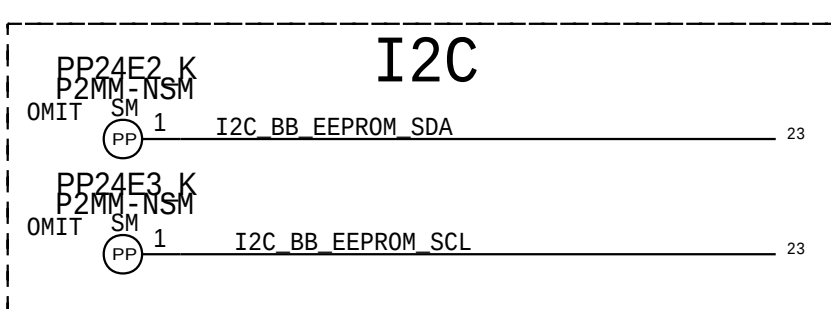
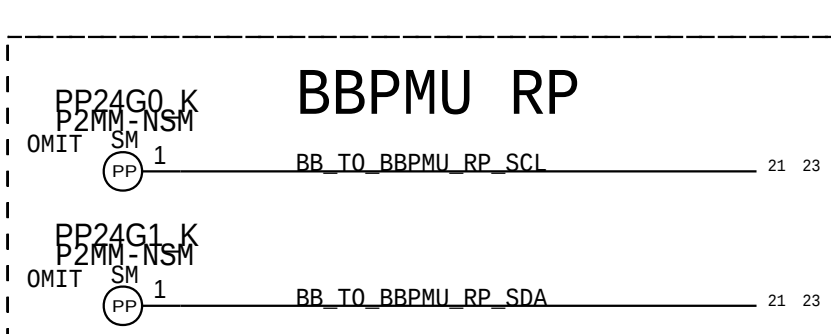
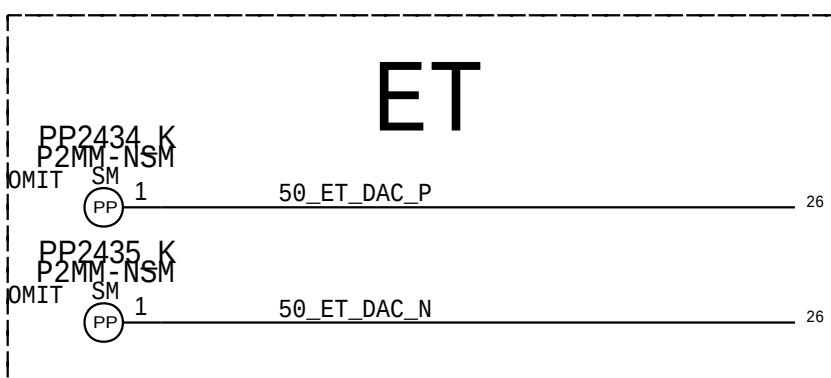
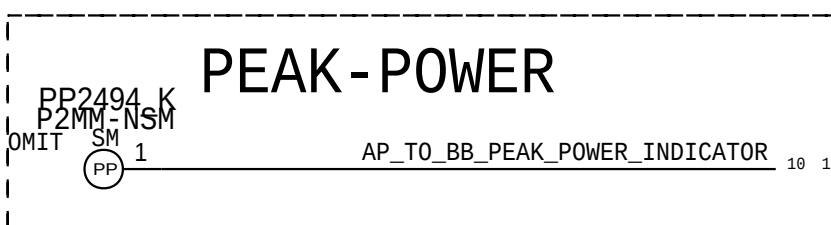
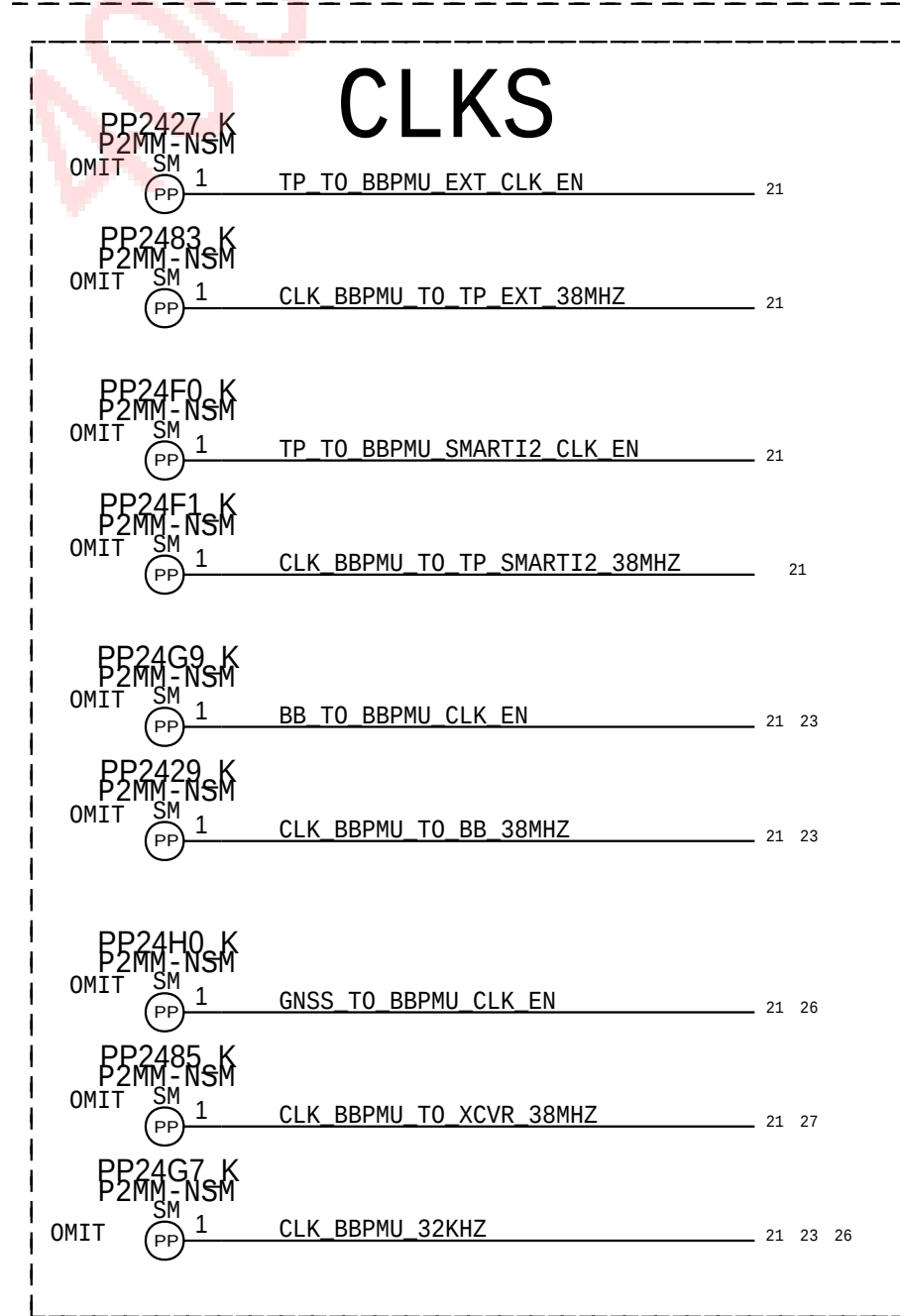
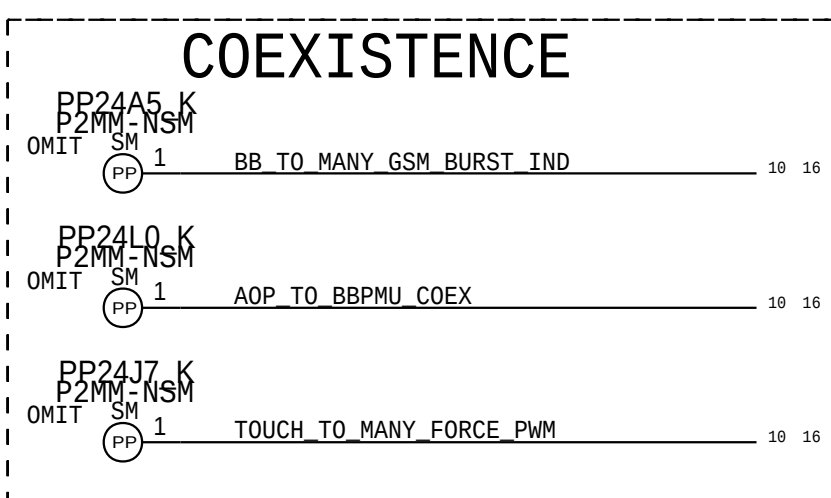
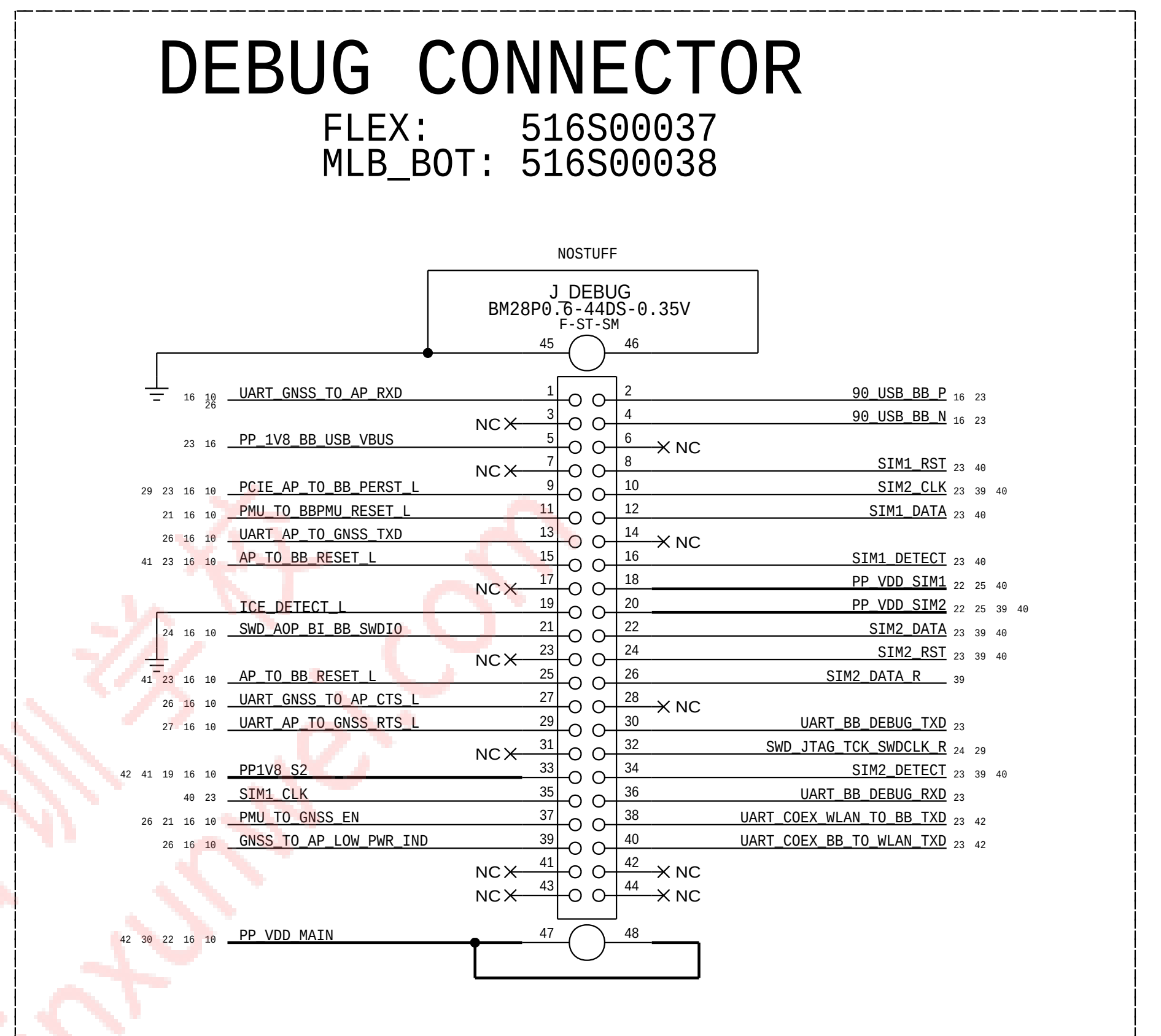
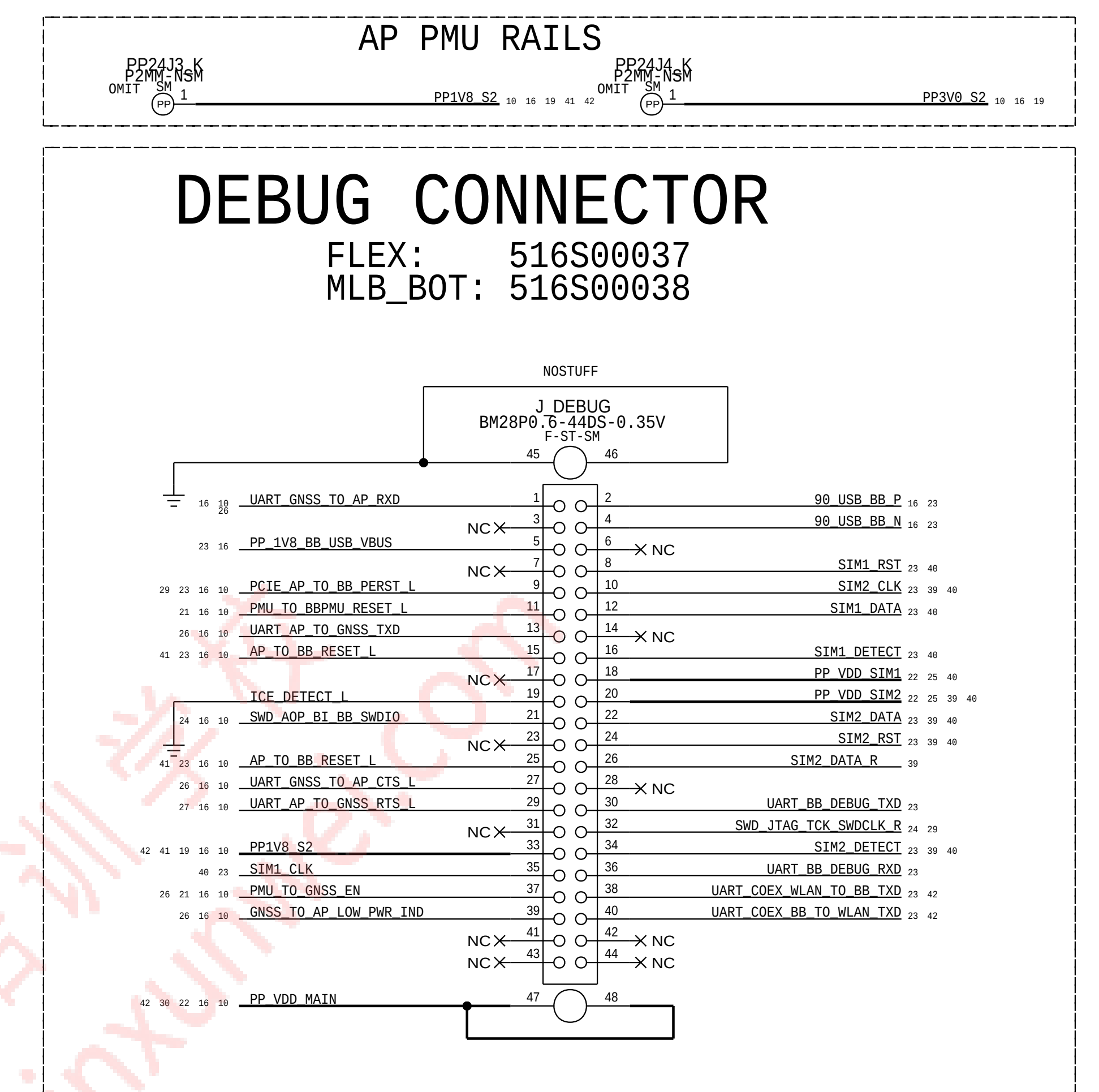
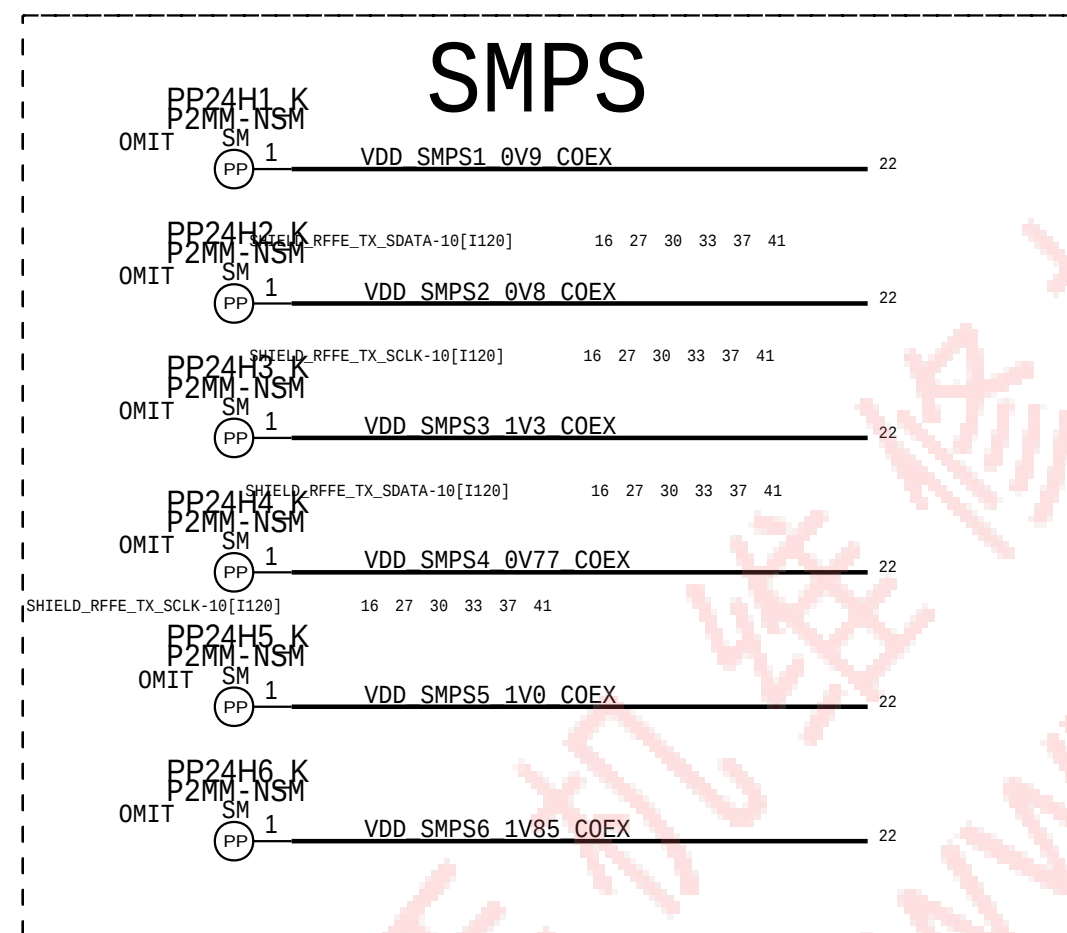
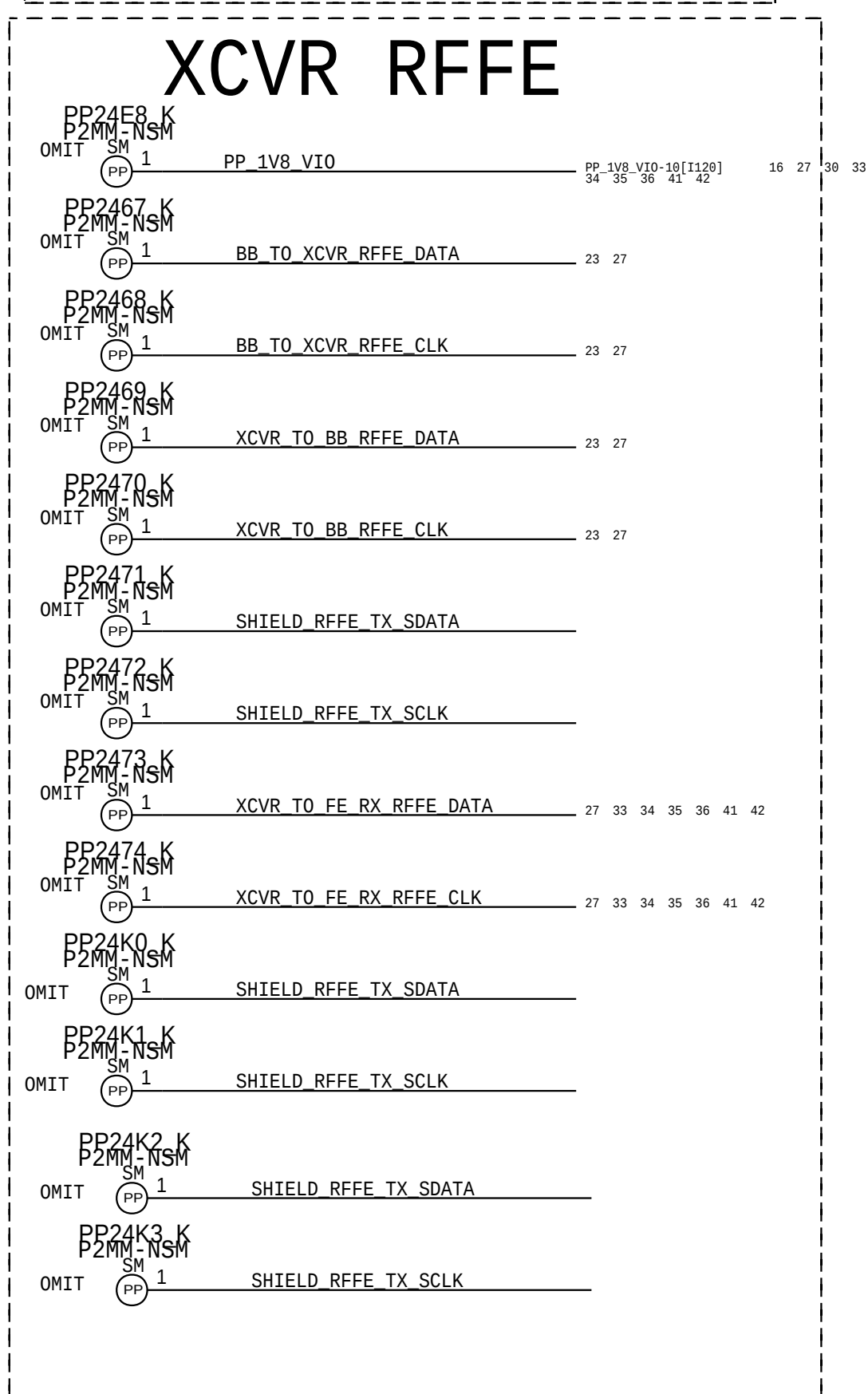
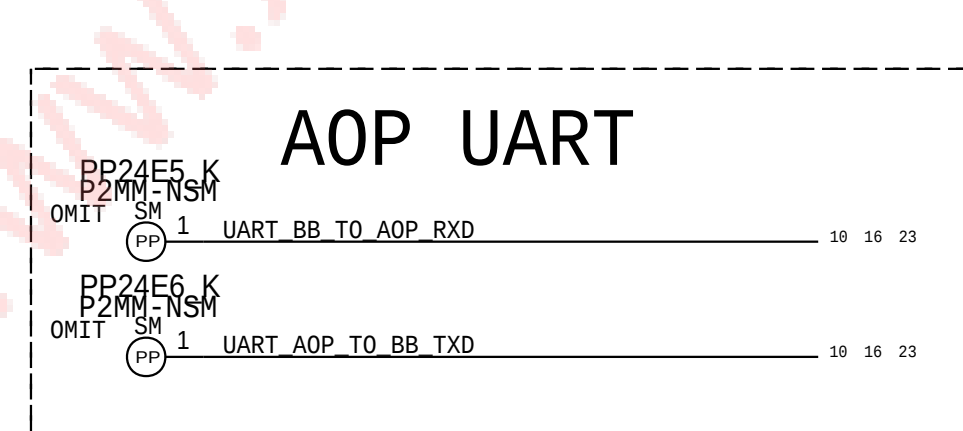
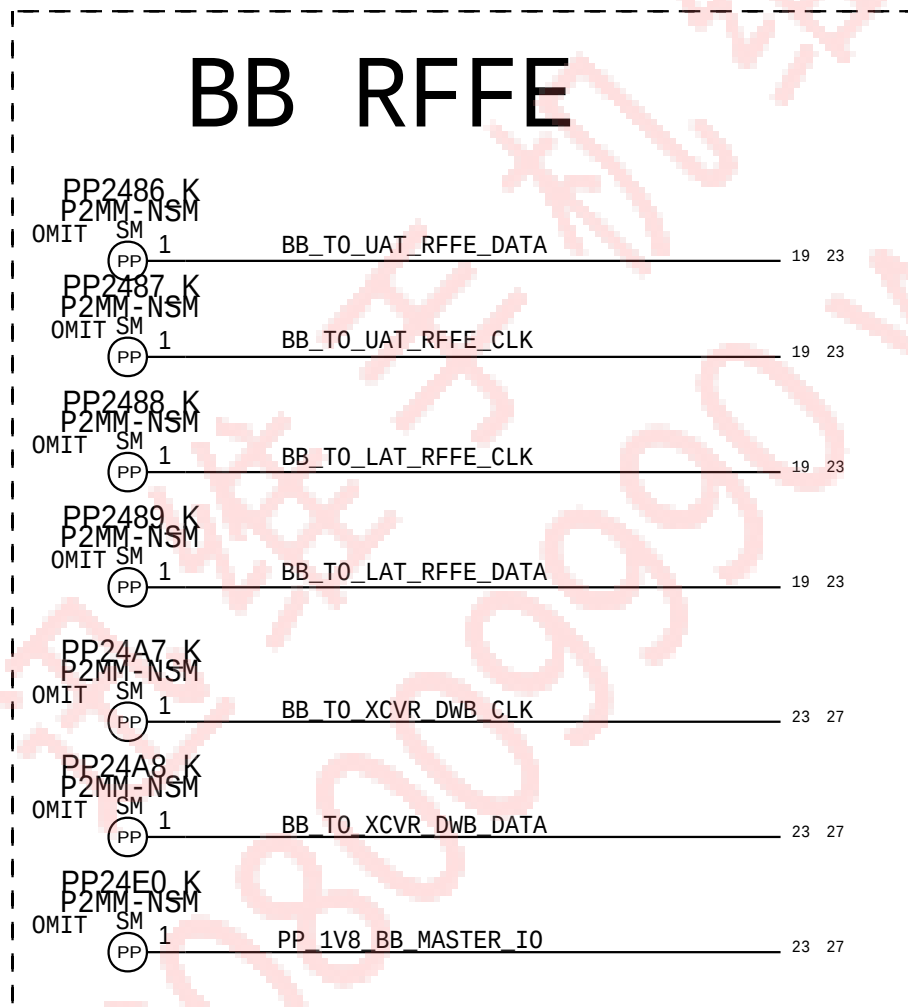
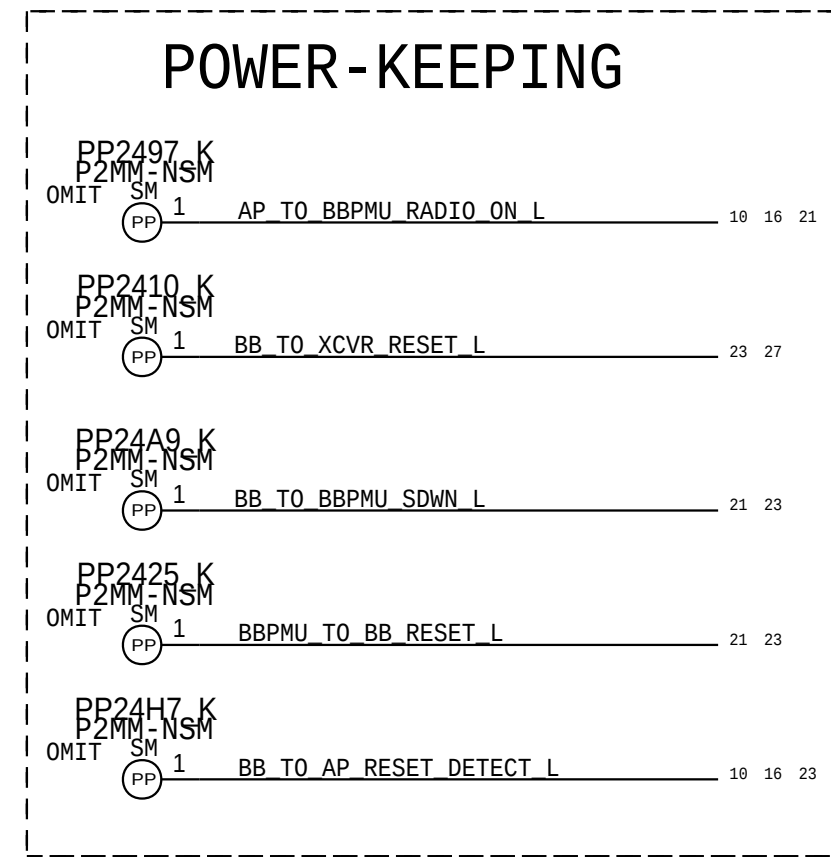
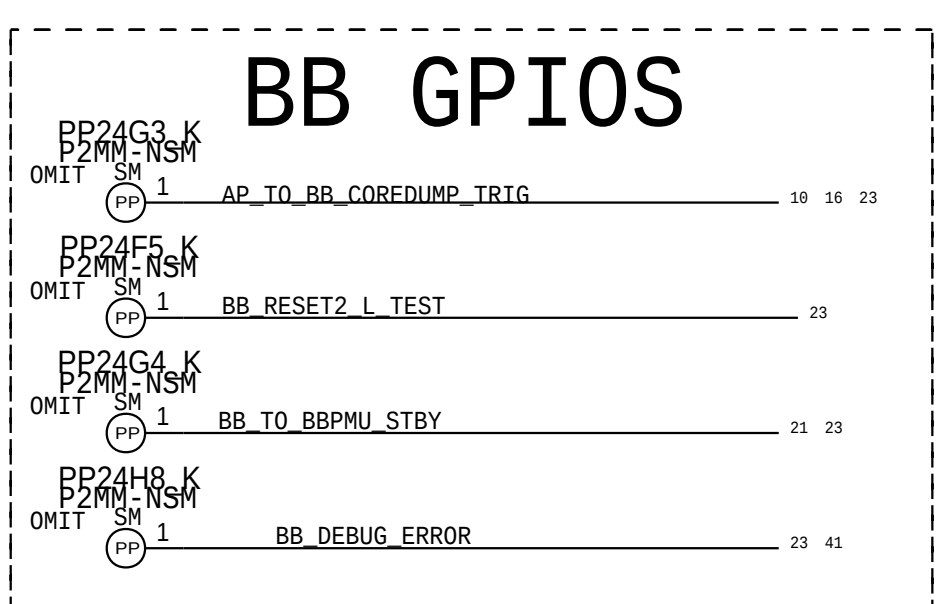
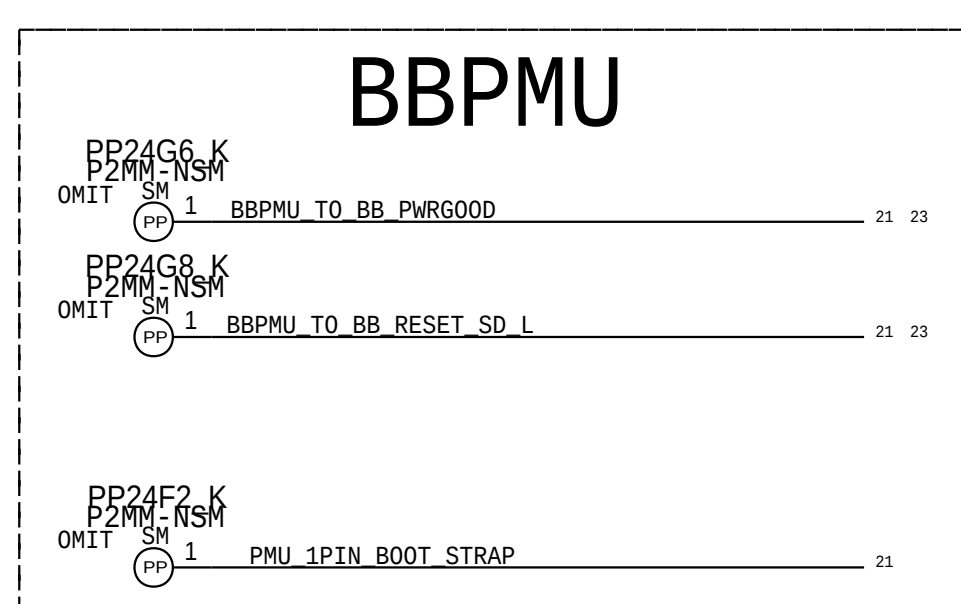
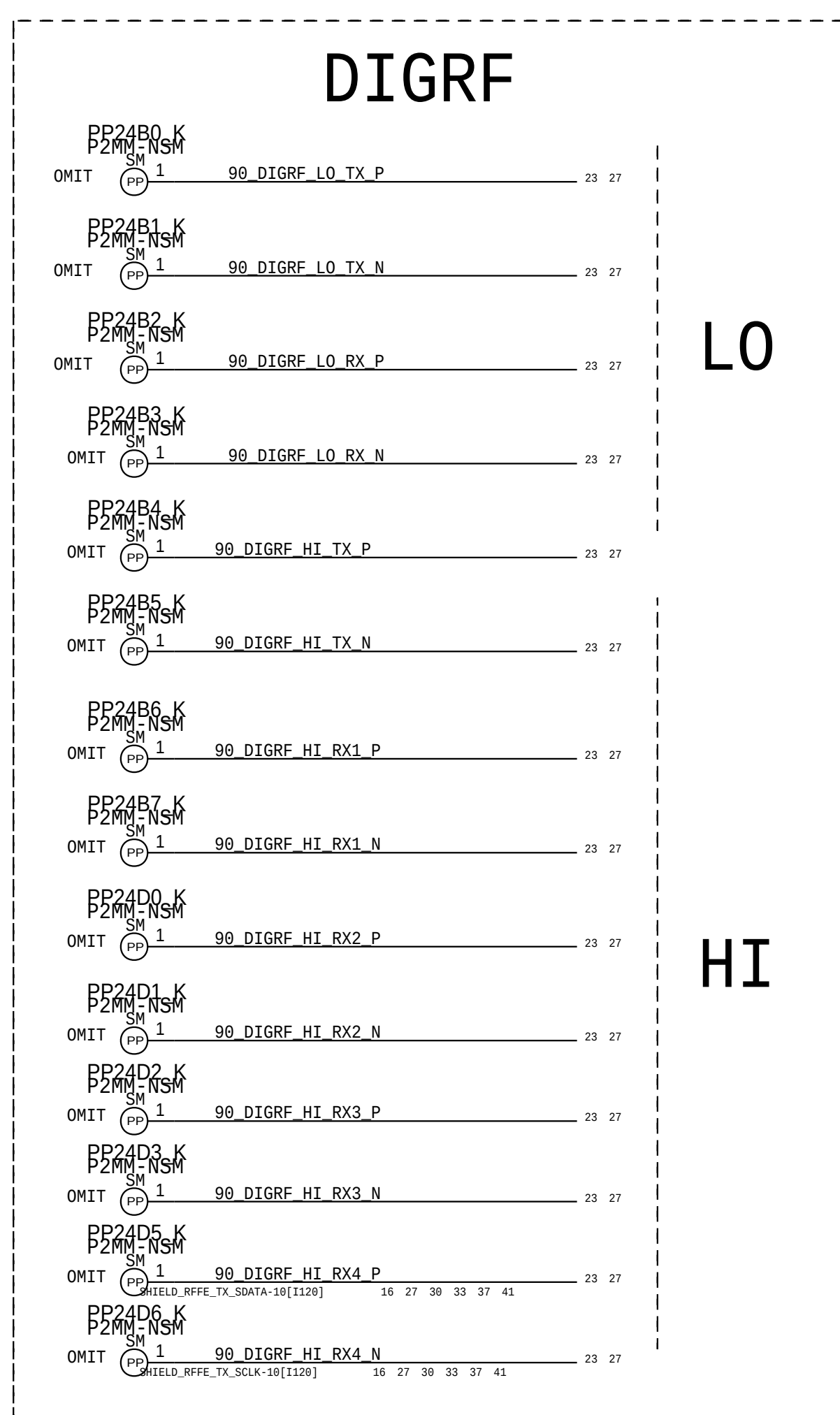
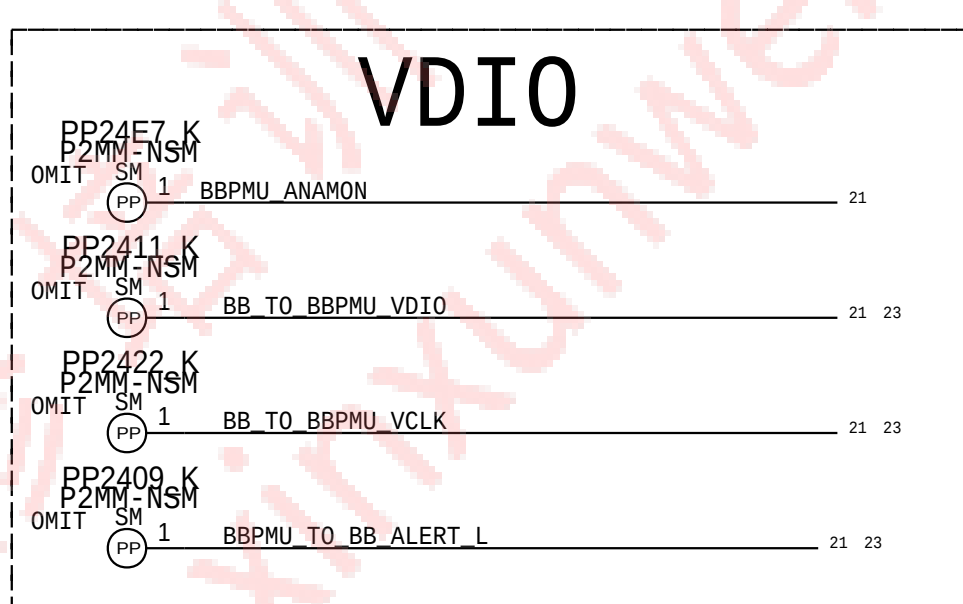
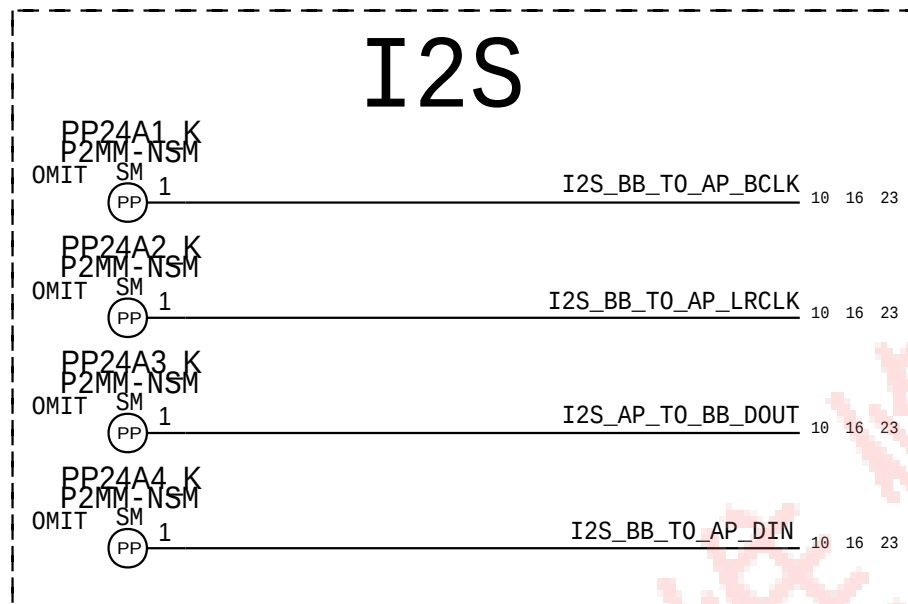
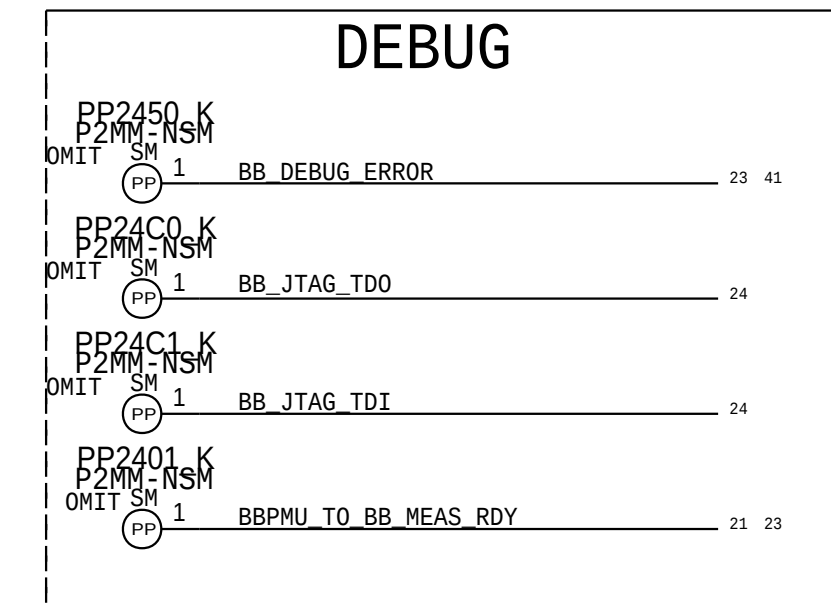
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41	40	25	PP_VDD_SIM2 MAKE_BASE=TRUE	==	PP_VDD_SIM2	39
41	40	25	SIM2_DATA_R MAKE_BASE=TRUE	==	SIM2_DATA_R	39



SIM: ESIM		
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	BRANCH	
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BRANCH			
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DEBUG: TEST POINTS



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BRANCH		
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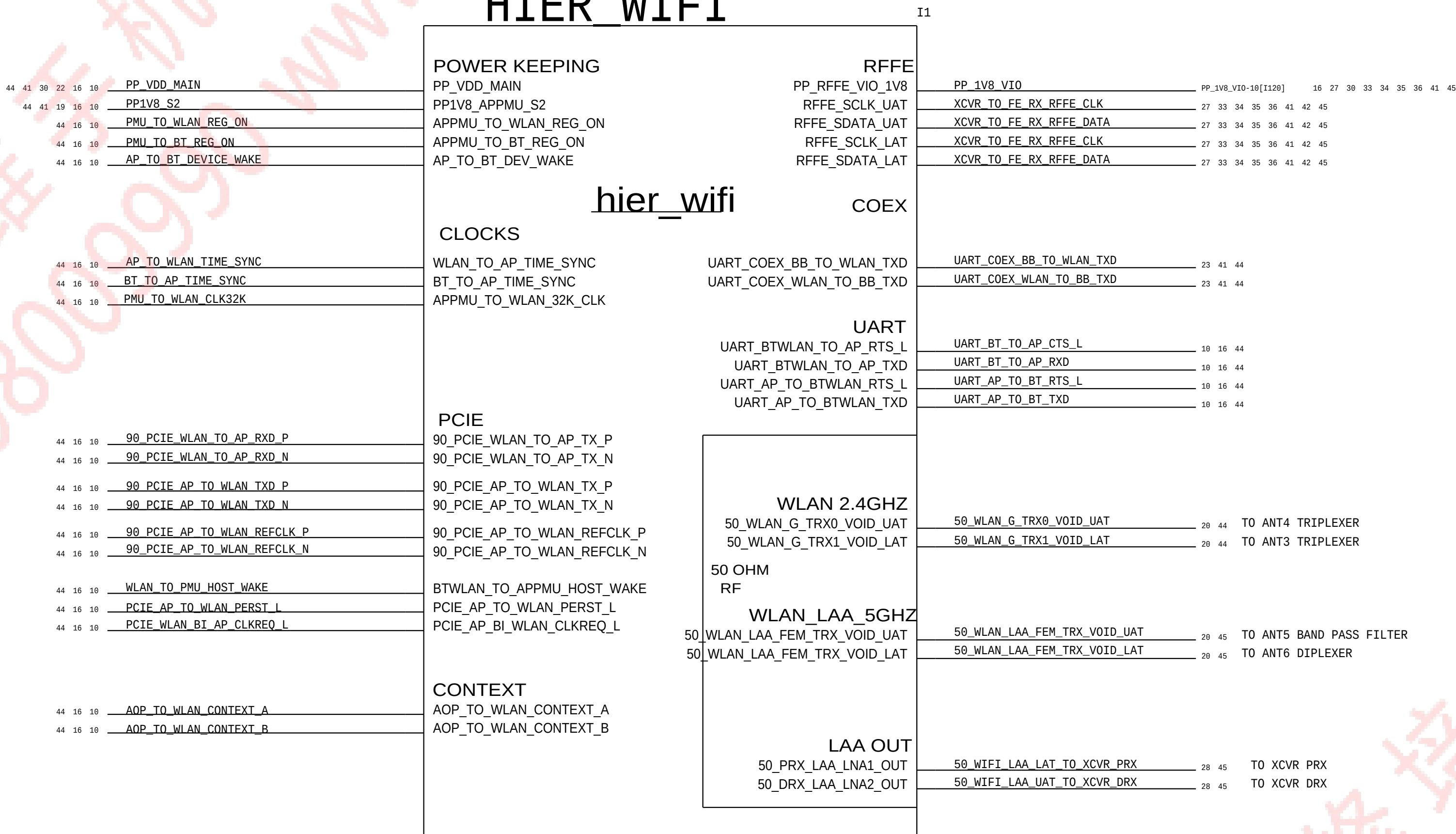


SYMBOL: WIFI

HIERARCHICAL SYMBOL FOR HIER_WIFI

DIETCOKE

HIER WIFI



PAGE TITLE		
SYMBOL: WIFI		
	DRAWING NUMBER	051-02695
	REVISION	4.0.0
	BRANCH	
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U_WLAN_W ALTERNATES

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998-14258	998-14229	BOM_TABLE_ALTS	U_WLAN_W	WIFI/BT MODULE

DRAWING TITLE

SCH, MLB, BOT, ICE, D33

DRAWING NUMBER

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SIZE

D

REVISION

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3

2

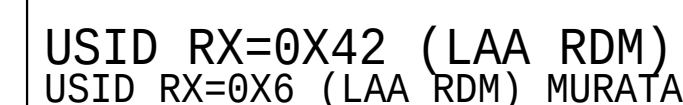
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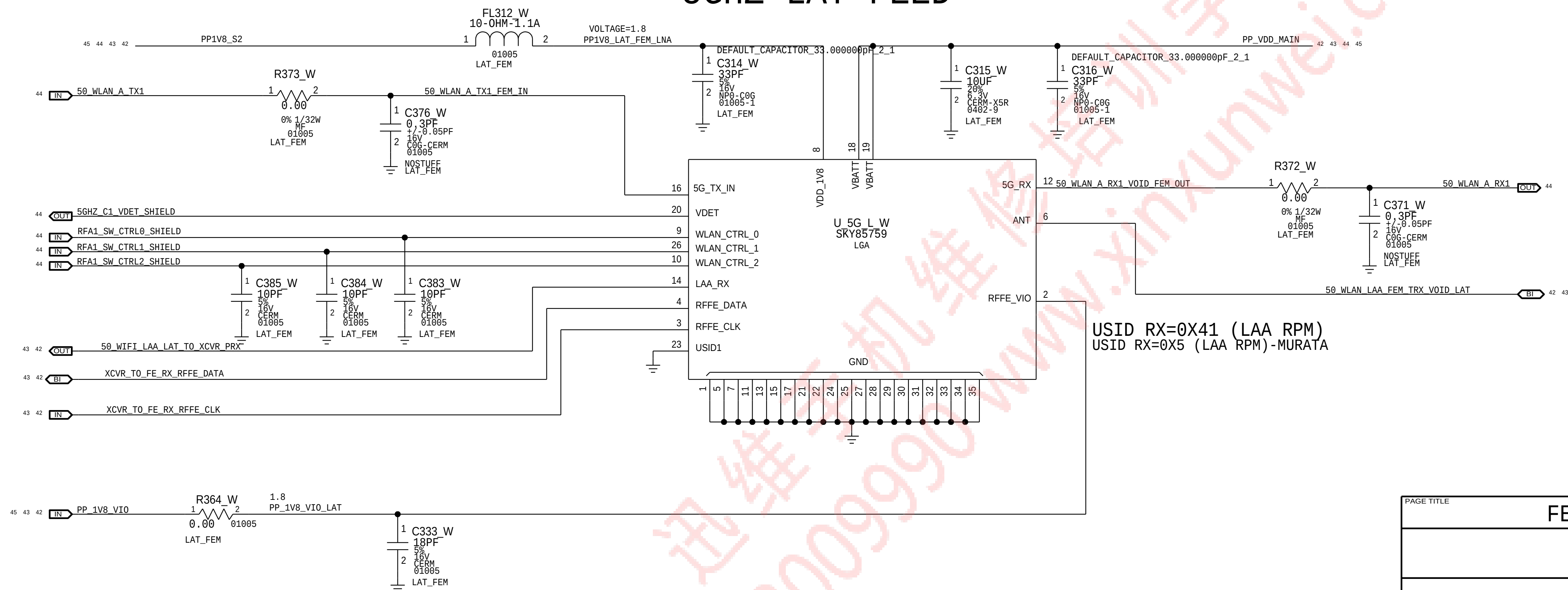
		SHEET 44 OF 47
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5GHZ UAT FEED



RFFE CLK AND DATA FILTERING CAPS ADDED TO UAT FEM ONLY (FURTHER FROM XCVR)

5GHZ LAT FEED



USID RX=0X41 (LAA RPM)
USID RX=0X5 (LAA RPM)-MURATA

PAGE TITLE		FEM MODULES	
	DRAWING NUMBER 051-02695		SIZE D
	REVISION 4.0.0		
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PART NUMBER

ALTERNATE FOR PART NUMBER

BOM OPTION

REF DES

COMMENTS:

197S00076

197S00060

BOM_TABLE_ALTS

Y7500_S

XTAL, 27P12 MHZ

197S00077

197S00060

BOM_TABLE_ALTS

Y7500_S

XTAL, 27P12 MHZ

DRAWING TITLE

SCH,MLB,BOT,ICE,D33

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D

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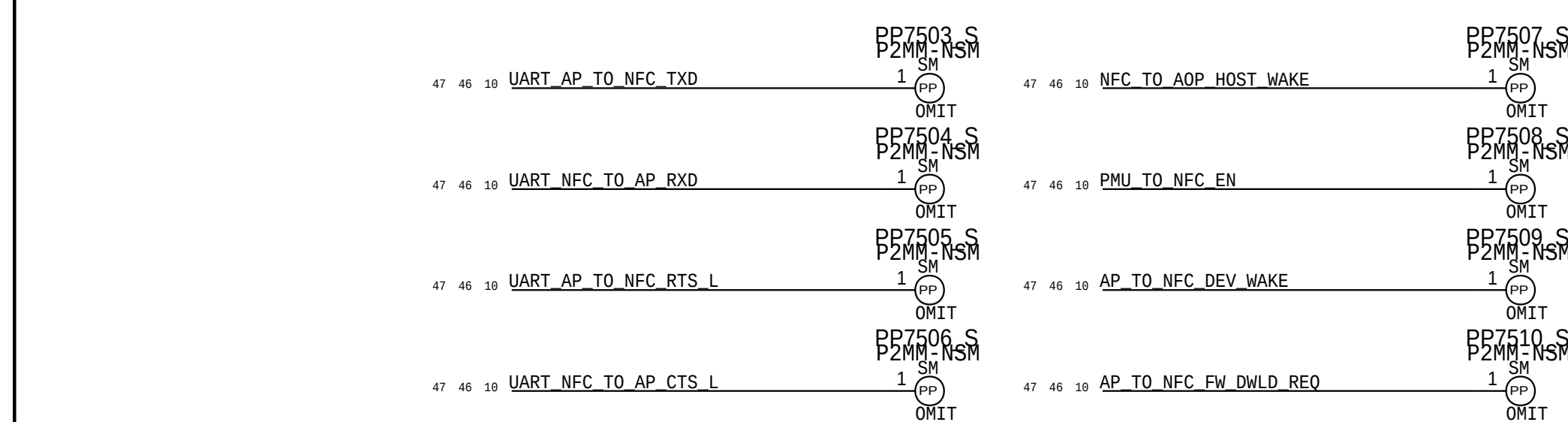
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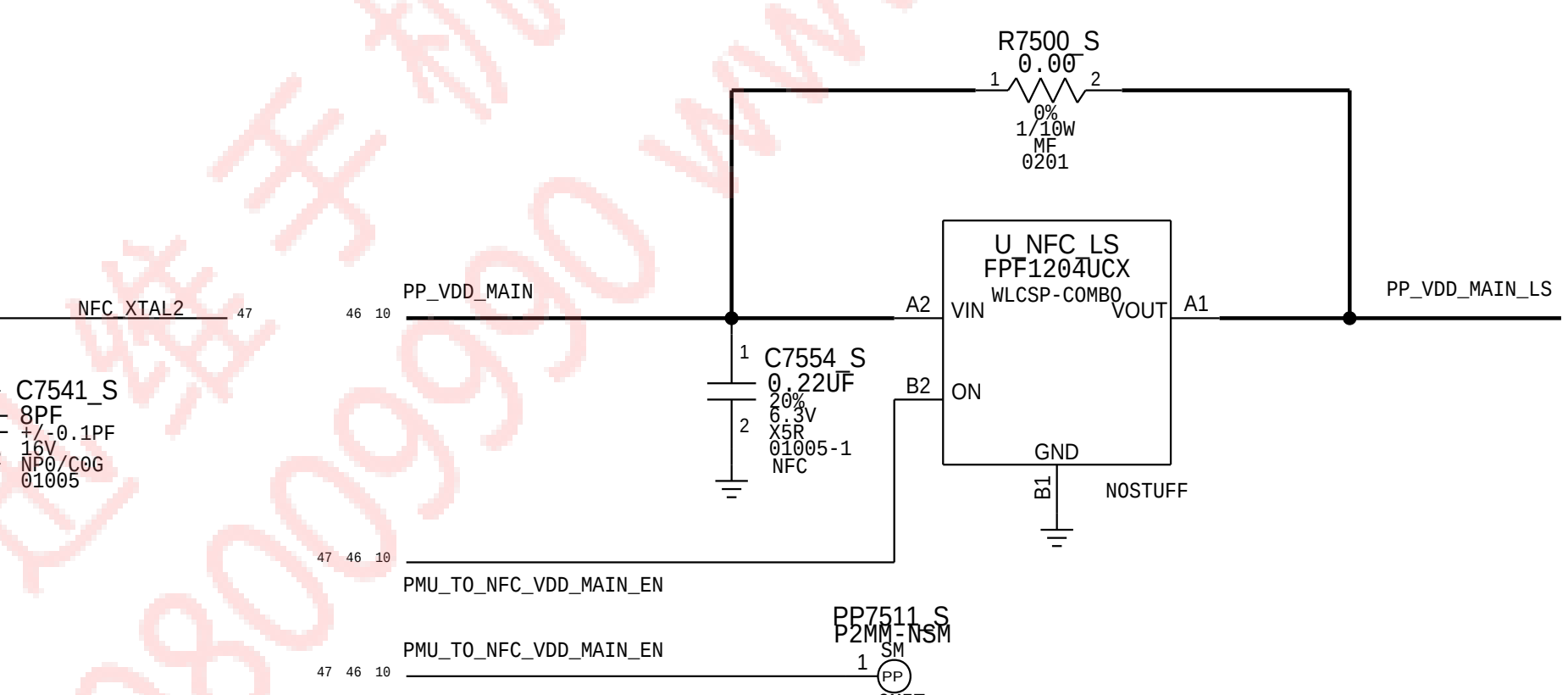


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NFC CONTROLLER



NFC FRONT END



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